

NetPoint® User Guide

NetPoint version 5.3

For an interactive version please visit <https://confluence.pmatechnologies.com/netpoint-user-guide/latest>.

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1) Getting Started

Welcome to the NetPoint® User Guide with NetRisk™ Risk Analysis. NetPoint® is a powerful and easy-to-use project planning and scheduling application. NetRisk™ Risk Analysis is a state-of-the-art GPM-based risk modeling and risk analysis module that runs within NetPoint®. Combined, NetPoint® with NetRisk™ offers seamless scheduling and risk analysis using the same graphical and intuitive interface. NetRisk™ Risk Analysis is available as a supplementary module to NetPoint®. If you'd like to add a NetRisk™ Risk Analysis license to NetPoint, please contact NetPoint® sales at <http://pma.sourcedigitalmarketing.com/contact>.

NetPoint® with NetRisk™ Risk Analysis makes use of the Graphical Path Method® to introduce a number of innovative concepts to the practice of scheduling and risk analysis. To ensure that all capabilities are fully understood and utilized, we encourage you to use this guide as a reference, to browse through online resources, and to consider signing up for training services. For more information about training, please visit: <http://pma.sourcedigitalmarketing.com/services>.

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About this Text

Within this guide, you'll find *steps* and *notes*. *Steps* are numbered instructions at the beginning of each chapter or section that indicate how to carry out a particular function. *Notes* serve to summarize the windows and fields, and they contain screenshots and important notes that expand upon the steps.

Certain categories of words in this user guide are color-coded to distinguish them from the rest of the text. **Blue** is used for the names of icons, commands, and buttons, such as **OK**. **Green** is used for menu paths, like **View > Toolbar > Dock to Top**. **Gray** is used for the names of windows, tabs, and text labels, like the **Codes Manager** or the **Properties** tab.

When additional or related information is available, a [*↗link*](#) will appear italicized and in blue. Clicking the link will jump you directly to that section.

NOTE: The features covered in this manual correspond with version 5.3.0.33. Please check the version and release date of your software by clicking **Help > About NetPoint...** from the menu bar, or contact **NetPoint support** if you need additional help or have other questions.

System Requirements

A successful collaborative planning session requires that participants remain engaged and focused. Using NetPoint on a well-equipped computer to drive the planning session will help to avoid costly delays in information processing, and will help maintain a smooth tempo of plan development and flow of information.

Installing/Updating

To install or upgrade NetPoint, administrator rights are required. Within 30 days of installation, license activation is required to grant continued access to NetPoint. If you do not have administrator rights, please contact your IT Department or system administrator. Once installed, administrator rights are NOT needed to run the NetPoint software. For more info, see [➔Installing/Upgrading NetPoint](#).

Operating System

NetPoint is compatible with Windows XP, Windows Vista, Windows 7, Windows 8, and Windows 10, including 64-bit versions.

Hard Disk

To install NetPoint, a minimum of 200MB of free storage space is recommended.

Virtual Memory

NetPoint requires a minimum of 1GB of RAM, but 4GB of RAM is suggested for better performance.

- Up to 250 total objects: at least 1GB
- 250 to 500 total objects: at least 2GB
- Over 500 total objects: at least 4 GB

Supported Schedule Size

Supported schedule size depends on the number of total objects (activities, links, milestones, etc.), the computer's processor/clock speed, and how much RAM it has. With a well-equipped machine, a delay of about 0.5 seconds can be expected for a schedule between 500-1,000 total objects, which is considered to be a tolerable upper limit. You can continue to work with a schedule until about 2,500 total objects with a delay of 1 second.

Tip: When working with larger schedules, we recommended hiding the resource histogram, as displaying it can add processing time.

NetRisk

With NetRisk, a delay of about 1 second can be expected in the **Schedule** view of the **NetRisk Manager** for a schedule with 500 total objects, which is considered to be a tolerable upper limit. A delay of 2 seconds can be expected for saving a schedule with a simulation of 1,000 iterations for 100 activities, for each simulation saved. For this reason, the number of saved simulations should be monitored on larger schedules.

Monitor & Projector Resolution

NetPoint requires a minimum screen resolution of 1024×768 and a 16-bit color setting. Higher resolution and a 32-bit color setting result in a better graphic image.

NOTE: A lower resolution may result in the inability to reach the [OK](#) and [Cancel](#) buttons in some dialog windows.

Concurrent License Monitoring

Concurrent license access is monitored by Sentinel HASP®, a third party product by SafeNet®, which is installed on your server and runs a service that monitors HASP licensing on your Network. In addition, each NetPoint workstation must be configured to the license server as described in the *NetPoint Workstation Activation & Workstation Configuration to License Server* document.

3rd Party Applications

For import/export, any necessary privileges will be set automatically upon installation.

Application	Supported Versions
Primavera P6	6.0 and higher
Microsoft Project	2007, 2010
Microsoft Excel	2010, 2013
Microsoft Outlook	2007, 2010, 2013

NOTE: To benefit from Microsoft Outlook integration, NetPoint requires 32-bit installation. To check in Outlook 2010, click “File > Help”, and then look for the bit number in the right-most sections under “About Microsoft Outlook” after the “Version”. To check in Outlook 2013, click “File > Account”, click the big “About Outlook” icon, and then look for the bit number on the first line at the very end.

NOTE: To benefit from Microsoft Excel integration, NetPoint requires 2010 or higher.

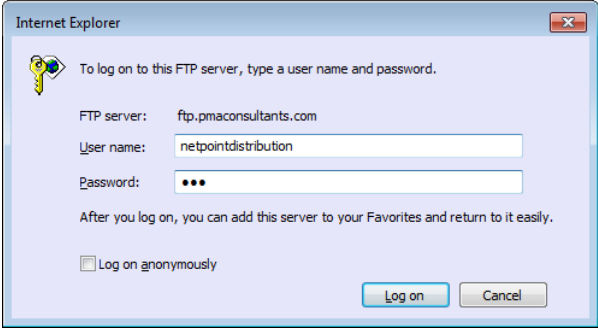
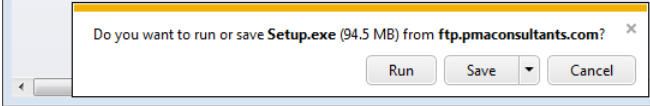
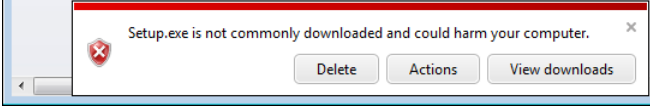
Installing/Upgrading NetPoint

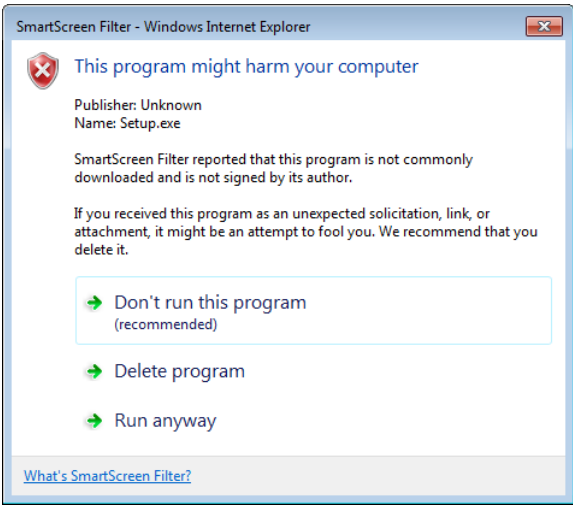
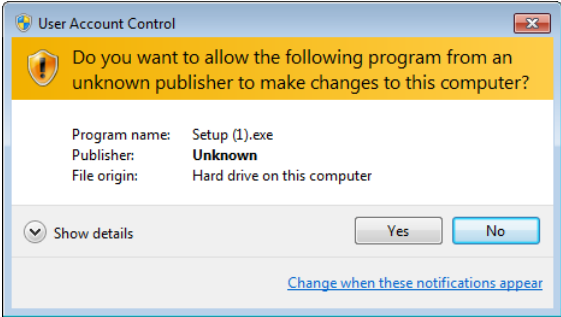
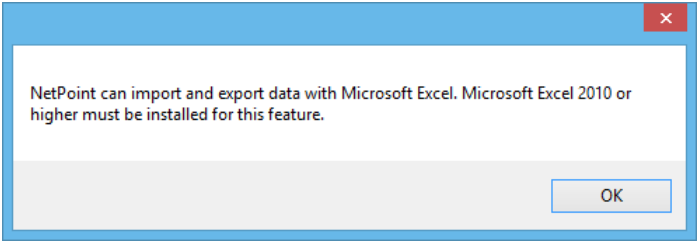
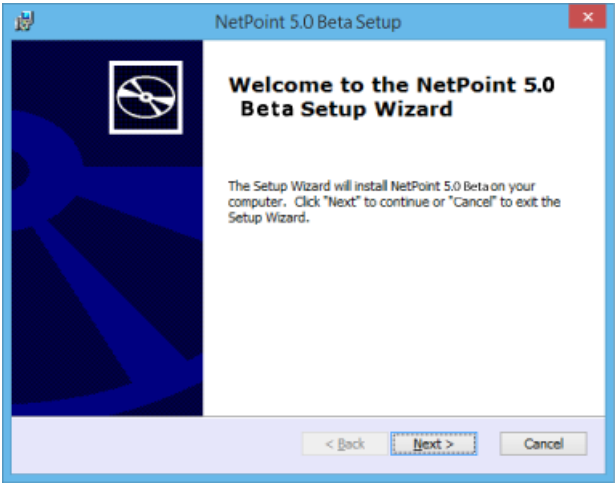
To install or upgrade NetPoint, administrator rights are required. If you do not have administrator rights, either an administrator must login to perform the installation or you must be granted administrator rights on your local machine. If this is the case, please contact your IT department or system administrator.

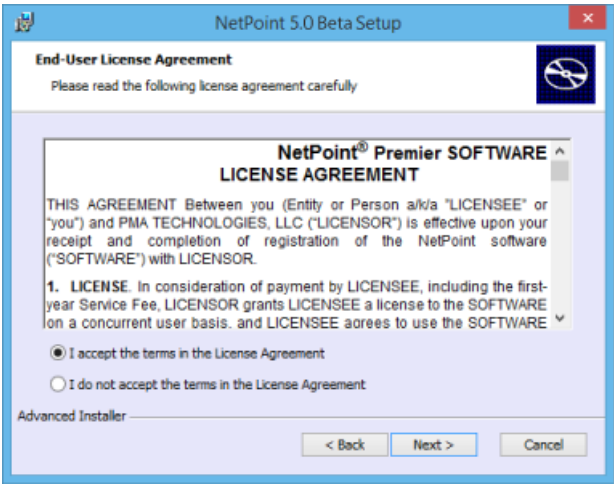
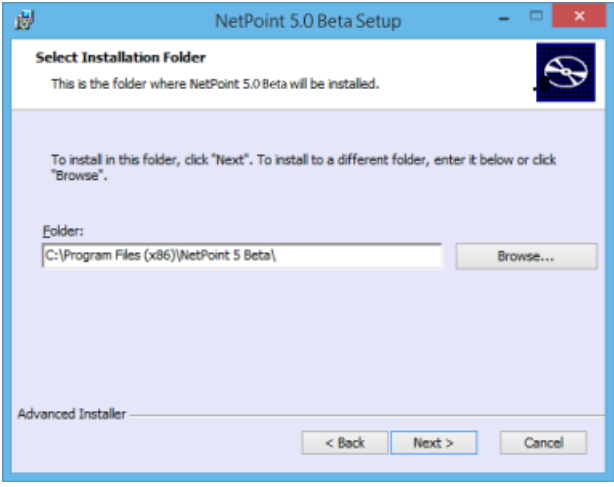
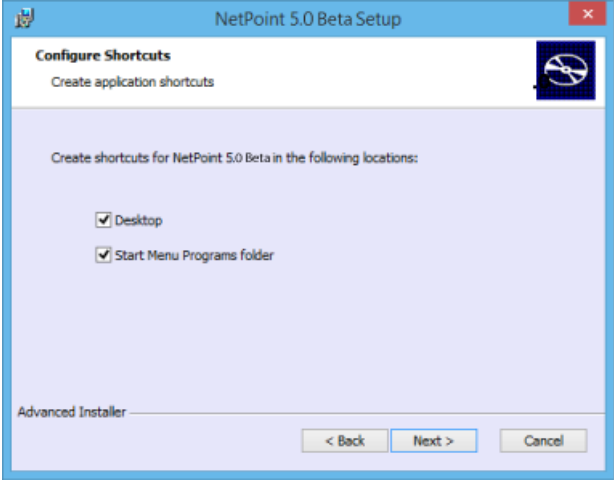
Once installed, administrator rights are NOT needed to run the software. For import/export, any necessary privileges will be set automatically upon installation. To check what version you're running, go to **Help > About NetPoint...** from the menu bar and check the release date at the top.

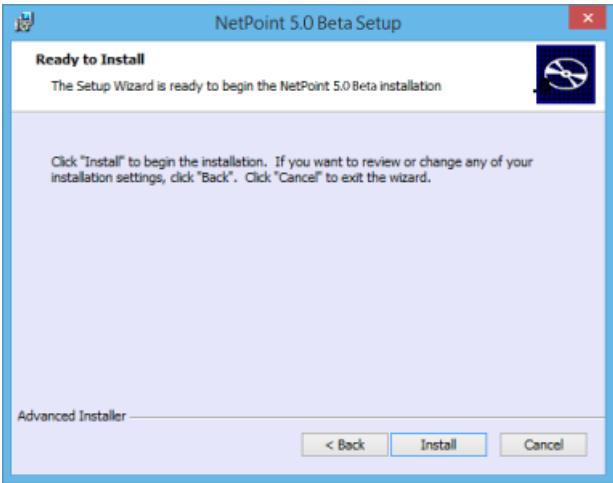
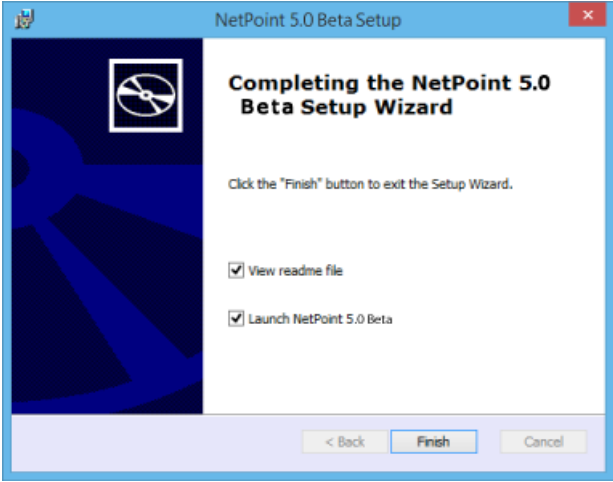
NOTE: For NetPoint to launch properly, Windows must have a “My Documents” folder defined within the user directory, as this is where NetPoint will stores recovery files. Otherwise, NetPoint will not be able to successfully initialize.

Steps

1.	Click or paste the download link into your system's web browser. For installing or upgrading a licensed version, enter the credentials in the security window and click Log On. NOTE: The trial does not require credentials and should begin downloading automatically.	 The screenshot shows an Internet Explorer security dialog box titled "Internet Explorer". It contains a key icon and the text: "To log on to this FTP server, type a user name and password." Below this, it lists "FTP server: ftp.pmaconsultants.com". There are input fields for "User name:" (containing "netpointdistribution") and "Password:" (masked with dots). Below the password field, it says "After you log on, you can add this server to your Favorites and return to it easily." There is a checkbox for "Log on anonymously" which is unchecked. At the bottom right are "Log on" and "Cancel" buttons.
2.	In the confirmation pop-up, click Run.	 The screenshot shows a Windows file download confirmation dialog box. It asks: "Do you want to run or save Setup.exe (94.5 MB) from ftp.pmaconsultants.com?". At the bottom are "Run", "Save", and "Cancel" buttons.
3.	When the download finishes, click Actions.	 The screenshot shows a Windows Security warning dialog box. It has a red shield icon with a white 'X' and the text: "Setup.exe is not commonly downloaded and could harm your computer." At the bottom are "Delete", "Actions", and "View downloads" buttons.

4.	When the filter window opens, click Run Anyway .	 SmartScreen Filter - Windows Internet Explorer This program might harm your computer Publisher: Unknown Name: Setup.exe SmartScreen Filter reported that this program is not commonly downloaded and is not signed by its author. If you received this program as an unexpected solicitation, link, or attachment, it might be an attempt to fool you. We recommend that you delete it. ☐ Don't run this program (recommended) ☐ Delete program ☐ Run anyway What's SmartScreen Filter?
5.	If the User Account Control window opens, click Yes .	 User Account Control Do you want to allow the following program from an unknown publisher to make changes to this computer? Program name: Setup (1).exe Publisher: Unknown File origin: Hard drive on this computer ☒ Show details Yes No Change when these notifications appear
6.	NetPoint setup will begin. A series of notifications will appear. Click OK for each one.	 NetPoint can import and export data with Microsoft Excel. Microsoft Excel 2010 or higher must be installed for this feature. OK
7.	In the NetPoint Setup window, click Next .	 NetPoint 5.0 Beta Setup Welcome to the NetPoint 5.0 Beta Setup Wizard The Setup Wizard will install NetPoint 5.0 Beta on your computer. Click "Next" to continue or "Cancel" to exit the Setup Wizard. Back <input checked="" type="button" value="Next"/> Cancel

8.	Check the option next to I accept the terms in the License Agreement and click Next.	 The screenshot shows the 'End-User License Agreement' window of the NetPoint 5.0 Beta Setup. The window title is 'NetPoint 5.0 Beta Setup'. The main heading is 'End-User License Agreement' with a subtext 'Please read the following license agreement carefully'. Below this is a scrollable text area containing the 'NetPoint® Premier SOFTWARE LICENSE AGREEMENT'. The agreement text includes: 'THIS AGREEMENT Between you (Entity or Person a/k/a "LICENSEE" or "you") and PMA TECHNOLOGIES, LLC ("LICENSOR") is effective upon your receipt and completion of registration of the NetPoint software ("SOFTWARE") with LICENSOR. 1. LICENSE. In consideration of payment by LICENSEE, including the first-year Service Fee, LICENSOR grants LICENSEE a license to the SOFTWARE on a concurrent user basis. and LICENSEE agrees to use the SOFTWARE'. At the bottom, there are two radio buttons: 'I accept the terms in the License Agreement' (which is selected) and 'I do not accept the terms in the License Agreement'. Below the radio buttons is the text 'Advanced Installer'. At the bottom right are three buttons: '< Back', 'Next >', and 'Cancel'.
9.	Leave the location as default or browse for a new one, and then click Next.	 The screenshot shows the 'Select Installation Folder' window of the NetPoint 5.0 Beta Setup. The window title is 'NetPoint 5.0 Beta Setup'. The main heading is 'Select Installation Folder' with a subtext 'This is the folder where NetPoint 5.0 Beta will be installed.' Below this is a text area with instructions: 'To install in this folder, click "Next". To install to a different folder, enter it below or click "Browse".' There is a text box labeled 'Folder:' containing the default path 'C:\Program Files (x86)\NetPoint 5 Beta\'. To the right of the text box is a 'Browse...' button. At the bottom, there is the text 'Advanced Installer' and three buttons: '< Back', 'Next >', and 'Cancel'.
10.	Configure the shortcuts or leave them as default, then click Next.	 The screenshot shows the 'Configure Shortcuts' window of the NetPoint 5.0 Beta Setup. The window title is 'NetPoint 5.0 Beta Setup'. The main heading is 'Configure Shortcuts' with a subtext 'Create application shortcuts'. Below this is a text area with instructions: 'Create shortcuts for NetPoint 5.0 Beta in the following locations:'. There are two checkboxes: 'Desktop' (which is checked) and 'Start Menu Programs folder' (which is also checked). At the bottom, there is the text 'Advanced Installer' and three buttons: '< Back', 'Next >', and 'Cancel'.

11.	Begin the installation by clicking Install .	 The screenshot shows the 'NetPoint 5.0 Beta Setup' window. The title bar says 'NetPoint 5.0 Beta Setup'. The main area has a heading 'Ready to Install' and a sub-heading 'The Setup Wizard is ready to begin the NetPoint 5.0 Beta installation'. Below this, it says: 'Click "Install" to begin the installation. If you want to review or change any of your installation settings, click "Back". Click "Cancel" to exit the wizard.' At the bottom, there are three buttons: '< Back', 'Install', and 'Cancel'.
12.	During installation, NetPoint will open momentarily for entering some system defaults. Enter your name and email and click OK .	 The screenshot shows the 'Set System Defaults' dialog box. It has a title bar 'Set System Defaults'. Inside, there's a section 'Default Properties' with the text: 'Please select your country and base time unit. All new activities will be assigned a calendar based on the workweek and holidays of the selected country.' Below this, there are two dropdown menus: 'Country:' set to 'United States' and 'Time Unit:' set to 'Days'. Further down, there's a paragraph: 'Each copy of NetPoint is associated with a name and email address. This information is local to the machine, and it is used to identify the sender when generating email notifications from within NetPoint.' Below this paragraph are two text input fields labeled 'Name:' and 'Email:'. At the bottom right is an 'OK' button.
14.	Check the option next to Launch NetPoint and click Finish .	 The screenshot shows the 'NetPoint 5.0 Beta Setup' window. The title bar says 'NetPoint 5.0 Beta Setup'. The main area has a heading 'Completing the NetPoint 5.0 Beta Setup Wizard'. Below this, it says: 'Click the "Finish" button to exit the Setup Wizard.' There are two checked checkboxes: 'View readme file' and 'Launch NetPoint 5.0 Beta'. At the bottom, there are three buttons: '< Back', 'Finish', and 'Cancel'.

Notes

Choosing a Default Calendar

By default, all activities and milestones in NetPoint are assigned a calendar when they are created. Click the down arrow  button and choose a country to use as the basis for the workweek and holidays for the default

calendar. From within NetPoint, custom calendars can be created with custom workweeks and holidays, and any of these can be set as default for the schedule. See [➔Project Calendars](#) and [➔Setting Default Properties](#) for more details.

Choosing a Default Time Unit

If your version of NetPoint includes PolyTime, you will also be able to set a default time unit for new schedules. Click the down arrow  button and choose a time unit to apply to all objects in the schedule. Unlike calendars, the time unit cannot differ from activity to activity. From within NetPoint, a different time unit may be chosen at any time during planning. See [➔PolyTime](#) for more details.

Entering Default User Information

NetPoint has the capability to send email notifications. Enter the name and email address that will be used to identify the sender. From within NetPoint, a different name and/or email may be used each time an email notification is generated. See [➔Resource Notifications](#) for more details.

Licensing

NetPoint uses Sentinel HASP®, a third-party product developed by SafeNet®, to enforce software protection and concurrent licensing.

In the case where a company has opted for concurrent licensing, the HASP utility it is installed on a local server associated with the company's network to allow end-user IT administrators to monitor and manager the licenses. For example, they can see which licenses are in-use and by whom or detach a license from the network for off-line use.

This chapter details instructions for the following:

1. Activating your license
2. Concurrent licensing

Activating Your License

If your license is only used on one computer, please follow the steps below. If your license is shared between multiple users and you are an IT administrator, please visit [➔ Concurrent Licensing](#); otherwise, contact your IT department for any assistance with activation or troubleshooting.

If your license has already been activated and now you're getting an error, it may be due to one of the following reasons:

You've reinstalled, upgraded, or made changes to your operating system

You've migrated to a new computer

Your organization has reinstalled, upgraded, or made changes to their licensing server*

*Please contact your IT department for any assistance with activation or troubleshooting

Steps

Collecting Information

1. Click the Windows Start Screen icon (for Windows 7 and earlier, click the **Start** icon from the Windows taskbar).
2. Choose **All Apps** (for Windows 7 and earlier, choose **All Programs**).
3. Choose the **NetPoint 5** folder.
4. Choose **NetPoint Activation**. The **HASP** window will open.
5. Click the **Collect Information** button. An Explorer window will open.
6. Name the file and click **Save**.
7. Navigate to the file in Windows Explorer.
8. Zip the file by right-clicking on the file and choosing **Send to > Compressed (zipped) folder**.
9. Send the zip file using your email client to support@pma.sourcedigitalmarketing.com

NOTE: Once NetPoint support receives your zip file, they will email an updated activation file back to you.

Applying the License

1. Save the file to your computer
2. Navigate to the file in Windows Explorer.
3. Unzip the file by right-clicking on the file and choosing **Extract All**.
4. Follow steps 1-4 under *Collecting Information* above.
5. Click the **Apply License Update** tab.
6. Click the **...** button.
7. Browse for the unzipped file and click **Open**.
8. Click **Apply Update**.

Concurrent Licensing (Server-based)

NOTE: The following page is for IT administrators. If you are not an IT administrator, please contact your IT administrator to perform any of the following steps:

A concurrent user license grants access to the number of users up to the number of licenses at any given time. This allows organizations to purchase software licenses based on maximum users at one time instead based on the total number of users. For example, if a company anticipate 50 users, but 40 of which work during the day and 10 at night, the company can opt to purchase only 40 concurrent user licenses rather than a single “named-seat” license for each of the 50 individual users. This is because 40 is the maximum number of users that will need to use the software at once.

Once setup, the software can be installed on as many computers as needed. When NetPoint is launched from one of these locally installed machines, a request for a license will be made to the NetPoint license server. If the number of licensees in use at that point is less than the number of those granted to the company, NetPoint will launch. If not, the user will be prompted to try again later to check again if a license is available.

Steps

Activating

1. Open a web browser on the local machine you wish to activate and type `http://localhost:1947`.
2. Under the **Options** on the left hand side, click **Configuration**.
3. Choose the **Access to Remote License Managers** tab.
4. Make sure the box is checked next to **Allow Access to Remote Licenses**, **Broadcast Search for Remote Licenses**, and **Aggressive Search for Remote Licenses**.
5. Enter the name/address of the licence server next to **Specify Search Parameters**.
6. Click the **Submit** button.
7. Under **Options** on the left hand side, click **Sentinel Keys**.

NOTE: Discovery of the license server may take a few minutes.

Detaching

1. Open a web browser on the local machine you wish to detach and type `http://localhost:1947`.
2. Make sure the license has been activated by following steps 2-7 above.
3. Once activated, choose the **Detachable Licenses** tab.
4. Check the box next to **Enable Detaching of Licenses**.
5. Click **Products** under **Options**.
6. Click the **Detach** button.
7. Select an expiration date.
8. Click the **Detach & Attach** button.

NOTE: The license will be restored after the number of days next to Max. Detatch Duration.

Automated Software Updates

NetPoint comes equipped with the ability to notify users of software updates and give them the option to install them. By default, NetPoint is configured to check for updates through a PMA-hosted server; however, you can instead choose to host your own server or “disable” the feature.

Hosting Options

PMA-hosted Option

The default option is ideal for individual licensees and for organizations that do not need to control when and how updates are administered to their end users. For example, if end-users can elect to update as soon as a new version is available (at their own discretion), organizations should not feel the need to establish their own update server.

Self-hosted Option

Self-hosting is ideal for organizations that prefer to control exactly when and how updates are administered to their end-users. For example, if an organization prefers to evaluate new versions prior to distributing them to their end users and/or wants to ensure that end-users can't ignore updates.

Disabling

To disable update notifications, please send us an [email](#).

The Update Process

The default application shortcut is configured to launch the NetPoint updater executable, which checks the remote server for newer versions of NetPoint. If one is available, the user is given the option to update or ignore. If accepted, NetPoint is updated as follows:

1. Recovery files are saved
2. The release notes are downloaded and opened
3. The new installation files are downloaded
4. The new installation files overwrite the old installation's files

Once the process is complete, a final message opens that confirms the successful update and prompts the user to click OK to open the application.

If the update fails for any reason (files could not be written or the internet connection is interrupted), a message opens that alerts the user that the update has failed and prompts them to run the update again with administrator privileges.

When self-hosting, the server administrator can then decide when to put the updated files onto their self-hosted server for final distribution to their end-users.

Software Update Eligibility

If a customer has not paid their annual software maintenance fees, they will not be eligible to receive software updates. In that case, a message will open that prompts them to contact NetPoint support, and they will be re-invoiced if necessary.

NOTE: To improve its products and services, PMA may collect data from end-users through the software update system, including but not limited to: license keys, email addresses, update statistics, and usage statistics. By using this system, end-users agree to these terms.

Project Folders

NetPoint® makes use of a project/folder-based structure, which organizes your schedule files (such as reports, templates, simulation data, recovery files, and iObjects) automatically together. Instead of opening and closing schedule *files*, you will be opening and closing project *folders*, which then load into NetPoint whichever schedule file they contain.

All *new* schedules must have a project folder created for them. This is done automatically when saving a schedule for the first time. All *legacy* schedules need to be imported and converted to projects before they can be opened. Once converted, however, schedules are opened and saved just like before, with only minor differences to the interface.

This section details instructions for the following:

1. Navigating in windows explorer
2. Converting files to folders
3. Opening and unzipping
4. Saving and renaming
5. Sharing and sending

Navigating in Windows Explorer

When working with projects, keep in mind that the contents of a project folder are not locked. However, it is not recommended to drag in or out any files that were placed there by NetPoint.

Project folders CAN be moved in and out of parent directories. However, we recommend keeping them all in a single location, so that browsing in NetPoint is less cumbersome.

Steps

1. Open **Windows Explorer**.
2. Browse to the location where the project was saved.
3. Double-click the folder which contains the project name.

Notes

Folder Contents

File type	Icon	Example	Definition
.np4		NetPointFile.np4	The schedule file for the project. NOTE: Do not remove or rename.
.nzp		[PROJECT_NAME].nzp	The project zip file. Only available once the project is saved. This includes all contents inside the project folder, and is updated each time the schedule is saved.
.npl		NetPointLayout.npl	The layout file. Only available after saving a layout. For more info, see the Layouts chapter. NOTE: Do not remove or rename.
.Lock		NetPointFile.Lock	The lock file. Only available when the schedule is currently open.
.NREC		[#####].NREC	The recovery file. Only available when there exist 2 or more unsaved changes. For more info, see Understanding File Recovery .
.xml		NetPointData.xml	Only available after exporting to P6.
.S01 or .B01		SimulationResults.S01	NetRisk simulation file. Only available after running NetRisk. For more info, see Schedule Risk Analysis . NOTE: Do not remove or rename.
.xls		[PROJECT_NAME].xlsx	The linked excel file. Only available after exporting to Excel. For more info, see Exporting Linked Reports .

File type	Icon	Example	Definition
Sub-folder		Attachment	A folder containing any embedded iObjects. For more info, see ➔Info Objects/Attachments .

Converting Files to Folders

All schedules created before version 4.1, release 22 Mar 13 must be converted to projects before they can be opened. At this time, converting these schedules must be done one at a time. This can be done outside of NetPoint by browsing for the file in Windows Explorer, or inside NetPoint through the file menu. Instructions for both are provided.

Upon converting, a folder of the same name will be created, the file will be automatically moved inside it, and the filename will be changed to “NetPointFile.np4.” In the event that you change your mind or something goes wrong during the import, a back-up copy of the schedule will automatically be saved to the original location with “-backUp” appended to the end. For a list of other changes, see the notes end of this section.

Steps

Outside of NetPoint

1. In an explorer window, browse for the desired file.
2. Double-click the .npt or .np4 file, or right-click and choose **Open**.

NOTE: If a message opens, see [↗ Troubleshooting Opening](#) for more info.

From Within NetPoint

1. First, choose **File > Convert .np4 file to Project Folder** from the menu bar.
2. Browse for the desired file and click **Open**.

NOTE: If a message opens, see [↗ Troubleshooting Opening](#) for more info.

Notes

Descriptions as Annotations

Descriptions that were being displayed as annotations in legacy schedules will be converted to conventional annotations. We recommend re-displaying the descriptions for these objects and deleting the annotations to avoid any unexpected behavior.

Lags on Non-FS Links

Lags on link types other than finish-to-start are no longer supported. Therefore, upon opening a schedule that contains them, an error message will display, the lags will be automatically removed, and the affected links will be spotlighted on the canvas.

Custom Positions of Data Elements

Data elements (durations, dates, descriptions, etc.) that have been positioned more than 3” away from their host objects will be reset, and their host objects will be spotlighted on the canvas.

Opening and Unzipping

When working in NetPoint, projects are opened just like files, with only minor differences to the interface. On the left, both project folders and non-project folders will show up in the tree. On the right, only projects will be listed. Projects should always be chosen on the right, NOT on the left. The left is only for selecting the parent directory in which the project resides.

If you've selected a parent directory on the left, but the project is not showing up on the right, its schedule file may have been removed from it. To correct this, look for a file named "NetPointFile.np4" that no longer resides in a folder named after a project and move it back in to the proper folder. Until this is done, you will be unable to open the project from within NetPoint.

Likewise, if a folder which is certainly not a NetPoint project appears in the list on the right, such as "MyDocuments," it likely contains an orphaned NetPointFile.np4. To correct this, determine which project folder is missing a schedule file, and return it to its proper location

NOTE: To open the schedule from Windows Explorer, search for the folder which contains the project name, open it, and then double-click the file called NetPointFile.np4.

Steps

Opening Projects

1. Choose **File > Open Project** from the menu bar.
2. On the left, browse for the directory in which the project resides, without yet choosing the project itself.
3. On the right, choose the project itself and click **Open**. **NOTE: At any time, a project may be deleted or renamed by right-clicking on it in the list.**

NOTE: If a message opens, see [➔Troubleshooting Opening](#) for more info.

Unzipping Projects (.nzp/.zip)

1. Choose **File > Unzip and Open Project** from the menu bar. The **Open** window will default to the folder where the last project was zipped and saved.
2. If a different project is desired, browse for the directory in which it resides on the left.
3. On right right, select the zip file file from the list and click **Open**. NetPoint will automatically save the project to the same folder where the zip file is.

NOTE: A .nzp can also be opened by simply double-clicking it in an Explorer window.

Opening Projects from Emails (.nzp Files)

1. Right-click on the file from your preferred email application.
2. To view the schedule immediately, choose **Open**. The **Opening Mail Attachment** window will open..
 - a. Click the **Open** button.
3. To save the .nzp file first, choose **Save As**.
 - a. Browse for the desired location and click **Save**. **NOTE: If you make a change to the file name, make sure that "nzp" still shows up next to Save as type. If it does not, make sure to append ".nzp" at the end of the filename to avoid losing the file type.**
 - b. To open the schedule, navigate to the location in an explorer window and double-click the .nzp file.

NOTE: If a message opens after opening, see [➔Troubleshooting Opening](#) for more info.

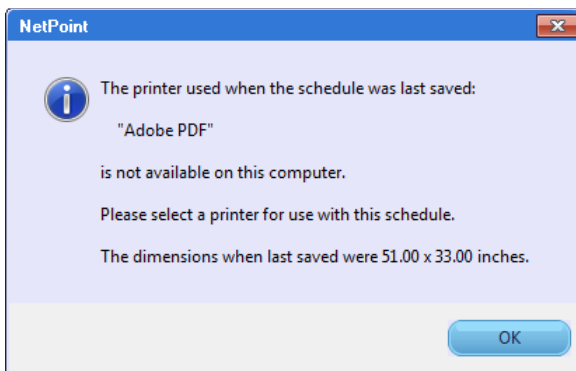
Opening Projects from Emails (.zip Files)

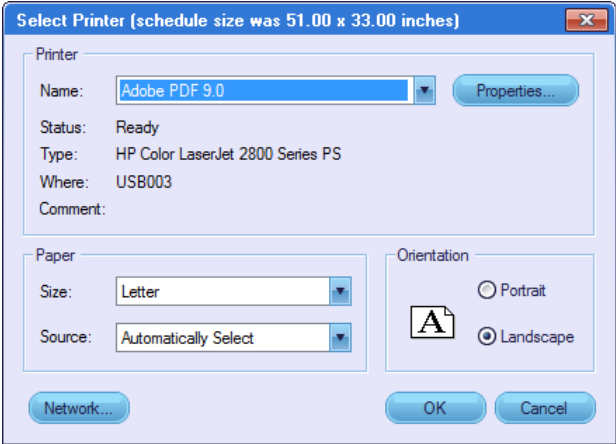
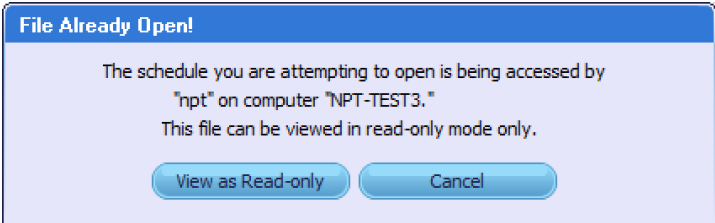
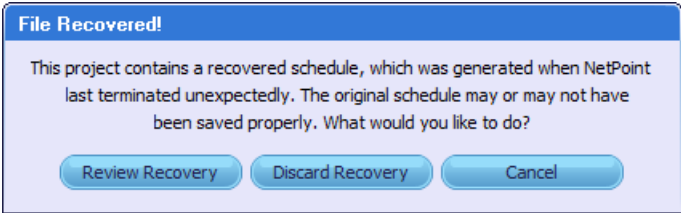
1. Right-click on the file from your preferred email application..
2. Choose **Save As** (recommended).
 - a. Browse for the desired location and click Save. **NOTE:** If you make a change to the file name, make sure that “nzp” still shows up next to Save as type. If it does not, make sure to append “.nzp” at the end of the filename to avoid losing the file type.
 - b. To open the schedule, see [➔Unzipping Projects](#).
3. To view the schedule without saving, choose **Open** (not recommended). The **Opening Mail Attachment** window will open.
 - a. Click **Open**. An explorer window will open. **NOTE:** The .zip file will be temporarily saved to “C:\Users\[your username]\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.Outlook[alphanumeric string]”.
 - b. Navigate up one level by clicking the name of the temporary location. **WARNING:** If you just open the folder and launch the *NetPointFile* project file without navigating up and unzipping, the .zip file will not go through the proper extraction process and the project may be missing a number of supporting files.
 - c. Right-click on the .nzp and choose **Extract All**.
 - d. If a Windows Security window opens, click **OK**.
 - e. To choose a new location, click **Browse**, select the desired location, and click **OK**.
 - f. Otherwise, just click **Extract**. A new explorer window will open.
 - g. Double-click the project folder.
 - h. Double-click the file called “NetPointFile.np4”.

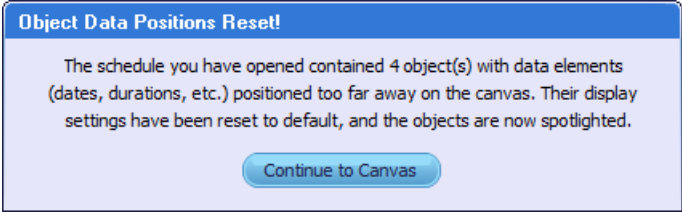
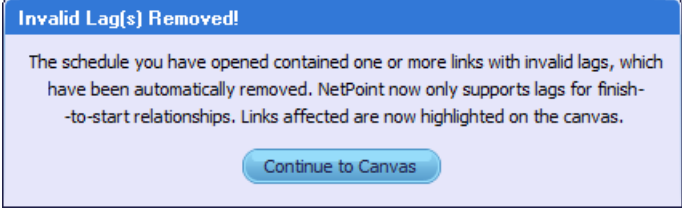
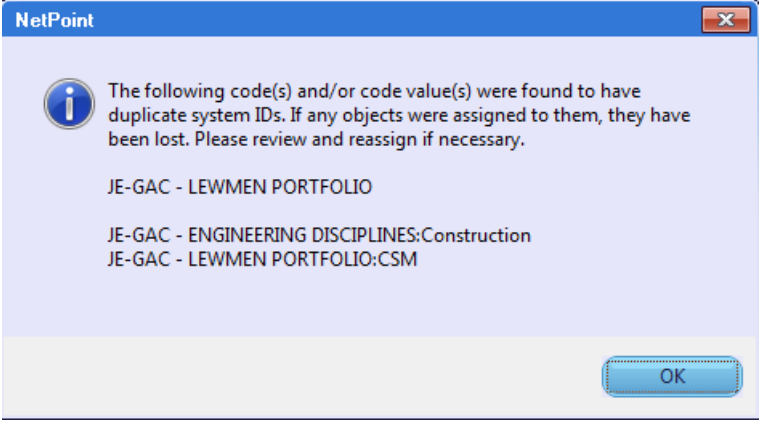
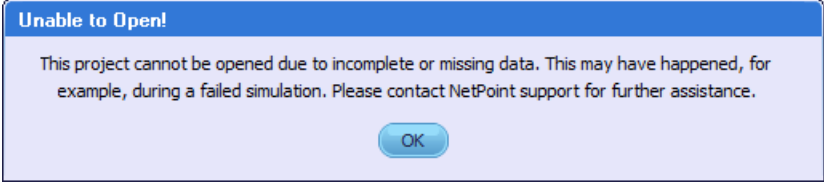
NOTE: If a message opens after opening, see [➔Troubleshooting Opening](#) for more info.

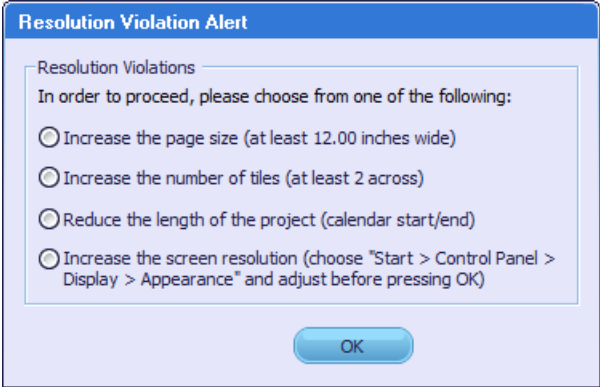
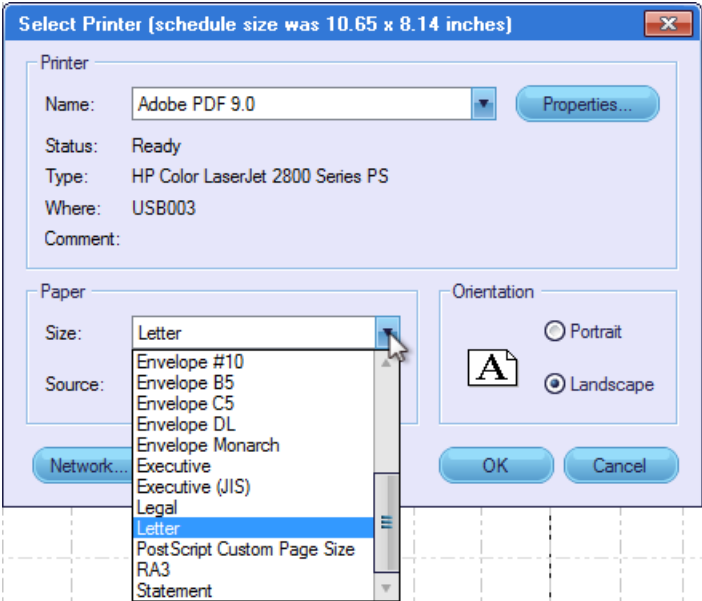
Notes

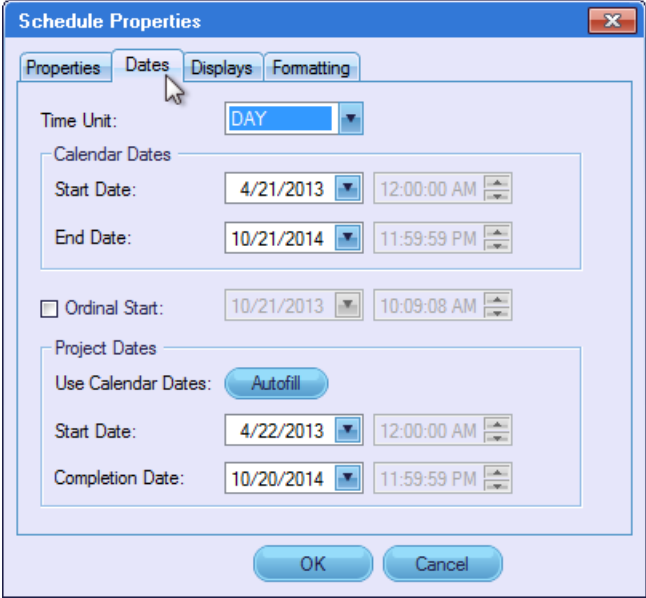
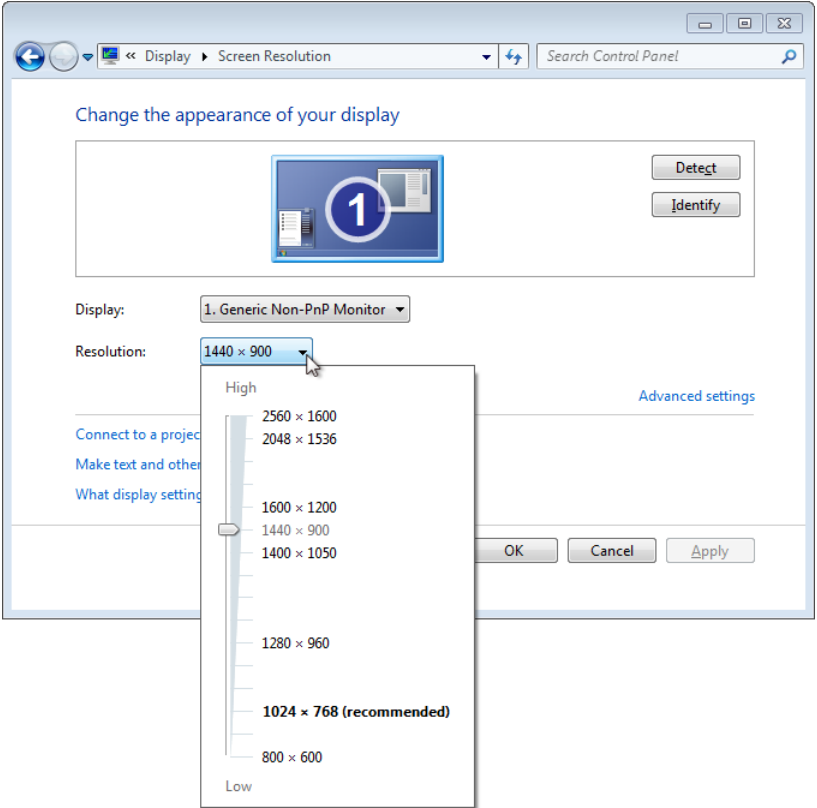
Troubleshooting Opening

Message	
Printer Unavailable	If the printer that was used when the schedule was last saved is unavailable or not installed, a warning will open. Click OK. 

Message	
	Choose a new printer next to Name and click OK. The schedule will then open. For more info on printers, see Print Setup. 
File Already Open	If the project is open on another machine, a lock may have been placed to protect the file. In this case, the schedule can only be opened in read-only mode, and the File Already Open! warning will open. Click View as Read-only. The schedule will then open. 
File Recovered	If an unexpected crash occurred, NetPoint® will automatically recover any schedules that may have been open and unsaved when the application terminated. For more info see Understanding File Recovery. 

Message	
Bad Data Positions	If any data elements are more than 3" away from their host objects, a warning will open. Click Continue to Canvas. No further action is required.  Object Data Positions Reset! The schedule you have opened contained 4 object(s) with data elements (dates, durations, etc.) positioned too far away on the canvas. Their display settings have been reset to default, and the objects are now spotlighted. Continue to Canvas
Invalid Lags	If a legacy schedule is opened where a non-finish-to-start link had a lag, a warning will open. Click Continue to Canvas. No further action is required. For more info on lags, see ➔ Adding Lags to Links.  Invalid Lag(s) Removed! The schedule you have opened contained one or more links with invalid lags, which have been automatically removed. NetPoint now only supports lags for finish-to-start relationships. Links affected are now highlighted on the canvas. Continue to Canvas
Duplicate Code/Value IDs	If a schedule is opened where a code or value is found to have the same system ID as another code or value, a warning will open and any assignments will be lost. Click OK. To review code assignments, see step 1 of Exporting an Existing Template and choose "Defaults – Codes & Code Assignments".  NetPoint  The following code(s) and/or code value(s) were found to have duplicate system IDs. If any objects were assigned to them, they have been lost. Please review and reassign if necessary. JE-GAC - LEWMEN PORTFOLIO JE-GAC - ENGINEERING DISCIPLINES:Construction JE-GAC - LEWMEN PORTFOLIO:CSM OK
Incomplete or Missing Data	If a schedule is opened which cannot be read, a warning will open. This may be because it's missing data, it's been corrupted, or it contains a bad  Unable to Open! This project cannot be opened due to incomplete or missing data. This may have happened, for example, during a failed simulation. Please contact NetPoint support for further assistance. OK

Message	
	format. This is rare and should not occur under normal circumstances. Click OK and contact NetPoint support.
Resolution Violation	If a legacy schedule is opened where one unit of time is less than the width of a pixel on the screen, the Resolution Violation Alert will open. Choose from the available options and click OK (we recommend choosing the second option to avoid additional steps). For more info on resolution, see Print Setup.  The dialog box titled "Resolution Violation Alert" contains a section "Resolution Violations" with the instruction: "In order to proceed, please choose from one of the following:". There are four radio button options: "Increase the page size (at least 12.00 inches wide)", "Increase the number of tiles (at least 2 across)", "Reduce the length of the project (calendar start/end)", and "Increase the screen resolution (choose 'Start > Control Panel > Display > Appearance' and adjust before pressing OK)". An "OK" button is at the bottom.
a.	If increasing the page size, a printer window will open. Click the dropdown button next to Size and choose a page size that meets the recommended width from the Resolution Violation Alert window.  The "Select Printer" dialog box shows the printer "Adobe PDF 9.0" with status "Ready", type "HP Color LaserJet 2800 Series PS", and location "USB003". The "Paper" section has a "Size" dropdown menu open, showing a list of paper sizes: Envelope #10, Envelope B5, Envelope C5, Envelope DL, Envelope Monarch, Executive, Executive (JIS), Legal, Letter (highlighted), PostScript Custom Page Size, RA3, and Statement. The "Orientation" section has "Portrait" and "Landscape" (selected) options. "OK" and "Cancel" buttons are at the bottom.
b.	If increasing the number of tiles, no

Message	
	further action will be required.
c.	If reducing the length of the project, the Schedule Properties window will open. Click the Dates tab, and under Calendar Dates, choose a start date that is later in time or an end date that is earlier in time. This process may have to be repeated until satisfactory. 
d.	If increasing the resolution, open the Control Panel from the Windows Start menu, choose Adjust Screen Resolution, and choose one that's higher. Go back to NetPoint and click OK. This process may have to be repeated until satisfactory. 

Saving and Renaming

When saving a project, the schedule itself will be named NetPointFile.np4, a folder will be created using the description you entered, and the schedule file will be automatically placed in it. The NetPointFile.np4 file should never be renamed; instead, to rename a project, always edit the folder itself.

In addition, each time a project is saved, a file with a .nzp extension will be saved in the project folder for easy emailing or uploading. For more info, see [➔ Sharing and Sending Projects](#).

NOTE: If you're downloading a project from the web through a browser, always right-click to specify where it will be saved, as some download locations may have limited access or permissions.

Steps

Saving

1. While viewing the desired schedule, choose **File > Save Project** or **File > Save Project As** from the menu bar.
2. On the left, browse for the location in which you want the new project to be saved, and select it in the list. **NOTE: To create a new folder, click the New Folder button. To edit or delete a folder, right-click and choose accordingly.**
3. After selecting the folder, a list of existing NetPoint projects will show up on the right. Enter a description next to **Save Project As**. This will be the name of the new project. **NOTE: At any time, a project may be deleted or renamed by right-clicking on it in the list.**
4. After entering a description, click **Save**.

Renaming

1. While viewing the desired schedule, choose **File > Save Project As** from the menu bar. By default, the **Save** window will default to the current project.
2. To choose a different project, browse on the left for the location in which the project resides.
3. On the right, right-click the project itself and choose **Rename Project**.
4. Begin typing and click **Enter**.
5. When finished, click **Cancel**.

Sharing and Sending

To copy or move projects between hard drive locations, make sure to copy/move the entire project folder and not just the NetPointFile.np4 file.

To upload or email a schedule, it is recommended to use the .nzp file that can be found within the project folder, as opposed to manually zipping the project or sending the NetPointFile.np4 file by itself. The .nzp file is a compressed version of the project folder, which is updated each time the schedule is saved. It includes ALL files placed in the project folder, regardless of whether or not they were generated by NetPoint or put there by the user.

Alternatively, NetPoint provides an email utility from within the application. The email utility, however, will ONLY include NetPoint-generated files, leaving out any files placed in the project folder by the user. For more information, see the notes at the end of this section.

NOTE: To benefit from Microsoft Outlook integration, NetPoint requires 32-bit installation. To check in Outlook 2010, click “File > Help”, and then look for the bit number in the right-most sections under “About Microsoft Outlook” after the “Version”. To check in Outlook 2013, click “File > Account”, click the big “About Outlook” icon, and then look for the bit number on the first line at the very end.

Steps

From within NetPoint

1. When viewing the desired schedule, choose **File > Send as Email Attachment** from the menu bar.
2. After a few seconds, the default email application will open, and a new message will be created with the NetPoint .nzp file attached.
3. To finish, compose the message as desired and click **Send**.

NOTE: To open a project from an email, see [➔ Opening and Unzipping Projects](#).

Notes

Email Utility

When using **File > Send as Email Attachment**, the contents of the project will be slightly different, according to the table below:

	Permanent .nzp file	File > Send as Email Attachment
Saved/ Unsaved Changes	Only represents the last saved state. Any changes, including iObjects that may have been added, will not be reflected when zipped unless the project has since been saved.	Represents the latest state, regardless of whether or not the project has been saved.

	Permanent .nzp file	File > Send as Email Attachment
Supporting Files	Includes ALL files placed in the project folder, whether generated by NetPoint or added by the user.	Includes ONLY files generated by NetPoint, including: simulation files, layout files, exported Excel templates, and iObjects that have been copied to the project folder.

About the Interface

The following aspects of the user interface are unique to NetPoint.

Autosave Windows

Many windows do not contain **OK** and **Cancel** buttons. Any changes made take effect when the window is closed do not require confirmation. To close these windows, click the **Exit** icon  and any changes will be automatically saved. The only way to cancel any changes is to use the **Undo** feature.

Undo Unavailable Icon

Some actions found in menus or windows are excluded from being undone. This is because some actions take place silently, and there would be no clear indication on the canvas if they were undone. To protect you from undoing an action and not realizing it, the **Undo** feature will skip over these actions. Any menu or window excluded from the **Undo** feature will be identified with the **Undo Unavailable** icon .

Menu Bar

Click on a menu title to show the menu contents. Ellipses (...) at the end of a menu item indicate that another interaction will follow that selection, such as a new window opening.

File Menu

Click the **File** menu to access the following commands:

	New	Opens a blank NetPoint canvas. Shortcut: Ctrl+N .
	Open...	Accesses saved project schedules. For detailed instructions, see ➔Opening and Unzipping Projects . Shortcut: Ctrl+O .
	Open .nzp...	Unzips a project's folder to a specified location and opens its schedule in NetPoint. For detailed instructions, see ➔Opening and Unzipping Projects .
	Open Recent	Lists the most recently opened projects. Choose to show 4, 10, or 16. Click on a specific plan name to open that plan.
	Save	Stores the latest changes to the file. Shortcut: Ctrl+S .
	Save As...	Saves a new schedule as a project or an existing schedule as a different project or to a different location. For detailed instructions, see ➔Saving New Projects .
	Close	Closes the current schedule without exiting NetPoint.
	Send Project as Email Attachment...	Zips the current project's folder, opens the default email client, and attaches the zip file. For detailed instructions, see ➔Sending and Zipping Projects .
	Convert .np4 file to Project...	Converts legacy NetPoint files into the folder-based structure and opens the project. For detailed instructions, see ➔Converting Legacy Schedule Files .
	Zip Project...	Zips a project's folder and saves it to a specified location. For detailed instructions, see ➔Sending and Zipping Projects .
	Print	Opens options for selecting a printer and adjusting print properties. Shortcut: Ctrl+P .
	Print Preview	Adjusts zoom and stretch so the entire plan fits in the screen and turns on page breaks. For detailed instructions, see ➔Printing and Presenting .
	Print Setup...	Opens options for changing the page size and orientation. For detailed instructions, see ➔Page Size .

	Page Tiling (a x b)	Opens options for tiling a schedule across multiple pages and hiding or showing page breaks. For detailed instructions, see ↗Page Size/Print Setup .
	Import/Export	Contains options for interfacing with Primavera P6, Open Plan, MS Project, XML, Excel, and for importing/exporting header/footer files. For detailed instructions, see ↗Inputs and Outputs .
	Insert Image File(s)...	Adds an image that will display on the canvas behind all objects except for shades. For detailed instructions, see ↗Image Files .
	Take Snapshot	Opens options for generating an image from the schedule. For detailed instructions, see ↗Capturing Snapshots .
	Exit	Closes down the application and all open projects.

Edit Menu

Click the **Edit** menu to access the following commands:

	Undo	Reverses the last actions. Undo does not apply to zoom, stretch, page setup, print setup, global text size, or calendar strip text size changes. Shortcut: Ctrl+Z .
	Redo	Reverses the previous undo actions. Redo does not apply to zoom, stretch, page setup, print setup, global text size, or calendar strip text size changes. Shortcut: Ctrl+Y .
	Cut	Removes selected item(s) from the canvas to the clipboard. Shortcut: Ctrl+X .
	Copy	Places selected item(s) onto the clipboard without removing them from the canvas. Shortcut: Ctrl+C .
	Paste	Inserts copied items from the clipboard. Shortcut: Ctrl+V .
	Paste Format	Applies only the display and formatting characteristics of copied items from the clipboard to selected objects of the same type. For detailed instructions, see Paste Format .
	Delete	Erases selected item(s) from the canvas without saving to the clipboard. Shortcut: Del .
	Default Format Copy Options...	Opens options for choosing which data elements and formatting settings are included in a given copy/paste format action. For detailed instructions, see Paste Format .
	Select All	Selects all items on the canvas. Shortcut: Ctrl+A .
	Find an Activity...	Opens a window for locating a specific activity. For detailed instructions, see Filters and Searching . Shortcut: Ctrl+F .
	Search and Filter Objects...	Opens options for filtering the schedule. For detailed instructions, see Filters and Searching . Shortcut: Ctrl+Shift+F .

View Menu

Click the **View** menu to access the following commands:

Un-Hide All Objects	Re-displays anything that was hidden on the canvas.
Horizontal Split	Divides the window into two landscape canvases. For detailed instructions, see ➔ Split-Screen/Multiple Canvases .
Vertical Split	Divides the window into two portrait canvases. For detailed instructions, see ➔ Split-Screen/Multiple Canvases .
Link Zoom Factor when Split	Maintains the same zoom percentage across split screens.
Zoom Cursor Text	Controls whether or not the date above the mouse cursor is zoomed in and out with the canvas zoom percentage.
Header	Turns on the header for viewing and/or printing.
Footer	Turns on the footer for viewing and/or printing.
Header/Footer Toolbar	Displays the toolbar for interacting with and modifying the header and/or footer.
Resource Key...	Displays or hides the resource key. For detailed instructions, see ➔ Reading the Resource Graphs and Curves .
Toolbar	Repositions, hides, or shows the toolbar.

Schedule Menu

Click the **Schedule** menu to access the following:

Modify Schedule Properties...	Opens a window for customizing and formatting the entire plan. For detailed instructions, see Schedule Properties .
Manage Schedule Progress...	Opens a window for actualizing objects with respect to a data date. For detailed instructions, see Actualizing Objects .
Set Password...	Opens options for settings a password to read or modify the schedule. For detailed instructions, see Password Permissions .
Shade Non-working Days	Adds vertical shading to all non-working days. For detailed instructions, see Displaying Holidays .
Prompt for Grid Numbers When Pasting	Enables a prompt each time an object is pasted from the clipboard to determine its vertical grid position. For detailed instructions, see Grid Options .
Open Layout Manager	Opens a window for saving, viewing, and deleting layouts and running the automated layout engine. For detailed instructions, see Layouts .
Animate When Running	Redraws the network in real-time while an automated layout is ran.
Always Fine Tune After Running	Applies a fine tune after running an automated layout. For detailed instructions, see Running Automated Layouts .
Run Automated Layout	Begins the layout processing. For detailed instructions, see Running Automated Layouts .
Layout Manager Settings	Contains options for resetting the Layout Manager settings. For detailed instructions, see Running Automated Layouts .

Objects Menu

Click the **Objects** menu to access the following commands:

Set Properties for Selected Objects...	Opens a window for modifying the properties of multiple objects simultaneously. For detailed instructions, see ➔Bulk-Editing Objects .
Set Default Properties for New Objects...	Opens options for setting properties before objects are drawn on the canvas. For detailed instructions, see ➔Setting Defaults .
Set Duration Display Format for Activities	Changes what time unit that durations are displayed in on the canvas. For detailed instructions, see ➔Setting Defaults .
Change Text Size	Changes the text size for the entire schedule at once.
Restore Default Positions of Data	Resets the position of any descriptions, IDs, dates, floats, drifts, buffers, gaps, and total floats back to their default locations. For detailed instructions, see ➔Repositioning Fields (GEM) .
Arrangement (Shades and Images)	Rearranges the order of shades, shapes, or images on the canvas. For detailed instructions, see ➔Rearranging Shades .
Line up Activities	Places the selected activities on the same horizontal line on the canvas. Whichever activity is selected first will become the target gridline. For detailed instructions, see ➔Grid Options .
Restore Faded Colors for All Objects	Restores the color of all objects that have been spotlighted in a yellow box or faded.
Highlight Driving Link Chains	Highlights activities and links with the smallest gaps and fades the rest. For detailed instructions, see ➔Highlighting Link Chains .
Redundant Links	Controls the display of redundant links on the canvas. For detailed instructions, see ➔Treating Redundancies .
Display Float and Drifts for Actualized Objects	Enables floats and drifts to be displayed for actualized objects. For detailed instructions, see ➔Forensic Floats for Actualized Objects .
Use Outlines for Critical Objects (non-actualized only)	Outlines all objects in red while maintaining their designated color (when critical). For detailed instructions, see Criticality .

Tools Menu

Click the **Tools** menu to access the following commands:

Manage Resources...	Opens a window for creating, modifying, deleting, assigning, and limiting resources. For detailed instructions, see Defining Resources .
Assign Resources...	Opens a window for assigning resources to multiple activities simultaneously. For detailed instructions, see Assigning Resources .
Modify Resource Display Settings...	Opens a window for adjusting the histogram calculation and plot displays. For detailed instructions, see Viewing the Resource Profile .
Open Codes Manager...	Opens a window for creating, modifying, deleting, and assigning codes. For detailed instructions, see The Codes Manager .
Manage Calendars...	Opens a window for creating, modifying, deleting, and assigning calendars. For detailed instructions, see Calendars .
Assign Calendars...	Opens a window for assigning calendars to multiple objects simultaneously. For detailed instructions, see Assigning Calendars .
Open Targets Manager...	Opens a window for capturing and comparing targets. For detailed instructions, see The Targets Manager .
Modify Visual Target Filters ...	Opens a window for adding, modifying, and deleting filters used in Visual Target Mode . For detailed instructions, see Visual Target Mode .
Manage Notification Recipients....	Opens a window for creating, modifying, and deleting recipients and for assigning them resources. For detailed instructions, see Noting Changes in Consumption .
Generate Resource Change emails...	Opens a window for notifying recipients of any changes in consumption of assigned resources. For detailed instructions, see Noting Changes in Consumption .

Risk Menu

Click the **Risk** menu to access the following commands:

Open NetRisk Manager...	Opens a window for managing the risk register, duration-ranging, floating, or pacing activities, and running the simulation and analyzing results (among others). For detailed instructions, see ➔The NetRisk Manager .
Run Simulation...	Opens the manager to the Run tab. For detailed instructions, see ➔Simulating the Schedule .
View Schedule at Selected Percentile...	Opens options for visualizing simulation data directly on the canvas. For detailed instructions, see ➔Viewing P-Values and Indices on the Canvas .
End Current NetRisk Session...	Disables the current session of NetRisk, allowing other uses to enable it (when running from a shared location). For detailed instructions, see ➔Enabling and Disabling NetRisk .

Metrics Menu

Click the **Risk** menu to access the following commands:

Open Metrics Manager...	Opens a window for managing the metrics and Schedule IQ™. For detailed instructions, see ➔Schedule IQ™ and Analytics .
View Schedule Statistics...	Opens a window for viewing basic schedule metrics and statistics (like number of activities, percentile floats, etc.). For detailed instructions, see ➔Schedule Statistics .

Windows Menu

Click the **Window** menu to access the following commands:

Cascade	Rearranges the current open files in a cascade view.
Tile	Rearranges the current open files in a tiled view.
List of Open Plans	Open NetPoint plans are listed here.

Help Menu

Click the **Help** menu to access the following commands:

About NetPoint...	Shows current release information and link to website.
Open User Guide...	Opens a PDF of the NetPoint User Guide for help.
Contact Support...	Opens the default email client and addresses a message to NetPoint support.
Collect Information for NetPoint Support...	Generates an HTML file containing recent actions taken and conditions of the software, which can then be emailed to NetPoint support. For detailed instructions, see ➔Generating Log Files .

Toolbar

The toolbar contains icons that access commonly used commands and objects. It includes system icons, object icons, and mode icons.

System Icons



System icons perform basic application functions, and are familiar to most users.

	New Project	Creates a blank NetPoint canvas. Shortcut: Ctrl+N .
	Open Project	Accesses saved NetPoint projects. For detailed instructions, see Opening and Unzipping Projects . Shortcut: Ctrl+O .
	Save Project	Stores the latest changes to the file. Shortcut: Ctrl+S .
	Cut to Clipboard	Removes selected item(s) from the canvas to the clipboard. Shortcut: Ctrl+X .
	Copy to Clipboard	Place selected item(s) onto the clipboard without removing them from the canvas. Shortcut: Ctrl+C .
	Paste from Clipboard	Inserts copied items from the clipboard. Shortcut: Ctrl+V . Click the down arrow to apply only the display and formatting characteristics of copied items from the clipboard to selected objects of the same type. For detailed instructions, see Paste Format .
	Print Project	Opens options for selecting a printer and adjusting print properties. Shortcut: Ctrl+P .
	Undo	Reverses the last actions. Undo does not apply to zoom, stretch, page setup, or print setup. Shortcut: Ctrl+Z .
	Redo	Reverses the previous undo actions. Redo does not apply to zoom, stretch, page setup, or print setup. Shortcut: Ctrl+Y .
	Select Tool	Grab, choose, select, or move activities and objects on the canvas. Shortcut: V .
	Multi-Object Select Tool	Grab, choose, or select multiple activities and objects without needing to use the SHIFT key. Shortcut: V .

	Pan Tool	Move around the canvas without needing to use the scroll bars. Shortcut: H/SPACE/MOUSEHWEEL.
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Object Icons



Objects are the building blocks of a plan.

	Add Activity	Activities are the building blocks of a schedule and represent work being done. They have a start date and a finish date, represented by end nodes, and they span a certain period of time. For detailed instructions, see Adding/Modifying Activities . Shortcut: A .
	Add Delay/ Gain	Delay/gains are a type of activity but whose duration can reach zero. Delay/ gains are useful for showing positive or negative impacts to the schedule or for atypical activities and do not support embedded nodes or resources. For detailed instructions, see Adding/Modifying Activities . Shortcut: D .
	Add Link	Links represent a relationship between objects, such as activities, milestones, and benchmarks. From them, float attributes are calculated, including gap, buffer, float and drift. Click the down arrow to choose a specific geometry. If object dates overlap, embedded nodes must be used. For detailed instructions, see Creating FS Links . Shortcut: CTRL+L .
	Link Multiple Activities (FS)	An FS link represents a relationship where the successor cannot start before the predecessor finishes. For detailed instructions, see Creating FS Links .
	Add Start Embed	A start embed is placed on a host activity to allow start-to-start (SS) relationships. It may be placed to coincide with its host activity's start or finish nodes, or it may be offset (to have a lead or lag) between the two. In a SS relationship, a successor cannot start before its predecessor starts. For detailed instructions, see Using Embeds to Create SS and FF Links . Shortcut: E .
	Add Finish Embed	A finish embed is placed on a host activity to allow finish-to-finish (FF) relationships. It may be placed to coincide with its host activity's start or finish nodes, or it may be offset (to have a lead or lag) between the two. In a FF relationship, a successor cannot finish before its predecessor finishes. For detailed instructions, see Using Embeds to Create SS and FF Links . Shortcut: E .
	Add Start Milestone	Milestones are zero duration events used to represent key dates or moments in the schedule. Like activities, milestones will track along with the network once they've been linked to any predecessors or successors. Start milestones

		align to the beginning of the day or time unit (by default, 9:00 am), and should be used for events which will precede an activity or represent an initiation. In addition, start milestones will snap to their earliest successor, always maintaining a zero gap link. For detailed instructions, see Adding/Modifying Milestones/Benchmarks . Shortcut: M .
	Add Finish Milestone	Milestones are zero duration events used to represent key dates or moments in the schedule. Like activities, milestones will track along with the network once they've been linked to any predecessors or successors. Finish milestones align to the end of the day or time unit (by default, 5:00 pm), and should be used for events which will succeed an activity or represent a termination. In addition, finish milestones will snap to their earliest predecessor, always maintaining a zero gap link. For detailed instructions, see Adding/Modifying Milestones/Benchmarks . Shortcut: M .
Start Benchmark Icon	Add Start Benchmark	Benchmarks are zero duration events used to represent a key date or moment in the schedule. Unlike activities and milestones, benchmarks will remain fixed at whatever date they're placed, regardless of whether or not they've been linked to any predecessors or successors. Start benchmarks align to the beginning of the day or time unit (by default, 9:00 am), and should be used for events which will precede an activity or represent an initiation. For detailed instructions, see Adding/Modifying Milestones/Benchmarks . Shortcut: B .
	Add Finish Benchmark	Benchmarks are zero duration events used to represent a key date or moment in the schedule. Unlike activities and milestones, benchmarks will remain fixed at whatever date they're placed, regardless of whether or not they've been linked to any predecessors or successors. Finish benchmarks align to the end of the day or time unit (by default, 5:00 pm), and should be used for events which will succeed an activity or represent an termination. For detailed instructions, see Adding/Modifying Milestones/Benchmarks . Shortcut: B .
	Add Hammock	Hammocks are a type of activity used to represent or summarize a group of other activities, milestones, and/or benchmarks. Hammocks, however, have no attributes of their own except for the descriptions/ID, and they don't accept resources, embedded nodes, or links. Click the down arrow for more options. For detailed instructions, see Hammocks . Shortcut: CTRL+H .
	Add Data Date	A data date is an object on the canvas used to represent the date to which the project has been statused. Data dates help provide a clear visual indicator of the progress of the schedule. For detailed instructions, see Data Dates . Shortcut: U .

	Add Info Object	Info objects are external files or attachments. Placing an info object onto the canvas is a way to embed a document within the schedule for viewing. For detailed instructions, see ➔Info Objects/Attachments . Shortcut: I .
	Add Shape	Shades are blocks of color used to highlight or delineate sections of the canvas. Click the down arrow to open a menu of other shapes. For detailed instructions, see ➔Shades . Shortcut: S .
	Add Notepad	Notepads are notes that do not show up on the canvas. Instead, they are represented by an icon and viewed by double-clicking the icon. For detailed instructions, see ➔Text Boxes and Notepads . Shortcut: N .
	Add Text Box	Text boxes are text objects that show up directly on the canvas. For detailed instructions, see ➔Text Boxes and Notepads . Shortcut: T .

Mode Icons



Mode icons provide quick access from the canvas to important functions.

	Open Gestural Interface	The gestural interface is a window for crashing, extending, or repositioning activities and other objects without the need for the keyboard. For detailed instructions, see Fine-Tuning/Gestural Controls . Shortcut: CTRL+G .
	Open Zoom/Stretch Controls	Zooming and stretching will only temporarily enhance readability on screen. They will have no effect on the printed page and they will reset when the project is printed or saved and closed. For detailed instructions, see Adjusting Zoom and Stretch . Shortcuts: CTRL+ +/-/1/0 (zoom) and SHIFT+ +/-/1/0 (stretch).
	Open Schedule Properties	Schedule Properties is a window for customizing and formatting the entire plan. For detailed instructions, see Schedule Properties .
	Run Automated Layout	Automated layouts enhance the organization of the schedule by assisting with, or completely taking control of, the placement and layout of objects in the network. Underneath the hood, dozens of factors are taken into account, including relationships between activities, link geometries, text and description lengths and positions, and grid and canvas dimensions. Click the down arrow for more options. For detailed instructions, see Running the Engine .
	Enable/Disable Logic	Logic is a mode whereby object relationships are enforced. When turned on, NetPoint will heal a negative-gap link as it forms to preserve the logic of the original relationship. For detailed instructions, see Overriding Logic .
	Change Planning Mode	In GPM Planning Mode, activities can be scheduled anywhere between their early and late dates without the need for setting a constraint and reducing total float. In Early Dates Mode, all <i>connected</i> activities are pulled back to their traditional CPM early-start dates. In Late Dates Mode, all <i>connected</i> activities are pushed forward to their traditional CPM late-start dates. Click the down arrow to jump to a specific mode. For detailed instructions, see GPM, Early, and Late Dates Modes .
	Enter/Exit Visual Target Mode	In Visual Target Mode, activities that are ahead of or behind schedule can be made either smaller, larger, or lighter or darker, resulting in a visual comparison. This formatting will have NO effect on colors or sizes specified outside of Visual Target Mode and the two will be stored separately. Visual Target Mode requires at least one target to have been captured by the user and a base and alternate designated. For detailed instructions, see Visual Target Mode . Shortcut: CTRL+T .

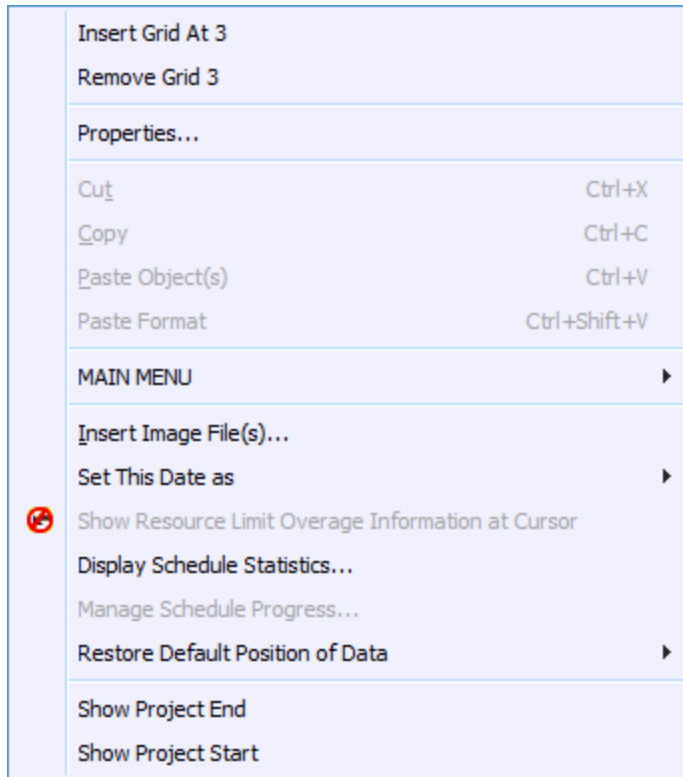
	Hide/Show Resource Profiles	The resource profile displays total and cumulative resource consumption in a chart at the bottom of the canvas. For detailed instructions, see ➔Viewing the Resource Profile. Shortcut: CTRL+R.
	Hide/Show Resource Limits	A limit is an object that restricts how much of a resource may be used at any given time. It is drawn on the canvas, like an activity, with a certain height and “duration”. The height at which the limit is drawn corresponds to the maximum value activities can consume of that resource when overlapping with the limit. For detailed instructions, see Limiting Resource Consumption.
	Change Display Mode	The Global Edit Mode (GEM) allows you to reposition object fields (e.g. descriptions, dates, durations, etc.) for multiple objects at a time. Upon entering GEM, all dates become fixed, and no object property boxes are accessible. However, object descriptions can still be modified by double-clicking them on the canvas. Date Lock mode prevents any changes to horizontal positions of objects on the canvas; however, activity dates can still be changed via their property boxes, and all objects can still be moved to different grids. For detailed instructions, see ➔Repositioning Fields and ➔Locking Relative Positions. Shortcut: CTRL+D, CTRL+E..

Right-Click Menus

Right-clicking on the canvas or on an object will bring up a menu with additional options, some of which may only be found in the menu itself.

Canvas Menu

Right-click on the canvas to see the following menu:



Insert Grid at [x]	Inserts a grid above the one designated. For detailed instructions, see Grid Options .
Remove Grid [x]	Removes the designated grid. For detailed instructions, see Grid Options .
Properties	Opens a window for customizing and formatting the entire plan. For detailed instructions, see Schedule Properties .
Cut	Removes selected item(s) from the canvas to the clipboard. Shortcut: Ctrl+X.
Copy	Places selected item(s) onto the clipboard without removing them from the canvas. Shortcut: Ctrl+C.
Paste Object(s)	Inserts copied items from the clipboard. Shortcut: Ctrl+V.
Paste Format	Applies only the display and formatting characteristics of copied items from the clipboard to selected objects of the same type. For detailed instructions, see Paste Format .

MAIN MENU	Contains the options from the main menu bar.
Insert Image File(s)	Adds an image that will display on the canvas behind all objects except for shades. For detailed instructions, see ➔Image Files .
Set This Date as	Changes the date of the project start or project completion. For detailed instructions, see Dates Tab .
Show Resource Limit Overage Information at Cursor	Display current and maximum resource consumption levels when the mouse cursor moves over the canvas. For detailed instructions, see Limiting Resource Consumption .
Display Schedule Statistics...	Opens a window for viewing basic schedule metrics and statistics (like number of activities, percentile floats, etc.). For detailed instructions, see ➔Schedule Statistics .
Manage Schedule Progress...	Opens a window for actualizing objects with respect to a data date. For detailed instructions, see ➔Actualizing Objects .
Restore Default Position of Data	Resets the position of any descriptions, IDs, dates, floats, drifts, buffers, gaps, and total floats back to their default locations. For more info, see ➔Repositioning Fields (GEM) .
Show Project End	Hides/shows the yellow project completion box.
Show Project Start	Hides/shows the yellow project start box.

2) Setting up a Schedule

Before drawing activities and building a schedule, it can be helpful to configure the plan appropriately.

Whether you're starting with a blank canvas or importing activities from another application, the page size, time unit, and project dates should all be optimized beforehand. Once set, they can be modified at any time during planning.

PolyTime/Variable Time Units

Time units are the intervals on which activities and other objects are based. For example, an activity's duration may span one day or any whole unit increment thereof. The schedule time unit may be set to a unit as small as minutes or as large as years. The time unit may not differ from activity to activity—once set, it applies to all objects within the schedule. Durations, dates, and float values are all calculated according to the schedule's time unit.

Small time units—like minutes and hours—are useful for projects with short durations or for showing an extremely high level of detail. Large time units—like months and years—are useful for projects with long durations or for showing an extremely broad level of detail. If the schedule is blank (i.e. no activities or other objects have been drawn), the time unit can be changed to a *higher or smaller* unit. The calendar dates and project dates will automatically update to a reasonable range. However, once an object has been placed on the canvas, the time unit may be changed *only to a smaller unit*. The calendar dates and project dates will remain fixed. This restriction is enforced because converting to a larger time unit would involve either rounding or truncation and would potentially result in changes to dates and durations (which would result in altered gaps, floats, drifts, etc.).

NOTE: Small time units can only be used if the duration of the project is reasonably short or if the size of the canvas is sufficiently large. Otherwise, the width of a time unit would be less than the width of a pixel on screen, and input from the mouse would no longer translate accurately on the canvas.

Steps

Changing the Default Time Unit for NetPoint

1. See step 12 of [➔Installing/Upgrading NetPoint](#).

Changing a Schedule's Time Unit

1. Click **Schedule > Modify Schedule Properties** from the menu bar and choose the **Dates** tab.
2. Click the down arrow next to **Time Unit** and choose accordingly.
3. When finished, click **OK**.

NOTE: When converting to time units smaller than days, some dates may appear to change; however, they are only realigning – all float calculations and values will remain constant. This is because with days, weeks, half-months, months, quarters, or years, dates are aligned to the *end of the prior unit*, whereas with hours, quarter hours, and minutes, dates are aligned to the *start of the next unit*. For example, a three-day activity that starts on May 1st will end on May 3rd at 11:59 pm. If converted to hours, however, the same three-day activity will now end on May 4th at 12:00 am. This is done to align with other applications as well as user expectations.

Notes

Resource Considerations

When converting plans from a higher time unit to a lower time unit, or when exporting a plan that is in any time unit other than days, resource assignments may not be accurate. For example if 48 widgets are assigned per *day* and the time unit is changed to hours, then 48 widgets would be assigned per *hour* instead of 2 widgets being assigned per *hour*. The following approaches may be taken to correct the outcome: 1) wait to

resource load the schedule until the file is converted or exported; 2) reassign every resource to every activity after converting or exporting to correct the values; or 3) delete the resource assignments prior to converting or exporting and start with a clean resource-assignment slate, but with resources defined.

Calendar Considerations

When working with time units of hours or smaller, NetPoint does not support “daily calendars” or “shifts”. For example, if a schedule in days is converted to hours, a working day is converted to 24 working hours. Similarly, a 1-day gap is converted to a 24 hour gap. As a result, hours, quarter-hours, and minutes are only recommended for schedules without non-work periods.

Changing Date Formats

NetPoint will display dates according to whichever format has been set for the “short date” in Windows. As a result, a schedule created on a computer in, for example, the U.S., will automatically be updated when opened on a computer in the U.K. and vice versa, assuming each user’s computer has been configured to their country.

Dates will be updated for objects on the canvas, in the calendar strips at the top, at the cursor, and in all lists and date controls. In addition, the names of days and months will also be updated to the correct language at the cursor and in the calendar strips. However, no matter what format is chosen, the year will never be displayed for an object’s start and finish dates on the canvas. To change the format of dates that you’re seeing, you must change the format of the “short date” chosen for Windows.

NOTE: Displaying days and months with letters can only be done for latin-based languages. If a non-latin based language is chosen (e.g. Armenian (Armenia)), and the short date includes months with letters (i.e. more than two ‘M’s’), question marks will appear for an object start and finish dates on the canvas. Question marks will also appear for the names of months in the calendar strip and days at the cursor.

Steps

1. Open the Windows Control Panel. Depending on your version of Windows, this can be done by clicking the **Start** button and selecting **Control Panel**, or by typing “control panel” in the search bar in the upper right-hand corner.
2. Click **Region**. The **Region** window will open.
3. First choose the desired language/country under the **Format** dropdown.
4. Next, choose the desired format next to **Short date**.
5. Click **Apply**.
6. Click with the mouse on the NetPoint canvas to confirm the changes.
7. When finished, click **OK**.
8. Click the **Close** icon to close the control panel.

Page Size/Print Setup

The page size should always be set specifically after a new document is created but before the network is drawn. This is because the page size will determine not just how the schedule prints, but also how it displays on-screen. Choosing a new page size will increase or decrease the horizontal width of the canvas by stretching it, and it will increase or decrease the vertical height of the canvas by adding or removing gridlines. It is recommended to choose the page size anticipated for printing, and if the schedule appears too small or too large, to use the zoom and stretch controls to enhance readability on-screen.

For some schedules, increasing the page size may help enhance readability; however, others may be too complex to view on a single page. In these cases, the plan may be tiled across multiple pages, horizontally and/or vertically. When printed, a letter and number will print on the lower left of each tile to convey tile sequence. The letter indicates the horizontal sequence; the number indicates the vertical sequence. These can be previewed, and the plan can be adjusted to automatically fit the screen using the print preview.

When opening a schedule that was created on another computer or network, the same page size may not be available, since page sizes are specific to printers. In this case, a warning will appear and the user will be prompted to change the page size automatically. If the printer is available but the driver doesn't match (likely with "software only" print drivers, e.g. Microsoft XPS Document Writer or Adobe PDF), a similar warning will open. In either case, the schedule will only open once a new printer and page size have been chosen. To assist the user, the warning lists both the printer and page size dimensions in which the schedule was last saved.

NOTE: A large page size can only be used if the time unit is reasonably high or if the duration of the project is sufficiently short. Otherwise, the width of a time unit would be less than the width of a pixel on screen, and input from the mouse would no longer translate accurately on the canvas. Similarly, horizontal tiles can only be removed if the same requirements are met.

Steps

Choosing a Printer/Page Size

1. Click **File > Print Setup** from the menu bar.
2. Next to **Name**, first choose the desired printer.
3. Next, in the **Paper** section, choose the desired **Size**.
4. If desired, change the **Orientation**.
5. When finished, click **OK**.

Adding/Removing Page Tiles

1. Click **File > Page Setup (Tiling: a x b)** from the menu bar.
2. Increase or decrease the numbers next to **Across** and **Down** as desired.
3. When finished, click **OK**.

NOTE: If the number of vertical tiles is reduced, any areas of the network falling below the new boundary will be cropped. Content can be restored, however, by returning to the original number of vertical tiles.

Viewing Page Breaks

1. Click **File > Page Setup (Tiling: a x b)** from the menu bar.

2. Under **Page Breaks**, check the box next to **Display on Canvas**.
3. When finished, click **OK**.

Print Preview

1. Click **File > Print Preview** from the menu bar.
2. To revert, click the **Reset** button next to **Zoom Factor**.
3. To continue in the preview, click the red **X** in the upper-right corner of the **Schedule Zoom & Stretch** window.

NOTE: Print preview can be cancelled at any time by clicking the **Zoom/Stretch**  icon from the toolbar and following step 2.

Notes

Resources

If resource histogram colors are not printing faithfully, first try printing to another printer. Depending on the printer, this may or may not resolve the issue. In the case that it doesn't, it is recommended to remove the histogram patterns entirely until a further solution is implemented.

Text

When text is too close to the margins, a warning will pop when printing and the text will be highlighted. The print operation will continue, but we recommend adjusting any text near page or tile borders to provide a buffer for the future.

Text strings may appear shorter on screen than when printed, depending on the monitor and resolution of your screen.

Images

If your schedule contains images that appear mirrored when printed, try printing to PDF first or printing to a different printer.

Headers/Footers

A header and/or footer can be activated at any time. Once activated, the header/footer toolbar will also show up automatically. The header/footer toolbar will update depending on which section is selected (shown as outlined in blue). A header or footer can be divided into as many sections as desired, and each section can be designated as an image, text, table, or legend.

Legends can be generated manually or automatically. The manual legend works by the user choosing a desired legend object to pop into the header/footer section. The automated legend works by scanning the scheduled and detecting what types of objects, shades, and/or resources are in-use as well as their colors, patterns and sizes.

Steps

Showing/Hiding

1. Click **View > Header** or **View > Footer** from the menu bar.

NOTE: If the header is shown, it will print. If it is hidden, it will not print, but it will be saved for the next time it is displayed.

Importing/Exporting

1. Click **File > Import/Export > Import Header/Footer** or **File > Import/Export > Export Header/Footer** from the menu bar.
2. Choose **Both**, **Header**, or **Footer** as desired.
3. Browse for the file and click **Open** (if importing) or enter a name and click **Save** (if exporting).

Defining an Image Section

1. Click **View > Header** or **View > Footer** from the menu bar.
2. Click the arrow in the upper-left corner of the section.
3. Click **Image**.
4. Browse for the desired file and click **Open**.
5. Click and drag to move or resize.
6. For the best size, right-click on the image and choose **Fit Image to Section**.

Defining a Text Section

1. Click **View > Header** or **View > Footer** from the menu bar.
2. Click the arrow in the upper-left corner of the section.
3. Choose **Text**.
4. Click once in the box and begin typing
5. For formatting, highlight the text and click the desired icon in the **Header/Footer Toolbar**.

Defining a Table Section

1. Click **View > Header** or **View > Footer** from the menu bar.
2. Click the arrow in the upper-left corner of the section.
3. Choose **Table**.
4. Right-click to add a row, remove a row, add a column, remove a column, high/show the borders, or insert "live" values.

NOTE: Values update live with the schedule. Values cannot be edited, but can be deleted by selecting and using the **Delete** or **Backspace** keys on the keyboard.

Defining a Legend Section

1. Click **View > Header** or **View > Footer** from the menu bar.
2. Click the arrow in the upper-left corner of the section.
3. Choose **Legend**.
4. Choose a legend object directly from the menu (manual) or choose **Automated Legend** to add all objects, shades, or resources automatically.
5. To edit the title of the legend, double-click the word “Legend”. Right-click anywhere and uncheck it to turn it off.
6. To edit an object’s label, double-click the text and type or right-click and choose **Edit Label**. Use the **Enter** key to put the label on multiple lines.
7. To realign all objects due to modified label lengths, right-click anywhere in the legend and choose **Align Legend Objects**.
8. To move an object, click it, drag it, and let go; or, right-click and choose accordingly.
9. To delete an object, select and click the **Delete** or **Backspace** keys on the keyboard or right-click and choose **Delete**.
10. To modify a legend objects properties, double-click the object itself or right-click and choose **Open Properties**.

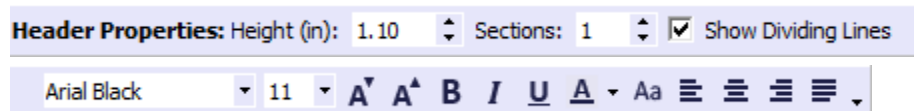
NOTE: Labels for data date, link, and iObject cannot be edited. Only one data date, link, and iObject can be added in total.

Adding/Removing/Resizing/Moving Sections

1. To add a section, click the up arrow next to **Sections** in the **Header/Footer Toolbar**.
2. To remove a section, click the down arrow next to **Sections** in the **Header/Footer Toolbar**.
3. To resize a section, move the cursor over the dividing line until it turns into a double-arrow and then click and drag with the mouse left or right.
4. To move a section left or right, click the arrow in the upper-left corner of the desired section and choose **Move Section Right** or **Move Section Left**.

Notes

Header/Footer Toolbar



Field	Definition
Height	The height of the header or footer area.
Sections	The number of sections in the header or footer.
Show Dividing Lines	Controls whether or not lines show up in between sections.

Field	Definition
Font	The font for text sections.
Font Size	The font size for text sections.
Decrement Size	An option for decreasing the font size for text sections.
Increment Size	An option for increasing the font size for text sections.
Bold	Applies bold formatting to text sections.
Italize	Applies italics formatting to text sections..
Underline	Applies underlines to text sections.
Font Color	Changes the color for text sections.
Text Case	Controls the case for text sections.
Left Align	Left aligns text sections.
Center Align	Center aligns text sections.
Right Align	Right aligns text sections.
Justify	Justifies text section alignment.

Schedule Properties

The schedule properties include settings and formatting options for the entire plan, including dates, colors, and timeline configuration. Instructions for a subset of functionality are provided below.

NOTE: At this time, pasting objects from one schedule to another such that one or more of the object's dates falls outside the calendar range of destination schedule is unsupported. The calendar range of the destination must be adjusted. Otherwise, an error message will appear and the paste will be cancelled.

Steps

Changing the Criticality Thresholds

1. Click **Schedule > Modify Schedule Properties** in the menu bar.
2. Increase or decrease the number next to **Criticality Threshold** or **Near-criticality Threshold**.
3. If desired, click the dropdown next to **Criticality Factor** and choose accordingly.
4. When finished, click **OK** to save and close the window.

NOTE: By default, critical activities are automatically turned red, but their original color may be preserved by choosing **Objects > Use Outlines for Critical Objects (non-actualized only)**.

Changing the Grid Spacing

1. Click **Schedule > Modify Schedule Properties** from the menu bar and choose the **Formatting** tab.
2. Next to **Grid Spacing**, click the up or down arrows to increase or decrease the value accordingly.
3. When finished, click **OK** to save and close the window.

NOTE: The maximum allowed value is determined by the current size of the page and the positions of existing objects within the canvas. The grid spacing can only be increased if the bottom grid is unoccupied. If objects already reside on the bottom grid, they must be moved higher in order to increase grid spacing.

Adjusting the Calendar Start/End Dates

1. Click **Schedule > Modify Schedule Properties** from the menu bar and choose the **Dates** tab.
2. Next to **Start Date**, click the down arrow to open the visual calendar picker, or type in a desired date manually.
3. Repeat step 2 for **End Date**.
4. Double check that the **Project Dates** are still within the new **Calendar Dates**. If no objects have yet been added to the canvas, this may be done automatically by clicking **Autofill** next to **Use Calendar Dates**.
5. When finished, click **OK** to save and close the window.

NOTE: The dates may also be modified graphically by moving objects on the canvas until the calendar warning is triggered. However, NetPoint will always enforce a minimum of at least 2 time units between the start of the network and the start of the calendar and a minimum of at least 4 time units between the end of the network and the end of the calendar. If there is a zero-gap link between the project start and any activities or milestones, pulling the project start earlier will also trigger the automatic calendar adjustment. The calendar will not adjust automatically, however, if moving it

earlier would place a zero-gap successor on a non-working day according to its calendar. These effects will become particularly noticeable with larger time units such a months and years.

Using an Ordinal Calendar

1. Click **Schedule > Modify Schedule Properties** from the menu bar and choose the **Dates** tab.
2. Check the box next to **Ordinal Start**.
3. Click the down arrow to open the visual calendar picker, or type in a desired date manually.
4. When finished, click **OK** to save and close the window.

Notes

Properties Tab

The **Properties** tab is where the general settings are changed.

The screenshot shows the 'Schedule Properties' dialog box with the 'Properties' tab selected. The dialog has four tabs: 'Properties', 'Dates', 'Displays', and 'Formatting'. The 'Properties' tab contains the following fields and controls:

- Project Title:** A text input field.
- Project Start:** A dropdown menu showing 'Project Start'.
- Project Completion:** A dropdown menu showing 'Project Completion'.
- Criticality Threshold:** A numeric input field with a value of 0.0 and a percentage dropdown set to '%'. There are up/down arrows next to the input field.
- Near-criticality:** A numeric input field with a value of 5.0 and a percentage dropdown set to '%'. There are up/down arrows next to the input field.
- Criticality Factor:** A dropdown menu showing 'Total Float (Float + Drift)' with a help icon (?) to its right.
- Show as Red:** A dropdown menu showing 'All Critical Activities'.

At the bottom of the dialog are 'OK' and 'Cancel' buttons.

Field	Definition
Project Title	A text object for describing the project. By default, it is positioned at the top center of the canvas (when displayed) but can be subsequently repositioned.
Project Start	A field for customizing the text of the project start box on the canvas.
Project Completion	A field for customizing the text of the project completion box on the canvas.

Field	Definition
Criticality Threshold	Determines at what point (e.g. days) an activity will be displayed as critical. % is calculated as the percent of remaining duration from the data date onwards. As a result, as the data date approaches the project finish date, the threshold will approach 0. If the data date equals to the project finish date and percent is chosen, the threshold will equal 0. For more info, see Criticality .
Near-criticality Threshold	Determines at what point (e.g. days) an activity will be displayed as near-critical (excludes critical activities). % is calculated as the percent of remaining duration from the data date onwards. As a result, as the data date approaches the project finish date, the threshold will approach 0. If the data date equals to the project finish date and percent is chosen, the threshold will equal 0. For more info, see Criticality .
Criticality Factor	Determines how criticality and near-criticality will be calculated. Float is defined as the amount of time an activity can be delayed before delaying the completion of the project. Drift is defined as the amount of time an activity can gain before advancing the start of the project. Total Float is defined as float + drift and is equal to the CPM value of Total Float. NOTE: Choosing Float will discount drift values. Choose Drift will discount float values. For more info, see Criticality .
Show as Red	Determines what to show as red on the canvas: all critical activities according to the threshold, the longest path according to latest early finish date, or none.

Dates Tab

The **Dates** tab contains all date related fields.

The screenshot shows the 'Schedule Properties' dialog box with the 'Dates' tab selected. The dialog has four tabs: 'Properties', 'Dates', 'Displays', and 'Formatting'. The 'Dates' tab contains the following settings:

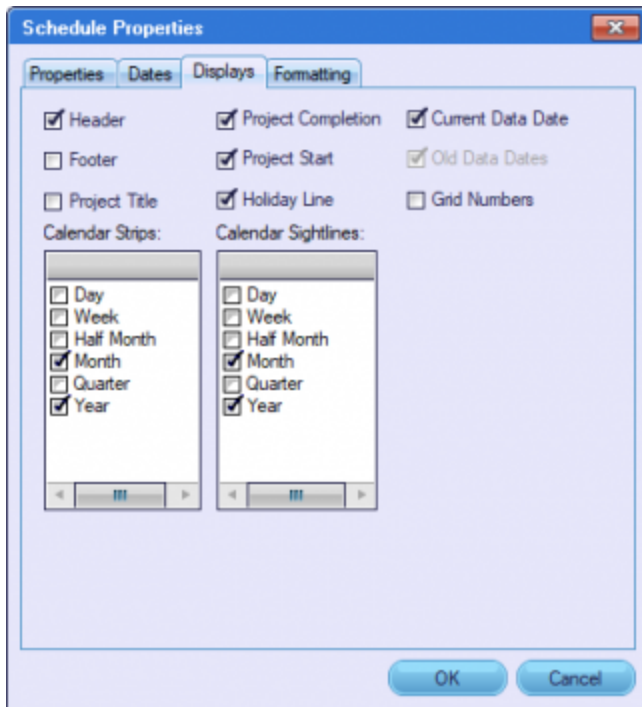
- Time Unit:** DAY (dropdown menu)
- Project Completion Type:** Fixed (dropdown menu)
- ☐ **Fiscal/Ordinal Start:** 6/30/2016 (calendar icon) 5:56:26 AM (time spinner)
- ☐ **Show Calendar as Ordinal**
- Calendar Dates**
 - Start Date:** 8/14/2008 (calendar icon) 12:00:00 AM (time spinner)
 - End Date:** 2/20/2013 (calendar icon) 11:59:59 PM (time spinner)
- Project Dates**
 - Use Calendar Dates:** Autofill (button)
 - Start Date:** 8/15/2008 (calendar icon) 12:00:00 AM (time spinner)
 - Completion Date:** 1/17/2013 (calendar icon) 11:59:59 PM (time spinner)

At the bottom of the dialog are 'OK' and 'Cancel' buttons.

Field	Definition
Time Unit	Intervals on which activities and other objects are based. If PolyTime is enabled, the user will have the option of scheduling in time units other than days. Changing the time unit may have a significant impact on the schedule. See PolyTime/Variable Time Units for more details.
Project Completion Type	Determines the behavior of the project completion box. If Fixed , the project completion will remain at whichever date it is entered or placed graphically. If Calculated , it will track automatically with the last <i>activity</i> in the network; it will not track along with milestones or benchmarks.
Fiscal/Ordinal Start	An option for shifting the quarter and year sightlines to correspond to a custom date, most often used to align to a fiscal year or ordinal scheme.
Show Calendar as Ordinal	An option for displaying the timeline, or calendar strips, by sequence instead of by time. The date chosen above becomes 0; to the right, increments of 1, and to the left, negative decrements of 1. Only available when the Fiscal/Ordinal Start is checked.
Start Date (Calendar)	The date that corresponds with the left edge of the canvas. By default, this will be set to 6 months prior to the current date and cannot be earlier than 1/1/1971 (if activities are near the project start, the recommended limit is 1/1/1974). Together with the End Date , this determines the maximum possible range in which the project itself may take place. Changing the calendar dates can affect the resolution of the schedule.
End Date (Calendar)	The date that corresponds with the right edge of the canvas. By default, this will be set to 12 months after the current date and cannot be any later than 12/31/2100 (if activities are near the project completion, the recommended limit is 12/31/2096). Together with the Start Date , this determines the maximum possible range in which the project itself may take place. Changing the calendar dates can affect the resolution of the schedule.
Use Calendar Dates	A shortcut for aligning the project dates 1 time unit within the calendar dates. This button will be disabled once any objects have been added to the canvas to prevent unintentional modification to the network.
Start Date (Project)	The actual start of the project. Displays as a yellow box on the canvas. To view the entire box, set the calendar's start well outside the project's start.
Completion Date (Project)	The actual finish of the project. Displays as a yellow box on the canvas. To view the entire box, set the calendar's end well outside the project's completion.

Displays Tab

The **Displays** tab contains checkboxes for hiding or showing different fields or elements on the canvas.

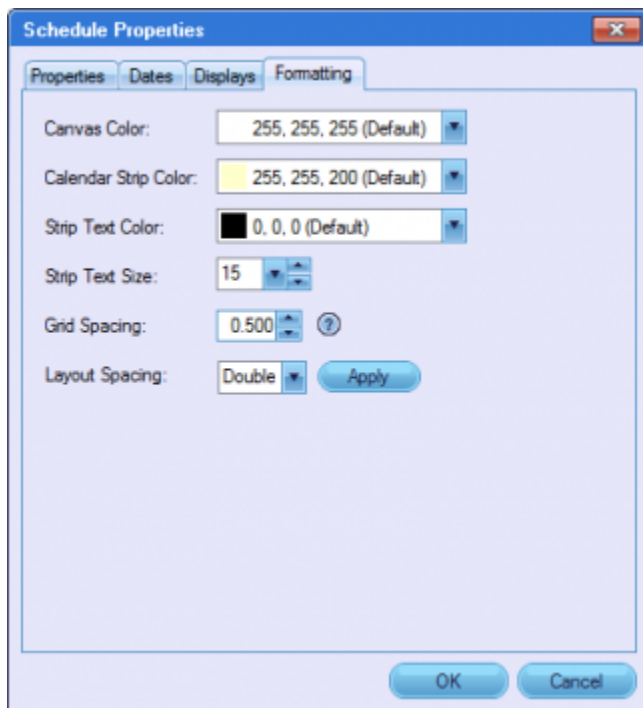


Field	Definition
Header	A designated area above the canvas for adding images, text, tables, or legends.
Footer	A designated area below the canvas for adding images, text, tables, or legends.
Project Title	A text object for describing the project. By default, it is positioned at the top center of the canvas (when displayed) but can be subsequently repositioned.
Project Completion	The actual finish of the project. Displays as a yellow box on the canvas. Can also be shown/hidden by right-clicking on the canvas, or on the yellow box itself.
Project Start	The actual beginning of the project. Displays as a yellow box on the canvas. Can also be shown/hidden by right-clicking on the canvas, or on the yellow box itself.
Holiday Line	A line at the top of the canvas for displaying non-working days. Holidays, weekends, and non-working periods will be displayed as red notches or rectangles; weather days will show up as blue notches or rectangles. The name of the holiday is also displayed, in blue, by hovering over it with the mouse cursor. The holiday line is based on whatever calendar is set as the default calendar for new activities. The holiday line cannot be displayed when working with time units larger than days and will be disabled accordingly. For more info, see Displaying Holidays .
Current Data Date	An object on the canvas used to represent the date to which the project has been statused. For more info, see Data Date .

Field	Definition
Old Data Dates	An object on the canvas used to represent the date to which the project has been statused. For more info, see Data Date .
Grid Numbers	Faded numbers for identifying gridlines. For more info, see Grid Options .
Calendar Strips	Horizontal rows at the top of the canvas for displaying the timeline or time units.
Calendar Sightlines	Vertical lines corresponding to the calendar strips for lining up dates on the canvas.

Formatting Tab

The Formatting tab contains controls for changing the look of the canvas.



Field	Definition
Canvas Color	The background color of the plan.
Calendar Strip Color	The color of the timelines at the top of the canvas.

Field	Definition
Strip Text Color	The color of the timeline text at the top of the canvas.
Strip Text Size	The size of the timeline text at the top of the canvas.
Grid Spacing	Determines how much vertical space there is between adjacent horizontal grid lines. For more info, see ➔Grid Options .
Layout Spacing	The number of inactive (empty) grids between active grids. An active grid is one that contains an object. An inactive grid is a grid that does NOT contain an object. Choosing Single will result in only active grids with no inactive grids; choosing Double will result in one inactive grid placed between every active grid; and choosing Triple will result in two inactive grids placed between every active grid. After choosing, click Apply for the changes to take effect.

Grid Options

In NetPoint, a horizontal grid system guides the placement of all activities, delays, milestones, benchmarks, and hammocks. Although they can be temporarily positioned between grids while selected, they will always snap to the nearest grid as soon as they're deselected. Info objects, shades and shapes, notepads, and text boxes, however, are free to be positioned anywhere on the canvas.

Grids provide an important reference point for objects on the canvas. In addition, they can be strategically used for improving the look and layout of the network. For example, the amount of space between grids can be adjusted, and grids can be added or removed.

Grid spacing determines how much vertical space there is between adjacent horizontal grid lines, but it will not actually stretch the canvas (increase or decrease its height). The maximum allowed value is determined by the current size of the page and the positions of existing objects within the canvas. The grid spacing can only be increased if the bottom grid is unoccupied. If objects already reside on the bottom grid, they must be moved higher in order to increase grid spacing.

NOTE: The default grid spacing of 0.500 inches allows activities with single line descriptions to be displayed with both the duration and the float block, without the chance of any text overlapping (using a small global text size). A grid spacing of 0.350 inches allows activities with single line descriptions to be displayed with *either* the duration *or* the float block (but not both), without the chance of any text overlapping (using a small global text size). Values smaller than 0.350 should only be used to customize advanced schedule layouts, because they may cause numerous text collisions unless activities are placed more than 1 grid apart.

Steps

Changing the Grid Spacing

1. With the **Select Tool**  chosen in the toolbar, click the small grey box at either the far right or left edge of the canvas of any grid.
2. Drag the cursor up (to decrease spacing) or down (to increase spacing) until it changes to a double-sided arrow and a decimal value appears.
3. When satisfied, let go of the mouse.

or,

1. Click **Schedule > Modify Schedule Properties** from the menu bar and choose the **Formatting** tab.
2. Next to **Grid Spacing**, click the up or down arrows to increase or decrease the value accordingly.
3. When finished, click **OK** to save and close the window.

Adding/Removing Grids

1. With the **Select Tool**  chosen in the toolbar, right-click *immediately above* the desired grid.
2. Choose **Insert Grid at #** or **Remove Grid #** accordingly.

NOTE: Grids that contain objects cannot be removed.

Lining Up Activities

1. With the **Select Tool**  chosen in the toolbar, click the activity that you want subsequently selected activities to align to.
2. Press and hold the **SHIFT** key on the keyboard and select additional activities. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all the activities, so long as they are next to each other.
3. Click **Objects > Line Up Activities** in the menu bar.

NOTE: If no activity is selected first (e.g.. a group is only selected by drawing a box around them), then the activities will align to the grid of whichever activity was *added to the canvas first*.

Hiding/Showing Grid Numbers

1. Click **Schedule > Modify Schedule Properties** from the menu bar and choose the **Displays** tab.
2. Next to **Grid Numbers**, check or uncheck the box to show or hide grid numbers respectively.
3. When finished, click **OK** to save and close the window.

Changing Layout Spacing

1. Click **Schedule > Modify Schedule Properties** from the menu bar and choose the **Formatting** tab.
2. Next to **Intermediate Grids**, choose **Single** for no intermediate grids between active grids, **Double** for one intermediate grid between active grids, and **Triple** for two intermediate grids between active grids.
3. Click **Apply**.
4. When finished, click **OK** to save and close the window.

Calendars

Calendars are a way for specifying which periods an activity, milestone, or benchmark may or may not be taking place due to national holidays, weekends, scheduled days off, and/or weather (blocked weather days).

If desired, a specific calendar may be chosen as the default calendar, but it will only apply within the active schedule. The only way to reset the default calendar permanently is to reinstall NetPoint, at which point a different country may be selected. For more info on changing the default calendar or reinstalling, see [↗ Default Properties](#) and [Installing/Upgrading NetPoint](#) respectively.

NOTE: If a milestone is assigned a calendar that's different from that of its predecessors or successors, and the network is pulled backwards, gaps may open up as the chain crosses a weekend or non-working days. As such, it is recommended to always keep milestones on the same calendar as their predecessors/successors.

NOTE: When moving 1 or more objects with different calendars, the behavior will be different depending on whether using the mouse or using the keyboard/gestural controls. For more info, see [↗ Fine Tuning/Gestural Controls](#).

This chapter includes details for the following:

1. Adding/modifying calendars
2. Configuring holidays
3. Assigning calendars
4. Displaying holidays
5. Shared calendars (global)

Adding/Modifying Calendars

NetPoint comes preloaded with two calendars for the user to start with. The first one is set upon installation of NetPoint. The second one called the **Calendar Day** calendar has every day as a working day. Additional calendars may be created, modified, and deleted through the **Project Calendars** window.

NOTE: If an object is copied from one schedule and pasted into another where its calendar is not defined, a warning will appear. By clicking **OK**, the object will be pasted, but its calendar will be reassigned to whichever calendar has been set as default for the schedule. To check or change the default calendar, see [➔ Default Properties](#) for more details.

Steps

Defining

1. Click **Tools > Manage Calendars** from the menu bar. The **Project Calendars** window will open.
2. Click the **New** button. The **Calendar Properties** window will open.
3. Enter a name.
4. To add a list of pre-defined holidays and/or non-working days, choose a country or pre-existing calendar from the dropdown next to **Templates** and click **Add to List**.
5. Check or uncheck any additional days and non-working.
6. Enter any weather days.
7. Click **Add**. The new calendar will be added to the **Project Calendars** window and the **Calendar Properties** window will reset.
8. To add additional calendars, repeat steps 3-7.
9. When finished, click the **Exit** icon.
10. Click **OK** to save the changes and close.

Modifying

1. Click **Tools > Manage Calendars** from the menu bar. The **Project Calendars** window will open.
2. Select a calendar in the list and click **Modify**. The **Calendar Properties** window will open.
3. Make desired changes.
4. Click **Update**. The **Calendar Properties** window will reset.
5. To modify additional calendars, repeat steps 2-4.
6. When finished, click the **Exit** icon.
7. Click **OK** to save the changes and close.

NOTE: The **Calendar Days** calendar, or any shared/global calendars, cannot be modified (for the **Calendar Days** calendar, the button will be disabled). If a calendar is modified which is assigned to an activity, then the start date and duration are held constant while the finish date is recalculated. However, if the activity has any constraints, then user input will be needed to resolve any violations. A calendar may never be modified if any of the objects it has been assigned to have been actualized. First de-actualize any objects before modifying the calendar. For more info, see [➔ Actualizing Objects](#).

Deleting

1. Click **Tools > Manage Calendars** from the menu bar. The **Project Calendars** window will open.
2. Select a calendar in the list and click **Delete**.

3. In the confirmation, click **Yes**.
4. To delete another calendar, repeat steps 2-3.
5. When finished, click **OK** to save and close.

NOTE: A calendar cannot be deleted if it has been assigned to any objects, or if it has been set as the default calendar for the schedule.

Notes

Calendar Properties Window

Calendar Properties

Name:

☐ Share to all projects ?

Templates: Add to List ?

Holidays

Description	Type

New... Modify... Delete

Non-working Days

☐ Mon ☐ Tues ☐ Wed ☐ Thu ☐ Fri ☐ Sat ☐ Sun

Weather Days ?

Jan	0	Apr	0	Jul	0	Oct	0
Feb	0	May	0	Aug	0	Nov	0
Mar	0	Jun	0	Sep	0	Dec	0

Add

Field	Definition
Name	A text field for describing the calendar.
Share to all Projects	A shared calendar will be copied to all new and existing schedules on the machine. For more info, see Shared Calendars (Global) .
Templates	A list of 23 countries (including 2 for the United States), complete with their national holidays and workweek, are provided as templates for creating a new calendar. Any user-defined calendars as well as the default calendars are also provided. This makes it possible

Field	Definition
	to start with an existing calendar and make changes rather than redefine all the holidays from scratch.
Holidays	Specific non-working periods.
Non-working Days	Non-working days that repeat every week.
Weather Days	A way to randomly assign non-working days due to external factors on a monthly basis. To prevent all of the weather days from falling on pre-existing non-working days (such as holidays or weekends), the number is adjusted by the pre-existing holidays and weekends. For example, if 7 weather days are entered for February (assuming a traditional US workweek and standard holidays), 5 weather days will fall on <i>working</i> days, and 2 weather days will fall on <i>pre-existing non-working</i> days.

Configuring Holidays

In NetPoint, the term holiday is used to represent national holidays as well as any user-defined non-working days and non-working periods. The latter can be useful for projects that require a non-working day more specific than weekends or weather days. When creating a holiday, you can choose between an annual, fixed-date, floating-date, or easter-relative. Instructions for adding, modifying, and deleting are provided below.

NOTE: If a holiday falls on a weekend (e.g. Christmas is on a Saturday), the following Monday will NOT be marked a holiday.

Steps

Adding

1. Click **Tools > Manage Calendars** from the menu bar. The **Project Calendars** window will open.
2. Click **New**, or select a calendar and click **Modify**. The **Calendar Properties** window will open.
3. Click the **New** button under the list of **Holidays**. The **Holiday Properties** window will open.
4. Enter a name.
5. Choose the desired type.
6. When finished, click the **Add** button. The holiday will be added to the **Calendar Properties** window and the **Holiday Properties** window will reset.
7. To add another holiday, repeat steps 4-6.
8. When finished, click the **Exit** icon to close the **Holiday Properties** window.
9. Click the **Add** or **Update** button, respectively.
10. Click the **Exit** icon to close the **Calendar Properties** window.
11. Click **OK** to save and close the **Project Calendars** window.

Modifying

1. Click **Tools > Manage Calendars** from the menu bar. The **Project Calendars** window will open.
2. Select a calendar and click **Modify**. The **Calendar Properties** window will open.
3. Select a holiday and click **Modify**. The **Holiday Properties** window will open.
4. Make the desired changed.
5. When finished, click the **Update** button. The **Holiday Properties** window will reset.
6. To modify another holiday, repeat steps 3-5.
7. When finished, click the **Exit** icon to close the **Holiday Properties** window.
8. Click the **Update** button and then click the **Exit** icon to close the **Calendar Properties** window.
9. Click **OK** to save and close the **Project Calendars** window.

Deleting

1. Click **Tools > Manage Calendars** from the menu bar. The **Project Calendars** window will open.
2. Select a calendar and click **Modify**. The **Calendar Properties** window will open.
3. Select the holiday and click **Delete**.
4. Repeat step 3 as desired.
5. When finished, click the **Update** button and then click the **Exit** icon to close the **Calendar Properties** window.
6. Click **OK** to save and close the **Project Calendars** window.

Notes

Holiday Properties

Type

Field	Definition
Fixed-Date	A fixed-date holiday occurs on a fixed <i>date and/or time</i> and can be either repeating or non-repeating. A fixed-date holiday is also known as a non-working period, as the duration may be less than a day. If a fixed-date holiday is <i>repeating</i> , it must repeat every year. Other time options will be disabled. To make a holiday non-repeating, leave the Repeat Every field blank or set it to zero. Fixed-date holidays, if repeating, will not take place over the entire duration of the schedule's calendar dates, but from the start date and on. NOTE: If the duration is set to less than a day but the time unit for the schedule is larger, the entire time unit will be considered a holiday. For example, if a holiday is created on December 20th with a duration of 1 hour but the time unit for the schedule is set to days, all of December 20th will be considered a holiday, even though the holiday line and shading will depict 1 hour.
Annual	An annual holiday is a type a fixed-date holiday but which must repeat every year (though not necessarily on the same day of the week) and must last for at least a day or more. Many national holidays, like New Years Eve and Independence Day in the USA are annual holidays. Annual holidays will take place over the entire duration of the schedule's calendar dates. Annual holidays have the additional feature where, if they fall on a day that is already a non-working day, they can be moved or not moved, according to the option selected.
Floating-Date	A floating-date holiday occurs on a fixed <i>day of the week</i> , though not necessarily on the same date, and must repeat every year. Many national holidays, like Thanksgiving and Labor Day in the USA are floating-date holidays. Floating-date holidays will take place over the entire duration of the schedule's calendar dates.
Easter-Relative	An Easter-relative holiday occurs a certain number of days before or after Easter. It must repeat every year and it must last for at least one day or more. Good Friday, for example, occurs on the Friday before Easter every year. Easter-relative holidays will take place over the entire range of the schedule's calendar dates.

Assigning Calendars

By default, all new activities, milestones, and benchmarks inherit a calendar derived from whichever country is chosen upon installation. An object's calendar may be changed, however, at any time. There are three ways to do so. The first method described is best for assigning a calendar to a *single* object. The second method is best for assigning the *same* calendar to *more than one* object. The last method is best for assigning *more than one* calendar to *more than one* object.

NOTE: If an activity's calendar is changed right after it's drawn (before pressing OK), then the start and finish dates are held constant while the duration is recalculated. If an activity's calendar is changed at any other point, then the start date and duration are held constant while the finish date is recalculated. If an activity has been limited by constraints, however, then user input will be needed to resolve any violations.

Steps

A Single Object

1. With the **Select Tool**  chosen in the toolbar, double-click the object or right-click and choose **Properties**.
2. For activities, click the **Dates** tab. For milestones or benchmarks, located the **Calendar** field on the main tab.
3. Click the dropdown to the right of **Calendar** and choose accordingly.
4. When finished, click **OK**.

Selected Objects

1. With the **Multi-Object Select Tool**  icon chosen in the toolbar, select the desired activities on-by-one on the canvas. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all the desired activities
2. Click **Objects > Set Properties for Selected Objects** from the menu bar.
3. Click the dropdown next to **Apply Calendar** and choose accordingly.
4. Repeat in the **Milestone** and/or **Benchmark** tabs as needed.
5. When finished, click **OK**.

NOTE: For more info, see the [Bulk-Editing](#) chapter.

Multiple Objects

1. Click **Tools > Assign Calendars** from the menu bar. The **Global Calendar Assignment** window will open.
2. Select the desired object in the list.
3. Once selected, click the dropdown under the **Calendar Assigned** column.
4. Choose a new calendar.
5. Repeat steps 2-4 as needed.
6. When finished, click **OK**.

NOTE: To filter this list, first selected the desired objects on the canvas.

Displaying Holidays

Holidays, weekends, and any user-defined non-workday days can be displayed tabularly in a list or graphically on the canvas using either vertical shading or a horizontal line. On the line, holidays, weekends, and non-working periods will be displayed as red notches or rectangles; weather days will show up as blue notches or rectangles. On the canvas, holidays and weekend will be displayed in light gray; weather days will be displayed in dark gray. Either way, all holidays will always be based on the default calendar for the schedule. For more info, see [➔Default Properties](#).

Steps

In a List

1. Click **Tools > Manage Calendars** from the menu bar.
2. Select the desired calendar and click the **Details** button.
3. When finished, click the **Exit** icon.
4. Click **OK** to save and close the **Project Calendars** window.

NOTE: The list of non-working days will include non-working days for the entire year, even if they fall outside of the calendar start and end.

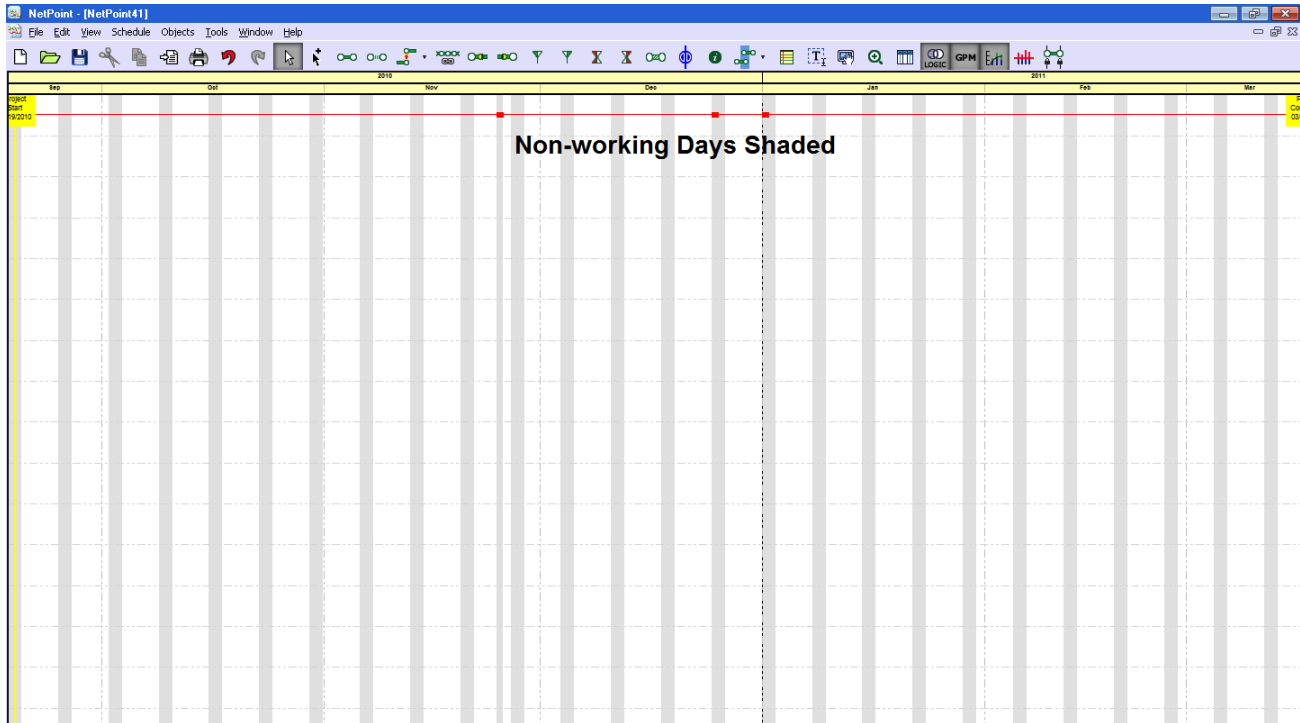
On the Canvas

1. Click **Schedule > Modify Schedule Properties** from the menu bar. The **Schedule Properties** window will open.
2. Click the **Displays** tab.
3. Check or uncheck the box next to **Holiday Line**.
4. Click **OK**.

Using Shading

1. Click **Schedule > Shade Non-working Days** from the menu bar.

Notes



Shared Calendars (Global)

NetPoint allows a calendar to be made available to all schedules, new and existing, on a given workstation. Once shared, a global calendar cannot be edited. If deleted, and the calendar had been assigned to any objects in a given project, a copy of the calendar will be retained in that schedule, and the entire calendar name will become italicized. When a deleted global calendar has no object assignments, it will be automatically removed.

If a schedule with a global calendar is opened on another workstation, the calendar will be inherited, but it will not be global. To edit an inherited global calendar, or to make it global on the new workstation, a new calendar must be created using the inherited calendar as a template.

If a schedule with a global calendar is opened on another workstation, but the same global calendar already exists, NetPoint will automatically recognize the existing calendar and re-assign it accordingly.

If a schedule with a global calendar is opened on another workstation and the same global calendar already exists but with different properties, the inherited one will be used and the existing one will be made unavailable for that schedule. Instructions for sharing are provided below.

Steps

Sharing an Existing Calendar

1. Click **Tools > Manage Calendars** from the menu bar. The **Project Calendars** window will open.
2. Select the calendar from the list and click **Modify**. The **Calendar Properties** window will open.
3. Check the box next to **Share to all Projects**.
4. Click the **Update** button. A confirmation will open.
5. Choose **Share, Disable Editing**.
6. To share another calendar, repeat steps 2-5.
7. When finished, click the **Exit** icon to close the **Calendar Properties** window.
8. Click **OK** to save and close the **Project Calendars** window.

Retaining an Inherited Global Calendar

1. Click **Tools > Manage Calendars** from the menu bar. The **Project Calendars** window will open.
2. Click the **New** button. The **Calendar Properties** window will open.
3. Enter a name.
4. Click the dropdown next to **Templates**.
5. Under **User Defined**, choose the desired global calendar.
6. Click the **Add to List** button.
7. Make any desired changes.
8. To save the calendar, click the **Add** button.
 - a. Click the **Exit** icon to close the **Calendar Properties** window.
 - b. Click **OK** to save and close the **Project Calendars** window.
9. To make the new calendar global, see step 5 above under *Sharing an Existing Calendar*.

Notes

For more info on calendars and their fields, see the notes of [➔Adding/Modifying Calendars](#).

Configuring/Using a Template

NetPoint does not offer any pre-defined templates; however, any schedule can be configured and then saved for future use.

Steps

Configuring a Template

1. Choose landscape or portrait, page size, and print tiling, if any. For more info, see [➔Page Size/Print Setup](#).
2. Configure settings like the background color of the canvas, what calendar strips and sightlines to display, and grid spacing. For more info, see [Schedule Properties](#).
3. Define any holidays or non-working days unique to your projects or company. For more info, see [➔Calendars](#).
4. Add cost, labor, material, or equipment requirements which are used across multiple projects. For more info, see [Resources](#).
5. Customize how all new activities will look when they're drawn on the canvas, including what data will be displayed (e.g. dates, float attributes, ID's, etc.) and how they're formatted (e.g. colors). For more info, see [➔Activity Defaults](#).
6. When ready, click **File > Save Project As** from the menu bar.
7. On the left, browse for the location in which you want the new project to be saved, and select it in the list.
8. After selecting the folder, a list of existing NetPoint projects will show up on the right. Enter a description next to **Save Project As**. This will be the name of the new template.
9. After entering a description, click **Save**.

Creating from Template

1. Click **File > Open Project** from the menu bar.
2. Browse for the location of the template on the left, select the template in the list on the right, and click the **Open** button.
3. Click **File > Save Project As** from the menu bar so as to not modify the template.
4. On the left, browse for the location in which you want the new project to be saved, and select it in the list.
5. After selecting the folder, a list of existing NetPoint projects will show up on the right. Enter a description next to **Save Project As**. This will be the name of the new project.
6. After entering a description, click **Save**.

Password Permissions

NetPoint provides two options for securing a schedule with a password. The first option (**Password required for READ**) will prevent the schedule from even being opened unless the password is entered. The second option (**Password required for Modify**) will allow the schedule to be opened without a password, but it will be in read-only mode. To gain full access, the password must be entered. If you have received a schedule but do not know the password, contact the creator of the schedule. If the password is lost or forgotten, please contact support@pma.sourcedigitalmarketing.com.

Steps

Setting a Password

1. Click **Schedule > Set Password** from the menu bar.
2. Choose the type of security from the dropdown.
3. Enter the password text.
4. Re-enter the password text.
5. Click **OK**.
6. Click **File > Save**.

Opening a Password Secured Schedule

1. Click **File > Open Project** from the menu bar.
2. Browse for the location of the project on the left, select the project in the list on the right, and click the **Open** button.
3. If the schedule was secured with a password for reading, you must enter the password in the text field.
 - a. When finished, click **OK**.
4. If the schedule was secured with a password for modifying, click **Use Password**.
 - a. Enter the password and click **OK**.
 - b. If you do not know the password, click **Read-Only** to open the schedule without editing permissions.

NOTE: In Read-Only mode, saving, and cut, copy, and paste will all be disabled.

Removing a Password

1. Open the schedule using steps 1-4 above.
2. Click **Schedule > Set Password** from the menu bar.
3. Click the dropdown and choose **No Password Required**.
4. Click **OK**.
5. Click **File > Save**.

GPM, Early, and Late Dates-Modes

NetPoint is based on the Graphical Path Method (GPM®), an innovative scheduling algorithm that governs the behavior of activities and events in a network-based schedule. However, activities, and their resource histograms and curves, can also be viewed on their CPM early or late dates. This gives NetPoint the ability to schedule forward from the project start or backward from the project completion.

In **GPM Planning Mode**, activities can be scheduled anywhere between their early and late dates (on what are called “planned dates”) without needing to *constrain* them (in CPM, constrained dates reduce total float).

In CPM **Early Dates Mode**, all *connected* activities are pulled back to their traditional CPM early-start dates (activities without any predecessors will remain on their GPM planned-dates). This mode is helpful for drawing and linking activities rapidly or push-planning from the project start. Effectively, activities can be drawn on the canvas as late as desired, without needing to think about exact dates or fine-tuning their positions.

In CPM **Late Dates Mode**, all *connected* activities are pushed forward to their traditional CPM late-start dates (activities without any predecessors will remain on their GPM planned-dates). This mode is helpful for pull-planning from the project completion.

In addition to switching between modes for the entire schedule, activities can be snapped to their early or late dates on an individual basis. Instructions are provided below.

NOTE: In CPM early and late dates modes, all link gaps and drifts equal zero, and all floats equal total floats.

NOTE: Do not change from GPM to early or late modes if there are any negative gaps in the schedule or any objects beyond the project completion. Otherwise, GPM dates may no longer match.

Steps

Changing Modes

1. Click the down arrow next to the **GPM/Early Dates/Late Dates** icon in the toolbar.
2. Select the desired mode from the list.

Snapping to Predecessors/Successors

1. With the **Select Tool** chosen in the toolbar, right-click on a desired activity.
2. Choose **Snap to Predecessors** or **Snap to Successors** accordingly.

Notes

Working in Early or Late Dates Modes

It is recommended that modifications to the network always be done in GPM Planning mode. If you modify the positions of activities in early or late-dates modes, negative gaps will temporarily form until the canvas has been clicked and activities return to their calculated positions.

Additionally, when switching between modes, you may observe unexpected results. For example, when switching between *early-dates* and GPM modes, the *later* of the two dates will be kept for all activities. When

switching between *late-dates* and GPM modes, the *earlier* of the two dates will be kept for all activities. In both cases, if any activity has zero total float, it will be maintained. Original dates and gaps will be restored only for those relationships that have gone unmodified. If the project start is modified, it will not revert.

Updating in Early or Late Dates Modes

During schedule updates in CPM early or late dates mode, all activities to the left of the data date (or crossing through it) must have start and finish constraints applied before a data date is placed. Otherwise, all connected activities and milestones will jump to the data date and be unable to be actualized. For instructions, see [➔Constraining Activities](#) and [➔Bulk-Editing Objects](#).

3) Building a Schedule

In NetPoint, a schedule is created by drawing or placing objects—such as activities, milestones, and benchmarks—on a time-scaled calendar called the canvas. Each object represents a real-world task or event. These objects can be related to one another by placing logic ties between them, which are represented by links.

Objects are scheduled wherever they are placed on the canvas, which can be anywhere within their total float range (at the time). When an activity is first added to the plan, it may be positioned on the early dates that a CPM forward pass would yield. However, once linked to predecessors and successors, stakeholders can re-position the activity to achieve a more desirable scenario. This is because GPM permits activities to be scheduled on an “as-planned” basis. This flexibility allows for a more realistic schedule.

In general, it is recommended that every GPM schedule have an uncorrupted critical path, defined as a continuous sequence of logically-connected activities terminating at, and carrying the least total float or negative total float relative to, a contractual benchmark or constrained milestone symbolizing a contract time.

Interacting with the Canvas

The NetPoint timescale is both interactive and instantaneous. When the mouse is moved over the canvas, the date and day of the week display above the cursor. As the cursor moves, the date changes to reflect the exact calendar date that the cursor is set at. If the date is a non-working period or holiday (according to the default calendar), the date text appears red.

When placing objects, the cursor will change to a cross-hairs as long as it is over the canvas. After placing an object, it will switch back to the select tool, except for with activities, delays, or shades.

When an object is selected, its dates and durations will enlarge, other fields will be hidden, and black boxes will appear at the ends and at the center. Instructions for selecting and modifying objects can be found below.

NOTE: Moving objects rapidly or suddenly with the mouse may cause negative gaps in rare instances. If this happens, click **Undo** and use the **Gestural Interface** or arrows on the keyboard instead. For more info, see [↗Fine-Tuning/Gestural Controls](#).

Steps

Selecting Multiple Objects

1. Click the **Select Tool**  icon from the toolbar.
2. Click the first object on the canvas.
3. Press the **SHIFT** key on the keyboard.
4. Without letting go, click the next object on the canvas.

or

1. Click the **Multi-Object Select Tool**  icon from the toolbar.
2. Click the first object on the canvas.
3. Click the second object on the canvas.

or

1. With either the **Select Tool**  or **Multi-Object Select Tool**  chosen in the toolbar, click on an empty portion of the canvas.
2. Without letting go, drag the cursor until a box is formed around the desired objects.
3. When finished, let go.

Selecting Overlapping Objects

1. With the **Select Tool**  chosen in the toolbar, right-click on the object you want to select.
2. Choose the desired object(s) until a checkmark appears to the left.
 - a. To select, choose **Select checked items**.
 - b. To open the right-click menu for a single item, choose **Context menu for selected item**.
 - c. To cancel, left-click outside of the menu.

Modifying Positions/Durations

1. With the **Select Tool**  chosen in the toolbar, click the desired object on the canvas, without releasing the mouse.
2. To change the grid position, move the mouse up or down.
 - a. When finished, release the mouse.
3. To change the start or finish, move the mouse left or right.
 - a. When finished, release the mouse.
4. To change the duration of an activity or width of a shade, release the mouse.
 - a. Move the cursor over the start or finish node or over the left or right edge respectively until a double-arrow appears
 - b. Click the mouse, and without letting go, drag the cursor to the left or right
 - c. When finished, release the mouse.
5. Click anywhere on the canvas to deselect the object.

NOTE: For other methods of modifying positions/duration, see [↗Fine-Tuning/Gestural Controls](#).

Setting Defaults

Default properties can be modified before objects are placed on the canvas to avoid having to modify them afterwards. For example, if you want all milestones to have a certain shape, the desired shape can be chosen ahead of time.

In addition, the duration display format can be changed from days (the default) to any other time unit larger than the schedule's time unit (for more info, see [➔PolyTime/Variable Time Units](#)).

Finally, NetPoint makes use of a scoring system for determining optimal link geometries. For a given link, each possible geometry is given point penalties based on how many grids it passes, and if it crosses over other objects or overlaps with other links. The more intersections and overlaps, the more point penalties and the higher the score. The geometry that receives the lowest possible score will be chosen when creating a link or running an automated layout. The default point penalties are intended to discourage intersection and overlap and favor shorter links. For instructions on optimizing links one-at-a-time, see [➔Optimizing Link Geometries](#).

Steps

For New Objects

1. Click **Objects > Set Default Properties** for New Objects from the menu bar.
2. Choose the desired tab. For embeds, links, shades, and text objects, click the **More** tab.
3. To return to defaults, click **Reset**.
4. When finished, click **OK**.

Duration Display Format

1. Click **Objects > Set Duration Display Format for Activities** from the menu bar.
2. In the sub-menu, choose the desired time unit.

NOTE: Only Days will calculate based on the calendar of the object. Any other time unit will calculate based on total calendar days.

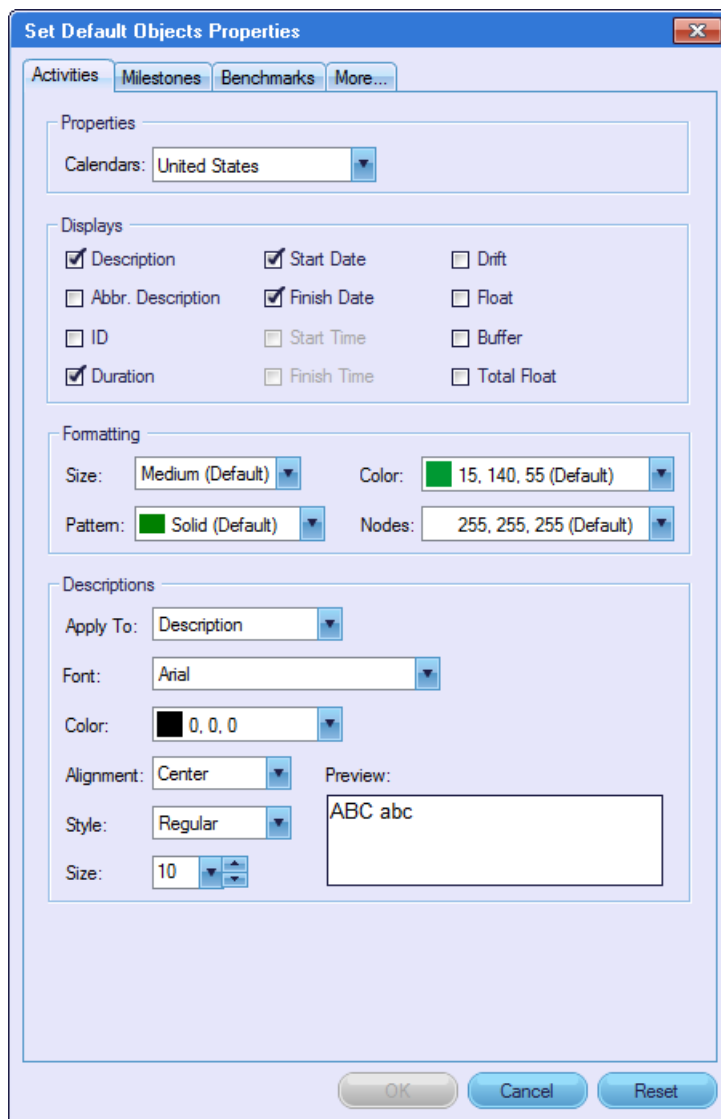
Link Scoring

1. Click **Schedule > Open Layout Manager** from the menu bar, click the arrow to the right of the **Links** tab, and choose **Penalties**.
2. Use the number controls to make any changes as desired.
3. Click **Save** to save the settings and close the manager.

Notes

Activities

For info on individual fields, see [➔Adding/Modifying Activities](#).



The dialog box is titled "Set Default Objects Properties" and has a close button in the top right corner. It features four tabs: "Activities", "Milestones", "Benchmarks", and "More...". The "Milestones" tab is currently selected.

Properties

Calendars: United States

Displays

☒ Description	☒ Start Date	☐ Drift
☐ Abbr. Description	☒ Finish Date	☐ Float
☐ ID	☐ Start Time	☐ Buffer
☒ Duration	☐ Finish Time	☐ Total Float

Formatting

Size: Medium (Default) Color: 15, 140, 55 (Default)

Pattern: Solid (Default) Nodes: 255, 255, 255 (Default)

Descriptions

Apply To: Description

Font: Arial

Color: 0, 0, 0

Alignment: Center Preview: ABC abc

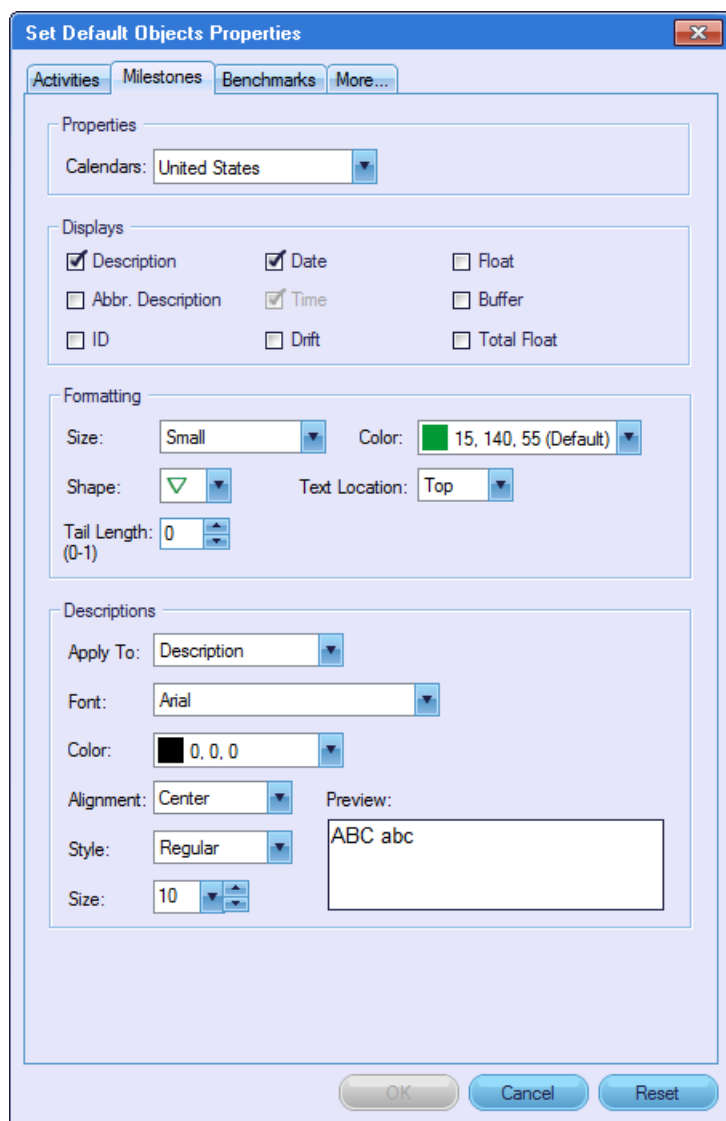
Style: Regular

Size: 10

Buttons: OK, Cancel, Reset

Milestones

For info on individual fields, see [Adding/Modifying Milestones/Benchmarks](#).



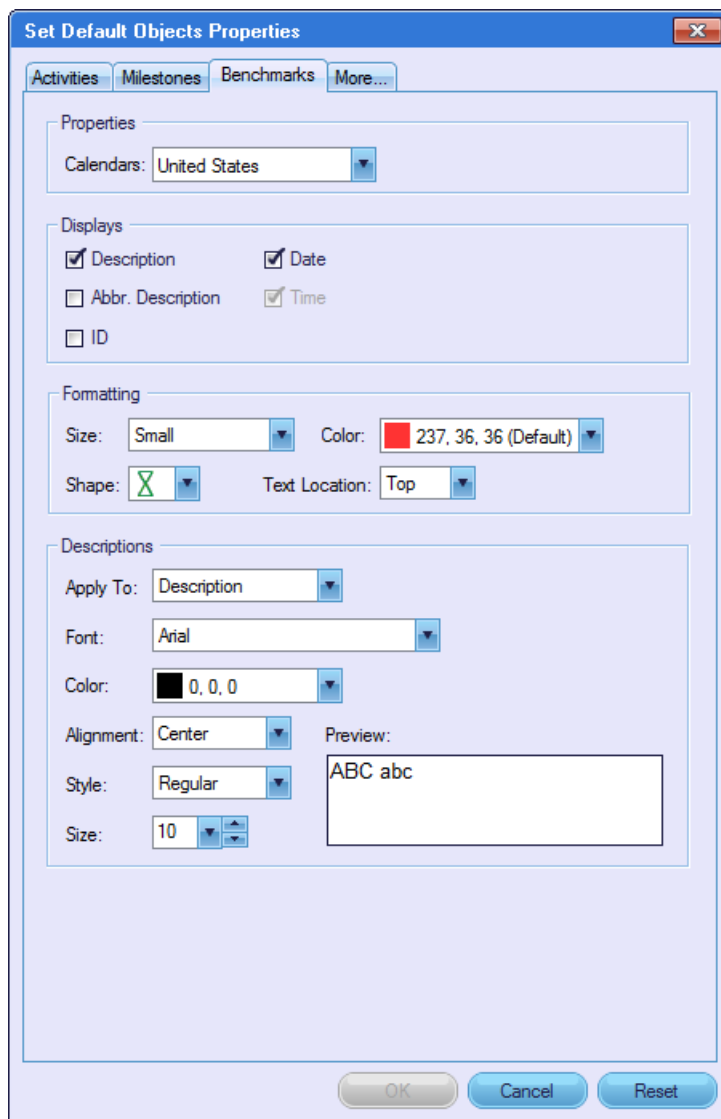
The dialog box is titled "Set Default Objects Properties" and has a close button in the top right corner. It contains four tabs: "Activities", "Milestones", "Benchmarks", and "More...". The "Benchmarks" tab is currently selected. The dialog is organized into four main sections: "Properties", "Displays", "Formatting", and "Descriptions".

- Properties:** A dropdown menu for "Calendars" is set to "United States".
- Displays:** A grid of checkboxes for various display options:
 - ☒ Description, ☒ Date, ☐ Float
 - ☐ Abbr. Description, ☒ Time, ☐ Buffer
 - ☐ ID, ☐ Drift, ☐ Total Float
- Formatting:** Controls for visual appearance:
 - Size:** A dropdown menu set to "Small".
 - Color:** A color picker showing a green square and the text "15, 140, 55 (Default)".
 - Shape:** A dropdown menu showing a green checkmark icon.
 - Text Location:** A dropdown menu set to "Top".
 - Tail Length:** A numeric input field set to "0" with a range of "(0-1)".
- Descriptions:** Controls for text formatting:
 - Apply To:** A dropdown menu set to "Description".
 - Font:** A dropdown menu set to "Arial".
 - Color:** A color picker showing a black square and the text "0, 0, 0".
 - Alignment:** A dropdown menu set to "Center".
 - Style:** A dropdown menu set to "Regular".
 - Size:** A numeric input field set to "10".
 - Preview:** A text box containing the text "ABC abc".

At the bottom of the dialog are three buttons: "OK", "Cancel", and "Reset".

Benchmarks

For info on individual fields, see [Adding/Modifying Milestones/Benchmarks](#).



The dialog box is titled "Set Default Objects Properties" and has a close button (X) in the top right corner. It contains four tabs: "Activities", "Milestones", "Benchmarks", and "More...". The "Activities" tab is selected.

Properties

Calendars: United States

Displays

☒ Description ☒ Date
☐ Abbr. Description ☒ Time
☐ ID

Formatting

Size: Small Color: 237, 36, 36 (Default)
 Shape: X Text Location: Top

Descriptions

Apply To: Description
 Font: Arial
 Color: 0, 0, 0
 Alignment: Center
 Style: Regular
 Size: 10

Preview: ABC abc

Buttons: OK, Cancel, Reset

More

In the **Embeds** section, each property can be set separately for start and finish embeds. To apply one set of selections to both types, click **Copy to Other Type**. For info on individual fields, see [Using Embeds to Create SS and FF Links](#), [Creating FS Links](#), and [Annotations/Markup Tools](#).

Set Default Objects Properties

Activities | Milestones | Benchmarks | More...

Embeds

Apply To: Start Copy To Other ☐ Date
 Behavior: CONSTANT ☐ Time
 Lag Display Position: Above Actv ☐ Lag from Start
☐ Lag to Finish
 Color: 255, 255, 0 (Default)

Links

Logic: Hard ☐ Gap
 Geometry: HVH ☐ Lag Notation

Shades

Orientation: Variable Color: 255, 255, 128 (Default)
 Edge Type: Snap to Period
 Behavior: Regular Object

Text Objects

Font: Arial
 Color: 0, 0, 0
 Alignment: Left Preview: ABC abc
 Style: Bold
 Size: 10
 Effects: ☐ Strikeout ☐ Underline

OK Cancel Reset

Link Scoring

Penalties

Layout Manager

Layouts | Structure | Selection | Placement | **Links** | Statistics

Link Scoring Penalties

Crosses an activity:	25
Crosses another link:	5
Contains a H leg > 50% of gap:	5
Contains a D leg > 50% of gap:	5
V or D leg crosses a grid, excluding intermediate grids:	2
Opposing H, V or D legs overlap:	10
Coinciding H, V or D legs overlap:	2

Field	Definition
Crosses an activity	If a link crosses an activity. Default is equal to 25 points.
Crosses another link	If a link crosses another link. Default is equal to 5 points.
Contains a H leg > 50% of gap	If the horizontal segment of a link has a gap greater than 50% of its duration. Default is equal to 5 points.
Contains a D leg > 50% of gap	If the diagonal segment of a link has a gap greater than 50% of its duration. Default is equal to 5 points.
V or D leg crosses a grid, excluding intermediate grids	If the vertical or diagonal segment of a link crosses a gridline, excluding “intermediate” grids (empty grids with no objects that are added to increase spacing). Default is equal to 2 points.
Opposing H, V, or D legs overlap	If any segment of a link overlaps with any segment of another link that's going in the opposite direction. Default is equal to 10 points.
Coinciding H, V, or D legs overlap	If any segment of a link overlaps with any segment of another link that's going in the same direction. Default is equal to 2 points.

Scores

Term	Definition
Link Score	The link score for a given geometry is equal to the sum of all its penalties. For example, if a link crosses one activity and two other links, its default score would be equal to 35 (25 penalty points for crossing the activity and 5 penalty points for crossing each link).
Sum Link Score	The sum link score is equal to the sum of the link scores for all links in the network.
Net Sum Link Score	The net sum link score is equal to the sum of the link scores for all links in the network excluding links between swim lanes if present
Mean Link Score	The mean link score is equal to the sum of the link scores for all links divided by the number of links in the network.
Activity Score	The activity score is equal to the sum of the link scores for every valid (not redundant) link connected to that activity divided by the number of links connected to that activity

Term	Definition
Mean Activity Score	The mean activity score is equal to the sum of the activity scores for all activities in the network divided by the number of activities in the network.
Layout Score	The layout score is a measure of layout optimization. The lower the score, the tidier the layout relative to link/activity crossings, link overlays, and link length in terms of number of grid crossings. The layout score is equal to the sum link score plus additional point penalties for any of the following: Activity/activity overlap: 50 points Activity/link overlap (H leg): 25 points Activity/milestone/benchmark overlap: 25 point

Activities and Delays

Activities are the building blocks of a schedule. They have a start date and a finish date, represented by end nodes, and they span a certain period of time. In addition to a regular activity, there is also a delay/gain object, which is unique in that it can have a duration of zero. If the duration is zero, both the start and end nodes align to the start of the day (or whichever time unit has been selected). Delays are useful for showing positive or negative impacts to the schedule or for atypical activities. Unlike regular activities, delays cannot have an embedded node or any resources assigned. Delays can be distinguished from regular activities by their bar, which includes vertical gaps from the top to the bottom.

This chapter includes details for the following:

1. Adding/modifying
2. Constraining activities
3. Text annotations
4. Splitting
5. Dissolving
6. Info objects/attachments

Adding/Modifying

Due to the visual nature of NetPoint, adding and modifying activities is best done graphically on the canvas. When the cursor is moved over the canvas, the date and day-of-the-week display above the crosshairs. As the cursor moves around the canvas, the date changes to reflect the exact calendar date that the cursor is at. If the date is a non-working period or holiday (according to the default calendar for the schedule), the text will appear red. If the default calendar were the **Calendar Day** calendar, the date will appear black.

Steps

Adding to the Canvas

1. Click the **Activity**  or **Delay/Gain**  icon in the toolbar.
2. Move the cursor to the desired date, indicated by the text above the crosshairs.
3. Left-click with the mouse.
4. Without releasing, drag the cursor to stretch the duration (displayed in blue) until the intended finish date.
5. Release the mouse. The **Activity Object** window will open.
6. Enter a **Description**.
7. Click **OK**.

Modifying Dates/Durations

1. Click the **Select Tool** icon  in the toolbar.
2. Select the desired activity on the canvas.
3. To change both the start and finish date simultaneously, left-click with the mouse and drag it to the right or left.
 - a. Release the mouse.
4. To change the duration, hover the mouse over the start or finish node until a double-arrow appears.
 - a. Click and drag the node earlier or later until it lands on the desired date.
 - b. Release the mouse.

NOTE: Dates and durations can also be modified in the **Dates** tab. To do so, double-click the desired object, or right-click and choose **Properties**.

Converting Between Activities to Delays

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired activity or delay.
2. Choose **Convert to Delay** or **Convert to Activity**, respectively.

NOTE: If an activity has an embedded node or resources have been assigned to it, the option to convert to a delay will be disabled.

Editing the Description

1. With the **Select Tool**  chosen in the toolbar, double-click the description directly on the canvas. A text box will open.
2. Edit as desired.
3. When finished, click anywhere outside the text box on the canvas.

Positioning Fields on the Canvas

1. Begin at step 1 of [Through GEM Mode](#).

Notes

Properties Tab

The screenshot shows the 'Activity Object' dialog box with the 'Properties' tab selected. The 'Description' field is a large text area. Below it are 'Abbr. Description' and 'ID' fields. The 'ID' field is labeled 'System ID # 1'. There are 'Assign...' buttons for 'Codes' and 'Resources', and an 'Actualize' button for 'Progress'. The 'Estimated Complete' field has a spinner set to 0 and a percentage sign. 'OK' and 'Cancel' buttons are at the bottom.

The **Properties** tab is where general settings are changed.

Field	Definition
Description	A text field for describing the activity. To stack, use the ENTER key. Descriptions must be less than 120 characters for P6 and less than 100 for Open Plan. Otherwise, they'll be truncated upon export.
Abbr. Description	A text field limited to 48 characters to be displayed on the canvas if the description is too long.
ID	A user-entered unique identifier. By default, the ID is not displayed, but it will display before the description, surrounded by brackets, if it is turned on. NetPoint internal ID displayed below.
Delay Type*	A field for categorizing the type of delay. *Only available for delay activities.
Codes	A window for adding or modifying code assignments.

Field	Definition
Resources	A window for adding or modifying resource assignments.
Progress	Whether or not the activity has been started. An activity can only be actualized if a data date has been set. For more info, see Data Dates .
Calculated Complete*	The amount of progressed duration divided by total duration. *Only available for actualized activities.
Estimated Complete	A user-entered estimate of work completed.

Dates Tab

Activity Object

Properties | **Dates** | Display | Format | Info Objects | Risk

Calendar: United States

Planned Duration/Dates

☐ Duration: 1

☐ Start: 7/11/2016 12:00:00 AM

☒ Finish: 7/11/2016 11:59:59 PM

Constraints

☐ Start no earlier than: 7/10/2016 12:00:00 AM

☐ Finish no later than: 7/11/2016 11:59:59 PM

Work Type: Undefined ☐ Enter Duration Formula

Formula

☐ Quantity: 1

☐ Rate: 1 per day

☒ Formula Duration: 1

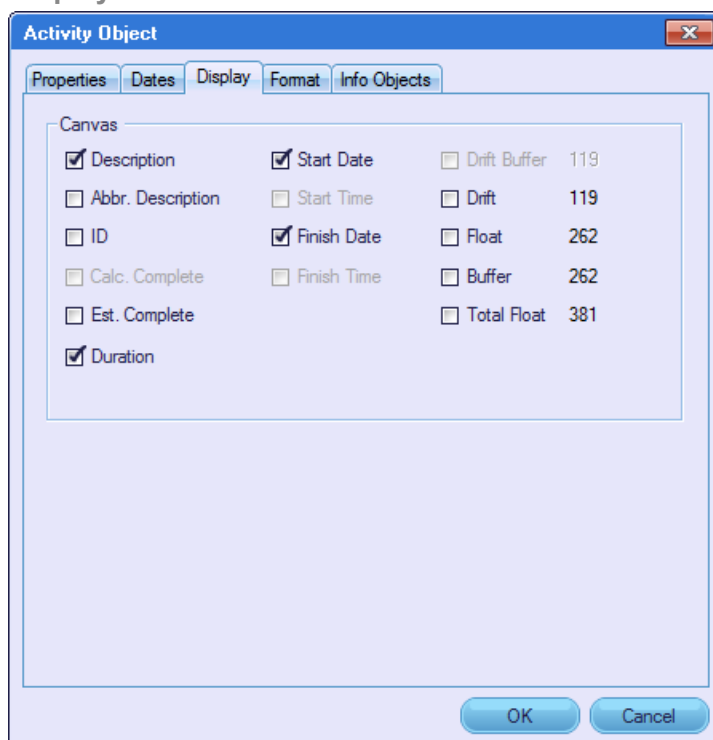
OK Cancel

The **Dates** tab contains all date related fields, such as the start and finish, duration, calendars, and constraints.

Field	Definition
Calendar	The schedule that work on the activity can take place. For more info, see Project Calendars .

Field	Definition
Duration	The amount of time the activity takes to complete.
Start	The date and/or time the activity begins.
Finish	The date and/or time the activity will be finished.
Start Constraint	An optional boundary preventing the activity from starting any earlier than the date selected. For more details, see step 1 of ↗Constraining Activities .
Finish Constraint	An optional boundary preventing the activity from finishing any later than the date selected. For more details, see step 1 of ↗Constraining Activities .
Work Type	A field for capturing the type of work the activity is representing. If physical work is chosen, a duration formula may be entered.
Quantity	The amount of units represented by the activity.
Rate	The number of units performed per day, or unit of time.
Formula Duration	The duration, computed as quantity x rate.

Displays Tab



The **Displays** tab contains checkboxes for hiding or showing different fields on the canvas.

Field	Definition
Description	The description of the activity. Only one of the description fields can be displayed at any given time.
Abbr. Description	A shorter description for viewing on the canvas if the description is too long.
ID	A user-entered unique identifier. If no description field is displayed but the ID is, then the ID will inherit the formatting of the abbreviated description.
Calc. Complete	The amount of progressed duration divided by total duration. Only available for actualized activities.
Est. Complete	A user-entered estimate of work completed.
Duration	The amount of time the activity takes to complete. When actualized, Duration is replaced by Progressed and Remaining durations.
Start Date	The date the activity begins.
Start Time	The time the activity begins; only enabled if the schedule time unit is smaller than days.
Finish Date	The date the activity will be completed.
Finish Time	The time the activity will be completed; only enabled if the schedule time unit is smaller than days.
Drift Buffer	The amount of time an activity can gain before advancing the finish of its nearest predecessor.
Drift	The amount of time an activity can gain before advancing the start of the project.
Float	The amount of time an activity can be delayed before delaying the completion of the project.
Buffer	The amount of time an activity can be delayed before delaying the start of its nearest successor.
Total Float	Drift + Float.

NOTE: Drift, float, buffer, and total float are displayed in that order below the activity bar on the same line. They may be distinguished from one another by the shape of their bounding boxes: drift points to the left, or project start; float and buffer point to the right, or project completion; and total float points in both directions. Either buffer or total float may be displayed, but not both at the same time. If too many fields are being displayed for the duration of the activity, the data may stack under and perpendicular to the bar. To customize the position of the fields on the canvas, see [Repositioning Fields \(GEM\)](#)

Format Tab

Activity Object

Properties Dates Display **Format** Info Objects

Bar and Nodes

Size: Medium (Default) Color: 15, 140, 55 (Default)

Pattern: Solid (Default) Nodes: 255, 255, 255 (Default)

Descriptions

Apply To: Description

Font: Arial

Color: 0, 0, 0 (Default)

Alignment: Center

Style: Regular

Size: 10

Preview: ABC abc

OK Cancel

The **Format** tab contains controls for changing the look of the object on the canvas.

Bar & Nodes

Field	Definition
Size	The size of the bar.
Pattern	The pattern of the bar. NOTE: Patterns display best when printed. Depending on your screen resolution, page size, zoom factor, and activity sizes, patterns may show up inconsistently on screen. For best results on screen, use the large bar size.
Color	The color of the bar. NOTE: Avoid red, since red is used to signal criticality. For more info, see Criticality.
Node	The color of the start and end nodes.

Descriptions

The description and the abbreviated description can be styled/formatted differently or the same. To make them different, choose **Description** from the dropdown, customize the options, and then choose **Abbr. Description** from the dropdown, and customize the options for it. The settings will be saved, and you can switch between the two to confirm.

To make them the same, choose **Both** from the dropdown, and then customize the options. Note that choosing both will only overwrite the options that receive a change. For example, imagine the description's font has been formatted to Arial and the abbreviated description's font has been formatted to Tahoma. If **Both** is chosen from the dropdown and the font is set to Calibri, then the font for both the description and the abbreviated description will be set to Calibri, but the color, alignment, style, and size will be unaffected, since they didn't receive any change. This can be confirmed by switching between the description and the abbreviated description and reviewing them without losing any of the changes.

Field	Definition
Font	The font for the description, abbreviated description, or both.
Color	The color of the description, abbreviated description, or both.
Alignment	The alignment of the description, abbreviated description, or both.
Style	The style of the description, abbreviated description, or both.
Size	The size of the description, abbreviated description, or both.

Constraining Activities

A constraint is a way to restrict the range that an object may move in time. If an object has been constrained, any attempt to move it beyond its constraint date will be prevented. Constraints are represented by exclamation marks to the left or the right of the date they're constraining.

NetPoint supports two types of constraints: a no-earlier-than (NET) constraint, also called a start-no-earlier-than (SNE) constraint, which defines the earliest date an object may backslide in time; and a no-later-than (NLT) constraint, also called a finish-no-later-than (FNL) constraint, which defines the latest date an object may slide forward in time. In the case of activities, no-earlier-than constraints are applied to start dates and no-later-than constraints are applied to finish dates. To set constraints for more than one activity or milestone at the same time, see [➔Bulk-Editing Objects](#).

NOTE: Once set for the first time, if the constraint is subsequently turned off, NetPoint continues to remember the date for future use.

Steps

1. With the **Select Tool**  chosen in the toolbar, double-click the desired activity, or right-click it and choose **Properties**. The **Activity Object** window will open.
2. Click the **Dates** tab.
3. To apply a NET constraint, check the box next to **Start no earlier than**. The constraint will default to the current start date.
4. To apply a NLT constraint, check the box next to **Finish no later than**. The constraint will default to the current finish date.
5. To change the constraint date, enter a new value or click the down arrow button. Selecting a date on the calendar closes the graphic calendar picker.
6. When finished, click **OK** to exit the window and apply the constraint. An exclamation mark will show up where the constraint was applied.

Text Annotations

Annotations can be used to add text directly to a specific object. Text can include a note, a comment, or a custom data field. When you add an annotation to an object, NetPoint is actually just creating a text box that is permanently tied to its host. As the object is moved, the annotation moves too, along with other object data, like dates. Only one annotation may be added to an object at a time. Because annotations can be re-positioned on the canvas without having to enter **Global Edit Mode**, it may be desirable to use them instead of descriptions.

NOTE: Annotations will also fade with their host objects.

Steps

Adding

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired activity.
2. Choose **Add Annotation at** and then **Start**, **Center**, or **Finish**. The cursor will change to the crosshairs, and the **Text Object** icon  in the toolbar will become pressed.
3. Left-click with the mouse at the desired date, indicated by the text above the crosshairs. (Dragging the crosshairs will have no consequence on the shape of the text box).
4. Release the mouse. The edit text window will open.
5. Begin typing.
6. When finished, click anywhere outside the text box on the canvas.

NOTE: Once an annotation is added, Add Annotation at will go disabled, as only one annotation may be added to an object, and the anchor point may never be changed.

Modifying Annotation Text

1. With the **Select Tool**  chosen in the toolbar, right-click on the text and choose **Edit Text**.
2. Edit as desired.
3. When finished, click anywhere outside the text box on the canvas.

Modifying Annotation Font

1. With the **Select Tool**  chosen in the toolbar, double-click the text or right-click and choose **Properties**. The **Text Object** window will open.
2. When finished making changes, click **OK**.

Hiding Annotations

1. With the **Select Tool**  chosen in the toolbar, right-click on the activity and choose **Hide Annotation**.
2. Click on the canvas to deselect the activity.

Deleting

1. With the **Select Tool**  chosen in the toolbar, select the text on the canvas and press the **Delete** key, or right-click on the host activity and choose **Delete Annotation**.
2. Click on the canvas to deselect the activity.

Splitting

Sometimes, it might be helpful divide activity into two separate activities. For example, an in-progress activity could be split at the data date, separating the finished portion from the remaining portion as two distinct activities. Any un-actualized activity, or un-actualized portion of an activity, can be split.

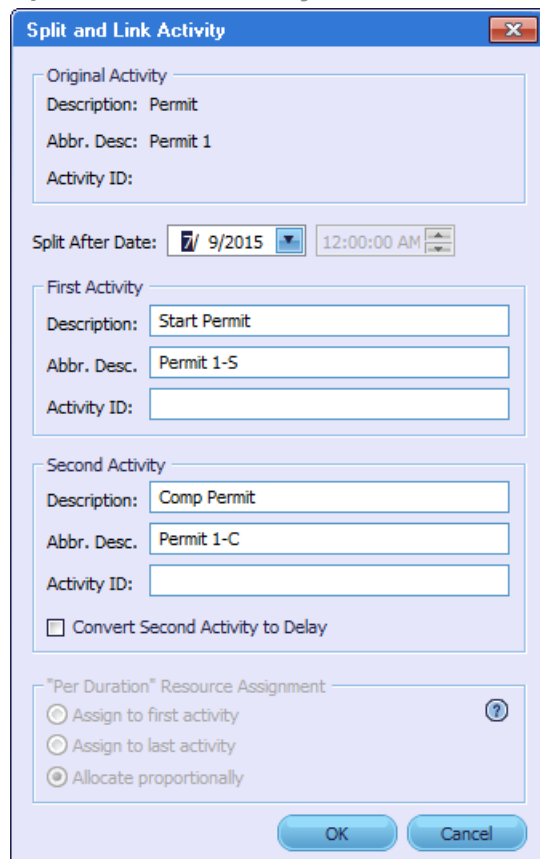
After splitting, NetPoint will automatically maintain any links directly into and out of the original activity, while creating a new finish-to-start link between the new activities. If the original activity had an embedded node before the split date, it will be part of the first activity. If the embed falls after the split date, it will be part of the second activity. Either way, the embed's date will always be maintained. The lag may or may not be recalculated, depending on its type and behavior. Any unlinked embeds will be deleted. Both activities will maintain the original bar color and node color, display settings, any code assignments, and any parent hammock. However, annotations and target information will only remain with the first activity. Any fields that were re-positioned will be reset for both.

Steps

1. With the **Select Tool**  chosen in the toolbar, move the cursor over a given activity, stopping at the date desired for the split.
2. Right-click and choose **Split Activity**. The **Split and Link Activity** window will open.
3. When finished, click **OK**.

Notes

Split and Link Activity Window



Split and Link Activity

Original Activity
 Description: Permit
 Abbr. Desc: Permit 1
 Activity ID:

Split After Date: 9/2015 12:00:00 AM

First Activity
 Description: Start Permit
 Abbr. Desc: Permit 1-S
 Activity ID:

Second Activity
 Description: Comp Permit
 Abbr. Desc: Permit 1-C
 Activity ID:
☐ Convert Second Activity to Delay

"Per Duration" Resource Assignment
☐ Assign to first activity
☐ Assign to last activity
☒ Allocate proportionally

OK Cancel

First Activity

Field	Definition
Description	The original activity's description, prefixed with "Start".
Abbr. Desc.	The original activity's abbreviated description, appended with "S".
Activity ID	The original activity's ID prefixed with "S".

Second Activity

Field	Definition
Description	The original activity's description, prefixed with "Comp"
Abbr. Desc.	The original activity's abbreviated description, appended with "C".
Activity ID	The original activity's ID prefixed with "C".

"Per Duration" "Resource Assignment

Field	Definition
Assign to first activity	When chosen, any resources assigned "per duration" are assigned to the first activity. Resources assigned "per day" are maintained on both.
Assign to last activity	When chosen, any resources assigned "per duration" are assigned to the second activity. Resources assigned "per day" are maintained on both.
Assign proportionally	When chosen, any resources assigned "per duration" are assigned to both activities. For example, if 10 carpenters were assigned per duration and it were split in 1/3, 4 would be assigned per duration to the first activity and 6 would be assigned to the second activity (if decimals arise, the first activity's quantity will be rounded-up and the second activity's will be rounded-down). Resources assigned "per day" are maintained on both.

Dissolving

If the need arises to delete an activity without deleting its logic relationships, dissolving can be useful. Dissolving an activity or delay will completely remove it from the network, while maintaining links to and from its predecessors and successors. Any resources or codes assigned to the dissolved activity, and any lags on links to or from the dissolved activity will be ignored. Any links to or from embedded nodes will be treated as FS links to or from the containing activity. If dissolving an activity results in an embedded node in either a predecessor or successor having no remaining links, that embed will be removed.

NOTE: No attempt will be made to correct any negative gaps resulting from the dissolve operation.

Steps

1. With the **Select Tool**  chosen in the toolbar, right-click on an activity or delay and select **Dissolve Activity** or **Dissolve Delay**, respectively.

Notes

Once dissolved, a new FS link will be created from each predecessor to each successor. None of these new links will have lags. The link geometry is chosen automatically to minimize link overlays on the display:

- For one-to-many relationships, VH links are used.
- For one-to-one and many-to-one relationships, HV links are used.
- For many-to-many relationships, VDV links are used.

Info Objects/Attachments

Info objects are external files or attachments. Adding an info object to an activity is a way to embed a document on the activity for viewing. An activity with one or more info objects will have an “i” icon placed over the bar to indicate a file has been attached. However, if the duration is reduced sufficiently to where the end nodes begin to overlap, the icon will be hidden. Info objects can be added, viewed, and deleted from the **Info Objects** tab of the **Activity Object** window.

Steps

Adding

1. With the **Select Tool**  chosen in the toolbar, double-click an activity, or right-click and choose **Properties**.
2. Click the **Info Objects** tab.
3. Click the **New** button. The **Info Object Properties** window will open.
4. Click the **Browse** button. (If you'd like to attach another schedule, first select **NetPoint project**).
5. Choose a file and click **Open**.
6. If desired, enter a new **Description**.
7. Click **OK** to save and close the **Info Object Properties** window.
8. To add another, repeat steps 3-7.
9. When finished, click **OK**.

NOTE: If the iObject is copied and linked to the project folder, the filepath shown will still display the original location, so that you can always remember where the original file was located.

Modifying

1. With the **Select Tool**  chosen in the toolbar, double-click an activity, or right-click and choose **Properties**.
2. Click the **Info Objects** tab.
3. Select the file in the list click the **Modify** button. The **Info Object Properties** window will open.
4. Modify as desired.
5. Click **OK** to save and close the **Info Object Properties** window.
6. To modify another, repeat steps 3-5.
7. When finished, click **OK**.

Deleting

1. With the **Select Tool**  chosen in the toolbar, double-click an activity, or right-click and choose **Properties**.
2. Click the **Info Objects** tab.
3. Select the file in the list and click the **Delete** button.
4. In the confirmation, click **Yes**.
5. To delete another, repeat steps 3-4.
6. When finished, click **OK**.

Opening/Viewing

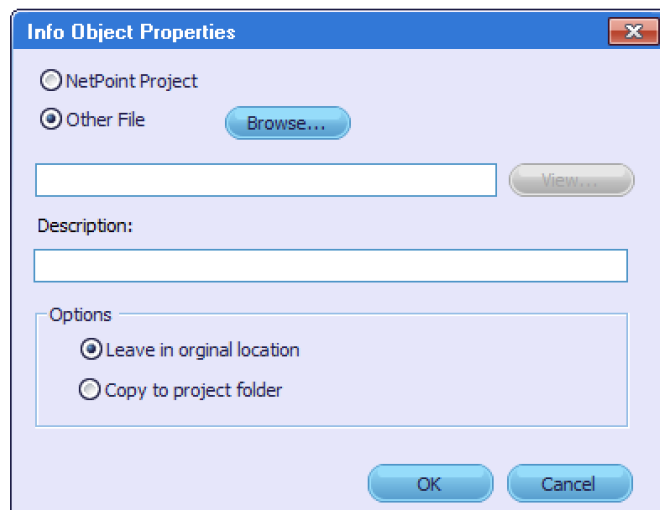
1. With the **Select Tool**  chosen in the toolbar, double-click an activity, or right-click and choose **Properties**.
2. Click the **Info Objects** tab.
3. Select the file in the list and click the **View** button. The file will open outside of NetPoint.
4. To view another, repeat step 3.
5. When finished, click **OK**.

NOTE: If the info object is a NetPoint schedule, the current schedule will be saved automatically. If you don't wish to save the schedule, click **Cancel** in the confirmation that opens.

Notes

Properties

The **Properties** window is where the general settings are changed.



Field	Definition
Source	Determines the type of file to attach.
Filepath	Displays the filepath of the original file, regardless of whether or not it's copied to to the project folder.
Description	A short description for displaying on the canvas. By default, the same as the filename.
Options	Provides the option to copy the file to the project folder so it's still viewable after zipping and sending the project.

Hammocks

Hammocks are a type of activity used to represent or summarize a group of other activities, milestones, and/or benchmarks. They are different from regular activities in that they have no attributes of their own (like dates or info objects), except for the descriptions/ID, and they don't accept resources or embedded nodes. If an embed is added, it will display as red. Furthermore, hammocks cannot be linked to other objects, and member objects can only belong to one hammock at a time. When exported to P6, any unconnected milestones or benchmarks will be connected to the LOE with a start-to-start or finish-to-finish relationship accordingly.

Hammocks are depicted on the canvas by a thick, diagonally hashed bar, beginning at the earliest start date and ending at the latest finish date of their member objects. The object itself will automatically reflect in real-time any changes in dates and durations to any of their member objects, and they will also display critical if any member objects are critical. Once created, hammocks can also be hidden on the canvas, and member objects can be added and removed. Instructions for all are provided below.

NOTE: If a hammock includes only delay/gain objects with total duration of zero, the hammock still displays the duration as one, because hammocks cannot have zero duration.

Steps

Creating Hammocks

1. With the **Select Tool**  chosen in the toolbar, select the first activity, milestone, or benchmark on the canvas.
2. To add more objects, press and hold the **Shift** key and select a second activity, milestone, or benchmark on the canvas to include. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all the objects to include in the hammock.
3. With all the objects still selected, click the **Hammock** icon  in the toolbar.
4. Use the up and down arrow buttons or enter the grid number to place the hammock at and click OK. **NOTE: A new gridline will be inserted for the hammock and all objects below it moved down.**
5. Enter a description and any other details.
6. When finished, click **OK** to close the window and finish creating the hammock.

Adding Objects to Hammocks

1. With the **Select Tool**  chosen in the toolbar, select the hammock on the canvas.
2. Press and hold the **Shift** key and select each activity, milestone, or benchmark to add to the hammock. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all the objects you want to add to the hammock.
3. Click the **Hammock** icon  in the toolbar. If one or more objects to be added already belong to a different hammock, a prompt will open. See 3a.
 - a. Choose **Add to Hammock** and click **Apply** for all objects you wish to remove and add. Otherwise, choose **Leave in Existing Hammock**.

NOTE: If multiple hammocks were selected when pressing the hammock icon in the toolbar, deselect all but one and retry the operation.

Removing Objects from Hammocks

1. With the **Select Tool**  chosen in the toolbar, select the first activity, milestone, or benchmark on the canvas to remove.
2. To remove more than one object, press and hold the **Shift** key and select each additional activity, milestone, or benchmark on the canvas. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all the objects you want to remove from the hammock.
3. Click the down arrow to the right of the **Hammock** icon  in the toolbar.
4. Click **Remove Object(s) from Hammock(s)**. A confirmation will open.
5. Press **Remove** to confirm. Hammock lengths will adjust accordingly

Hiding Hammocks

1. With the **Select Tool**  chosen in the toolbar, select the hammock you want to hide.
2. To hide multiple hammocks, press and hold the **Shift** key and select each additional hammock on the canvas. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all the hammocks you want to hide.
3. Click the down arrow to the right of the **Hammock** icon  in the toolbar.
4. Choose **Hide Hammock(s)**.

NOTE: If no member objects or hammocks have been selected, choosing this menu option will hide *all* hammocks in the schedule.

Showing Hammocks

1. With the **Select Tool**  chosen in the toolbar, select a member activity from the hammock you want to show.
2. To show more than one hidden hammock, press and hold the **Shift** key and select additional member objects on the canvas. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all the member objects of the hammocks you want to show.
3. Click the down arrow to the right of the **Hammock** icon  in the toolbar.
4. Choose **Show Hidden Hammock(s)**.

NOTE: To show *all* hidden hammocks, including hammocks which you may not remember the member objects of, choose this menu option without any objects selected on the canvas.

Including/Excluding from Resource Display

1. With the **Select Tool**  chosen in the toolbar, right-click the hammock.
2. Choose **Show Resources for Hammock ONLY** or **Hide Resources for Hammock ONLY**.

Notes

Properties Tab

The screenshot shows the 'Hammock Object' dialog box with the 'Properties' tab selected. The dialog includes the following elements:

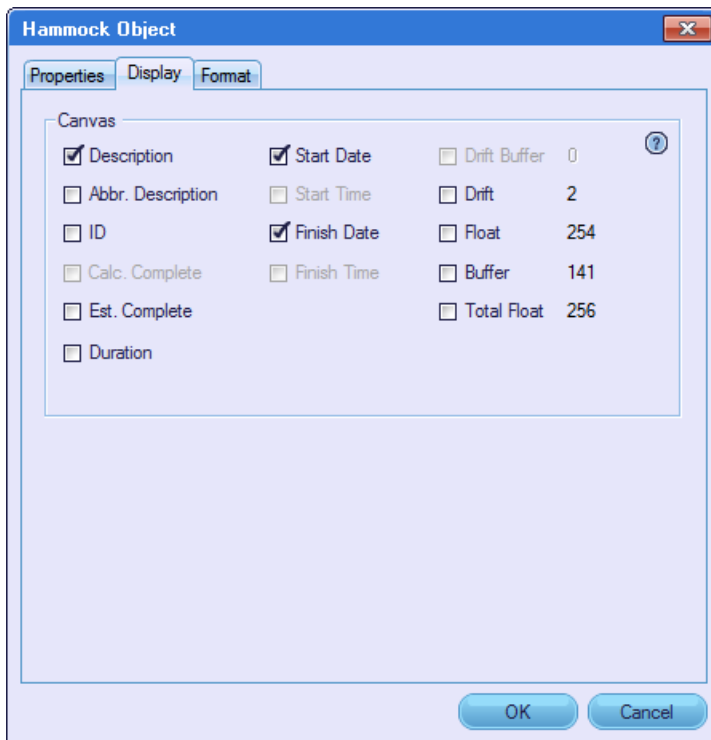
- Description:** A large text area for the main description.
- Abbr. Description:** A smaller text area for an abbreviated description.
- ID:** A text field for a unique identifier, with the text 'System ID # 113' displayed below it.
- Codes:** A button labeled 'Assign...'.
- Resources:** A button labeled 'Assign...'.
- Progress:** A button labeled 'Actualize'.
- Estimated Complete:** A spinner box set to '0' followed by a '%' symbol.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

The **Properties** tab is where general settings are changed.

Field	Definition
Description	A text field for describing the hammock. To stack, use the ENTER key. Descriptions must be less than 120 characters for P6 and less than 100 for Open Plan. Otherwise, they'll be truncated upon export.
Abbr. Description	A text field limited to 48 characters to be displayed on the canvas if the description is too long.
ID	A user-entered unique identifier. By default, the ID is not displayed, but it will display before the description, surrounded by brackets, if it is turned on. NetPoint internal ID displayed below.
Codes	A window for adding or modifying code assignments.
Resources	Hammocks can not be assigned resources.
Progress	A hammock cannot be actualized.

Field	Definition
Calculated Complete	The amount of progressed duration divided by total duration. Only available once any of its member objects are actualized. A warning will be shown if some member objects are not yet actualized as they will throw off the calculations.
Estimated Complete	A user-entered estimate of work completed.

Displays Tab



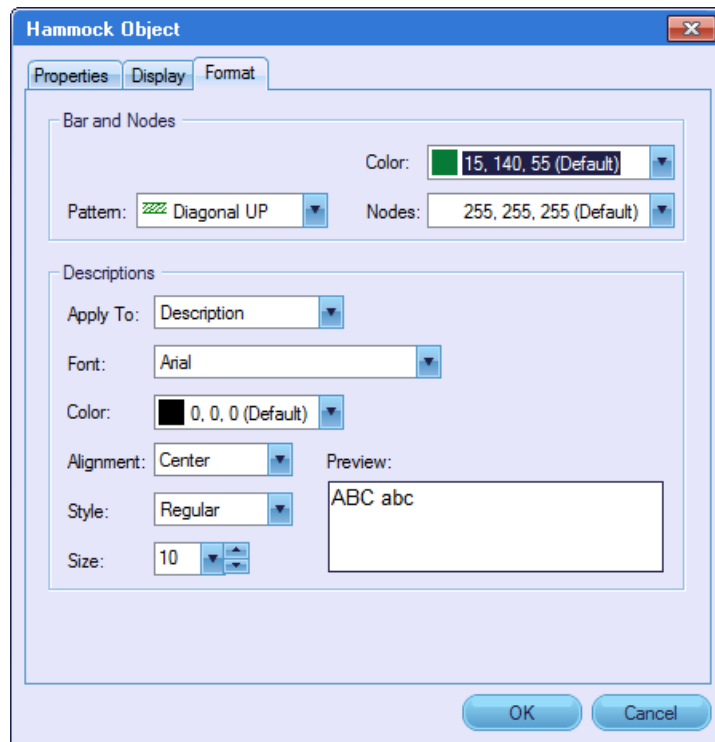
The **Displays** tab contains checkboxes for hiding or showing different fields on the canvas.

Field	Definition
Description	The description of the hammock. Only one of the description fields can be displayed at any given time.
Abbr. Description	A shorter description for viewing on the canvas if the description is too long.
ID	A user-entered unique identifier. If no description field is displayed but the ID is, then the ID will inherit the formatting of the abbreviated description.

Field	Definition
Calc. Complete	The amount of progressed duration divided by total duration. Only available if at least one member has been actualized.
Est. Complete	A user-entered estimate of work completed.
Duration	The amount of time the hammock takes to complete. When actualized, Duration is replaced by Progressed and Remaining durations.
Start Date	The date the earliest member object begins.
Start Time	The time the earliest member object begins; only enabled if the schedule time unit is smaller than days.
Finish Date	The date the latest member object will be completed.
Finish Time	The time the latest member object will be completed; only enabled if the schedule time unit is smaller than days.
Drift Buffer	The amount of time the hammock can gain before advancing the finish of its nearest predecessor. Equal to the lowest of all member objects' drift buffers.
Drift	The amount of time the hammock can gain before advancing the start of the project. Equal to the lowest of all member objects' drifts.
Float	The amount of time the hammock can be delayed before delaying the completion of the project. Equal to the lowest of all member objects' floats.
Buffer	The amount of time the hammock can be delayed before delaying the start of its nearest successor. Equal to the lowest of all member objects' buffers.
Total Float	Drift + Float. Equal to the lowest of all member objects' total floats.

NOTE: Drift, float, buffer, and total float are displayed in that order below the hammock bar on the same line. They may be distinguished from one another by the shape of their bounding boxes: drift points to the left, or project start; float and buffer point to the right, or project completion; and total float points in both directions. Either buffer or total float may be displayed, but not both at the same time. If too many fields are being displayed for the duration of the activity, the data may stack under and perpendicular to the bar. To customize the position of the fields on the canvas, see [Repositioning Fields](#).

Format Tab



The **Format** tab contains controls for changing the look of the object on the canvas.

Bar and Nodes

Field	Definition
Pattern	The pattern of the bar. NOTE: Patterns display best when printed. Depending on your screen resolution, page size, zoom factor, and activity sizes, patterns may show up inconsistently on screen. For best results on screen, use the large bar size.
Color	The color of the bar.
Node	The color of the start and end nodes.

Descriptions

The description and the abbreviated description can be styled/formatted differently or the same. To make them different, choose **Description** from the dropdown, customize the options, and then choose **Abbr. Description** from the dropdown, and customize the options for it. The settings will be saved, and you can switch between the two to confirm.

To make them the same, choose **Both** from the dropdown, and then customize the options. Note that choosing both will only overwrite the options that receive a change. For example, imagine the description's font has been formatted to Arial and the abbreviated description's font has been formatted to Tahoma. If **Both** is chosen from the dropdown and the font is set to Calibri, then the font for both the description and the abbreviated description will be set to Calibri, but the color, alignment, style, and size will be unaffected,

since they didn't receive any change. This can be confirmed by switching between the description and the abbreviated description and reviewing them without losing any of the changes.

Field	Definition
Font	The font for the description, abbreviated description, or both.
Color	The color of the description, abbreviated description, or both.
Alignment	The alignment of the description, abbreviated description, or both.
Style	The style of the description, abbreviated description, or both.
Size	The size of the description, abbreviated description, or both.

Copy/Paste

Format options can be copied and pasted between hammocks and activities/delays, and hammocks can be used as sources for target data element positioning in Global Edit Mode (and vice versa). For more info, see [↗Repositioning Fields](#).

Milestones and Benchmarks

Milestones and benchmarks are zero duration events that are used to represent key dates or moments in the schedule. Two types are available: a start milestone/benchmark, and a finish milestone/benchmark. The start type aligns to the beginning of the day or time unit (by default, 9:00 am), and should be used for events which will precede an activity or represent an initiation; the finish type aligns to the end of the day or time unit (by default, 5:00 pm), and should be used for events which will succeed an activity or represent a termination. In addition, both milestones and benchmarks can be represented by a variety of shapes; however, each one has its own set of behaviors.

Milestones, like activities, will track along with the network, once they've been linked to any predecessors or successors. In addition, a start milestone will snap to its earliest successor and a finish milestone will snap to its latest predecessor, always maintaining a zero gap link. If a start milestone has no successors (for example, if it's been linked to an activity that precedes it in time), it will not snap and erase any gap, but it will continue to track. If a finish milestone has no predecessors, it will also not snap but continue to track. While such relationships are discouraged and illogical, they may serve as a workaround snapping, where desired.

Benchmarks, unlike activities and milestones, will remain fixed at whatever date they are placed, regardless of whether or not they've been linked to any predecessor or successors. This means that even as an activity is pushed earlier or later than it, a negative gap link will open up as it passes. Although by default, benchmarks show up red, they are not and cannot be critical. This is because they do not have any float or drift themselves. In fact, any activities connected to the benchmark will have their float and drift calculated to the benchmark rather than to the project start and completion. This allows benchmarks to be used to divide up float and allocate it to different stages instead of having it accumulate undisturbed from schedule completion.

NOTE: If you're planning on running a GPM risk analysis, benchmarks should be converted to milestones for best practice. For more info, see [Converting Between Milestones and Benchmarks](#).

This chapter includes details for the following:

1. Adding/modifying
2. Constraining milestones
3. Text annotations
4. Info objects/attachments

Adding/Modifying

Due to the visual nature of NetPoint, adding and modifying milestones and benchmarks is best done graphically on the canvas. When the cursor is moved over the canvas, the date and day-of-the-week display above the crosshairs. As the cursor moves around the canvas, the date changes to reflect the exact calendar date that the cursor is at. If the date is a non-working period or holiday (according to the default calendar for the schedule), the text will appear red. If the default calendar were the **Calendar Day** calendar, the date will appear black.

Steps

Adding to the Canvas

1. Click either the **Start Milestone** , **Finish Milestone** , **Start Benchmark Icon**, or **Finish Benchmark**  icon in the toolbar.
2. Move the cursor to the desired date, indicated by the text above the crosshairs.
3. Left-click with the mouse. The **Milestone Object** or **Benchmark Object** window will open.
4. Enter a **Description**.
5. Click **OK**.

Modifying Dates

1. Click the **Select Tool** icon  in the toolbar.
2. Select the desired event on the canvas.
3. Left-click with the mouse, and without releasing, drag it to the right or left until it lands on the desired date.
4. Release the mouse.

NOTE: Dates can also be modified using the arrow keys on the keyboard or in the **Dates** tab. To do so, double-click the desired object, or right-click and choose **Properties**.

Converting Between Milestones and Benchmarks

1. Click the **Select Tool** icon  in the toolbar.
2. Select the desired event on the canvas.
3. To convert between start and finish events, choose **Convert to Start** or **Convert to Finish** accordingly.
4. To convert between milestones and benchmarks, choose **Convert to Milestone** or **Convert to Benchmark** accordingly.

NOTE: Any existing constraints will be removed before converting. Once converted, the date is adjusted to maintain the current position of the object. For example, when a start milestone at 3/30/2010 is converted to a finish milestone, the date is adjusted to 3/29/2010, as the start of 3/30 and the end of 3/29 are at the same position on the schedule. This also allows the type of a benchmark to be changed without affecting float or drift calculations.

Editing the Description

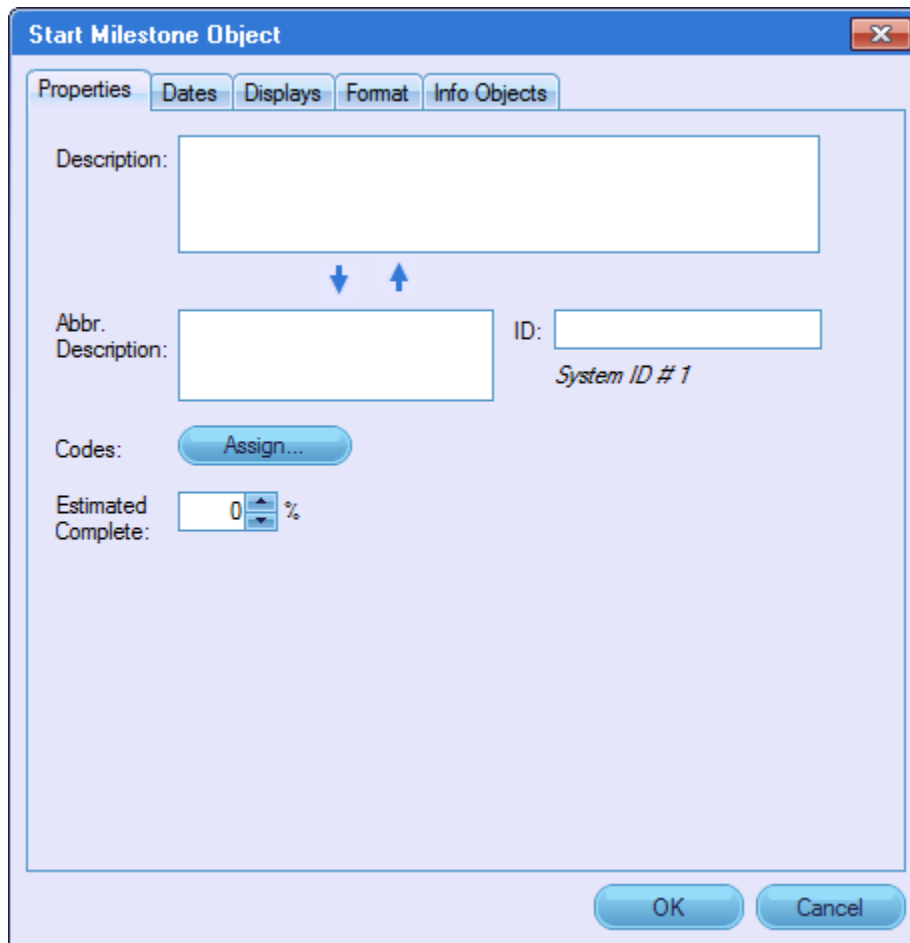
1. With the **Select Tool**  chosen in the toolbar, double-click the description directly on the canvas. A text box will open.
2. Edit as desired.
3. When finished, click anywhere outside the text box on the canvas.

Positioning Fields on the Canvas

1. Begin at step 1 of [Through GEM Mode](#).

Notes

Properties Tab



The screenshot shows the 'Start Milestone Object' dialog box with the 'Properties' tab selected. The dialog has a title bar with a close button. Below the title bar are five tabs: 'Properties', 'Dates', 'Displays', 'Format', and 'Info Objects'. The 'Properties' tab contains the following fields and controls:

- Description:** A large text input field.
- Abbr. Description:** A text input field.
- ID:** A text input field with the value 'System ID # 1'.
- Codes:** A button labeled 'Assign...'.
- Estimated Complete:** A spinner control showing '0' followed by a '%' sign.

At the bottom of the dialog are 'OK' and 'Cancel' buttons.

The **Properties** tab is where general settings are changed.

Start Benchmark Object

Properties | Dates | Display | Format | Info Objects

Description:

Abbr. Description: ID:
System ID # 2

Codes:

Estimated Complete: %

Field	Definition
Description	A text field for describing the milestone/benchmark. To stack, use the ENTER key. Descriptions must be less than 120 characters for P6 and less than 100 for Open Plan. Otherwise, they'll be truncated upon export.
Abbr. Description	A text field limited to 48 characters to be displayed on the canvas if the description is too long.
ID	A user-entered unique identifier. By default, the ID is not displayed, but it will display before the description, surrounded by brackets, if it is turned on. NetPoint internal ID displayed below.
Codes	A window for adding or modifying code assignments.
Estimated Complete	A user-entered estimate of work completed.

Dates Tab

The screenshot shows the 'Start Milestone Object' dialog box with the 'Dates' tab selected. The dialog has a title bar with a close button. Below the title bar are five tabs: 'Properties', 'Dates', 'Displays', 'Format', and 'Info Objects'. The 'Dates' tab contains the following elements:

- Calendar:** A dropdown menu showing 'United States'.
- Date/Time:** Two input fields showing '7/ 9/2016' and '12:00:00 AM', followed by an 'Actualize' button.
- Constraints:** A section with two checkboxes and corresponding date/time inputs:
 - ☐ Start no earlier than: 7/ 9/2016 12:00:00 AM
 - ☐ Start no later than: 7/ 9/2016 12:00:00 AM
- Contract Date:** A checkbox and a date/time input field showing '7/ 9/2016' and '12:00:00 AM'.
- Change Log:** A table with the following headers: 'Description', 'Contra...', 'Variance', 'Date L...', 'Associa...', and 'Associa...'. The table body is empty.

At the bottom of the dialog are 'OK' and 'Cancel' buttons.

The **Dates** tab is where date-related settings are changed.

Start Benchmark Object

Properties Dates Display Format Info Objects

Calendar: United States

Date/Time: 10/30/2016 12:00:00 AM Actualize

Contract Date

☐ 10/30/2016 12:00:00 AM

Change Log

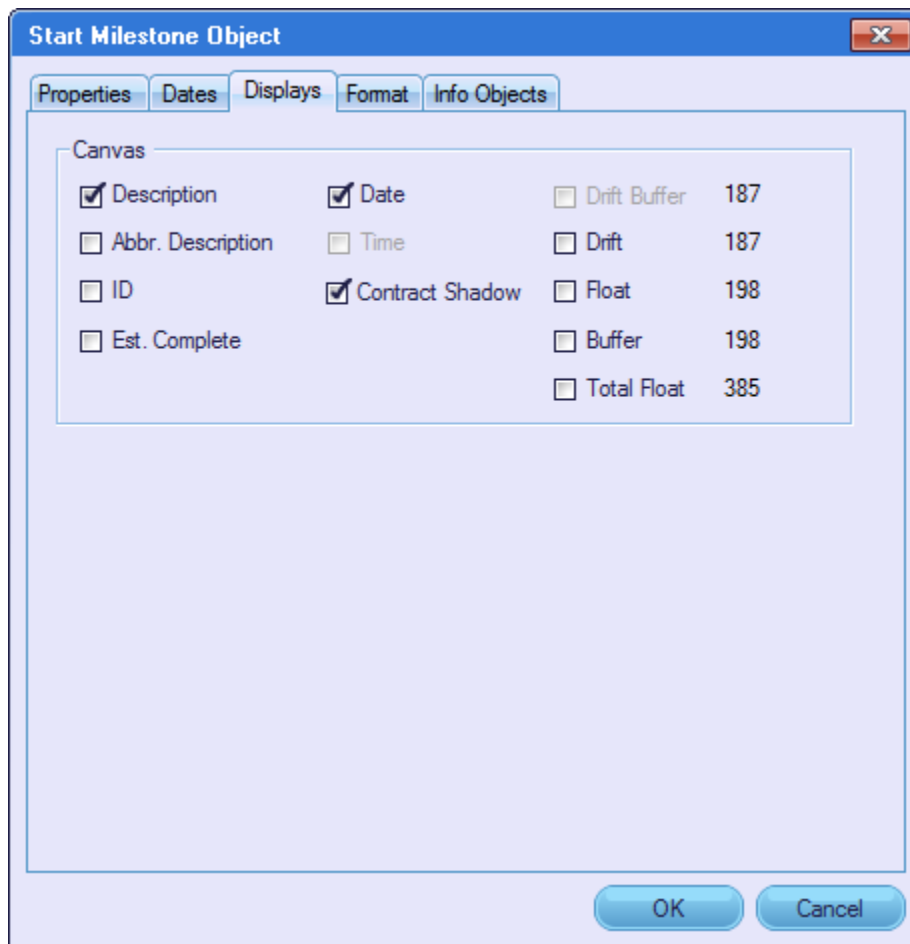
Description	Contract D...	Variance	Associated...	Associated ...

OK Cancel

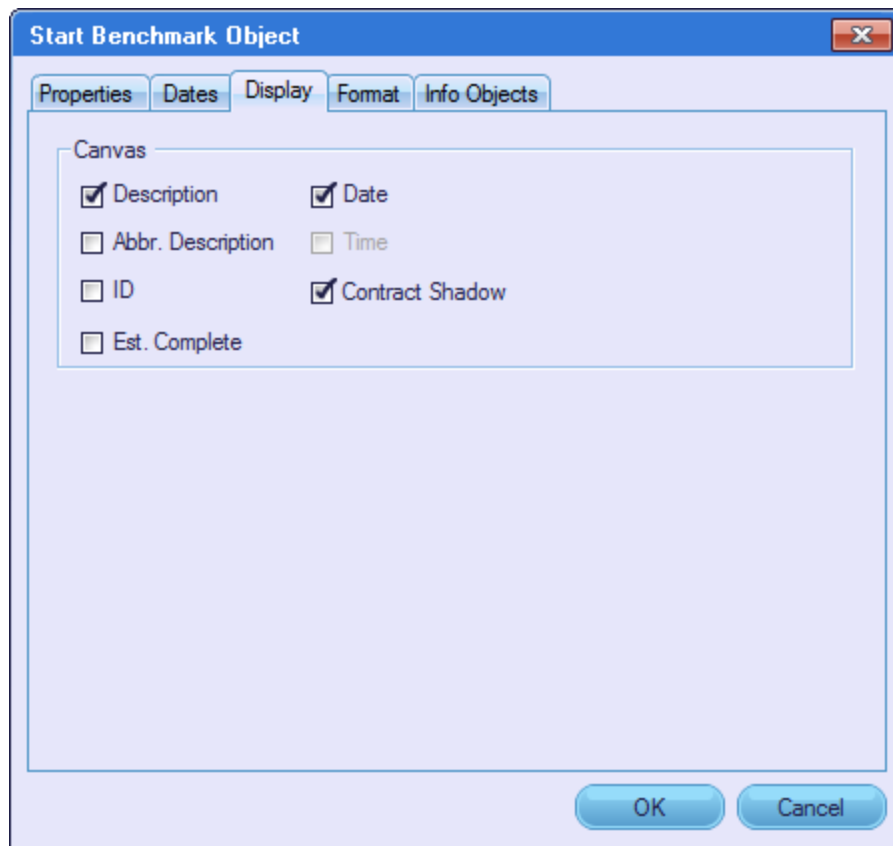
Field	Definition
Calendar	The schedule that work on the event can take place. For more info, see Project Calendars .
Date/ Time	The date and/or time the event starts or finishes (depending on the type). The date can be made actual by clicking Actualize , but only if a Data Date has been set. For more info, see Data Dates .
Start no earlier than/ Finish no earlier than*	*Only available for milestones. An optional boundary preventing the event from starting/finish any earlier than the date selected. For more details, see step 1 of Constraining Milestones .
Start no later than/ Finish no later than*	*Only available for milestones. An optional boundary preventing the event from starting/finish any later than the date selected. For more details, see step 1 of Constraining Milestones .

Field	Definition
Contract Date	The date of the event according to a contract.
Change Log (Contract Date)	A history of changes to the contract date, with the following columns – Description : a text field for naming the version of the contract date. Contract Date : the date of the version of the contract date. Variance : the difference between the planned date on the canvas and the version of the contract date. Date Logged : the date/time the version of the contract date was modified. Associated Target : the most recent target created after the version of the contract date was modified. Associated Data Date : the most recent data date that existed on the canvas at the time the version of the contract date was modified.

Displays Tab



The **Displays** tab contains checkboxes for hiding or showing different fields on the canvas.



Field	Definition
Description	The description of the milestone/benchmark. Only one of the description fields can be displayed at any given time.
Abbr. Description	A shorter description for viewing on the canvas if the description is too long.
ID	A user-entered unique identifier. If no description field is displayed but the ID is, then the ID will inherit the formatting of the abbreviated description.
Est. Complete	A user-entered estimate of work completed.
Date	The date the event starts or finishes
Time	The time the event starts or finishes; only enabled if the schedule time unit is smaller than days.
Contract Shadow	A visual depiction of the most recent contract date, if any, that was entered.

Field	Definition
Drift Buffer*	*Only available for milestones. The amount of time a milestone can gain before advancing the finish of its nearest predecessor.
Drift*	*Only available for milestones. The amount of time a milestone can gain before advancing the start of the project.
Float*	*Only available for milestones. The amount of time a milestone can be delayed before delaying the completion of the project.
Buffer*	*Only available for milestones. The amount of time a milestone can be delayed before delaying the start of its nearest successor.
Total Float*	*Only available for milestones. Drift + Float.

NOTE: Drift, float, buffer, and total float are displayed in that order below the milestone on the same line. They may be distinguished from one another by the shape of their bounding boxes: drift points to the left, or project start; float and buffer point to the right, or project completion; and total float points in both directions. Either buffer or total float may be displayed, but not both at the same time. To customize the position of the fields on the canvas, see [↗Repositioning Fields \(GEM\)](#).

Format Tab

The screenshot shows the 'Start Milestone Object' dialog box with the 'Format' tab selected. The dialog has a title bar with a close button. Below the title bar are five tabs: 'Properties', 'Dates', 'Displays', 'Format', and 'Info Objects'. The 'Format' tab is active, showing two main sections: 'Icon' and 'Descriptions'.

Icon Section:

- Size:** A dropdown menu set to 'Medium (Def)'.
- Color:** A color selection box showing a green square and the values '15, 140, 55 (Default)'.
- Shape:** A dropdown menu showing a green checkmark icon.
- Text Location:** A dropdown menu set to 'Top'.
- Tail Length:** A numeric input field set to '0' with a range of '(0-1)'.

Descriptions Section:

- Apply To:** A dropdown menu set to 'Description'.
- Font:** A text input field set to 'Arial'.
- Color:** A color selection box showing a black square and the values '0, 0, 0'.
- Alignment:** A dropdown menu set to 'Center'.
- Style:** A dropdown menu set to 'Regular'.
- Size:** A numeric input field set to '10'.
- Preview:** A text box displaying 'ABC abc'.

At the bottom right of the dialog are two buttons: 'OK' and 'Cancel'.

The **Format** tab contains controls for changing the look of the object on the canvas.

Icon

Field	Definition
Size	The size of the event on the canvas.
Shape	The shape of the event. A start event will be filled in on the left (or point to the left if it's a flag); a finish event will be filled in on the right (or point to the right if it's a flag). NOTE: Any pattern in-use by a milestone will be unavailable for use by a benchmark, and vice-versa.
Color	The color of the event. NOTE: Avoid red for milestones, since red is used to signal criticality. For more info, see Criticality.
Text Location	The default position for the ID, description, and date/time
Tail Length (0-1)*	*Only available for milestones. An optional line extending from the bottom of the milestone down the canvas, with 0 being no line and 1 being to the bottom of the canvas.

Shape	
	Hourglass
	Vertical Diamond
	Horizontal Diamond
	5-point Star
	4-point Star
	Flag
	Triangle

Descriptions

The description and the abbreviated description can be styled/formatted differently or the same. To make them different, choose **Description** from the dropdown, customize the options, and then choose **Abbr.** **Description** from the dropdown, and customize the options for it. The settings will be saved, and you can switch between the two to confirm.

To make them the same, choose **Both** from the dropdown, and then customize the options. Note that choosing both will only overwrite the options that receive a change. For example, imagine the description's font has been formatted to Arial and the abbreviated description's font has been formatted to Tahoma. If **Both** is chosen from the dropdown and the font is set to Calibri, then the font for both the description and the abbreviated description will be set to Calibri, but the color, alignment, style, and size will be unaffected, since they didn't receive any change. This can be confirmed by switching between the description and the abbreviated description and reviewing them without losing any of the changes.

Field	Definition
Font	The font for the description, abbreviated description, or both.
Color	The color of the description, abbreviated description, or both.
Alignment	The alignment of the description, abbreviated description, or both.
Style	The style of the description, abbreviated description, or both.
Size	The size of the description, abbreviated description, or both.

Constraining Milestones

A constraint is a way to restrict the range that an object may move in time. If an object has been constrained, any attempt to move it beyond its constraint date will be prevented. Constraints are represented by exclamation marks to the left or the right of the date they're constraining.

NetPoint supports two types of constraints: a no-earlier-than (NET) constraint, also called a start-no-earlier-than (SNE) constraint, which defines the earliest date an object may backslide in time; and a no-later-than (NLT) constraint, also called a finish-no-later-than (FNL) constraint, which defines the latest date an object may slide forward in time. In the case of milestones and benchmarks, no-earlier-than constraints are applied to start types and no-later-than constraints are applied to finish types. To set constraints for more than one activity or milestone at the same time, see [➔Bulk-Editing Objects](#).

Tip: Instead of constraining a milestone, consider using a benchmark. Benchmarks are constrained to the date that they're set at and allow negative-gap to measure movement beyond the set date.

NOTE: If a milestone is constrained to a non-working day based on its calendar, it may be subject to having extra float or drift that can never be zeroed out. For example, a start milestone whose earliest successor starts on Monday but whose no-later-than constraint is set to Saturday will be forced to start on Friday and have one day of float even though it's snapped.

Steps

1. With the **Select Tool**  chosen in the toolbar, double-click the desired milestone, or right-click it and choose **Properties**.
2. To apply a NET constraint, check the box next to **Finish no earlier than** (or **Start no earlier than** for a start milestone). The constraint will default to the current date.
3. To apply a NLT constraint, check the box next to **Finish no later than** (or **Start no later than** for a start milestone). The constraint will default to the current date.
4. To change the constraint date, enter a new value or click the down arrow button. Selecting a date on the calendar closes the graphic calendar picker.
5. When finished, click **OK** to exit the window and apply the constraint. An exclamation mark will show up where the constraint was applied.

Text Annotations

Annotations can be used to add text directly to a specific object. Text can include a note, a comment, or a custom data field. When you add an annotation to an object, NetPoint is actually just creating a text box that is permanently tied to its host. As the object is moved, the annotation moves too, along with other object data, like dates. Only one annotation may be added to an object at a time. Because annotations can be re-positioned on the canvas without having to enter **Global Edit Mode**, it may be desirable to use them instead of descriptions.

NOTE: Annotations will also fade with their host objects.

Steps

Adding

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired event.
2. Choose **Add Annotation at** and then **Start**, **Center**, or **Finish**. The cursor will change to the crosshairs, and the **Text Object** icon  in the toolbar will become pressed.
3. Left-click with the mouse at the desired date, indicated by the text above the crosshairs. (Dragging the crosshairs will have no consequence on the shape of the text box).
4. Release the mouse. The edit text window will open.
5. Begin typing.
6. When finished, click anywhere outside the text box on the canvas.

NOTE: Once an annotation is added, Add Annotation will go disabled, as only one annotation may be added to an object, and the anchor point may never be changed.

Modifying Annotation Text

1. With the **Select Tool**  chosen in the toolbar, right-click on the text and choose **Edit Text**.
2. Edit as desired.
3. When finished, click anywhere outside the text box on the canvas.

Modifying Annotation Font

1. With the **Select Tool**  chosen in the toolbar, double-click the text or right-click and choose **Properties**. The **Text Object** window will open.
2. When finished making changes, click **OK**.

Hiding Annotations

1. With the **Select Tool** chosen in the toolbar, right-click on the event and choose **Hide Annotation**.
2. Click on the canvas to deselect the event.

Deleting

1. With the **Select Tool**  chosen in the toolbar, select the text on the canvas and press the **Delete** key, or right-click on the host event and choose **Delete Annotation**.
2. Click on the canvas to deselect the event.

Info Objects/Attachments

Info objects are external files or attachments. Adding an info object to a milestone or benchmark is a way to embed a document on the event for viewing. An event with one or more info objects will have an “i” icon placed over the bar to indicate a file has been attached. However, if the duration is reduced sufficiently to where the end nodes begin to overlap, the icon will be hidden. Info objects can be added, viewed, and deleted from the **Info Objects** tab of the event’s **Object** window.

Steps

Adding

1. With the **Select Tool**  chosen in the toolbar, double-click an event, or right-click and choose **Properties**.
2. Click the **Info Objects** tab.
3. Click the **New** button. The **Info Object Properties** window will open.
4. Click the **Browse** button. (If you’d like to attach another schedule, first select **NetPoint project**).
5. Choose a file and click **Open**.
6. If desired, enter a new **Description**.
7. Click **OK** to save and close the **Info Object Properties** window.
8. To add another, repeat steps 3-7.
9. When finished, click **OK**.

NOTE: If the iObject is copied and linked to the project folder, the filepath shown will still display the original location, so that you can always remember where the original file was located.

Modifying

1. With the **Select Tool**  chosen in the toolbar, double-click an event, or right-click and choose **Properties**.
2. Click the **Info Objects** tab.
3. Select the file in the list click the **Modify** button. The **Info Object Properties** window will open.
4. Modify as desired.
5. Click **OK** to save and close the **Info Object Properties** window.
6. To modify another, repeat steps 3-5.
7. When finished, click **OK**.

Deleting

1. With the **Select Tool**  chosen in the toolbar, double-click an event, or right-click and choose **Properties**.
2. Click the **Info Objects** tab.
3. Select the file in the list and click the **Delete** button.
4. In the confirmation, click **Yes**.
5. To delete another, repeat steps 3-4.
6. When finished, click **OK**.

Opening/Viewing

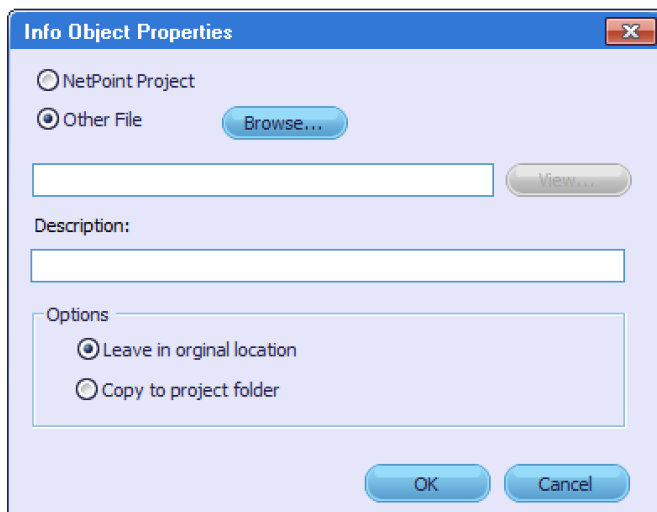
1. With the **Select Tool**  chosen in the toolbar, double-click an event, or right-click and choose **Properties**.
2. Click the **Info Objects** tab.
3. Select the file in the list and click the **View** button. The file will open outside of NetPoint.
4. To view another, repeat step 3.
5. When finished, click **OK**.

NOTE: If the info object is a NetPoint schedule, the current schedule will be saved automatically. If you don't wish to save the schedule, click **Cancel** in the confirmation that opens.

Notes

Properties

The **Properties** window is where the general settings are changed.



Field	Definition
Source	Determines the type of file to attach.
Filepath	Displays the filepath of the original file, regardless of whether or not it's copied to to the project folder.
Description	A short description for displaying on the canvas. By default, the same as the filename.
Options	Provides the option to copy the file to the project folder so it's still viewable after zipping and sending the project.

Links, Embeds, and Logic

Different objects, such as activities, milestones, and benchmarks, may be related or unrelated to one another. In NetPoint, these relationships are represented by links, and they form the backbone of a GPM schedule. From them, float attributes are calculated, including gap, buffer, float and drift. A link is represented as a yellow line with red arrows pointing from the first object to the second object.

In a relationship, the first object is called the predecessor; the second object the – successor. If an object is moved earlier than its predecessor or later than its successor, it creates a negative-gap link, which corrupts the logic of the original relationship. As long as **Logic Mode** is turned on, however, NetPoint will heal negative-gap links as they form, moving the predecessor or successor along as it's modified in real-time. This automatic correction is known as self-healing.

This section details instructions for the following:

1. Creating FS links
2. Using embeds to create SS and FF links
3. Link geometries
4. Adding link lags
5. Text annotations
6. Spotlighting/highlighting link chains
7. Treating redundancies
8. Overriding logic

Creating FS Links

The simplest kind of link, the finish-to-start (FS), represents a relationship where the successor cannot start before the predecessor finishes. If the predecessor and successor overlap on the timescale, an embedded node must be placed first. For more information, see [➔Using Embeds to Create SS & FF Links](#).

FS links can be created between activities, milestones, and benchmarks, either in pairs or for multiple activities. However, the regular **Link** icon can only link two objects at a time. If more than two objects are selected for linking, an error message will open. When linking multiple activities, the **FS Link Chain** icon must be used. Instructions for both are provided.

NOTE: For diagonal links, only the vertical or horizontal portions may be selected.

Steps

Two Activities

1. With the **Select Tool**  chosen in the toolbar, select the first activity, milestone, or benchmark on the canvas
2. Press and hold the **Shift** key and select the second activity, milestone, or benchmark on the canvas. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around both objects, so long as it includes only the two desired objects for linking.
3. With both objects still selected, click the **Link** icon  in the toolbar.
4. Once linked, both objects remain selected. To de-select, click an empty portion of the canvas.

Multiple Activities

1. With the **Multi-Object Select Tool**  chosen in the toolbar, select the first activity, milestone, or benchmark on the canvas.
2. Select the second activity, milestone, or benchmark on the canvas. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all objects, so long as none of them overlap on the timescale.
3. Repeat step 2 for each object, as needed.
4. With all the objects still selected, click the **FS Link Chain** icon  in the toolbar.
5. Once linked, all the objects remain selected. To de-select, click an empty portion of the canvas.

Notes

Link Object

Properties

Logic:

Geometry:

Offset:

Displays

☐ Gap 0 ☒ Offset Notation

Attributes

Predecessor:

Successor:

Relationship:

Score:

VALID

Properties

Field	Definition
Logic	A label for categorizing links. The designation is for supplemental information; it has no bearing on actual schedule logic or calculations. For a breakdown, see below.
Geometry	The physical shape of the link. A link can bend vertically, horizontally, and diagonally, in eight different combinations. NetPoint makes use of a scoring system to identify the best possible geometry for a link. For more information, see Link Geometries .
Lag	A way to enforce a minimum gap. For more info, see Adding Link Lags .

Logic Types

Type	Definition
Hard	A relationship that is inherent to the nature of the work being done.
Soft	A relationship that is a preferred choice.
Resource-Flow	A relationship that is based on resource usage.

Type	Definition
External	A relationship that is mandated or dictated by an external source, such as a governmental agency.
Contractual	A relationship that is mandated or dictated by a contract.
Special	A relationship that is unique to the schedule.

Displays

Field	Definition
Gap	The duration of the link.
Lag Notation	The value of the lag. If the gap is equal to zero, it will automatically turn off, even if the box is checked for display.

Attributes

Field	Definition
Predecessor	The object the link is emanating from. This field is just an indicator and cannot be edited.
Successor	The object the link is merging into. This field is just an indicator and cannot be edited.
Relationship	The type of link. This field is just an indicator and cannot be edited.
Score	A point value based on how many grids the link passes through, or if it crosses over other objects or overlaps with other links, for example. For more information on scoring, see ➡ Link Geometries .
Validity	An indicator of whether or not the link is valid or redundant. For more information, see ➡ Treating Redundancies .

Using Embeds to Create SS and FF Links

An embedded node (or embed), represented by single yellow circle, is placed on a host activity to allow finish-to-finish (FF) and start-to-start (SS) relationships. It may be placed to coincide with its host activity's start or finish nodes, or it may be offset (to have a lead or lag) between the two. In a FF relationship, a successor cannot finish before its predecessor finishes. In a SS relationship, a successor cannot start before its predecessor starts.

If an embed is placed outside of an activity, it will be displayed in red until it is moved onto an activity. Any such orphan embed not associated with an activity cannot be linked and will not be stored when the schedule is saved or closed.

Steps

Start-to-Start

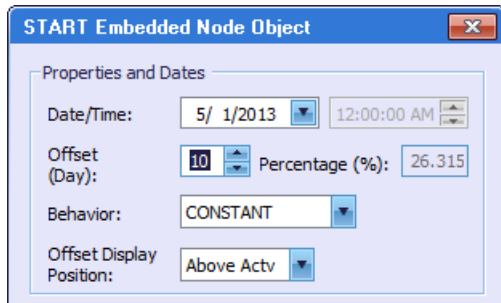
1. Click the **Start Embed** icon  in the toolbar.
2. Move the cursor over the desired activity to be the predecessor in the relationship.
3. Click the mouse, and while keeping it depressed, slide the cursor along the bar of the activity to adjust the offset (or lag) noted in blue.
4. Release the mouse to place the embed. Once an embed has been successfully placed on an activity, the **Embed Object** window opens automatically.
5. Make any desired changes and click **OK**.
6. Click the **Select Tool** icon  in the toolbar.
7. Press and hold the **Shift** key and select a valid successor. For a list of valid successors, see [Rules for Placing Embeds](#)
8. Select the embed on the canvas.
9. With both objects still selected, click the **Link** icon  in the toolbar.
10. Once linked, both objects remain selected. To de-select, click an empty portion of the canvas

Finish-to-Finish

1. Click the **Finish Embed** icon  in the toolbar.
2. Move the cursor over the desired activity to be the successor in the relationship.
3. Click the mouse, and while keeping it depressed, slide the cursor along the bar of the activity to adjust the offset (or lag) noted in blue.
4. Release the mouse to place the embed. Once an embed has been successfully placed on an activity, the **Embed Object** window opens automatically.
5. Make any desired changes and click **OK**.
6. Click the **Select Tool** icon  in the toolbar.
7. Press and hold the **Shift** key and select a valid predecessor. For a list of valid predecessors, see [Rules for Placing Embeds](#)
8. Select the embed on the canvas.
9. With both objects still selected, click the **Link** icon  in the toolbar.
10. Once linked, both objects remain selected. To de-select, click an empty portion of the canvas

Notes

Properties



The **Properties** section is where general settings are changed.

Field	Definition
Date/Time	The date and/or time the event starts or finishes (depending on the type).
Lag	The offset from the start or finish date.
Behavior	One of four different types. For more info, see the table below.
Lag Display Position	The location for displaying the lag on the canvas.

Behavior

	Start	Finish
Constant	<i>Fixed to Start.</i> The lag of this embed will remain <i>constant</i> to the <i>start date</i> of its host activity. In effect, the embed's date will change along with the start date but not if the finish date changes.	<i>Fixed to Finish.</i> The lag of this embed will remain <i>constant</i> to the <i>finish date</i> of its host activity. In effect, the embed's date will change along with the finish date but not if the start date changes.
Reverse	<i>Fixed to Finish.</i> The lag of this embed will remain <i>constant</i> to the <i>finish date</i> of its host activity. In effect, the embed's date will change along with the finish date but not if the start date changes.	<i>Fixed to Start.</i> The lag of this embed will remain <i>constant</i> to the <i>start date</i> of its host activity. In effect, the embed's date will change along with the start date but not if the finish date changes.
Proportionate	<i>Proportional to Start.</i> The lag of this embed will remain <i>proportional</i> to the <i>start date</i> of its host activity. In effect, the embed's date will always be changing to maintain its percentage from the start. If the duration of the host activity is	<i>Proportional to Finish.</i> The lag of this embed will remain <i>proportional</i> to the <i>finish date</i> of its host activity. In effect, the embed's date will always be changing to maintain its percentage from the finish. If the duration of the host activity is

	Start	Finish
	changed, the proportion may not be perfectly maintained because the number of days available may no longer allow the embed to fall on a day at exactly the same proportion away from the start as before; i.e., a rounding error because of the number of days available.	changed, the proportion may not be perfectly maintained because the number of days available no longer allows the embed to fall on a day at exactly the same proportion away from the finish as before; i.e., a rounding error because of the number of days available.
Variable	<i>Fixed Symmetrically.</i> The lag of this embed will remain <i>constant</i> to the <i>whichever</i> date is changed—if the start date is changed, it will remain constant to the start; if the finish date is changed, it will remain constant to the finish. In effect, the embed's date will change along with both the start date <i>and</i> the finish date.	<i>Fixed Inversely.</i> The lag of this embed will remain <i>constant</i> to the <i>opposite of whichever</i> date is changed—if the start date is changed, it will remain constant to the finish; if the finish date is changed, it will remain constant to the start. In effect, the embed's date will <i>never</i> change with either the start date or the finish date.

Displays

The Displays section contains four checkboxes arranged in a 2x2 grid:

- ☐ Date
- ☐ Offset from Start
- ☐ Time
- ☐ Offset to Finish

The **Displays** section contains checkboxes for hiding or showing different fields on the canvas

Field	Definition
Date	The date the event starts or finishes (depending on the type).
Time	The time the event starts or finishes (depending on the type).
Lag from Start	The number of days (or time units) offset from the start.
Lag to Finish	The number of days (or time units) offset from the finish.

Formatting

The Formatting section includes a color selection dropdown menu. The dropdown is currently set to a yellow color with the text "255, 255, 0 (Default)". Below the dropdown are two buttons: "OK" and "Cancel".

The **Formatting** section contains controls for changing the look of the object on the canvas

Field	Definition
Color	The color of the embed.

Rules for Placing Embeds

The rules for placing embeds are as follows: for activities and embeds, a start embed may only be placed on a predecessor activity and linked to the start of a *successor* activity or a *finish* embed on a successor activity. For milestones and benchmarks, a start embed can be placed on a predecessor activity or successor activity and linked to either start or finish milestones and benchmarks. But in the case where the start embed is on a predecessor activity and linked to a start milestone, the milestone will not snap. If the start embed is on a successor activity and linked to a finish milestone, the milestone will not snap.

A finish embed may only be placed on a successor activity and linked to the start of a *predecessor* activity or a *start* embed on a predecessor activity. For milestones and benchmarks, a finish embed can be placed on a successor activity or successor activity and linked to either start or finish milestones and benchmarks. But in the case where the finish embed is on a successor activity and linked to a finish embed, the milestone will not snap. If the finish embed is on a predecessor activity and linked to a start milestone, the milestone will not snap.

Link Geometries

When creating a link for the first time, each possible geometry is ranked by a scoring system based on how many grids it passes through, or if it crosses over other objects or overlaps with other links, for example. The more intersections and overlaps, the higher the score. Whichever geometry receives the lowest possible score will be chosen. The default scoring system is intended to discourage intersection and overlap and favor shorter links. To modify the default scoring penalties, see [↗Link Scoring](#). To disallow diagonal geometries, see the [Placement Tab](#) section of *Running the Engine*.

As more objects or relationships are added, once-optimized links may soon become convoluted. In these cases, the user can choose to optimize a link based on its link score. Alternatively, a geometry can be chosen manually without considering link score. Instructions for both are provided. To optimize geometries across the entire network at once, see [Running The Engine](#).

Steps

Choosing Manually

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired link.
2. Click **Change Link to** from the menu and choose a new geometry accordingly.
3. The menu will close and the link will update automatically.

Optimizing Automatically

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired link.
2. Choose **Optimize Link Shape** from the menu.
3. The menu will close and the link will update automatically.

Notes

Geometries

Abbreviation	Definition
V	Vertical
H	Horizontal
VH	Vertical-Horizontal
HV	Horizontal-Vertical
VHV	Vertical-Horizontal-Vertical
HVH	Horizontal-Vertical-Horizontal
VDV	Vertical-Diagonal-Vertical

Abbreviation	Definition
HDH	Horizontal-Diagonal-Horizontal
HDV	Horizontal-Diagonal-Vertical
VDH	Vertical-Diagonal-Horizontal

Adding Link Lags

A lag, also known as a link offset, is a way to prevent a link from collapsing to zero gap; or, in other words, to enforce an absolute minimum gap. Lags may be used, for example, to more accurately model the curing time for concrete, where using an activity may incorrectly suggest that there is work taking place.

The value of the lag equals the number of days from the predecessor's finish node and adopts the calendar of the predecessor activity. If a link had a lag of 7 days, and its successor were pulled back as far as it could go, the link would cease collapsing once the gap equaled 7. If a negative value is entered, this allows the start of the successor to push before the finish of the predecessor activity even when logic is on.

A lag displays as a red dot on top of the link, but it is not visible if the lag is too close to an activity node. You can also tell when a link has a lag in place by a red dot placed next to its gap value. This gives a cue that a lag is affecting the link gap value. Both the gap, and the dot next to the gap, however, will automatically be hidden if the gap reaches zero.

NOTE: Negative lags can only be applied to finish-to-start relationships. Upon opening a legacy schedule with negative lags on relationships other than finish-to-start, they will be removed automatically, a message will display, and the links with lags removed will become highlighted on the canvas.

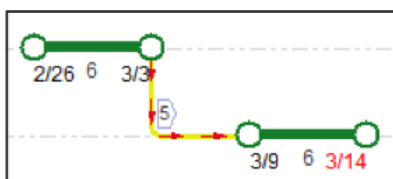
Steps

1. With the **Select Tool**  chosen in the toolbar, double-click on a desired link, or right-click and choose **Properties**.
2. Under **Properties**, enter a number other than **0** next to **Lag**.
3. To display a red dot next to the gap value, check the box next to **Lag Notation**.
4. Click **OK** to close the window and apply the lag.

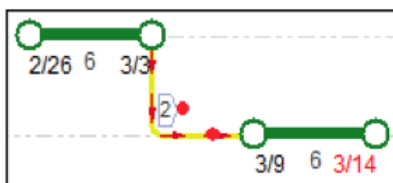
Notes

Lag Gap Calculation

Once a lag is defined, it affects the gap calculation. The gap of a link with a lag is calculated as the gap of the predecessor to the successor minus the lag value. In the following example, the original gap between the predecessor and successor is 5 calendar days.



Below, a lag of 3 calendar days is placed.



As you can see, the new gap of 2 days is equal to the original gap of 5 days minus the lag of 3 days.

Text Annotations

Annotations can be used to add text directly to a specific object. Text can include a note, a comment, or a custom data field. When you add an annotation to an object, NetPoint is actually just creating a text box that is permanently tied to its host. As the object is moved, the annotation moves too, along with other object data, like dates. Only one annotation may be added to an object at a time.

NOTE: Annotations will also fade with their host objects.

Steps

Adding

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired link.
2. Choose **Add Annotation**. The cursor will change to the crosshairs, and the **Text Object** icon  in the toolbar will become pressed.
3. Left-click with the mouse at the desired date, indicated by the text above the crosshairs. (Dragging the crosshairs will have no consequence on the shape of the text box).
4. Release the mouse. The edit text window will open.
5. Begin typing.
6. When finished, click anywhere outside the text box on the canvas.

NOTE: Once an annotation is added, Add Annotation will go disabled, as only one annotation may be added to an object, and the anchor point may never be changed.

Modifying Annotation Text

1. With the **Select Tool**  chosen in the toolbar, right-click on the text and choose **Edit Text**.
2. Edit as desired.
3. When finished, click anywhere outside the text box on the canvas.

Modifying Annotation Font

1. With the **Select Tool**  chosen in the toolbar, double-click the text, or right-click and choose **Properties**. The **Text Object** window will open.
2. When finished making changes, click **OK**.

Hiding Annotations

1. With the **Select Tool**  chosen in the toolbar, right-click on the link and choose **Hide Annotation**.
2. Click on the canvas to deselect the link.

Deleting

1. With the **Select Tool**  chosen in the toolbar, select the text on the canvas and press the **Delete** key, or right-click on the host link and choose **Delete Annotation**.
2. Click on the canvas to deselect the link.

Spotlighting/Highlighting Link Chains

As more and more relationships are added to the canvas, link geometries will continue to be optimized to minimize crossing and overlapping. Nevertheless, it can sometimes become difficult to discern exactly which two objects a given link is connecting and to do so quickly.

Similarly, a user may wish to fade all links that don't have a driving relationship in order to clarify the canvas. To accomplish these tasks, NetPoint provides a feature for identifying the pre and post objects of a link and for highlighting driving link chains. Instructions for both are provided.

Steps

Spotlighting Pre/Post Objects

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired link.
2. Choose **Spotlight Pre-Object** or **Spotlight Post-Object**.
3. The menu will close and the object will be highlighted accordingly.
4. To restore the canvas, choose **Objects > Restore Faded Colors for All Objects** from the menu bar.

Highlighting Driving Link Chains

1. Click **Objects > Highlight Driving Link Chains** from the menu bar.
2. Any links with positive gap (and the objects they connect) will be faded on the canvas.
3. To restore the canvas, choose **Objects > Restore Faded Colors for All Objects** from the menu bar.

Treating Redundancies

The simplest case of a redundant link is a start-to-start (SS) or finish-to-finish (FF) link between two activities that are already finish-to-start (FS) connected. The more complex case of a redundant link is an SS link, FF link, or no-lag FS link connecting an activity to a successor, which, topology constant, cannot acquire a zero gap under any combination of durations and lags, because the successor is inevitably driven by a successor to the activity. The reverse of a redundant link is a valid link.

Redundant links can be hidden, displayed, faded, or shown. You can also tell if a link is redundant by opening its object property window. In the bottom left, it will display if it's redundant or valid.

NOTE: Redundancy options through the menu bar are continually checked; rather, these are one-time operations, which must be done periodically as the schedule evolves.

Steps

All at once

1. Click **Objects > Redundant Links** from the menu bar.
 - a. To hide all redundant links on the canvas, click **Hide**.
 - b. To show all redundant links on the canvas, click **Show**.
 - c. To fade all redundant links on the canvas, click **Fade**.
 - d. To spotlight all redundant links on the canvas, click **Spotlight**.

One-by-one

1. Click **Schedule > Open Layout Manager...** from the menu bar.
2. Click the down arrow next to the **Statistics** icon and choose **Links**.
 - a. To spotlight a link on the canvas, select it in the list.
 - b. To hide a redundant link on the canvas, check the box under the **Hide** column next to the link.
 - c. To fade a redundant link on the canvas, check the box under the **Fade** column next to the link.
 - d. To delete a redundant link on the canvas, select it in the list and click **Delete**.
3. Click **Save** or **Close** to close the manager. The list will be saved automatically regardless.

Overriding Logic

Logic is a mode whereby object relationships are enforced. When turned on, NetPoint will heal a negative-gap link as it forms to preserve the logic of the original relationship. For example, if an activity is moved *earlier* in time, NetPoint will move any zero-gap predecessors along with the activity in real-time (until reaching an actualized object or constraint). Additionally, if the activity has any zero-gap successors, they will be moved earlier to maintain the zero-gap link (unless they were actualized or constrained).

Likewise, if an activity is moved *later* than in time, NetPoint will move any zero-gap successors later along with the activity (until reaching an actualized object or constraint). In this case, however, all predecessors will remain fixed, even if connected by a zero-gap link, allowing the activities to break open. The same rules also apply when crashing or extending an activity's duration. In short, logic mode will propagate a change throughout a network. Without logic mode on, a change will be localized to the affected object itself.

This process by which predecessors and successors are moved automatically is known as self-healing. By default, logic mode is turned on but can be turned off if desired. It is helpful to save the plan before using this option.

NOTE: If a milestone is assigned a calendar that's different from that of its predecessors or successors, and the network is pulled backwards, gaps may open up as the chain crosses a weekend or non-working days, despite logic being on. As such, it is recommended to always keep milestones on the same calendar as their predecessors/successors.

Steps

1. Click the **Logic Mode**  icon in the toolbar.
2. Adjust the network as desired.
3. When finished, turn **Logic Mode** back on.

NOTE: Negative-gap links will not be corrected by the system until Logic Mode is turned back on and one or more of the activities impacting these links are moved (either the predecessor or successor of the link with negative-gap or an activity which falls on the same path as the link with negative-gap).

Notes

Logic Loops

NetPoint does not allow an activity to have the same activity as a predecessor and successor (or in the predecessor and successor chains (for example, activity A linked to activity B, linked to activity A). This is known as a logic loop. However, it is still possible to open a schedule that was created before loops were prohibited. In this case, a window will open, the links in the logic loop will be highlighted, and one of the links must be deleted before proceeding. If multiple loops are present, the window will open for each loop.

Markup Tools

NetPoint provides a number of non-planning objects for adding extra information to the canvas. Such objects do not contribute to the logic of the network but are used merely for providing supplemental information.

Shades

Shades are custom-drawn rectangles of color. They are more flexible than the basic shapes and can be re-sized more naturally. Shades can be used to create a key or legend, show horizontal lanes, show vertical windows, or even spotlight an activity or area of the network. Both shapes and shades are added through the same toolbar icon.

Steps

Adding a Shade

1. Click the **Shapes** icon  on the toolbar.
2. Move the cursor to the desired location.
3. Left-click with the mouse.
4. Without releasing, drag the cursor to create the desired shape.
5. Release the mouse. The **Shade Object** window will open.
6. To make the shade entirely horizontal or vertical, click the dropdown next to **Orientation** and choose accordingly.
7. When finished, click **OK**.

Locking/Unlocking a Shade

1. Click the **Select Tool** icon  in the toolbar.
2. Double-click the shade or right-click and choose **Properties**.
3. Click the dropdown next to **Behavior** and choose **Canvas Background** to lock or **Regular Object** to unlock.
4. When finished, click **OK**.

NOTE: Even when unlocked, a shade will never be included when dragging to select other objects. To include one or more shades, first select the other objects, and then press the **SHIFT** key on the keyboard, clicking each shade individually.

Rearranging a Shade

1. Click the **Select Tool** icon  in the toolbar.
2. Left-click with the mouse on the desired shade.
3. Click **Objects > Arrangement (Shades and Images)** from the menu bar.
4. Choose **Bring to Front** or **Send to Back** accordingly.

NOTE: Shades can only be moved in front of/behind one another, but not in front of/behind other shapes or images.

Moving/Resizing a Shade

1. Click the **Select Tool** icon  in the toolbar.
2. Left-click with the mouse on the desired shade.
3. To move, drag the shade without letting go to the new desired location.
4. To resize, move the cursor over any of the black boxes until a double-arrow appears.
 - a. Left-click with the mouse.
 - b. Without letting go, drag the box until the new desired size is achieved.

- c. Release the mouse.

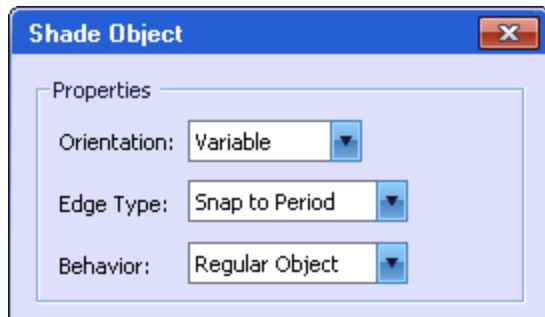
NOTE: If a horizontal or vertical shade is resized so that one of the edges no longer spans the entire width or length of the canvas, the shade's orientation will automatically adjust to be **Variable**.

Notes

Shades

Properties

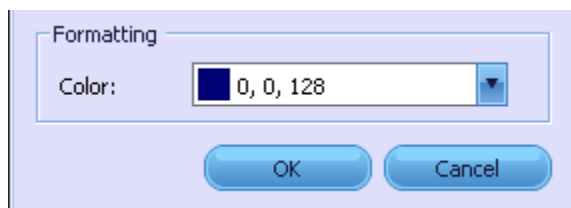
The **Properties** tab is where general settings are changed.



Field	Definition
Orientation	Determines the overall shape of the shade. Variable shades take the exact shape that is drawn on the canvas. Horizontal shades span the entire width of the canvas, maintaining their upper and lower edges that are drawn. Vertical shades span the entire height of the canvas, maintaining the left and right edges that are drawn.
Edge Type	Determines what the edges of the shade correspond to. Snap to Period will align the left edge and the right edge with the beginning and end of each day (or time unit, when using PolyTime). Absolute will allow the left edge and right edge to fall in between the beginning and end of each day (or time unit).
Behavior	Determines whether or not a shade can be selected via left-click on the canvas. Regular Object enables selection via left-click on the canvas. Canvas background will lock it, and thus prevent selection.

Formatting

The **Formatting** section contains controls for changing the look of the object on the canvas.



Field	Definition
Color	The color of the shade.

Text Annotations

Annotations can be used to add text directly to a specific shade. Text can include a note, a comment, or a custom data field. When you add an annotation to an object, NetPoint is actually just creating a text box that is permanently tied to its host at the user-selected reference point. As the shade is moved, the annotation moves too. Only one annotation may be added to a shade at a time. Because annotations can be re-positioned on the canvas without having to enter **Global Edit Mode**, it may be desirable to use them as descriptions.

NOTE: Annotations will also fade with their host shade.

Steps

Adding

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired shade.
2. Choose **Add Annotation** and then choose a reference point. The cursor will change to the crosshairs, and the **Text Object** icon  in the toolbar will become pressed.
3. Left-click with the mouse at the desired date, indicated by the text above the crosshairs. (Dragging the crosshairs will have no consequence on the shape of the text box).
4. Release the mouse. The edit text window will open.
5. Begin typing.
6. When finished, click anywhere outside the text box on the canvas.

NOTE: Once an annotation is added, the chosen anchor point is indicated by a check mark, and it can be changed at any time by right-clicking and choosing another option.

Modifying Annotation Text

1. With the **Select Tool**  chosen in the toolbar, right-click on the text and choose **Edit Text**.
2. Edit as desired.
3. When finished, click anywhere outside the text box on the canvas.

Modifying Annotation Font

1. With the **Select Tool**  chosen in the toolbar, double-click the text or right-click and choose **Properties**. The **Text Object** window will open.
2. When finished making changes, click **OK**.

Hiding Annotations

1. With the **Select Tool**  chosen in the toolbar, right-click on the shade and choose **Hide Annotation**.
2. Click on the canvas to deselect the shade.

Deleting

1. With the **Select Tool**  chosen in the toolbar, select the text on the canvas and press the **Delete** key, or right-click on the host shade and choose **Delete Annotation**.
2. Click on the canvas to deselect the shade.

Shapes

Shapes are low-fidelity, black and white symbols for annotating the canvas. These include arrows, brackets, lines, rectangles, and a circle, for example. Both shapes and shades are added through the same toolbar icon.

Steps

Adding Shapes

1. Click the down arrow to the right of the Shapes icon  and choose the desired shape.
2. Move the cursor to the desired location.
3. Left-click with the mouse.

Locking/Unlocking Shapes

1. Click the **Select Tool** icon  in the toolbar.
2. Right-click on the desired shape.
3. Choose **Set as canvas background** to lock or unlock accordingly.

NOTE: Even when unlocked, a shape will never be included when dragging to select other objects. To include one or more shades, first select the other objects, and then press the **SHIFT** key on the keyboard, clicking each shade individually.

Rearranging Shapes

1. Click the **Select Tool** icon  in the toolbar.
2. Left-click with the mouse on the desired shape.
3. Click **Objects > Arrangement (Shades and Images)** from the menu bar.
4. Choose **Bring to Front** or **Send to Back** accordingly.

NOTE: Shapes and images can be moved in front of or behind other shapes or images, but not in front of or behind shades.

Moving/Resizing Shapes

1. Click the **Select Tool** icon  in the toolbar.
2. Left-click with the mouse on the desired shape.
3. To move, drag the shape without letting go to the new desired location.
4. To resize, move the cursor over any of the black boxes until a double-arrow appears.
 - a. Left-click with the mouse.
 - b. Without letting go, drag the box until the new desired size is achieved.
 - c. Release the mouse.

NOTE: To resize a shape asymmetrically, right-click on the shape and uncheck the option **Preserve Aspect Ratio While Resizing** before attempting to resize.

Text Boxes and Notepads

Text boxes are notes that show up directly on the canvas. Notepads are notes that do not show up on the canvas. Instead, they are represented by an icon and viewed by double-clicking the icon. Text boxes and notepads are unique in that they can exist on the canvas irrespective of dates. This is useful for creating a header, footer, or watermark—the text box will not move with the network when the page size is changed or the canvas is stretched. Only text boxes and notepads possess this capability.

Steps

Adding

1. Click the **Text Box**  icon or **Notepad** icon  from the toolbar.
2. Move the cursor to the desired location or date on the canvas.
3. Left-click with the mouse at the desired location or date, indicated by the text above the crosshairs. (Dragging the crosshairs will have no consequence on the shape of the text box or notepad).
4. Release the mouse. The edit text window will open.
5. Begin typing.
6. When finished, click anywhere on the canvas outside the text box or notepad.

Modifying Text

1. With the **Select Tool**  chosen in the toolbar, right-click on the text box or notepad and choose **Edit Text**.
2. Edit as desired.
3. When finished, click anywhere on the canvas outside the text box or notepad.

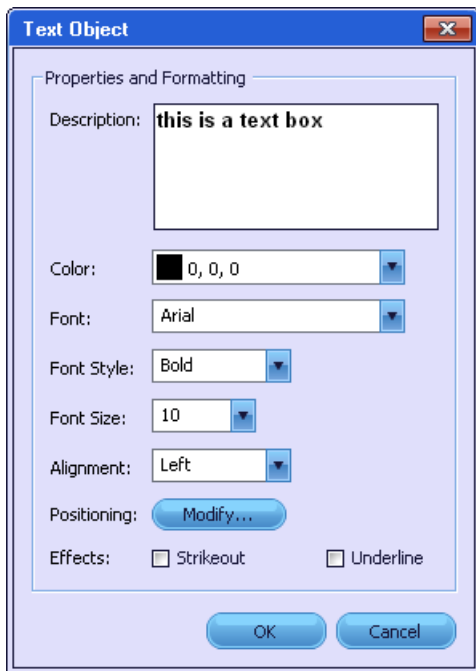
Modifying Font

1. With the **Select Tool**  chosen in the toolbar, double-click the text box or notepad or right-click and choose **Properties**. The **Text Object** window will open.
2. When finished making changes, click **OK**.

Notes

Properties and Formatting

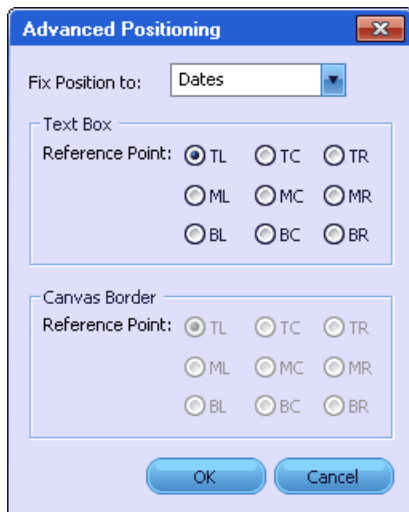
The **Properties and Formatting** section is where general settings as well as controls for changing the look are changed.



Field	Definition
Description	A text that will show up on the canvas or in the notepad.
Color	The color of the text.
Font	The font for the text.
Font Style	The style of the text.
Font Size	The size of the text.
Alignment	The alignment of the text.
Positioning	Determines how the text box or notepad will behave when the page size is changed or the canvas is stretched.
Effects	Options for striking or underlining the text.

Positioning

Positioning options determine how the text box or notepad will behave when the page size is changed or the canvas is stretched. This is useful for creating a header, footer, or watermark.



Field	Definition
Fix Position to	Determines whether or not the text object or notepad is tied to dates (like all NetPoint objects), or not. To create a header, footer, or watermark, choose Canvas Border. The Canvas Border Reference Point section will become enabled. See below.
Text Box Reference Point	Determines which corner of the text box is anchored to a date (or the canvas border, if chosen above). Options include Top Left , Middle Left , Bottom Left , Top Center , Middle Center , Bottom Center , Top Right , Middle Right , or Bottom Right . For example, if Dates and TR was chosen for a text box and the page size was changed or the canvas was stretched, the top right corner would move along to maintain the same date. Likewise, if additional text was added, the text box would increase down and to the left.
Canvas Border Reference Point	Determines which corner of the canvas the text box is anchored to (only when Canvas border is chosen above). Options include Top Left , Middle Left , Bottom Left , Top Center , Middle Center , Bottom Center , Top Right , Middle Right , or Bottom Right . For example, if Canvas Border and TR was chosen for a text box and the page size was changed or the canvas was stretched, the distance between the text box and the top right corner of the schedule would remain fixed. To create a header, set the Text Box Reference Point and the Canvas Border Reference Point to TC. To create a footer, set the Text Box Reference Point and the Canvas Border Reference Point to BC. To create a watermark, set the Text Box Reference Point and the Canvas Border Reference Point to MC.

Info Objects/Attachments

Info objects are external files or attachments. Adding an info object to the canvas is a way to embed a document for viewing. The toolbar icon is only for placing info objects onto the canvas. To attach info objects to activities, milestones, and benchmarks, open their object windows and choose the **Info Objects** tab.

Steps

Adding iObjects

1. Click the **Info Object** icon  from the toolbar.
2. Move the cursor over the canvas to the desired location.
3. Left-click with the mouse. The **Info Object Properties** window will open.
4. Click the **Browse** button. (If you'd like to attach another schedule, first select **NetPoint project**).
5. Choose a file and click **Open**.
6. If desired, enter a new **Description**.
7. Click **OK** to save and close the **Info Object Properties** window.

NOTE: If the iObject is copied and linked to the project folder, the filepath shown will still display the original location, so that you can always remember where the original file was located.

Modifying iObjects

1. With the **Select Tool**  chosen in the toolbar, double-click the desired iObject, or right-click and choose **Properties**.
2. Modify as desired.
3. Click **OK** to save and close the **Info Object Properties** window.

Opening/Viewing iObjects

1. With the **Select Tool**  chosen in the toolbar, double-click the desired iObject, or right-click and choose **Properties**.
2. Click the **View** button. The file will open outside of NetPoint.
3. When finished, click **OK**.

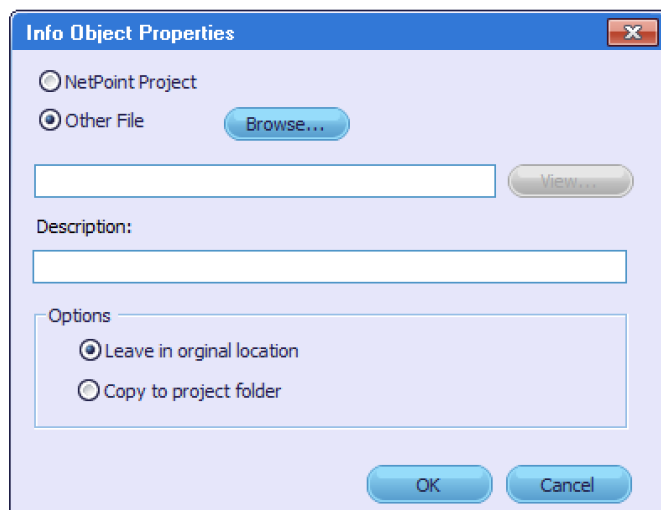
NOTE: If the info object is a NetPoint schedule, the current schedule will be saved automatically. If you don't wish to save the schedule, click **Cancel** in the confirmation that opens.

Notes

Properties

Info Objects

The **Properties** window is where the general settings are changed.



The dialog box is titled "Info Object Properties" and has a close button (X) in the top right corner. It contains two radio buttons: "NetPoint Project" and "Other File". The "Other File" option is selected, and next to it is a "Browse..." button. Below this is a text input field and a "View..." button. Underneath is a "Description:" label followed by another text input field. At the bottom left, there is an "Options" section with two radio buttons: "Leave in original location" (selected) and "Copy to project folder". At the bottom right are "OK" and "Cancel" buttons.

Field	Definition
Source	Determines the type of file to attach.
Filepath	Displays the filepath of the original file, regardless of whether or not it's copied to to the project folder.
Description	A short description for displaying on the canvas. By default, the same as the filename.
Options	Provides the option to copy the file to the project folder so it's still viewable after zipping and sending the project.

Images

Images may be inserted onto the canvas from Windows Explorer or other applications that are an OLE compatible drag source. The following formats are supported: JPEG (.jpg), PNG, TIFF (.tif), GIF, EMF, and BMP.

Steps

Adding Images

1. Click File > Insert Image Files from the menu bar (or drag and drop an image into NetPoint directly from an open Windows Explorer window).
2. Browse for the file and click **Open**.

Rearranging Images

1. Click the **Select Tool** icon  in the toolbar.
2. Left-click with the mouse on the desired image.
3. Click **Objects > Arrangement (Shades and Images)** from the menu bar.
4. Choose **Bring to Front** or **Send to Back** accordingly.

NOTE: Images can only be moved in front of/behind one another, but not in front of/behind shades or other shapes.

Moving/Resizing an Image

1. Click the **Select Tool** icon  in the toolbar.
2. Left-click with the mouse on the desired image.
3. To move, drag the image without letting go to the new desired location.
4. To resize, move the cursor over any of the black boxes until a double-arrow appears.
 - a. Left-click with the mouse.
 - b. Without letting go, drag the box until the new desired size is achieved.
 - c. Release the mouse.

Work Breakdown Structure (WBS) and Codes

A work breakdown structure (WBS) is a hierarchical decomposition of work that breaks down the schedule into manageable sections. It allows for more than just one subcategory, or level, and can be thought of as an overall outline of the project.

Codes are also used for organizing or sorting but have limitations that the WBS does not. For example, codes can only have 1 level of sub-categories, or “values”. Also, values can be assigned to activities, but not codes themselves. Take a project with a code called “Area” with the values “North,” “South,” “East,” and “West”. An activity could then be assigned to “North” but not to “Area” itself. (If desired, an activity could be assigned to “North” AND “South” by choosing the assignment behavior **Multiple**). In general, codes are not necessarily related to one another and are optional; the WBS is a single cohesive hierarchy which effectively summarizes the project.

Both the WBS as well as codes can be used as a basis for swimlanes to lay out the network, and they can be exported or imported from Excel. This chapter includes details for the following:

1. The Codes Manager
2. Configuring the WBS
3. Adding/modifying codes
4. Making Assignments
5. Creating swimlanes
6. Exporting/importing from Excel

The Codes Manager

The **Codes Manager** is a hub for codes functionality, such as defining the work breakdown structure (WBS), adding codes, and making assignments. The window itself is non-modal, meaning that you can navigate to other menus or make modifications on the canvas without having to close it.

In each view, the bottom list can be filtered using the search field provided above it, and both lists can be customized by having their columns sorted, added and removed. In the bottom list, the canvas will be scrolled and the item spotlighted as you select different items, but this can be disabled. In addition, both lists can be individually exported to Excel. Instructions provided below.

NOTE: Sorting is not saved and will refresh when changing views or leaving and coming back.

Steps

Sorting, Adding, and Removing Columns

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. To sort by a certain field, click once on the column's label to sort from top to bottom, or click twice to sort from bottom to top.
3. To add or remove a column, right-click on any column.
 - a. Choose the desired field. A checkmark will appear or disappear and the list will refresh.
 - b. To close the menu, click anywhere outside of it.
4. Repeat step 3 as desired.
5. To remove a column graphically (bottom list only), left-click on the desired column.
 - a. Without releasing the mouse, drag it up or down until a black "X" appears.
 - b. Release the mouse.
6. Repeat step 5 as desired.
7. To rearrange a column, left-click on the desired column
 - a. Without releasing the mouse, drag it left or right until two red arrows appear to indicate where it will be placed.
 - b. Release the mouse
8. When finished, click **Close**. Any changes will be automatically saved.

NOTE: Any customization is tied to the computer (not to the schedule) and will be reset upon re-installation of NetPoint.

Exporting a List to Excel

1. Click **Tools > Open Codes Manager** from the menu bar.
2. Right-click on the columns at the top of either list.
3. Choose either **Export Displayed Rows & Columns** or **Export All Rows & Columns**.

Filtering Lists

1. Click **Tools > Open Codes Manager** from the menu bar.
2. Click in the search field above the lower list.
3. Begin typing, and the list will adjust to display any matches.
4. Select an activity in the list and it will be indicated on the canvas.
5. When finished, click **Close**.

NOTE: Any alphanumeric characters can be entered, and the results will include anything that matches, starts with, or contains what has been entered, pulled from every column currently being shown. For example, if searching for a duration, make sure the **Duration** column is being displayed.

Enabling/Disabling Canvas Scroll

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. Click the **List** icon  in the upper-right corner
3. Click **Scroll Canvas with Manager Selections**. A checkmark will appear or disappear accordingly.

NOTE: This setting will also apply to the **NetRisk** and **Targets** managers, and will be saved after the manager is closed.

Notes

Manager Views

Icon	Definition
	The Codes view is for adding codes and values and assigning them to activities. For instructions, see Adding/Modifying in the <i>Codes</i> chapter.
	The WBS view is for organizing WBS levels and assigning them to activities. For instructions, see Adding/Modifying in the <i>Codes</i> chapter.

Configuring the WBS

The work breakdown structure can have an unlimited number of levels, or nodes. Instructions for adding, renaming, reordering, changing levels, and deleting are included below.

Tip: All nodes and their children can be expanded and collapsed by right-clicking on a node with at least 2 levels of children.

Steps

Adding Nodes

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. Click the **Add** button. The new node will be added at the same level of whichever node was selected.
3. Begin typing to rename the level.
4. When finished, press **ENTER**.
5. To add a node at a different level, select a node and click the **Add Child** button. The new node will be added 1 level deeper.
6. When finished, click **Close**. Any changes will be automatically saved.

Tip: Nodes can be added directly above or below the selected node by right-clicking and choosing accordingly.

Importing/Exporting Nodes

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. Click the list icon  in the upper-right corner and choose **Import WBS from XML File** or **Export WBS to XML File**.
3. Browse and click **Open** or **Save** accordingly.

Renaming Nodes

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. Double-click the desired node in the list.
3. Begin typing to rename the node.
4. When finished, press the **ENTER**.
5. When finished, click the **Close** button. All changes will be automatically saved.

Reordering Nodes

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. Left-click on the desired node.
3. Without releasing the mouse, drag it up or down until a red line with two arrows appears to indicate where it will be placed.
4. Release the mouse.
5. When finished, click the **Close** button. All changes will be automatically saved.

NOTE: Nodes cannot be reordered to different parents or to different levels. To do so, see [Changing Node Levels](#) below.

Changing Node Levels

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.

2. Left-click on the desired node.
3. Click the **Move** button.
4. Choose **In Level** or **Out Level** accordingly.
5. When finished, click the **Close** button. All changes will be automatically saved.

Modifying Node Colors

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. Click the **Settings** icon  above the upper list.
3. Click the down arrow next to the desired level.
4. Choose a preset or click **Other**.
 1. To enter an RGB value, click the **Custom** tab.
 2. Enter the desired values and click **OK**.
 5. Click **OK** to save the settings.

Deleting Nodes

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open
2. Select the desired node in the list. To delete multiple nodes, press and hold the **SHIFT** or **CTRL** key on the keyboard.
 1. Select additional nodes.
 2. Release the key
 3. Click the **Delete** button.
 4. When finished, click the **Close** button. All changes will be automatically saved.

Viewing the Roll-up

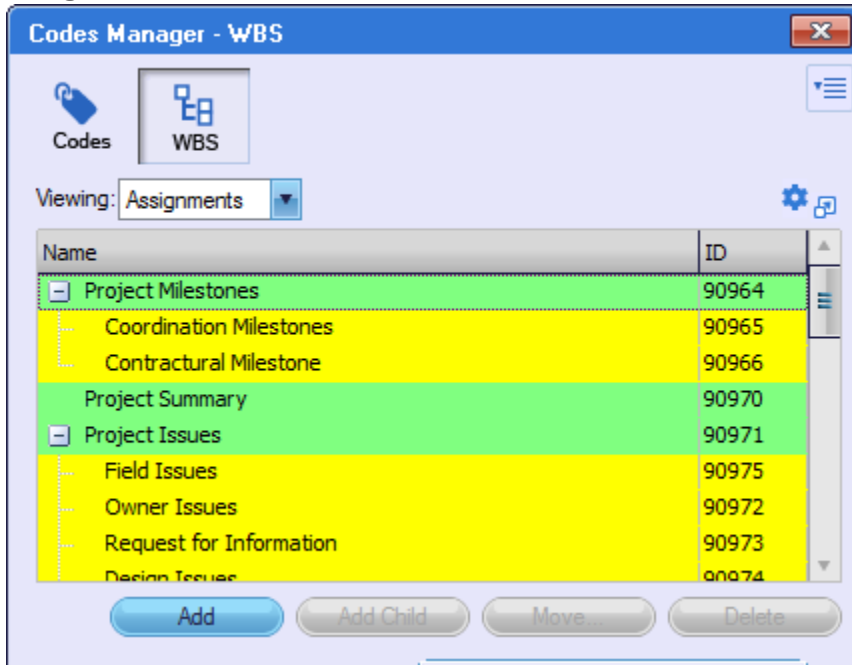
1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open
2. Click the dropdown next to **Viewing** and choose Roll-up.
3. When finished, click the **Close** button. All changes will be automatically saved.

Notes

WBS View

The WBS view contains two sub-views.

Assignments

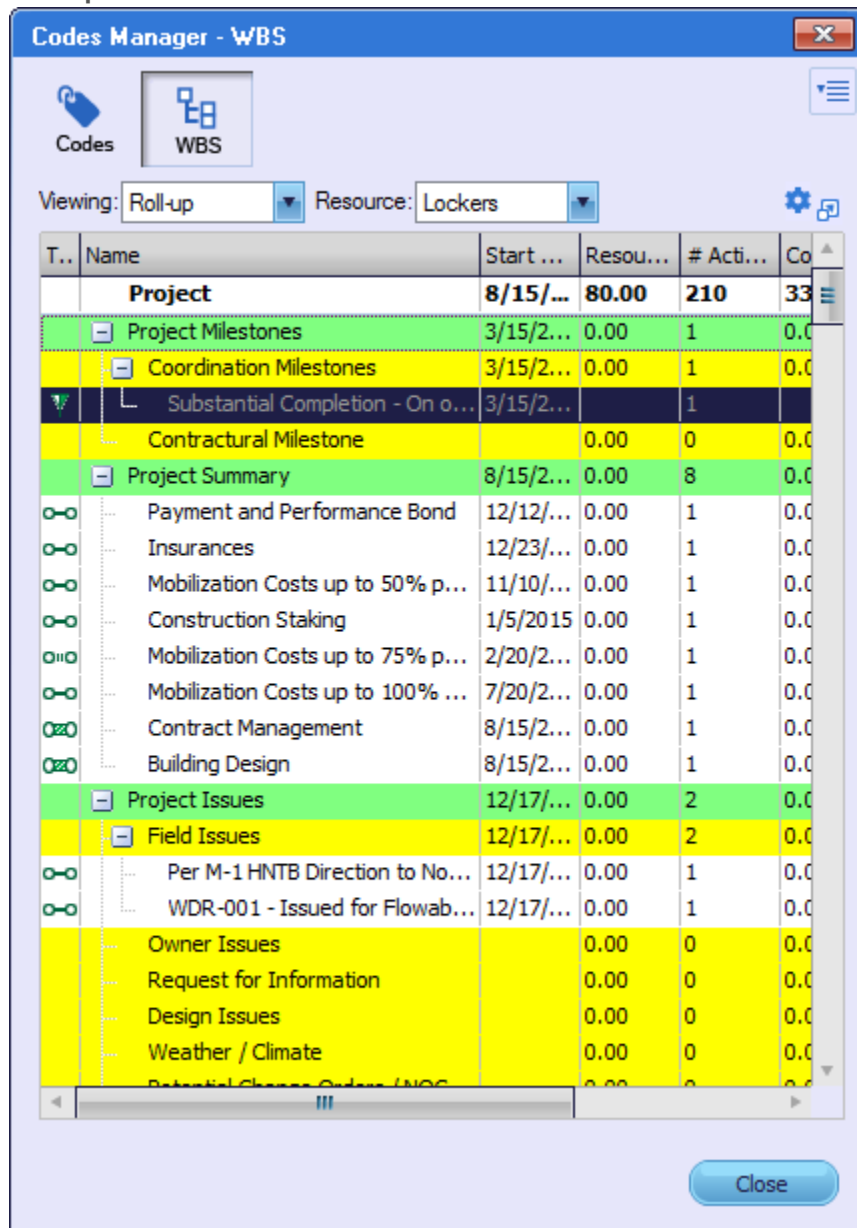


The **Assignments** sub-view is for adding and editing WBS nodes and making assignments.

*Tip: to view a larger version, click the **Expand** icon  above the list. Any customization of the columns will be lost upon closing and reset to that of the Codes Manager.*

Column	Definition
Name	The text field for describing the node.
ID	An automatically generated identifier for each node.

Roll-up



The **Roll-up** sub-view is for analyzing and displaying summaries of WBS nodes and their children. Any activities, milestones, or benchmarks that have not been assigned will show up under an unassigned node at the bottom.

*Tip: to view a larger version, click the **Expand** icon  above the list. Any customization of the columns will be lost upon closing and reset to that of the Codes Manager.*

Column	Definition
Type	An image representing the type of object.
Name	The text field for describing the object.

Column	Definition
ID	The user-entered unique identifier.
Start Date	The date and/or time the object begins.
Finish Date	The date and/or time the object will be finished.
Duration	The amount of time the activity takes to complete.
Resource Consumption	The amount consumed of the chosen resource.
# Activities	The number of activities assigned to the node.
Cost	The cost according to the chosen resource.
Total Cost	The cost across all resources.
Drift	The amount of time the object can gain before advancing the start of the project.
Float	The amount of time the object can be delayed before delaying the completion of the project.
Total Float	Drift + Float.
Grid	The gridline or row number that the object is drawn on.
Calendar	The calendar assigned to the object..

Adding/Modifying Codes

Codes and their values are added and modified through the **Codes Manager**.

Steps

Adding Codes

1. Click **Tools > Manage Codes...** from the menu bar. The **Codes Manager** will open.
2. Click the **Codes** icon.
3. Click the **New Code** button.
4. Begin typing to rename the code and press **ENTER**.
5. Double-click the default value and begin typing to rename it.
6. When finished, click **Close**. Any changes will be automatically saved.

Adding Values

1. With the **Codes Manager** open, select the desired code (or one of its values).
2. Click the **New Value** button.
3. Begin typing to rename the code and press **ENTER**.
4. When finished, click **Close**. Any changes will be automatically saved.

Modifying Codes/Values

1. Click **Tools > Manage Codes...** from the menu bar. The **Codes Manager** will open.
2. Click the **Codes** icon.
3. To rename a code or value, double-click its name in the list.
4. To modify a code's assignment behavior, select it in the list and click the **Behavior** button. The **Advanced Code Properties** window will open.
 - a. Choose accordingly and click **Save**.
5. When finished, click **Close**. Any changes will be automatically saved.

NOTE: Once an assignment has been made, the **Advanced Code Properties** window will still open but the assignment behavior will not be able to be modified.

Deleting Codes/Values

1. With the **Codes Manager** open, select the desired code or value in the list.
2. Click the **Delete** button.
3. When finished, click **Close**. Any changes will be automatically saved.

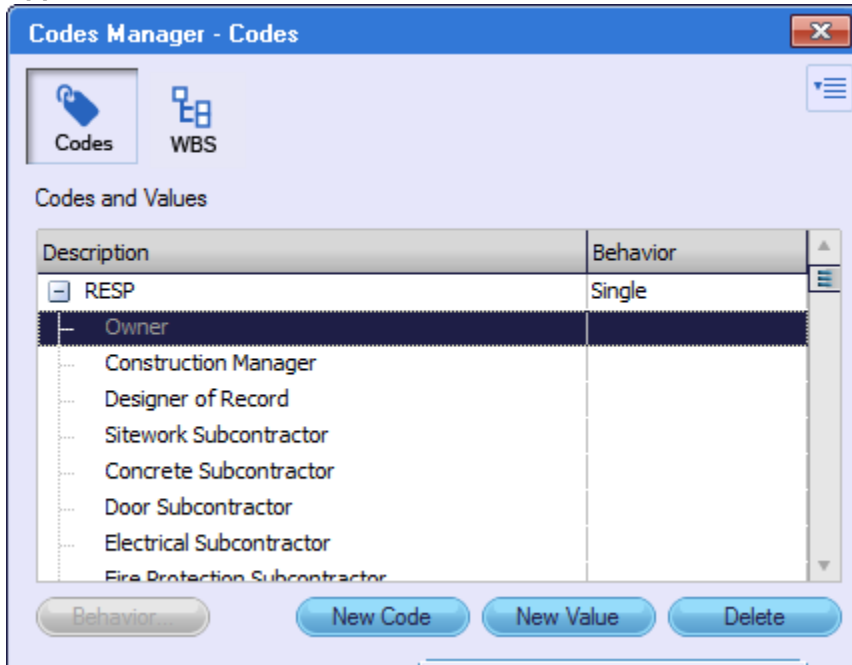
NOTE: If an attempt is made to delete a code or value that has already been assigned, the system will prompt you first.

Notes

Codes View

The **Codes** view contains two list boxes. The main list on top displays all codes and values; the secondary list on bottom displays all activities, milestones, and benchmarks. As you click through the list on top, the bottom list refreshes and shows which objects are assigned to which values. If the item in the list on top is a code, then the bottom list will aggregate all objects assigned to ALL of its values. For more info, see [➔ Making Assignments](#).

Upper List



Column	Definition
Description	The text field for describing the codes and values.
Behavior	Whether or not more than one value of a code can be assigned to the same activity (Multiple). Default as Single . For more info, see the <i>Advanced Code Properties</i> at the end of this section.

Making Assignments

An activity, milestone, or benchmark can only be assigned one WBS node at a time. Making a new assignment will therefore erase any prior assignment, which can be verified in the **Assignment** column.

For codes, assignments actually take place at the value level. Property windows allow you to see which *code values* have been assigned to a specific object. The **Codes Manager** allows you to see which *objects* have been assigned to a different code values. And finally, the **Set Global Properties** window allows you to edit assignments in-bulk for multiple objects. Instructions for each are provided below.

Steps

Assigning a WBS Node

1. Click **Tools > Manage Codes** from the menu bar.
2. In the list on top, select the desired node.
3. In the list on bottom, check or uncheck the desired objects.
4. When finished, click **Close**. Any changes will be saved automatically.

Assigning a Code via Property Window

1. With the **Select Tool**  chosen in the toolbar, double-click the desired object, or right-click and choose **Properties**.
2. Click the **Assign** button next to **Codes**. The **Assign Codes** window opens.
3. Check or uncheck the desired values in the list.
4. When finished, click **Save**.
5. Click **OK**.

NOTE: Single-assignment codes are highlighted in green. Multiple-assignment codes are highlighted in orange. Codes may be clicked to hide or show their values.

Assigning a Code via Codes Manager

1. Click **Tools > Manage Codes** from the menu bar.
2. Click the **Codes** icon.
3. In the list on top, select the desired value.
4. In the list on bottom, check or uncheck the desired objects.
5. When finished, click **Close**. Any changes will be saved automatically.

NOTE: If an attempt is made to assign an object to another value of the same single-assignment code, the system will prompt you first.

Assigning a Code to Multiple Selected Objects

1. With the **Multi-Object Select Tool**  chosen in the toolbar, select the desired activities on-by-one on the canvas. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all the desired activities.
2. Click **Objects > Set Properties for Selected Objects** from the menu bar.
3. Click the dropdown next to **Change Codes** and choose a desired code.
4. Click the next dropdown to the right.
 - a. To assign a value, click twice until a black check appears.

- b. To unassign a value, click once until blank.
 - c. To leave unchanged, click until the minus sign appears. For more info, see [Setting Properties with Check boxes](#) in the *Bulk-Editing* chapter.
5. Repeat steps 3-4 as desired in either the **Activity**, **Milestone**, or **Benchmark** tabs.
6. When finished, click **OK**.

NOTE: When a value of a single-assignment code is checked, all other values of the same code will be automatically unchecked.

Notes

WBS View

Lower List

By default, the list on bottom will be automatically ordered with any objects that have been assigned on top. This can be changed by clicking any of the columns. As you click through the list, the canvas will be scrolled and the item spotlighted. For more info on how to turn off scrolling, see [Enabling/Disabling Canvas Scroll](#) in *The Codes Manager* section

A.	T.	Description	Assignment	Start ...	Finish ...	Duration
☐	▼	Substantial Comp...	Coordination Mile...	3/15/2...	3/15/2...	0
☐	▼	Parking Lots Rea...	Parking Lot, Hard...	2/12/2...	2/12/2...	0
☐	▼	Maintenance Are...		1/29/2...	1/29/2...	0
☐	▼	Construction Cor...		3/4/2016	3/4/2016	0
☐	▼	Men's and Wome...		1/8/2016	1/8/2016	0
☐	▼	Substantial Comp...	PROJECT CLOSE-...	3/15/2...	3/15/2...	0
☐	▼	Final Completion	PROJECT CLOSE-...	5/10/2...	5/10/2...	0

*Tip: to view a larger version, click the **Expand** icon  above the list. Any customization of the columns will be lost upon closing and reset to that of the Codes Manager.*

Column	Definition
Assign	Controls whether or not the object is assigned to the value from above. NOTE: This column will be disabled if the item selected in the list at the top is not a value.
Type	An image representing the type of object.
Description	The text field for describing the object.

Column	Definition
Assignment	The currently-assigned node, if any.
Abbr. Description	The text field limited to 48 characters to be displayed on the canvas if the description is too long
ID	The user-entered unique identifier.
Start Date	The date and/or time the object begins.
Finish Date	The date and/or time the object will be finished.
Duration	The amount of time the activity takes to complete.
Drift	The amount of time the object can gain before advancing the start of the project.
Float	The amount of time the object can be delayed before delaying the completion of the project.
Total Float	Drift + Float.
Grid #	The gridline or row number that the object is drawn on.

Codes View

Lower List

Behavior New Code New Value Delete

Objects assigned to "Owner" Type anything to begin...

A..	T..	Description	Start Date	Finish Date	Duration
☒	▼	1st Floor Acceptance - Ar...	5/19/2014	5/19/2014	
☒	▼	1st Floor Acceptance - Ar...	2/4/2014	2/4/2014	
☒	▼	1st Floor Acceptance - Ar...	1/29/2014	1/29/2014	
☒	▼	1st Floor Acceptance - Ar...	6/19/2014	6/19/2014	
☒	▼	1st Floor Acceptance - Ar...	7/18/2014	7/18/2014	
☒	▼	1st Floor Acceptance - Ar...	7/17/2014	7/17/2014	
☒	▼	1st Floor Acceptance - Ar...	1/20/2014	1/20/2014	
☒	▼	2nd Floor Acceptance - A...	6/26/2014	6/26/2014	

Select Checked Highlight Restore All Close

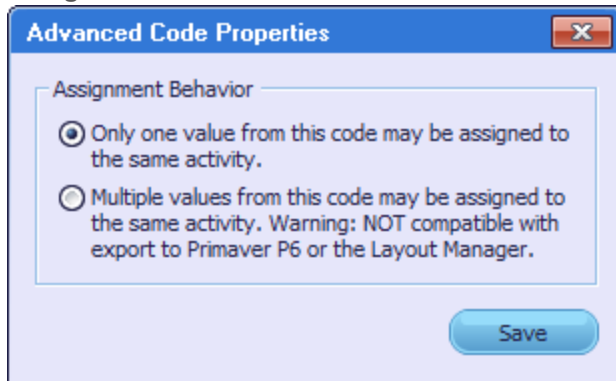
By default, the list on bottom will be automatically ordered with any objects that have been assigned on top. This can be changed by clicking any of the columns. As you click through the list, the canvas will be scrolled and the item spotlighted. For more info on how to turn off scrolling, see [➔Enabling/Disabling Canvas Scroll](#) in *The Codes Manager* section.

*Tip: to view a larger version, click the **Expand** icon  above the list. Any customization of the columns will be lost upon closing and reset to that of the Codes Manager.*

Column	Definition
Assign	Controls whether or not the object is assigned to the value from above.
Type	An image representing the type of object.
Description	The text field for describing the activity.
Abbr. Description	A text field limited to 48 characters to be displayed on the canvas if the description is too long.
ID	The user-entered unique identifier.
Start Date	The date and/or time the activity begins.
Finish Date	The date and/or time the activity will be finished.
Duration	The amount of time the activity takes to complete.
Drift	The amount of time an activity can gain before advancing the start of the project.
Float	The amount of time an activity can be delayed before delaying the completion of the project.
Total Float	Drift + Float.
Grid #	The row number of grid location of the object on the canvas.

Advanced Code Properties

Assignment Behavior



By default, all codes have an assignment behavior of **Single**, meaning that only one value of the same code can be assigned to the same activity. This is required for exporting to P6 or using for swimlanes in the **Layout Manager**. If desired, a code can have an assignment behavior of **Multiple**, meaning that more than one value of the same code can be assigned to the same activity.

NOTE: Behavior cannot be changed unless all assignments are removed.

Creating Swimlanes from Codes

Once assignments have been made, a code or the WBS can be used to organize a schedule into lanes, with one lane being created for each of the code's values or WBS nodes. A lane is represented by a horizontal shade extending the entire width of the canvas, and a name (which may or may not be displayed). The height of the lane is determined by the number of activities in the lane and if and how they overlap.

Steps

1. See [Setting up Swimlanes](#) in the *Automated Layouts* chapter.

Exporting/Importing from Excel

Codes can be exported as part of a linked Excel report or as a single standalone spreadsheet. The WBS can be exported as a standalone spreadsheet. Instructions for both are provided below

Steps

Exporting the Linked Excel Report

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. Click the **List** icon  in the upper-right corner of the **Codes Manager**.
3. Choose **Export Linked Excel File (Default Report)**.

NOTE: For a detailed explanation of this report, see *Object Data* and *Tables* in the section *Exporting Linked Reports*.

Exporting a Standalone Spreadsheet

1. Click **Tools > Open Codes Manager** from the menu bar. The **Codes Manager** will open.
2. Right-click on the columns at the top of any list.
3. Choose either **Export Displayed Rows & Columns** or **Export All Rows & Columns**.

Costs and Resources

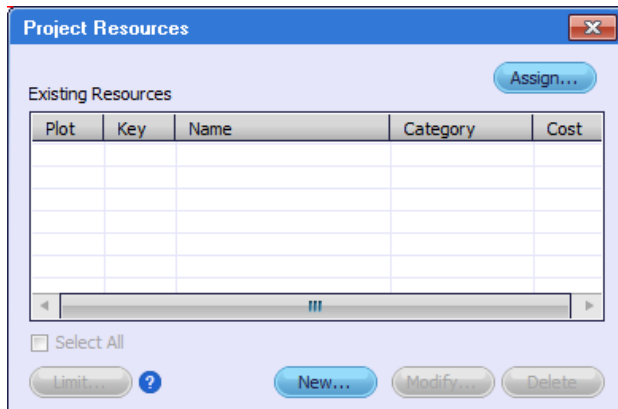
NetPoint allows users to define the resources needed to execute the planned project. A resource can take the form of a skilled human resource (individuals, groups, or crews), equipment, capital/costs, or material. Cost-loading a schedule is done through the resource feature. Follow the steps below for instructions.

Any NetPoint plan may be resource-loaded to calculate resource usage, to level resource overages, and to visually illustrate how resources are used throughout a project. Resources are displayed as a set of histograms and curves at the bottom of the schedule.

There are four steps related to adding resources: defining the project resources, assigning resources to activities, adjusting resource display settings, and applying resource limits.

Defining Resources

To define project resources, click **Tools > Manage Resources...** from the menu bar. The **Project Resources** window opens. To define resources via Excel, see [Adding and Modifying Resources via Excel](#).

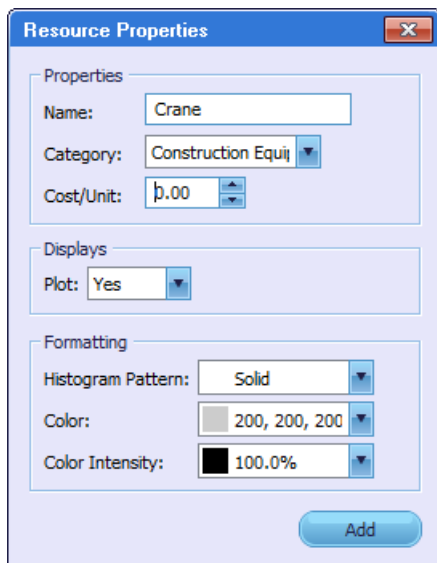


Click the **New...** button.

The **Resource Properties** window opens, offering several resource fields.

Properties

The **Properties** section is where basic settings are changed.



Name

The name is a description of the resource. If cost-loading the schedule, enter a name like **Budget** or **Cost**. This field is required.

Category

The category describes the type of resource. All categories have the same functionality in NetPoint, but they export to external software with different functionality. To choose a category, click the down arrow button to display a dropdown menu. If cost-loading the schedule, choose **Cost**.

Cost/Unit

Cost/unit is the cost of the resource per unit of resource. This field is not currently used in NetPoint, but can be used when exporting to Excel. If cost-loading the schedule, enter the cost as **1**. This field is required.

Displays

The **Displays** section contains aesthetic properties that can be turned on or off. Control is achieved via the dropdown.

Plot

Plot refers to whether or not the resource will be displayed in the profile at the bottom of the schedule. This setting may also be changed directly from the **Project Resources** window under the plot column by checking or un-checking the boxes next to the desired resources.

Formatting

The **Formatting** tab contains aesthetic properties that are not strictly on/off controls, like colors.

Histogram Pattern

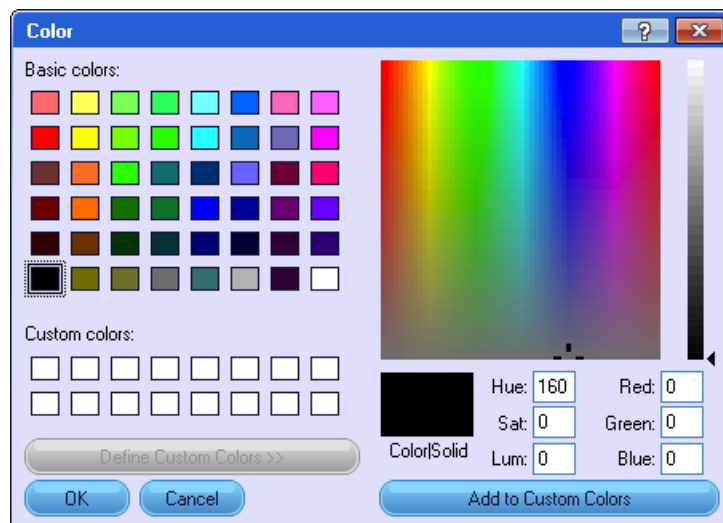
A pattern can be added to differentiate between resources. By default, resources display as solid colors. Click the down arrow button next to **Histogram Pattern** to display a dropdown menu and then click to select the desired pattern. This is very effective for distinguishing between different resources.

Color

This field represents the color of the resource in the histogram. By default, it will be gray, but choosing a unique color is recommended.

To change a color, click the down arrow button anywhere the **Color** field shows up and choose from the list. Any colors already used in the schedule will appear. To choose from a palette or define a custom color, select **More**.

After clicking **More**, the custom color palette window will open.



To create a custom color, click in the rainbow color area; the selected color displays below in the **Color/Solid** box.

To select a darker or lighter tone, move the arrow up and down along the vertical slider bar to the right.

Click the **Add to Custom Colors** button. The new color displays in the **Custom Colors** area and is ready to be selected just the same as the basic colors above.

Click **OK** to apply the color and exit the function.

Once chosen for the first time, any basic or custom color will be displayed in the dropdown list for future selections.

Color Intensity

This number represents the darkness of the histogram bars. Click the down arrow button to display a dropdown menu of percentages.

Finalizing Resources

After entering the fields, click the **Add** button. The new resource will be added to the **Project Resources** window and the **Resource Properties** window will reset.

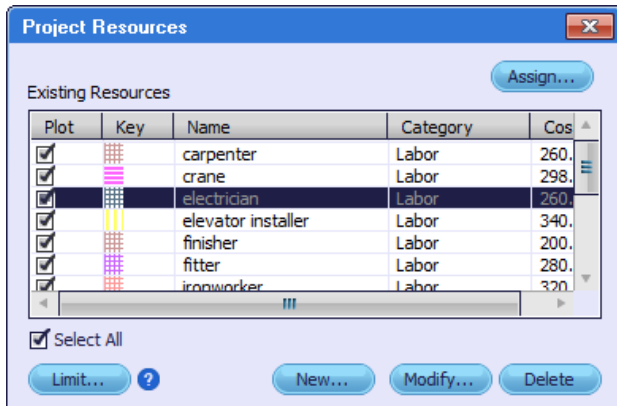
Additional resources may be defined consecutively by repeating the steps above. When finished, click the **Exit** icon  to close the **Resource Properties** window.

NOTE: The **Add** button must be clicked each time to define a resource before clicking the **Exit** icon . Otherwise, the resource will not be saved to the list of Project Resources.

Modifying an Existing Resource

To modify a resource that has already been defined, click **Tools > Manage Resources...** from the menu bar. The **Project Resources** window opens.

Select the resource from the list of **Existing Resources**. At this point, the **Modify...** and **Delete** buttons become enabled.



Click the **Modify...** button.

The **Resource Properties** window opens, showing the several resource fields filled in according to the current properties of the resource.

After making any changes, click the **Update** button and the **Resource Properties** window will reset.

Click the **Exit** icon  to close the **Resource Properties** window.

NOTE: The **Update** button must be clicked before clicking the **Exit** icon . Otherwise, the change will not be saved.

Deleting an Existing Resource

To delete any resource, open the **Project Resources** window, select a resource from the list of **Project Resources**, and click the **Delete** button. Click the **Exit** icon  to close the **Project Resources** window.

Assigning Resources to Activities

Once resources have been defined for the project, the next step is to assign them to specific activities. Each resource assignment approach requires the activity to be selected and the planned units and distribution for the activity to be entered.

There are two ways to assign resources. The first method described is best for assigning multiple resources to a single activity. The second method is best for assigning a single resource to multiple activities.

Steps

A Single Activity

1. Double-click on the activity or right-click on the activity and select **Properties**.
2. Click the **Assign** button next to **Resources**.
3. Click the checkbox next to the desired resource.
4. Click in the **Distribution** column to change to **per Duration**; otherwise, leave as **per Day**.
5. Click in the **Planned Units** column and enter the amount of resource required.
6. When finished, click **OK**.

NOTE: If the Planned Units column is left blank, no assignment will take place.

Notes

Activity Resource Assignment

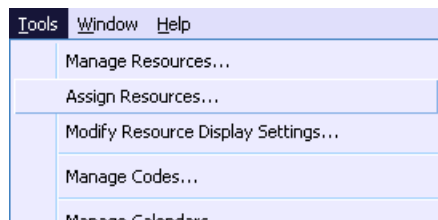
A..	Description	Category	Distribution	Planned Units
☒	Electrician	Labor	per Day	10
☐	Plumber	Labor		
☐	Carpenter	Labor		
☐	Painter	Labor		
☐	Concrete	Material		
☐	Equipment	Equipment		
☐	Drywall	Material		

Column	Definition
Assign	Controls whether or not the resource is assigned.
Description	The text field for describing the resource.

Column	Definition
Category	The type of resource.
Distribution	How the resource is used by the activity. Options include per time period (e.g. per Day if the schedule is in days) or per the entire activity duration (i.e. Per Duration). If a resource is assigned per period, the <i>rate of consumption</i> will remain fixed while the <i>total amount</i> will fluctuate with the activity duration. For example, if activity A has been assigned 10 carpenters per day with a duration of 10 days, then activity A will require 100 carpenter-days. If the duration is reduced to 5 days, activity A will still require 10 carpenters per day but the <i>total amount</i> will be reduced to 50 carpenter-days. If a resource is assigned per duration, the <i>total amount</i> will remain fixed while the <i>rate of consumption</i> will fluctuate with the activity duration. For example, if activity B has been assigned 10 carpenters per duration with a duration of 10 days, then activity B will require 1 carpenter per Day. If the duration is reduced to 5 days, activity B will still require 10 carpenter-days total but the <i>rate of consumption</i> will increase to 2 carpenters per day. NOTE: If any resources have been assigned per period and the schedule time unit is changed, they may not properly convert. For example, 48 widgets per DAY would become 48 widgets per HOUR instead of 2 widgets per HOUR.
Planned Units	The amount of the resource required. Resources may be assigned using up to 5 decimals. Note, however, that they will be rounded to the nearest 100th in Excel and for display purposes only, to the nearest 10th place in the Global Resource Assignment window..

Multiple Activities

To assign a resource, one at a time, to multiple activities, click **Tools > Assign Resources...** in the resource section of the dropdown menu.



The **Global Resource Assignment** window opens. Every activity in the schedule will be listed in the box. (If any objects are selected on the canvas, only those will be listed in the box).

Activities may be ordered by description, ID, start date, finish date, or grid number by selecting from the **Order List By** dropdown button.

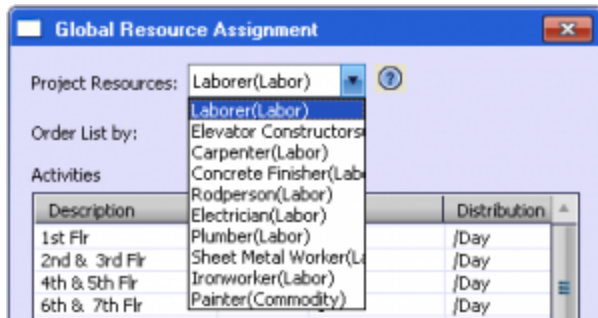
The first column lists the description of the activities.

The second column lists the activity ID.

The third column lists the rate.

The fourth column lists the distribution.

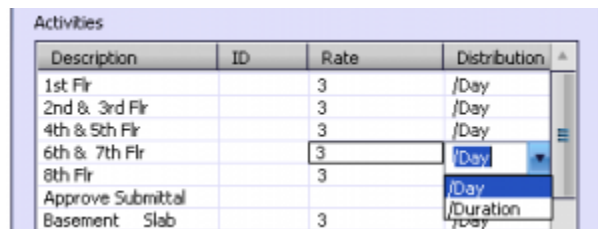
At the top is a dropdown button containing all the resources in the project. To populate the rate and distribution columns, click the down arrow button and select the desired resource.



The rate and distribution columns will populate according to the resource selected. Only one resource may be viewed at a time.

To modify resource assignments, click on the row of the desired activity to enable the rate and distribution columns.

Enter the amount or number of the resource required in the rate column. In the distribution column, choose from the dropdown button.



When finished, click the **Exit** icon

NOTE: If the rate column is left blank, no assignment will take place.

Viewing the Resource Profile

Once resources are defined and assigned to activities, the histogram and curves (i.e. resource profile) will be displayed on the canvas.

Choosing Resources to Display

From the menu bar, click **Tools > Manage Resources...** from the resource section of the dropdown menu. The **Project Resources** window opens.

To control which resources are being displayed, check or uncheck the box under the plot column next to the desired resource.

NOTE: By default, all resources will be displayed unless the plot is turned off during definition of that resource.

Removing Activities

Individual activities can be removed from the resource profile by selecting them on the canvas and then clicking **Objects > Set Properties for Selected Objects**.

In the **Set Global Properties** window, under the **Displays** section, click twice to uncheck the box next to **Resource Profile**. When finished, click **OK**.

The selected activities will be removed from both the histogram and cumulative curve calculations.

Hiding or Showing

To hide the histograms and curves on the canvas, click the **Resource Display Mode** icon in the toolbar.

Modifying Individual Display Settings

To adjust how the histogram is calculated, how to display the plots, or which plots to display, the resource display settings may be modified.

Click **Tools > Modify Resource Display Settings...** from the menu bar. The **Resource Display Settings** window opens with a number of fields.

Calculate Histogram By

This field determines the period over which the histogram is distributed. To change it, click the down arrow button to display a dropdown menu.



The histogram may be calculated based on **Daily**, **Weekly**, **Monthly**, **Quarterly** or **Yearly** totals. If PolyTime is enabled, the histogram may only be calculated based on the schedule time unit or above. For example, if the schedule time unit is set to months, then only **Monthly**, **Quarterly** or **Yearly** totals are available.

The histogram will display the resource usage based on the time unit selected.

Position of Plots

The resource profile can be anchored to the bottom of the canvas or slide automatically as the canvas scrolls.

Position of Plots:	Auto Scroll
Height of Plots (0.1-1):	Fixed at the bottom

If **Auto-Scroll** is selected, then the resource profile is always visible. Scrolling to any specific vertical point on the chart keeps the display of resources constant.

If **Fixed at the bottom** is selected, then the resource profile always anchors at the bottom of the chart, which might not be seen in the current displayed page but can be seen by scrolling to the bottom of the chart. This option will cause the screen display of resources to exactly match with the printed schedule.

Height of Plots

The resource profile can cover as much of the vertical height of the window as desired, with 1 being the maximum.

Height of Plots (0.1-1):	0.25
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Limits

If this box is checked and resource limits have been applied, then any resource which is being consumed over its allowable maximum will be displayed at the cursor when hovering over the histogram on the canvas.

Plots

There are four available plots: the **Histogram plot**, the **Baseline plot**, the **Early Dates plot**, and the **Late Dates plot**.

Displays		Plots	Plot Rulers	Increment
	Histogram:	☒	☒	2
	Baseline:	☒	☐	25
	Early Dates:	☐	☐	1
	Late Dates:	☒	☐	25

Click the checkbox under the plot column to display the corresponding graph on the canvas.

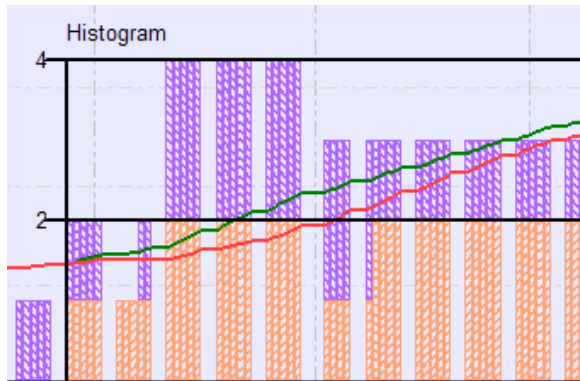
NOTE: By default, all graphs are plotted unless turned off.

The **Histogram** plot displays as a color-coded bar chart.

The **Baseline** curve displays as a green (cumulative) resource curve.

The **Early Dates** curve displays as a blue (cumulative) resource curve.

The **Late Dates** curve displays as a red (cumulative) resource curve.



Histograms represent the *daily consumption* of resources (or weekly, monthly, etc. as long as it's greater than the schedule's time unit) for each activity.

Baseline represents the *total cumulative consumption* of resources for every activity. Baseline can be calculated as $RES_n = RES_{n-1} + R_i$, where "n" denotes every day between the planned dates for activity "I" and R_i denotes the resource daily-rate for activity "I".

Early Dates represents what the *total cumulative consumption* of resources would look like if all activities were on their early-dates.

Late Dates represents what the *total cumulative consumption* of resources would look like if all activities were on their late-dates.

Plot Rulers

Each plot has its own plot ruler. The plot rulers correspond with the list of plots to the left. They may only be displayed if their corresponding plot is displayed.

To display a desired ruler, check the corresponding checkbox.

When checked, the increment of the ruler can also be adjusted.

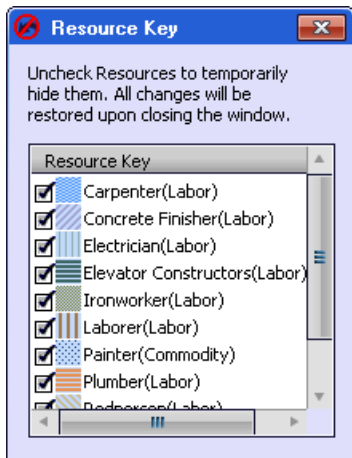
Plot Ruler Increments

To adjust the increment of the plot ruler, enter the increment value desired in the field. This value refers to the amount of resources between each horizontal line. For example, with an increment of 50, every line represents an additional 50 resources. If the increment were 25, there would be twice as many ruler lines, with the difference between each line being only 25.

Reading the Resource Graphs & Curves

When measuring consumption for individual resources, make sure only that resource is displayed in the resource profile. For example, to measure the cost for a particular schedule, all resource except for **Budget** would be turned off. This is most easily done by opening the resource key.

To view the resource key for resources being displayed, choose **View > Resource Key...** The **Resource Key** window opens, displaying all resources currently on screen.

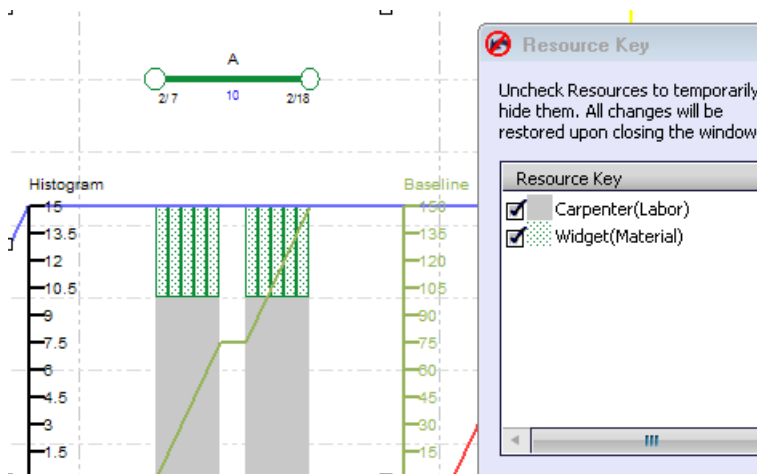


To temporarily display or hide a resource's histogram, uncheck the box to the left of the resource name.

Any displayed resources hidden through the resource key only remain hidden until the **Resource Key** window is closed.

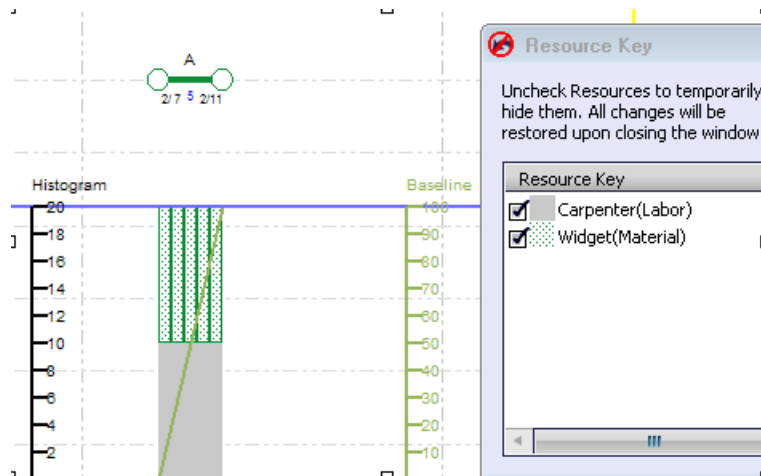
At that point, all previously hidden resources will display as set in **Tools > Manage Resources....**

In the following example, Activity A requires 15 units *each day* of a combination of resources: about 10 of those are carpenters and about 5 of those are widgets.



We know this by measuring the height at which each resource coincides with the histogram plot ruler. This is what the histogram does: it makes it quick and easy to see graphically which resources have been assigned to which activities and how much are required each day.

In this example, the Activity's duration has been changed from 10 days to 5 days.



Note that it still requires 10 carpenters per day even after the duration has changed, which means that it must have been assigned 10 carpenters *per day*. The amount of widgets, however, has increased from 5 to 10, which means that Activity A must have been assigned 50 widgets *per duration*.

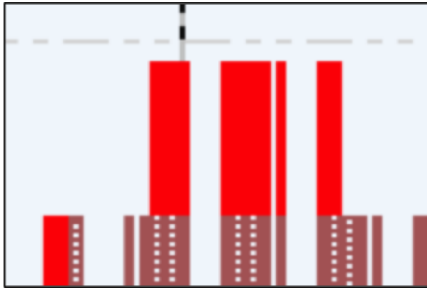
The resource curves show the *total cumulative consumption* of resources for every activity. For Activity A, the total consumption is derived by multiplying the histogram (i.e. daily consumption) by the total duration of Activity A. From example 1 to example 2, you can see that the total consumption drops from 150 to 100 as the duration decreases.

Add the total consumption for every activity together and the result is the baseline curve, colored green. The early-dates curve, colored blue, shows what the total consumption would look like if every activity were on its early-dates. The late-dates curve, colored red, shows what the total consumption would look like if every activity were on its late dates.

Limiting Resource Consumption

A limit is an object that restricts how much of a resource may be used at any given time. It is drawn on the canvas, like an activity, with a certain height and “duration”. The height at which the limit is drawn corresponds to the maximum value activities can consume of that resource when overlapping with the limit.

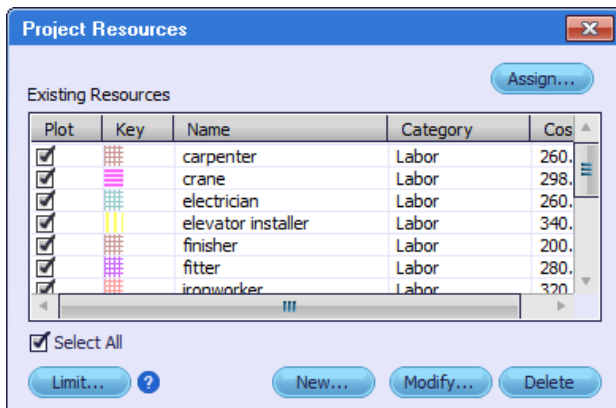
After defining resources, assigning resources to activities, and adjusting the resource display settings, the user may set resource consumption limits for specific time periods. Any resource overage appears as a red portion of the histogram over each of the displayed resources.



NOTE: A limit can never overlap on the timescale with another limit. As a result, a new limit cannot be added to a period that already has a limit. The original limit must be deleted, moved, or crashed in duration before a new one can be drawn.

Adding Limits

First, make sure that at least one resource plot is displayed on the canvas. To set a resource limit, click **Tools > Manage Resources...** from the menu bar. The **Project Resources** window opens.



Click the **Limit** button. The **Resource Limits** window opens.

First, select the resource for limit definition by clicking the resource name in the list. The following fields will fill accordingly. They may be modified or left as default.

Resource Limits

- 1) Select a Resource by clicking the name in the list.
- 2) Adjust the Canvas Ruler or leave as default.
- 3) Click "Start" and the Canvas Ruler will be laid over the Histograms.
- 4) Click the Limit icon to toggle the cursor to the crosshairs. Click and drag the crosshairs to place the Limit at the desired maximum.
- 5) When finished, click "Done" to finalize the Limit.

Existing Resources

Name	Group
Backhoe 4wd	Labor
elec crew	Labor
hyd excavator 35klbs	Labor
landscape crew	Labor
site crew	Labor
test	Commodity
track dozer jd 650h	Labor

Canvas Ruler

Ceiling: Increment: Segments=0

Height of Plots (0.1-1):

Start

Ceiling

This is the ceiling, or upper bound of the ruler, up to which resource limits may be drawn. It cannot be greater than 2x the highest daily assignment of that resource to any given activity. In other words, for any given day, the ceiling cannot be more than twice the requirement for that day. Limits may only be drawn beneath this ceiling. This is a required field.

Increments

This value refers to the amount of resources between each horizontal line. For convenience, the number of segments this corresponds to is also displayed below.

Height of Plots

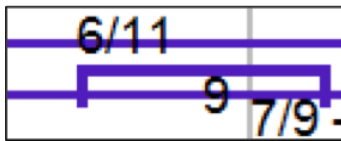
This is how much vertical height the resource profile will cover, with 1 being the maximum.

After modifying the fields or leaving them as default, click **Start** and the canvas ruler will be laid over the histograms. This ruler is provided as a system of measurement for drawing.

Click the **Limit** icon to toggle the cursor between the **Select Tool** and the crosshairs.

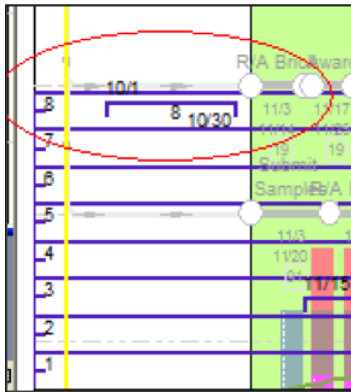
To add a limit object, click and drag the crosshairs, paying attention to both the grid level, or height, as well as the time period, or duration of the limit object. Limits may be drawn at any height beneath the maximum.

Once placed, the limit start date and end date are shown and the limit quantity displays under the bar.



Click the **Done** button to finalize the limit. Additional limits may be assigned by repeating the steps above or concurrently before finalizing.

In this example a maximum resource limit of **8** is set for a selected resource from 10/1 to 10/30.



If any resource limits were assigned before for the selected resource, the limits corresponding to previous assignments are shown as purple bars.

Modifying or Deleting Limits

To modify or delete a resource limit, open the **Assign Resource Limits** window by clicking **Tools > Manage Resources...** from the menu bar and clicking the **Limit** button.

Select the desired resource from the list.

Click the **Start** button.

At this point, a purple ruler spanning the canvas, or gridlines, will be laid over the histograms as well as any existing limit objects.

To graphically modify a limit, hover over the end of the bar to access the double arrows or select the bar and use the arrow keys.

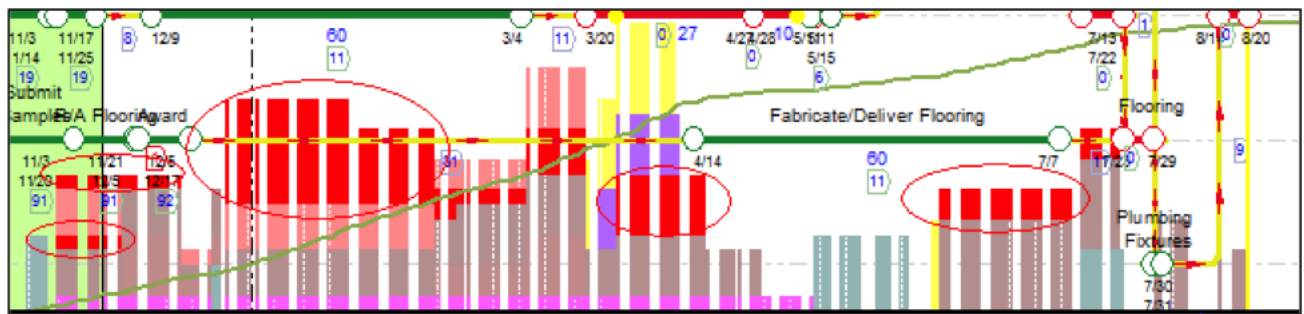
To delete a limit, select it using the **Select Tool** and click **Delete**.

When finished, click the **Done** button.

Displaying Limit Overages

After applying resource limits to selected resources, the out-of-limit portions can be highlighted. To display this warning, click the **Resource Limit Mode** icon on the toolbar. Overages are displayed in red.

NOTE: Limit overages will only be displayed when the histogram is plotted in the same time unit as the schedule. To adjust how the histogram is calculated, see [➔Modifying Individual Display Settings](#).

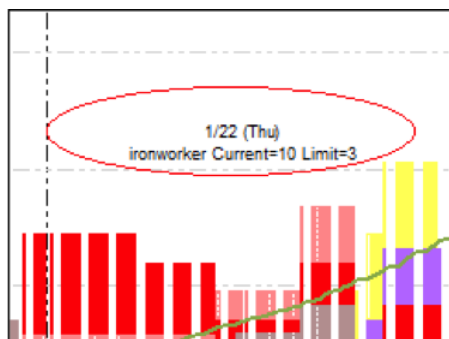


To see specific overages, select **Tools > Modify Resource Display Settings...** from the resources section of the dropdown menu.

The **Resource Display Settings** window opens.

Select the checkbox under limits and click the **Save** button.

Once this option is selected, it will display current and maximum resource consumption levels when the mouse cursor moves over any red portion of the Histogram.



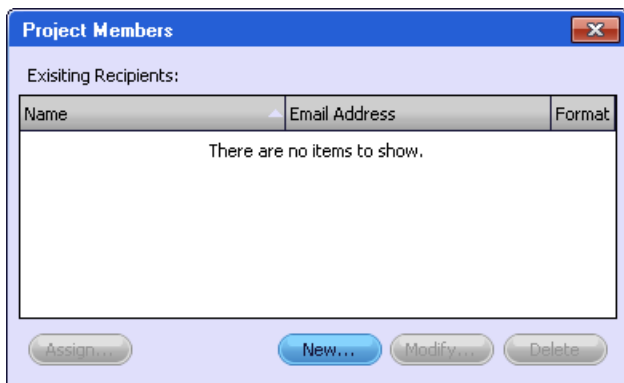
Noting Changes in Consumption

NetPoint has the capability to send email notifications to designated users regarding changes in resource consumption. These notifications may be sent using HTML (recommended) or in TEXT (provided to allow reading on devices which do not support HTML formatting).

Defining Email Recipients

Before this feature can be used, a list of potential recipients must be created. For each recipient, a full name, email address, and email format must be specified.

To define email recipients, click **Tools > Manage Notification Recipients...** from the menu bar. The **Project Members** window opens.



Click the **New...** button.

The **Recipient Info** window opens with the following fields.



Name

Enter a person in the **Name** field. This field is required.

Email Address

NetPoint will only check for a valid format. As a result, be sure to enter the **Email Address** correctly.

Email Format

This field determines the format of the email message sent. **HTML** is the recommended format. **Text** should only be used when the email client does not support HTML. Choosing **Both** creates a message containing both formats, and allows the email client to determine which format should be displayed.

Text Format

Resource notifications for Richard Hewens

Resource: Concrete

[changed] Distribute Schedule
01/31/2011(was: 01/04/2011) - 04/06/2011(was: 03/10/2011) (48 days)
rate: 200.000 per activity

[changed] Review NetPoint Schedule
09/03/2010 - 12/10/2010(was: 11/23/2010) (69 days (was: 57))
rate: 60.000 per activity

Resource: Hershey Bar

[changed] Distribute Schedule
01/31/2011(was: 01/04/2011) - 04/06/2011(was: 03/10/2011) (48 days)
rate: 12.000 per activity

Resource: Limestone

[changed] Review NetPoint Schedule
09/03/2010 - 12/10/2010(was: 11/23/2010) (69 days (was: 57))
rate: 15.000 per activity

HTML Format

Resource notifications for Richard Hewens

Status	Activity	Start Date	Finish Date	Duration	Res. Usage
Resource: Concrete					
changed	Distribute Schedule	12/22/2010 01/04/2011	02/25/2011 03/10/2011	48	200.000 per activity
changed	Review NetPoint Schedule	09/03/2010	11/12/2010 11/23/2010	50 57	60.000 per activity
Resource: Hershey Bar					
changed	Distribute Schedule	12/22/2010 01/04/2011	02/25/2011 03/10/2011	48	12.000 per activity
Resource: Limestone					
changed	Review NetPoint Schedule	09/03/2010	11/12/2010 11/23/2010	50 57	15.000 per activity

Finalizing Recipients

After entering in the fields, click the **Add** button. The new recipient will be added to the **Project Members** window and the **Recipient Info** window will reset.

Additional recipients may be defined consecutively by repeating the steps above. When finished, click the **Exit** icon  to close the **Recipient Info** window.

NOTE: The **Add** button must be clicked each time to define a recipient before clicking the **Exit** icon . Otherwise, the recipient will not be saved to the List.

Modifying and Deleting Recipients

To modify a recipient that has already been defined, click **Tools > Manage Notification Recipients...** from the menu bar. The **Project Members** window opens.

Select a recipient from the list of **Existing Recipients**. At this point, the **Modify...** and **Delete** buttons become enabled.



Click the **Modify...** button.

The **Recipient Info** window opens, showing the several info fields filled in accordingly.

After making any changes, click the **Update** button and the **Recipient Info** window will reset.

Click the **Exit** icon  to close the **Recipient Info** window.

NOTE: The **Update button must be clicked before clicking the **Exit** icon . Otherwise, the change will not be saved.**

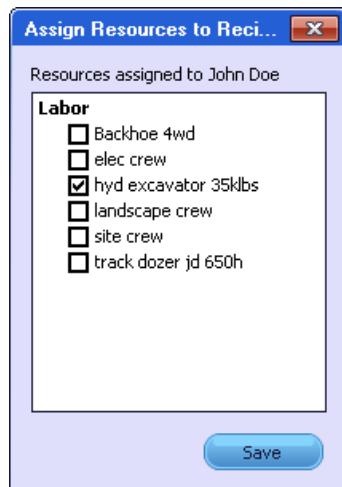
To delete any recipient, open the **Project Members** window, select a recipient from the list of **Existing Recipients**, and click the **Delete** button. Click the **Exit** icon  to close the **Project Members** window.

Assigning Resources to Recipients

Assigning resources to recipients is a way to designate an individual recipient responsible for tracking its usage.

To assign resources to individual recipients, click **Tools > Manage Notification Recipients...** from the menu bar. The **Project Members** window opens.

Select a recipient and click the **Assign...** button. The **Assign Resources to Recipients** window opens.



All existing resources will be listed, grouped by resource category, even if they haven't been assigned to any activities yet. Resource categories may be minimized by clicking their title. Any resources already assigned to the selected recipient will contain a check.

To assign or unassign resources, check or uncheck their boxes respectively.

When finished, click **Save** to save and close the window.

Notifying Recipients

To generate and email resource notifications, click **Tools > Generate Resource Change emails...** from the menu bar. The **Send Notification Emails** window opens.



The name and email of the sender will be filled in by default. They may be changed, if desired.

When finished, click **Send**. All recipients assigned resources will get an email.

The email notifications will show ALL resources assigned to the recipient and their corresponding activities with the following status options:

No change

There have been no changes since the last email.

Added

Something has been added since the last email. For example:

- A new resource was assigned to the recipient
- A new activity was assigned a resource which the recipient is responsible for

Deleted

Something has been removed since the last email. For example:

- A resource was unassigned from the recipient
- An activity was unassigned a resource for which the recipient is responsible
- A resource assigned to the recipient was deleted
- An activity was deleted which was assigned a resource for which the recipient was responsible

Changed

Something has changed since the last email. For example:

- An activity has been changed (e.g. its start date, finish date, or duration) which was assigned a resource for which the recipient is responsible
- An activity's assignment has changed (e.g. the quantity or distribution) for a resource which the recipient is responsible

4) Optimizing a Schedule

NetPoint's synchronous interface, which allows users to build schedules graphically, makes the user responsible for the layout of the network. This differs greatly from traditional CPM applications, where the software authors the layout of the network with little-to-no input from the user. Although this greatly enhances the presentation of the schedule, such flexibility can become overwhelming when working with larger schedules, and intimidating when importing schedules from CPM applications, such as Primavera or MS Project, or from Excel.

In these situations, we recommend running the automated layout engine. This tool greatly enhances the layout of the schedule by assisting with, or completely taking control of, the placement and layout of objects in the network. Underneath the hood, dozens of factors are taken into account, including relationships between activities, link geometries, text and description lengths and positions, and grid and canvas dimensions. For example, how many finish-to-start chains are there? Which links overlap with other links or activities? Are descriptions clashing? Answers to answer these questions and more are found in a matter of seconds.

In addition to layout, descriptions and IDs may be repositioned with the Global Edit Mode to minimize clashing; dates and durations can be adjusted by using the gestural interface to level resources; or the schedule may be analyzed for integrity or other factors. NetPoint provides a number of tools to facilitate these processes.

Layouts

In NetPoint, it's possible to save multiple potential layouts for the same network or schedule. For example, you could have one layout where activities are organized based on work trades; another, using lanes based on WBS categories; and yet another optimized to avoid overlaps using the automated layout engine.

Layouts only store positions of objects, geometries of links, and visibility of text boxes and shades. Any other changes made to a layout – including the positioning, hiding, or showing of data elements; the adding, deleting, fading, or colors of objects; or any customization of the canvas – will apply across all layouts.

This section details instructions for the following:

1. Creating/modifying
2. Analyzing links
3. Viewing statistics
4. Exporting lists to Excel

The Layout Manager

The **Layout Manager** itself is non-modal, meaning that you can navigate to other menus or make modifications on the canvas without having to close it. The upper list of the **Layouts** view can be customized by having its columns sorted, added and removed. As you click through various lists, the canvas will be scrolled and the item spotlighted. In addition, all lists can be individually exported to Excel.

NOTE: Sorting, position, and selection are not saved and will refresh when changing views or leaving and coming back.

Sorting, Adding, and Removing Columns

1. Click **Schedule > Open Layout Manager** from the menu bar. The **Layout Manager** will open.
2. Click the desired icon at the top.
3. To sort by a certain field, click once on the column's label to sort from top to bottom, or click twice to sort from bottom to top.
4. To add or remove a column, right-click on any column.
 - a. Choose the desired field. A checkmark will appear or disappear and the list will refresh.
 - b. To close the menu, click anywhere outside of it.
5. Repeat step 4 as desired.
6. To remove a column graphically (Layouts only), left-click on the desired column.
 - a. Without releasing the mouse, drag it up or down until a black "X" appears.
 - b. Release the mouse.
7. Repeat step 6 as desired..
8. To rearrange a column (Layouts only), left-click on the desired column
 - a. Without releasing the mouse, drag it left or right until two red arrows appear to indicate where it will be placed.
 - b. Release the mouse
9. When finished, click **Close**.

NOTE: As new columns are added, existing columns will be shrunk to make more space, except for in the Layouts view, where a scroll bar will be added. Any customization is tied to the computer (not to the schedule) and will be reset upon re-installation of NetPoint.

Exporting a List to Excel

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. In any view, right-click on the columns at the top of any list.
3. Choose either **Export Displayed Rows & Columns** or **Export All Rows & Columns**.
4. Repeat one by one for all lists desired.

Notes

Manager Views

Icon	Definition
 Layouts	The Layouts view is for creating, modifying, and deleting layouts. For instructions, see ➔ Creating/Modifying in the <i>Layouts</i> section..
 Structure	The Structure view is for applying swim lanes and locking objects on grids. For instructions, see Setting up Swimlanes and ➔ Locking Relative Positions , respectively.
 Selection	The Selection view is for determining how activities are selected for automated layout processing. For instructions, see Running the Engine .
 Placement	The Placement view is for configuring how activities are arranged during automated layout processing. For instructions, see Running the Engine .
 Links	The Links view is for viewing and correcting links in the network, including redundancies. For instructions, see ➔Analyzing Links .
 Statistics	The Statistics view is for analyzing layout metrics. For instructions, see ➔Viewing Statistics .

Creating/Modifying

Layouts are created in the **Layout Manager**. Once created, a layout cannot be modified and then saved again. This means any changes made to a layout can only be captured by creating a new layout (which will automatically be based on what is seen on the canvas), and then deleting the original layout. To assist you, NetPoint keeps track if any changes were made, and it will prompt you to capture a new layout, when necessary.

If an object is deleted from one layout, it will also be deleted from every layout. If an object is added to one layout, it's position will be the same (or as close as possible, depending on the number of grids) across all layouts. Once a layout is created, a separate file with a .NPL extension is also created within the schedule's project folder containing supporting data.

Steps

Creating New Layouts

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. Under the list of **Saved Layouts**, click the **New** button. A new layout will be added to the list based on the network currently being viewed on the canvas.
3. To rename the layout, select it in the list under the **Description** column and enter a new name.
 - a. When finished, click outside the layout or hit **Enter**.
4. Click **Save** or **Close** to both save the list of layouts and close the manager.

NOTE: At this time, a layout cannot be updated after it has been saved. To effectively update a layout, first apply the layout to the canvas that you want to update, make the desired changes on the canvas, save a new layout (which will automatically be based on the current network), and then delete the layout you originally wanted to update.

Switching Between Layouts

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. Choose a layout in the list and click **Apply**. Alternatively, you may right-click on the layout to apply it.
3. If the layout on the canvas contained unsaved changes, a warning will open. To save click **Cancel** and see *Creating New Layouts* above. Otherwise, click **Apply, Discard Changes**. The canvas will update with the network of the selected layout.
4. Click **Save** or **Close** to both save the list of layouts and close the manager.

NOTE: When switching between layouts, pay special attention to shades and text boxes, because they may be positioned optimally in one layout but not make sense in another. For example, if a layout was ran with lanes, previously created shades and text boxes may clash with the lanes' shades and text boxes. For this reason, the Layouts view includes a section at the bottom for turning on and off shades and text boxes specific to certain layouts.

Deleting Layouts

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. Choose a layout in the list and click **Delete**. After deleting the layout, the canvas will still remain the same.
3. Click **Save** or **Close** to both save the list of layouts and close the manager.

NOTE: Layouts will be deleted without being asked for prior confirmation. To restore a deleted layout, click the **Undo** icon in the toolbar immediately following the delete, and then click anywhere in the **Layout Manager** to refresh the window.

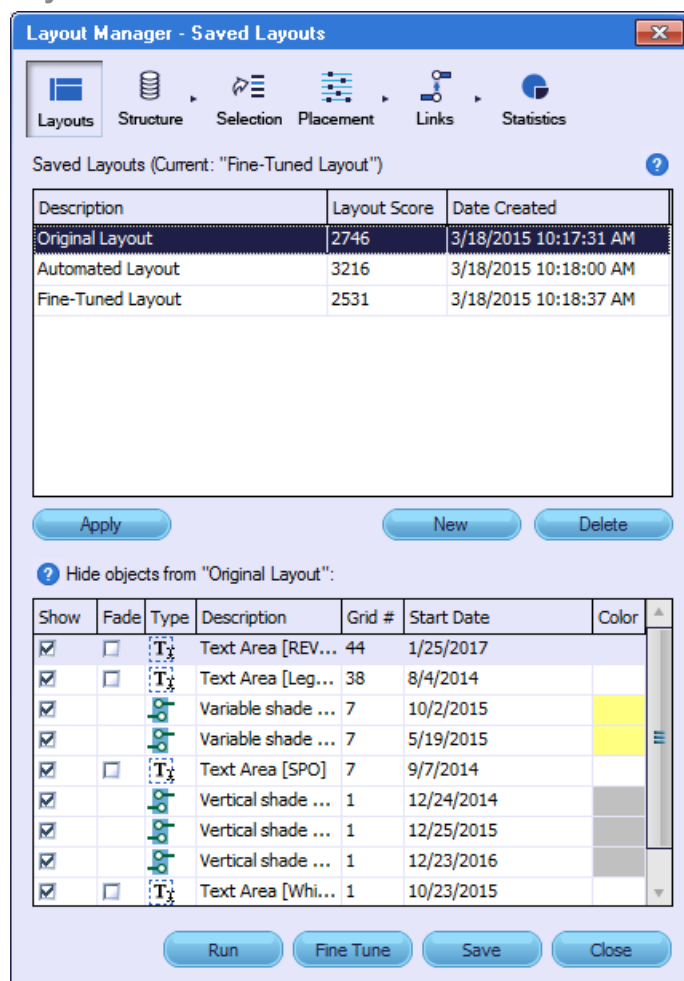
Managing Shades/Text

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. Choose a layout in the list. The list on the bottom will refresh.
3. To hide a shade or text object, select it in the list and uncheck the box under **Show**. The canvas will update in real-time.
4. To fade a shade or text object, select it in the list and check the box under **Fade**. The canvas will update in real-time.
5. Click **Save** or **Close** to both save the list of layouts and close the manager.

NOTE: The display of shades and text will be unique to each layout; however, the position in one layout will apply to all layouts.

Notes

Layouts View



The **Layouts** view contains two list boxes.

Upper List

The main list on top displays all saved layouts. As you click through the top list, the bottom list refreshes and shows which shades and text objects are being displayed in a given layout.

Column	Definition
Description	A text field for describing the layout.
Layout Score	A measure of layout optimization. The lower the score, the tidier the layout relative to link/activity crossings, link overlays, and link length in terms of number of grid crossings. For more info, see ➔Link Scoring .
Date Created	The date and time the layout was stored.
Structure Criteria	Checkboxes displaying whether or not certain structure criteria was used when the layout was generated (for automated layouts). For more info, see ➔ Locking Objects on Grids and Setting up Swimlanes .
Selection Criteria	Displays the type of selection criteria used when the layout was generated (for automated layouts). For more info, see Selection View from <i>Running the Engine</i> .
Placement Criteria	Displays the type of placement criteria used when the layout was generated (for automated layouts). For more info, see the Placement View sections from <i>Running the Engine</i> .
Link Penalty Criteria	Displays the link penalty values when the layout was generated (for automated and manual layouts). For more info, see ➔Link Scoring .
Statistics	Displays statistics for when the layout was generated (for automated and manual layouts). For more info, see ➔Viewing Statistics .

Lower List

The secondary list on bottom displays all user-created shades and text objects on the canvas. As you click through the list on the bottom, the canvas will scroll to their location and they will spotlighted in yellow.

Column	Definition
Show	Controls whether or not the object will be displayed on the canvas.
Fade	Controls whether or not the object is faded on the canvas.
Type	Displays whether the object is a shade or text box.

Column	Definition
Description	Displays a description of the object, and the text (if it's a text box).
Grid #	The grid number or row the object is located on.
Start Date	The date the object is located at on the canvas.
Color	The color of the object (shades only).

Analyzing Links

If activities and other objects are the backbone of the schedule, then the links can be said to be the muscles – they're the ones that make everything possible, including float calculations. Analyzing the health of your links can be done in the **Links** view of the **Layout Manager**. For example, links that are redundant can be viewed, hidden, or faded on the canvas, and links that are convoluted (i.e. intersecting or overlapping) can be identified and filtered for. Instructions for both are provided below.

Steps

Redundant Links

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. To choose a different layout, select it in the list and click **Apply**.
1. If the layout on the canvas contained unsaved changes, a warning will open. To save, click **Cancel** and see [➔Creating New Layouts](#). Otherwise, click **Apply, Discard Changes**. The canvas will update with the network of the selected layout.
3. Choose the **Links** view.
 - a. To spotlight a link on the canvas, select it in the list.
 - b. To hide a redundant link on the canvas, check the box under the **Hide** column next to the link.
 - c. To fade a redundant link on the canvas, check the box under the **Fade** column next to the link.
 - d. To delete a redundant link on the canvas, select it in the list and click **Delete**.
4. Click **Save** or **Close** to both save and close the manager.

NOTE: Deleting a link is permanent to the schedule and will apply across all layouts.

Convoluted Links

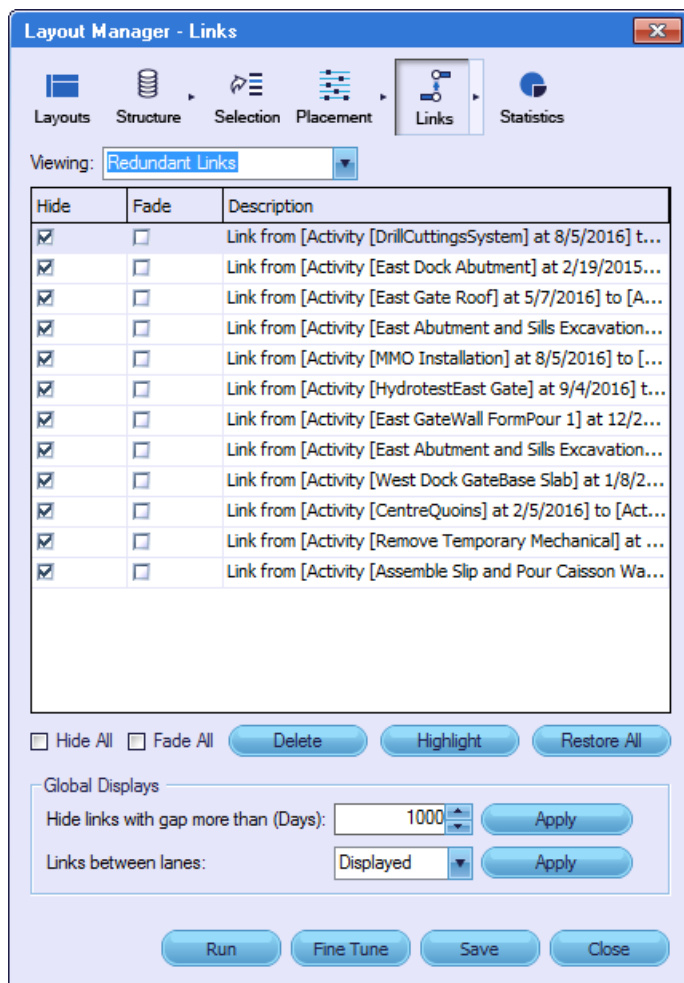
1. Click **Schedule > Open Layout Manager...** from the menu bar.
2. To choose a different layout, select it in the list and click **Apply**.
1. If the layout on the canvas contained unsaved changes, a warning will open. To save, click **Cancel** and see [➔Creating New Layouts](#). Otherwise, click **Apply, Discard Changes**. The canvas will update with the network of the selected layout.
3. Click the **Links** view.
4. Click the down arrow button next to **Viewing** and choose **Convoluted Links**.
5. Click the up and down arrows next to **Filter our scores <** to filter the list of convoluted links.
 - a. To spotlight a link on the canvas, select it in the list.
6. Click **Save** to close the manager. Otherwise, any threshold entered will not be saved.

Notes

Links View

Redundant Links

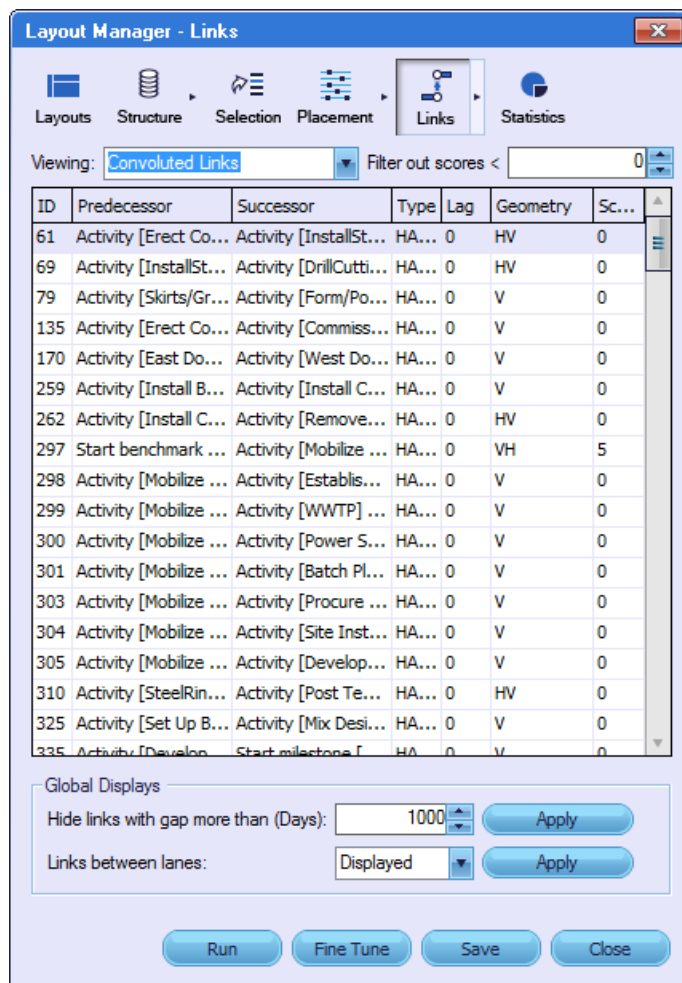
Any and all redundant links will show up in the list. As links are selected in the list, the canvas will scroll to their location and they will be spotlighted in yellow. For more info, see [➔Treating Redundancies](#).



Column	Definition
Hide	Controls whether or not to hide or show the link on the canvas.
Fade	Controls whether or not to fade the link on the canvas
Description	A description of the link defined by the object it's emanating from and merging into.

Convolved Links

A convoluted link is simply a link with a geometry that yields a high or less-than-perfect score. As links are selected in the list, the canvas will scroll to their location and they will be spotlighted in yellow. The **Filter out scores** will determine which links show up in the list below. Any links with a score above the threshold will show up. By default, the threshold is equal to 0, which means all links whether convoluted or not will show up at first. As the number increases, the number of links displayed will likely decrease. For more info, see [Link Scoring](#). The value set here for the threshold will also be used in the **Statistics** view.



Column	Definition
ID	A number used internally to identify the link.
Predecessor	The object the link emanates from.
Successor	The object the link merges into.
Type	A label for categorizing links. The designation is for supplemental information; it has no bearing on actual schedule logic or calculations. For more info, see the Logic Types section in <i>Creating FS Links</i> .
Lag	A way to enforce a minimum gap. For more info, see Adding Link Lags .
Geometry	The physical shape of the link. A link can bend vertically, horizontally, and diagonally, in eight different combinations. NetPoint makes use of a scoring system to identify the best possible geometry for a link. For more information, see Link Geometries .

Column	Definition
Score	A point value based on how many grids the link passes through, or if it crosses over other objects or overlaps with other links, for example. For more information on scoring, see Link Scoring .

Global Displays

Any changes made here will ONLY apply to the layout on the canvas currently being viewed, NOT to any previously stored layouts.

Column	Definition
Hide links with gap more than (Days)	An option for hiding links based on how much gap they have.
Links between lanes	An option for hiding, showing, or fading links that span between swim lanes. For more info, see Setting up Swimlanes.

Viewing Statistics

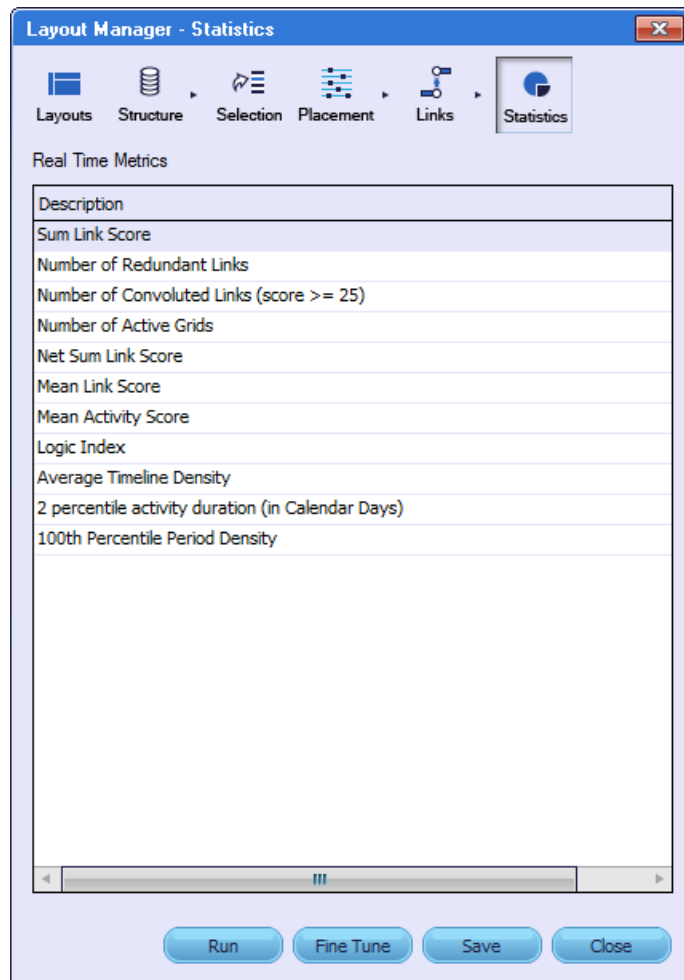
Whether or not one layout is “better” than another is oftentimes subjective. With the introduction of layouts, a number of new statistics are available to help you objectively analyze different aspects of your schedule. For example, the number of activities divided by the number of weeks can tell you how busy your schedule is; and the number of links divided by the number of activities can tell you how dense the logic is. Characteristics like these and other “vital-signs” can be found in the **Statistics** view of the **Layout Manager**. Instructions are provided below.

Steps

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. To choose a different layout, select it in the list and click **Apply**.
 - a. If the layout on the canvas contained unsaved changes, a warning will open. To save, click **Cancel** and see [➔Creating New Layouts](#). Otherwise, click **Apply, Discard Changes**. The canvas will update with the network of the selected layout
3. Choose the **Statistics** view.
4. Click **Save** or **Close** to both save and close the manager.

Notes

Statistics



The following will always correspond to the layout currently being viewed on the canvas.

Term	Definition
Sum Link Score	The sum of the link scores for all links in the network. For more info, see ↗Setting Defaults .
Number of Redundant Links	The number of redundant links in the network. For more info, see ↗Treating Redundancies .
Number of Convolved Links	The number of links with a score beneath the indicated threshold. By default, the threshold will be set at 25 until it's set in the Links view, at which point whatever threshold was last used there will also be used here. For more info, see ↗Setting Defaults .

Term	Definition
Number of Active Grids	The number of grids with objects placed on them by the layout engine.
Net Sum Link Score	The sum of the link scores for all links in the network excluding links between lanes if present. For more info, see ➔Setting Defaults .
Mean Link Score	The sum of the link scores for all links in the network divided by the number of links in the network. For more info, see ➔Setting Defaults .
Mean Activity Score	The sum of the activity scores for all activities in the network divided by the number of activities in the network. For more info, see ➔Setting Defaults .
Logic Index	The number of valid links between activities (i.e. excluding redundant links) divided by the total number of activities.
Average Timeline Density	The number of activities divided by the number of weeks (for a schedule in days) from the first object to the last object in the schedule. For a schedule in hours, it would be per day; for a schedule in months, it would be per quarter; etc.).
2 percentile activity duration	The value which 2% of all activities have a duration less than or equal to. The width of these activities on the canvas can be set in the Dimensions view of the Placement view.
100th Percentile Period Density	The highest number of activities scheduled per week (for a schedule in days). For a schedule in hours, it would be per day; for a schedule in months, it would be per quarter; etc.).

Exporting Lists to Excel

Any list in the **Layout Manager** can be exported as a single standalone spreadsheet. Instructions provided below

Steps

Exporting a Standalone Spreadsheet

1. Click **Schedule > Open Layout Manager** from the menu bar. The **Layout Manager** will open.
2. Choose the desired view by clicking its icon.
3. Right-click the columns at the top of any list.
4. Choose either **Export Displayed Rows & Columns** or **Export All Rows & Columns**.

Automated Layouts

Automated layouts are generated by running a schedule through the layout engine. Each time the layout engine is run, a new layout is automatically created and stored in the **Layouts** view of the **Layout Manager**, alongside any other manual or automated layouts.

Running the layout engine can be as complex or as simple as needed. If you use the default or most recently saved settings, it can be run with just the click of a button. This may be useful for a novice user or for an already-optimized schedule, where, over the course of the schedule life-cycle, a particular strategy has proven to yield a consistently desirable layout. Alternatively, a variety of settings can be adjusted in the **Layout Manager**, isolating the effects of the minutest of changes and giving advanced users even greater control over the resulting layouts.

Any settings that are saved using the **Save** button will be shared across schedules and across sessions of NetPoint.

NOTE: The time it takes to run an automated layout depends on the number of objects (activities, links, milestones, etc.), the computer's processor/clock speed, how much RAM it has, and the settings in the Layout Manager. Once a schedule reaches 250 objects, we suggest tweaking some of these settings to reduce the time required. An "Out of Memory" error may occur on schedules with over 2100 objects (700 activities + 1400 links). This should be considered an approximate upper limit. For more details, see [Running On Large Schedules](#).

This section details instructions for the following:

1. Running the engine
2. Locking objects on grids
3. Setting up swimlanes

Running the Engine

When you run an automated layout, every object in the schedule is analyzed, indexed, and placed back on the canvas. This analysis begins by identifying clusters of activities and other objects within the network based on certain characteristics. For example, objects may be grouped by float chains, step chains, or logic steps (defined below). Once all of the objects in the network have been identified, they're then indexed and sorted. To determine their order, a series of rules for breaking ties is used. For example, if three step-chains have been identified, the chain with the most activities will be ordered first. In the case that two of them share the same number of activities, then the chain with the longest duration will be ordered first, and so on. These settings are located in the **Selection** view.

After all objects have been analyzed and indexed, they will start to be placed back on the canvas. For example, placement can begin at the top or in the middle of the canvas, and grids can be added to make more space. These settings are located when viewing **Rules** in the **Placement** view. Minimum dimensions can also be changed to affect how and when activities are placed next to one another, and these settings are located when viewing **Thresholds** in the **Placement** view.

There are actually two different processes that can be performed: a "run" and a "fine-tune," depending on the state of the schedule. "Running" will completely rearrange the network, using the settings from the **Structure**, **Selection**, **Placement**, and **Penalties** views to improve the schedule's layout. This is best for schedules that are being imported from another application or where a significant overhaul of the layout is desired. "Fine-tuning" will only make small adjustments based on activity and link scores from the **Placement** and **Penalties** views. This is typically best when attention has already been given to the layout, or when an automated layout has already been run before. If the resulting fine-tune cannot improve the layout (i.e. increases the layout score by more than 5%), the fine-tune will be canceled and the canvas will revert to the previous layout.

NOTE: Whether "running" or "fine-tuning," all activities and planning objects have the potential to be rearranged unless they have been assigned to a lane, locked on a grid, or actualized left of the data date. For more info, see [Setting Up Swimlanes](#) and [Actualizing Objects](#).

Steps

Run from Default Settings

1. Click **Schedule > Run Automated Layout** from the menu bar, or click the **Run Automated Layout**  icon in the toolbar.
2. The canvas will update and a new layout will automatically be saved in the **Layouts** view of the **Layout Manager**.

Run from Custom Settings

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. Click through the **Structure**, **Selection**, and **Placement** views to modify the available settings.
3. When ready, click **Run**.
4. The canvas will update, the manager will switch to the **Layouts** view, and a new layout will automatically be saved.
5. To save the settings for future use, click **Save**. Settings that are not unique to the schedule will carry through when accessed from other schedules or across sessions. Otherwise, click **Close**.

Fine-Tune Only

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. Click through the **Placement** and **Penalties** views to modify the available settings.
3. Click **Fine-Tune**.
4. The canvas will update, the manager will switch to the **Layouts** view, and a new layout will automatically be saved.
5. To save the settings for future use, click **Save**. Setting that are not unique to the schedule will carry through when accessed from other schedules or across sessions. Otherwise, click **Close**.

NOTE: To fine-tune a layout with swim lanes, make sure **> Always Fine-Tune After Running** is selected in the **Schedule** menu. Otherwise, clicking **Fine-Tune** will not maintain lane configurations.

Running On Large Schedules (>250 activities, links, milestones, etc.)

1. Click **Schedule** from the menu bar and make sure **Animate When Running** and **Always Fine Tune After Running** are turned off (they should not have a check next to them).
2. Choose **Open Layout Manager** from the **Schedule** menu, click the **Structure** view, and make sure lanes are turned off (the box next to **Derive Lanes from** should be unchecked).
3. Click the **Placement** view.
4. Next to **Start at**, make sure **Halfway Grid** is chosen.
5. Click **Run**.
6. OPTIONAL: If you want to run with swim lanes, click the **Structure** view, check the box next to **Derive Lanes from**, and click **Run** again.
7. Click **Fine-Tune**. Note: this step may take a few minutes with overly large schedules.
8. In the **Layouts** view, click **New**.
9. Select it in the list and type a new name.
10. To save the settings for future use, click **Save**. Setting that are not unique to the schedule will carry through when accessed from other schedules or across sessions. Otherwise, click **Close**.

Resetting the Layout Manager

1. Click **Schedule > Layout Manager Settings** from the menu bar.
2. Choose **Load Last Saved** to reset to the last saved state or **Load Defaults** to reset to the default state.

NOTE: All settings that are not unique to the schedule will be reset; notable exclusions include lists, such as layouts, codes, links, or statistics, for example.

Notes

Selection View

Layout Manager - Selection

Layouts Structure **Selection** Placement Links Statistics

Select Activities by

☒ Float Chains ?

☐ Step Chains ?

☐ Logic Step ? ☒ Consider FS Strings

Selection Order

I/O	Rule	Direction
☒	Number of Locked Activities	Least
☒	Number of Activities	High to Low
☒	Total Float	Low to High
☐	Chain Duration	Low to High
☐	Logic Index	Low to High
☐	Start Date	Early to Late
☐	Finish Date	Early to Late

Run Fine Tune Save Close

Select Activities by

Objects may be identified and clustered based on float chains, step chains, or logic steps.

Field	Definition
Float Chains	A float chain is a path of activities through the network where the total float of each successor is the same as the total float of its immediate predecessor (the critical path is an example of a float chain whose activities share a total float of zero). When chosen, a chain of activities who share the same total float with one another is selected and placed (for example, the critical path), followed by another chain of activities who share the same total float with one another, and so on. Choose this option when you want your layout to be organized in terms of float values.
Step Chains	A step chain is a path of activities through the network where the logic step of each successor is exactly one higher than its immediate predecessor. When chosen, a chain of activities with consecutive logic steps is selected and placed, starting with an object that has a logic step of 1, followed by an object with a logic step or 2, and so on, ending with an object that has the

Field	Definition
	highest logic step. Once complete, another chain of activities with consecutive logic steps is identified in a similar manner, and so on. Choose this option when you want your layout to be organized in terms of relationships.
Logic Step	A logic step is the numbered position of an activity in a logic chain (if in multiple logic chains, the highest of those numbers). For example, any activity in the network that is connected to the project start will have a logic step of 1, regardless of how late in time the activity is scheduled. When chosen, first all the activities with a logic step of 1 are selected and placed, then all the activities with a logic step of 2, and so on. Choose this option when you want your layout to be drawn from left to right.
FS Strings	An FS string is path of three or more activities (either a float chain or step chain) that are connected in a finish-to-start relationship (each intermediate activity having only one predecessor and one successor). FS Strings can be chosen if either Float Chains or Step Chains is chosen. When selected, the chains containing FS strings are sorted after all the chains without FS strings but before individual objects are sorted. Check this option when you want FS strings to be drawn on the outside (top and bottom) of the network.

Selection Order

After clusters have been identified, they are ordered based on the following rules, which can be turned on or off or can be rearranged by clicking and dragging them up or down in the list. In any case, the clusters will always be ordered first followed by individual objects. By default, these rules have already been configured to yield the best result for the majority of schedules. It's difficult to suggest when to apply or not apply certain rules, or to change the direction, due to the individuality of every schedule. **We suggest using trial and error until you are satisfied with the layout.** The following options are available for both float chains and step chains. The direction can be reversed by clicking in the cell and using the drop-down button.

Rule	Definition
Number of Locked Activities	The number of locked activities in the chain.
Number of Activities	The number of activities in the chain.
Total Float*	The total float of the chain. *Only available for float chains.
Chain Duration	The length of the chain in calendar days.
Logic Index	The number of links between activities (excluding redundant links) divided by the total number of activities in the chain.

Rule	Definition
Start Date	The start date of the earliest activity in the chain.
Finish Date	The finish date of the last activity in the chain.

The following options are available for logic steps. Since logic steps are not chains, ordering is applied on the individual object level.

Rule	Definition
Start Date	The start date of the object.
Duration	The duration of the object.
Finish Date	The finish date of the object.

Placement View(Rules)

Layout Manager - Rules

Layouts Structure Selection **Placement**

Rules
Thresholds

Grid Placement

Start at: Halfway Grid Proceed: Down Before Up

☒ Choose active over inactive ☐ Replenish inactive grids when used

Aesthetic

Layout Spacing: Double

☒ Stack activity descriptions if needed ☒ Allow links with a D leg

☐ Hide links with gap > 1000

☒ Show links between lanes: Displayed

☐ Include redundant links in link score calculations

Run Fine Tune Save Close

Grid Placement

After the starting grid is utilized, whichever grid yields the lowest score for that object will always be chosen. In the case that no single grid yields the lowest score, the settings for **Proceed** and **Choose active over inactive** will be used as tiebreakers. For more info on activity and link scores, see [➔Link Scoring](#).

Field	Definition
Start at	The starting grid is where the first object (and any subsequent objects with no predecessors) will be placed when the layout engine is run. You can choose the top, middle, or bottom grid of the canvas. Choose the top grid if you want most milestones and benchmarks toward the top, and a generally top-heavy layout; the halfway grid if you want most milestones and benchmarks toward the middle, and an evenly balanced layout; and the bottom grid if you want most milestones and benchmarks toward the bottom, and a generally bottom-heavy layout. For large schedules or to reduce processing time, choose Top. For more details, see Running On Large Schedules.
Proceed	If an object cannot be placed on the grid with the lowest score – for example, if the preceding relationship is start-to-start or finish-to-finish; or, if there is already an object or link in that location; or, if the grid above and below yield the same score – the direction will come into play. In these cases, you can choose to proceed up first or down first. Choose Up Before Down when starting at the bottom grid.
Choose Active Over Inactive	An active grid is one that already contains an object. An inactive grid is one that is totally empty. If an object cannot be placed on the grid with the lowest score, objects will be placed on grids which already have objects (active) by default. Uncheck this rule if you have tight grid spacing or layout spacing to create a taller layout.
Replenish inactive grids when used	During processing, inactive grids are added in-between active grids. In the event that an object gets placed on one of these inactive grids, two inactive grids are added to maintain the proper spacing, so long as the checkbox is selected.

Aesthetic

The following settings are visual only:

Field	Definition
Layout Spacing	Layout Spacing will determine how many inactive (empty) grids are added in-between active grids <i>after</i> processing. Regardless of what's chosen (Single , Double , or Triple), one and only one grid will be added <i>during</i> processing. Single spacing will result in a higher layout score. For more info, see ➔Link Scoring . Increase this setting if you want more padding in between rows of activities or a taller layout.

Field	Definition
Stack activity descriptions if needed	By default, activity descriptions will be stacked if they begin clashing with other text or objects. Uncheck this rule if you have tight grid spacing or layout spacing and prefer to shorten descriptions manually where necessary.
Allow links with a D leg	By default, diagonal links will be used when necessary. This setting also applies when creating links in NetPoint. Uncheck this rule if you do not want any diagonal links.
Hide links with gap >	Controls whether or not long links (links spanning a wide period of time) will be hidden, depending on how many days of gap they have. Check this rule if you do not want links spanning a wide period of time to show up, and adjust the threshold accordingly.
Show links between lanes	Controls whether or not tall links (links spanning multiple lanes) will be displayed, hidden, or faded. Only relevant if the layout is being run with lanes activated.
Include redundant links in link score calculations	By default, redundant links will not be included in link score calculations, and will therefore not affect the resulting layout. Check this rule if you want the position and geometry of redundant links to be taken into account when laying out the network. For more info, see ➔Treating Redundancies

Placement View (Thresholds)

Layout Manager - Thresholds

Layouts Structure Selection **Placement** Rules Thresholds

Grid Spacing
Value (inches): 0.35

Zero-gap Linked Activities
Allow on same grid if 1 activity is at least (inches): 0.50
or
If both activities are at least (inches): 0.25

Non-linked Activities
Allow on same grid if separated by at least (inches): 0.50 ?

Displays & Formatting
☐ Add page tiles so shortest activities are at least (inches): 0.50
Max. width for milestone and benchmark descriptions (inches): 1.00

Run Fine Tune Save Close

Grid Spacing

Field	Definition
Value (inches)	Grid spacing is the distance in inches between gridlines (by default, 0.35 inches). Increase this number if you want more padding in between rows of objects or a taller layout.

Zero-gap Linked Activities

Field	Definition
Allow on same grid...	If the durations of two zero-gap linked activities are too small, it can be difficult to tell where one activity begins and another one ends (when they're on the same grid). The Zero-gap Linked Activities thresholds are for deciding when such activities can be placed on the same grid. Increase these numbers if you want to increase the frequency with which zero-gap linked activities are placed on separate gridlines.

Non-linked Activities

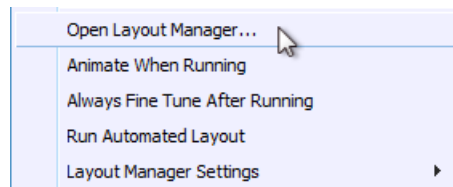
Field	Definition
Allow on same grid...	If non-linked activities are too close to one another on the same grid such that any link between them wouldn't be visible, it could appear as though they were linked. The Non-Linked Activities threshold is for deciding what distance is too little for placing such activities on the same grid. Increase this number if you want more space between non-linked activities.

Displays & Formatting

Field	Definition
Add page tiles...	Controls whether or not page tiling will be added (but not removed) to ensure that 98% of all activities are at least the width specified, when zoom and stretch are 100%. Decrease this number if you have activities with extremely short and extremely long durations together in the same schedule (e.g. 1 day activities with 100 day activities).
Max width...	If milestone and benchmark descriptions exceed the Max. width for milestone & benchmark names, additional lines will be added accordingly. Decrease this number if your milestones and benchmarks have long descriptions and are close to other objects.

Automated Layouts Menu

The automated layouts menu is located in both the **Schedule** menu as well as underneath the **Run Automated Layout** icon in the toolbar.



Field	Definition
Open Layout Manager	This menu item opens the Layout Manager to the Layouts view.
Animate When Running	If selected, the layout will be animated as it's redrawn in real-time. For large schedules or to reduce processing time, make sure it is turned off. For more details, see Running On Large Schedules.
Always Fine Tune After Running	By default, "running" and "fine-tuning" are applied separately, and the results are stored in different layouts. If selected, running will always include a fine-tune, and the results will be saved in a single layout in the Layouts view. For large schedules or to reduce

Field	Definition
	processing time, make sure it is turned off. For more details, see Running On Large Schedules .
Run Automated Layout	Select to start processing.
Layout Manager Settings	Within the Layout Manager , there's a difference between "lists" and "settings". In short, "lists" are schedule-specific (except for the one in the Selection view), whereas "settings" can carry through from one project to another. If Load Defaults is selected, all "settings" will reset to their states when NetPoint was installed. If Load Last Saved is selected, all "settings" will reset to their states when the manager was most recently saved. To load the settings from a previously run layout, see Resetting The Layout Manager .

Locking Relative Positions

Locking is a way to freeze an object relative to other objects. For example, if you have activity A locked on grid 1, B locked on grid 10, and C locked on grid 20, then A will be above B, and B will be above C, but not necessarily on the same gridlines. If lanes are used and an activity is both locked and assigned to a lane, the lane assignment will override the locking. By default, actualized objects are locked, but this can be deselected.

Locking can be done on an activity-by-activity basis, or it can be done automatically by entering a threshold and locking all of the activities whose activity scores fall below that threshold. All activities and planning objects will be rearranged unless they have been assigned to a lane or locked on their grids. Instructions for locking individual activities and for locking multiple activities are provided below. For more info on lanes, see [Setting Up Swimlanes](#).

Steps

Individual Objects

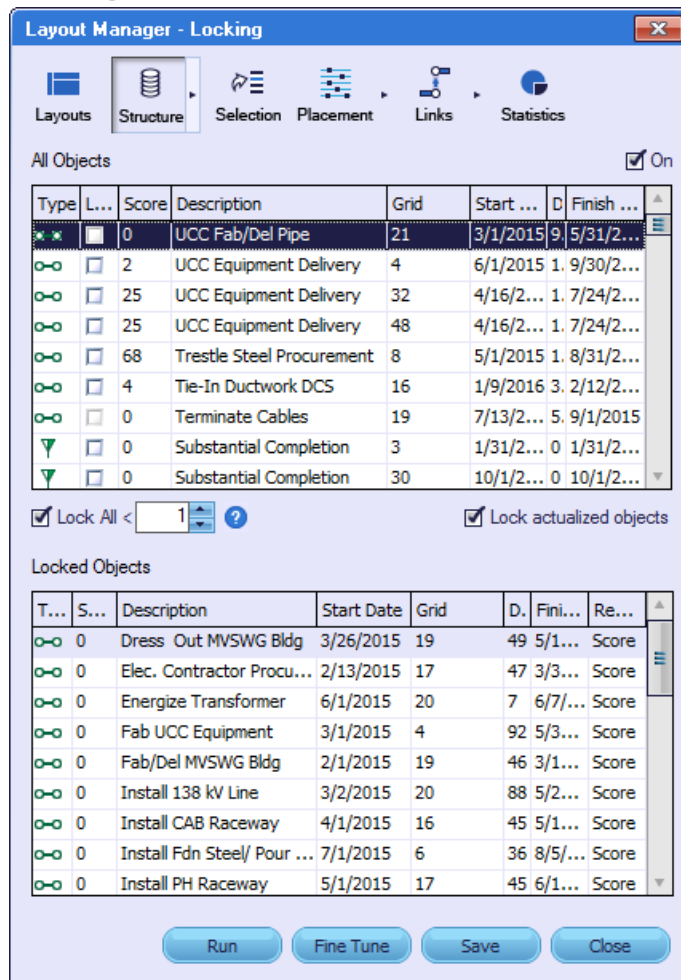
1. Click **Schedule > Open Layout Manager** from the menu bar.
2. Click the arrow to the right of the **Structure** view, and choose **Locking**.
3. Check the box next to **On**. **NOTE: If the box is un-checked, the most recent locking setup will be stored internally for future use.**
4. To lock an object, check the box next to it under the **Lock** column.
5. To apply immediately, see [Running the Engine](#). Otherwise, click **Save** to store the changes.

Multiple Activities

1. Click **Schedule > Open Layout Manager** from the menu bar.
2. Click the arrow to the right of the **Structure** view, and choose **Locking**.
3. Check the box next to **On**. **NOTE: If the box is un-checked, the most recent locking setup will be stored internally for future use.**
4. Check the box next to **Lock All**.
5. Type a number and press the **TAB** key on your keyboard (or click the up and down arrows with the mouse). All activities whose score is less than this value will be locked (milestone and benchmarks will not be included with this option). **NOTE: If any objects were locked individually, they will be combined with the activities locked here.**
6. To apply immediately, see [Running the Engine](#). Otherwise, click **Save** to store the changes.

Notes

Locking View



The locking view consists of two list boxes.

Upper List

The main list on top shows all activities, delays, milestones, benchmarks, and hammocks in the schedule. As you click through the list on the top, the canvas will scroll to their location and they will be spotlighted in yellow. As any of the objects from the list above are locked, they are copied to the list below.

Column	Definition
Type	The type of object.
Lock	Controls whether or not the object will be locked to its respective grid.
Score	The sum of the link scores for every valid (not redundant) link connected to that activity divided by the number of links connected to that activity. For more information on scoring, see Link Scoring .

Column	Definition
Description	The text field for describing the activity.
Abbr. Description	A text field limited to 48 characters to be displayed on the canvas if the description is too long.
Grid	The gridline or row number that the object is drawn on.
Start Date	The date the object begins.
Duration	The amount of time the object takes to complete.
Finish Date	The date and/or time the object will be finished.

Lower List

As you click through the list on the bottom, the canvas will scroll to their location and they will be spotlighted in yellow.

Column	Definition
Type	The type of object.
Score	The sum of the link scores for every valid (not redundant) link connected to that activity divided by the number of links connected to that activity. For more information on scoring, see ➡Link Scoring .
Description	The text field for describing the activity.
Abbr. Description	A text field limited to 48 characters to be displayed on the canvas if the description is too long.
Grid	The gridline or row number that the object is drawn on.
Start Date	The date the object begins.
Duration	The amount of time the object takes to complete.
Finish Date	The date and/or time the object will be finished.

Column	Definition
Reason	The reason the object was locked, whether by the user, from the score, or because it was actualized.

Locking Schedule Dates

To prevent unintentional changes to the horizontal positions of objects on the canvas, the **Date Lock** mode can be used. **Date Lock** mode prevents any mouse-based changes to horizontal positions of objects on the canvas; however, if need be, dates can still be changed via property boxes, and all objects can still be moved to different grids.

Steps

1. Click the down arrow to the right of the **Change Display Mode**  icon from the toolbar.
2. Choose **Date Lock** from the menu.

NOTE: Pressing the ALT key while moving an object will temporarily enable **Date Lock**.

Criticality

In GPM (and CPM), the critical path is the longest continuous path to the project completion (or an interim contractual milestone), when anticipating on-time or early completion. If anticipating late completion, the critical path is any continuous path that extends beyond the contract date for a completion milestone.

In NetPoint, criticality (if based on total float) is a signal of whether an activity's total float is at or below a threshold (often zero, except in the case of early or late completion). For special circumstances where criticality is to be based on drift or float (rather than total float), that option is allowed within NetPoint, and the threshold can be changed to any number, positive or negative.

For example, if a project were already significantly behind schedule, a large number of activities might already be critical. To reduce the noise and identify only the most critical, the threshold could be changed to -5 instead of 0. Conversely, if a project were ahead of schedule, there might not be any critical activities. In this case, the threshold could be set to positive 5 so that even activities with total float were signaled out. By default, critical activities and hammocks are automatically turned red, but they may be outlined instead to preserve their original color.

When using outlines, actualized activities will only have their nodes outlined in red to preserve the blue outline on the bar. In-progress activities will have their progressed portion outlined in blue and their remaining portion outlined in red. Actualized milestones and benchmarks will not be outlined in red, despite the setting.

NOTE: A hammock will display as critical if at least 1 member is critical.

Steps

Changing the Criticality Thresholds

1. Click **Schedule > Modify Schedule Properties** in the menu bar.
2. Increase or decrease the number next to **Criticality Threshold** or **Near-criticality Threshold**.
3. If desired, click the dropdown next to **Criticality Factor** and choose accordingly.
4. When finished, click **OK**.

Using Outlines for Criticality

1. Click **Objects > Use Outlines for Critical Objects (non-actualized only)** from the menu bar.

Repositioning Fields (GEM)

Fields such as descriptions, dates, and float values, also known as data elements, can be repositioned on the canvas to eliminate clashing or simply better coincide with the geometry of the network. Supported objects include activities, delays, milestones, benchmarks, links, data dates, and hammocks. If any fields end up being more than 3" away from their host objects, they will be reset to default the next time the schedule is opened. Repositioning can be done individually for a single object or for the entire schedule using the **Global Edit Mode**, or **GEM**.

Upon entering **GEM**, most toolbar icons as well as the menu bar will be disabled (except for select actions within the **Edit** menu). In addition, most actions via right-click will be disabled, except for actualizing, fading, or hiding/showing, for example. GEM also includes a tool for aligning text boxes.

NOTE: GEM is a mode distinct from (and behaves as a toggle with) the traditional “build mode” of NetPoint, where objects and relationships are added to the canvas. Upon entering GEM, all dates become fixed, and no object property boxes are accessible. However, object descriptions can still be modified by double-clicking them on the canvas. In addition, objects can still move between grids by using the up and down arrows on the keyboard or in the Gestural Window. No canvas objects can be selected, modified, or moved, except for text objects (including annotations).

Steps

By Right-Clicking

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired activity, delay, milestone, benchmark, link, data date, or hammock.
2. Choose the **Customize Positions of** or **Customize Data Positions** option. The object will be spotlighted in purple.
3. Left-click with the mouse the field you wish to move.
4. Without letting go, drag the field to its new position.
5. Release the mouse.
6. When finished, right-click anywhere on the canvas and choose **Save Modified Positions**.

Through GEM Mode

1. Click the **Display Mode (GEM/Date Lock/Normal)**  icon in the toolbar, or click the down arrow and choose **Global Edit Mode (GEM)**. All fields will be highlighted in orange.
2. Left-click with the mouse the field you wish to move.
 - a. Without letting go, drag the field to its new position. Alternatively you can use the arrow keys on the keyboard to move in increments of 1/20th of an inch.
 - b. Release the mouse.
3. To modify multiple objects of the same type, select the first object on the canvas.
 - a. Press and hold the **SHIFT** key on the keyboard.
 - b. Select one or more objects of the same type (activities, delays, and hammocks will all be considered as activities). Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all of the desired objects, or use the **Multi-Object Select Tool** .
 - c. Repeat step 2-2b. All matching fields will jump to match the new position.
4. When finished, exit **GEM** by clicking the **Display Mode (GEM/Date Lock/Normal)**  icon in the toolbar.

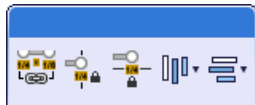
Aligning Text Objects

1. Click the **Display Mode (GEM/Date Lock/Normal)**  icon in the toolbar.
2. Identify a text object that you'd like to align others to, and select it on the canvas.
3. Press and hold the **SHIFT** key on the keyboard.
4. Select one or more additional text objects. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around one or more text objects or use the **Multi-Object Select Tool** .
5. In the auxiliary toolbar, click the down arrow next to the **Align Text Horizontal**  or **Align Text Vertical**  icons.
6. Choose from the sub menu.
7. When finished, exit **GEM** by clicking the **Display Mode (GEM/Date Lock/Normal)**  icon in the toolbar.

Notes

GEM Toolbar

Upon entering GEM, an auxiliary toolbar will also open, offering the following options:



Icon	Name	Definition
	Keep Dates Symmetric	When selected, this icon will tie an activity's start and finish dates together so that any changes to positions mirror each other symmetrically.
	Vertical Only	When selected, this icon will lock data elements to the vertical axis, preventing them from being moved horizontally.
	Horizontal Only	When selected, this icon will lock data elements to the horizontal axis, preventing them from being moved vertically.
	Align Text Horizontally	When expanded, this menu allows all selected text objects (excluding descriptions) to be aligned to their tops, centers, or bottoms. Text objects will align to whichever object was first selected. If they were selected using the rectangular drag box, they will align to whichever object was placed on the canvas first.
	Align Text Vertically	When expanded, this menu allows all selected text objects (excluding descriptions) to be aligned to their lefts, centers, or rights. Text objects will align to whichever object was first selected. If they were selected using the rectangular drag box, they will align to whichever object was placed on the canvas first.

Paste Format

The paste format feature allows you to copy the styling of one object and paste it to another. This includes the positioning of data elements (descriptions, dates, etc.), element displays (whether or not data are being displayed on the canvas), and formatting (size, colors, etc.). This feature is especially useful for applying the styling of one object that has already been customized to your specifications to another set of objects.

Styling can be pasted for all object types, but it can only be applied to other objects of the same type at any given time. For example, if you copy the styling of an activity, it can be pasted to other activities, but not to milestones (they will be ignored if selected). Activities, delays, and hammocks are all considered objects of the same type. If you want to edit the formatting of different types of objects simultaneously, see [Bulk-Editing Objects](#) for more info. If you want to edit the positioning of different types of objects simultaneously, see [Repositioning Fields \(GEM\)](#).

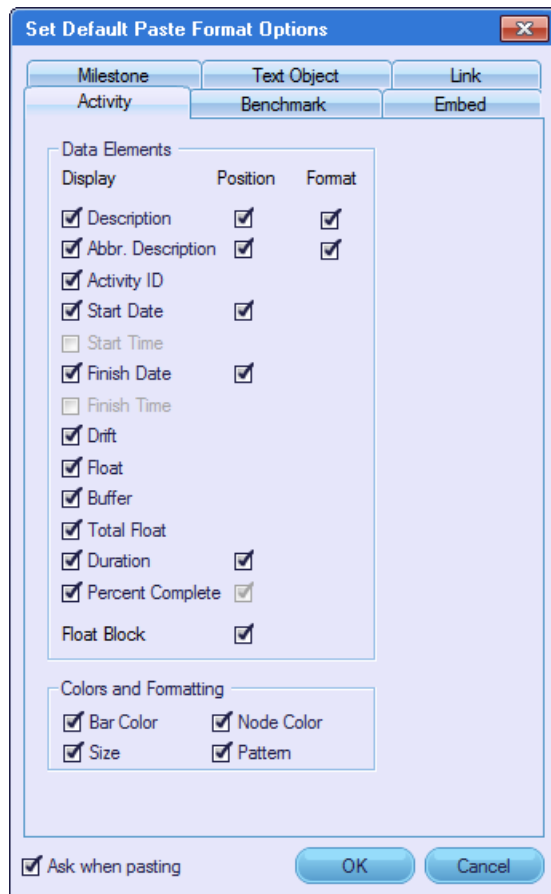
Steps

1. With the **Select Tool**  chosen in the toolbar, select the object you want to copy the styling from on the canvas.
2. In the toolbar, click the **Copy** icon (or press **CTRL + C** on the keyboard)
3. Select one or more objects of the same type that you want to paste the styling to on the canvas.
4. In the toolbar, click the down arrow next to the **Paste** icon and choose **Paste Format** (or press **SHIFT + CTRL + V** on the keyboard).
5. Turn on or off the application of displays, positioning, or formatting on an element-by-element basis in the tab for the type of object selected.
6. When finished, click **OK**. After pasting, the objects will remain selected.
7. To de-select and verify the results, click away in an empty portion of the canvas.

NOTE: If more than one object was selected when copying, **Paste Format** will be disabled.

Notes

Default Paste Format Options

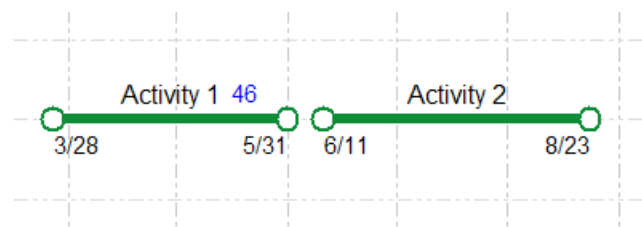


The **Set Default Paste Format Options** window can be accessed at any time via **Edit > Default Format Copy Options....** This window allows you to turn on or off the application of displays, positioning, or formatting on an element-by-element basis. All of these options are available in separate tabs for each object type.

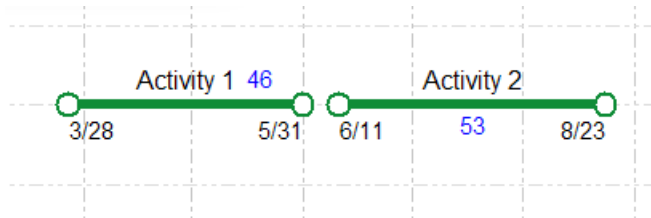
It also provides a checkbox at the bottom in order to control the display of this window after every paste format operation. If it is set to display after a paste, the checkbox will instead allow the modified options to be saved as the new default options.

Examples

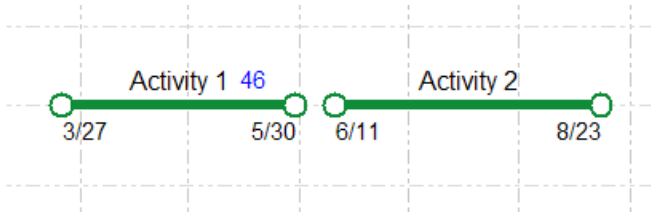
Allowing for individual control over display and position aspects of data elements gives a lot of flexibility when pasting the format to target objects. In the example below, the duration of Activity 1 has been moved to the right of the description, and the duration display has been turned off for Activity 2.



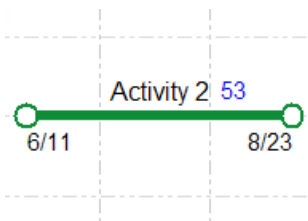
If only the display column is checked for the duration, pasting from Activity 1 to Activity 2 will yield the following result.



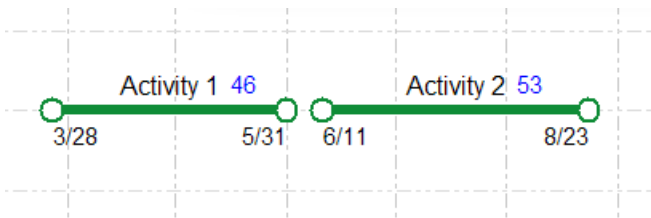
But if only the *position* column is checked, pasting from Activity 1 to Activity 2 will instead yield this result.



Even though the duration is not displayed for Activity 2, its position has been updated behind the scenes. If it were turned on, it would display accordingly.



In the final example, both columns have been checked, and the paste yields the following result as expected.



NOTE: Durations will not paste between partially actualized and non-actualized activities.

Fine-Tuning/Gestural Controls

When working on devices with touchscreens, or if you have to zoom way out for particularly large schedules, using your fingers or the mouse to change dates or durations can be difficult or imprecise. To make fine-tuning easier in these cases, NetPoint provides on-screen gestural controls for decreasing and increasing the duration of an activity (known as crashing and extending, respectively), and for moving one or more objects horizontally (known as pushing and pulling) and vertically (known as shifting).

Keep in mind that when you use the gestural controls (or arrow keys on the keyboard) to push or pull *multiple* objects, each object will move according to the working days (or time units) defined in its own calendar. If all objects have the same calendar, then using the gestural controls or keyboard arrow keys to move multiple objects can achieve expected results. However, if some objects have different calendars, those based on calendars with more working days will appear to move ‘faster’ on the canvas. In this case, using the mouse may yield results closer to your expectations. For more info, see the *Notes* section below.

NOTE: If a milestone is assigned a calendar that’s different from that of its predecessors or successors, and the network is pulled backwards, gaps may open up as the chain crosses a weekend or non-working days. As such, it is recommended to always keep milestones on the same calendar as their predecessors/successors.

Steps

Crashing/Extending (Activities)

1. Click the **Gestural** icon  in the toolbar.
2. Select the desired activity on the canvas. To select multiple activities, select the desired activities one-by-one on the canvas using the **Multi-Select Tool**, or click on a blank portion of the canvas and drag a rectangular marquee box around all of the desired activities.
3. Enter a number next to **By** or click the up and down arrows to modify the magnitude of the change.
4. To crash or extend by percent duration (as opposed to by time), click the dropdown arrow and choose **Percent**.
5. To reduce the duration (crash) of the selected activities from the *start*, click the innermost arrow at the *start* (or press **L** on the keyboard + the **RIGHT** arrow key). To reduce the duration from the *finish*, click the innermost arrow at the *finish* (or press **R** on the keyboard + the **LEFT** arrow key).
6. To increase the duration (extend) of the selected activities from the *start*, click the outermost arrow at the *start* (or press **L** on the keyboard + the **LEFT** arrow key). To increase the duration from the *finish*, click the outermost arrow at the *finish* (or press **R** on the keyboard + the **RIGHT** arrow key).
7. When finished, click the red **Exit** icon.

Pushing or Pulling Objects

1. Click the **Gestural** icon  in the toolbar.
2. Select the desired object on the canvas. To select multiple objects, select the desired objects one-by-one on the canvas using the **Multi-Select Tool**, or click on a blank portion of the canvas and drag a rectangular marquee box around all of the desired objects.
3. Enter a number next to **By** or click the up and down arrows to modify the magnitude of the change. Positions may only be moved in increments of time.
4. If percent is chosen, click the dropdown arrow and choose **Day** (or, if PolyTime is enabled, the corresponding time unit). Otherwise, pushing/pulling will be disabled.

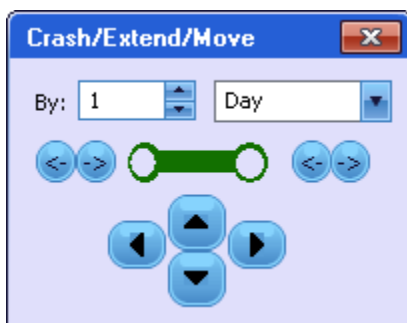
5. To move the selected objects *later* in time, click the **RIGHT** arrow button below the activity (or press the **RIGHT** arrow key on the keyboard).
6. To move the selected objects *earlier* in time, click the **LEFT** arrow button below the activity (or press the **LEFT** arrow key on the keyboard).
7. When finished, click the red **Exit** icon.

NOTE: If optimizing or performing what-if scenarios, it is recommended to use an increment of no greater than 10. Large increments should only be used in specific situations (e.g. if a window is missed and the network must be delayed a year).

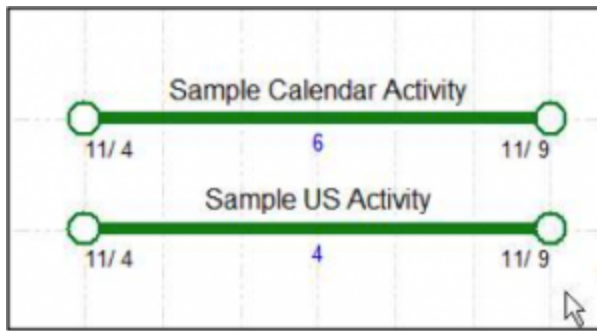
Shifting Objects Vertically

1. Click the **Gestural** icon  in the toolbar.
2. Select the desired object on the canvas. To select multiple objects, select the desired objects one-by-one on the canvas using the **Multi-Select Tool**, or click on a blank portion of the canvas and drag a rectangular marquee box around all of the desired objects.
3. To move the selected objects *up* a grid, click the **UP** arrow button below the activity (or press the **UP** arrow key on the keyboard).
4. To move the selected objects *down* a grid, click the **DOWN** arrow button below the activity (or press the **DOWN** arrow key on the keyboard).
5. When finished, click the red **Exit** icon.

Notes



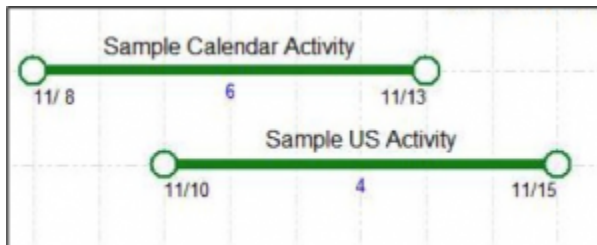
When multiple objects are moved with the arrow keys or gestural controls, the starting *date* of each object is moved by the specified number of *working days* in the calendar of the object. This means that objects assigned calendars with more non-working days will move 'faster' than objects assigned calendars with fewer non-working days. When multiple objects are moved with the mouse, the starting *position* of each object is moved by the same distance, and the *dates* are then recalculated from the new starting positions. For example, imagine two activities assigned different calendars that start and finish on the same dates:



If both "Sample Calendar Activity" and "Sample US Activity" are selected and dragged with the mouse until the starting date of the first activity is 11/8 the following is the result..



However, if the move is performed using the arrow key four times the result is:



The difference occurs because the "Sample Calendar Activity" is moved by 4 calendar days, while "Sample US Activity" is moved by 4 working days (equivalent to 6 calendar days because the starting point of each activity moves past a weekend).

Adjusting Zoom & Stretch

Zooming and stretching can help to reduce data clashing, clarify logic relationships, or just better fit the user's monitor size. The zoom function magnifies or shrinks the canvas proportionally. The stretch function increases or decreases the horizontal width of the canvas. Zooming and stretching will only temporarily enhance readability on a digital display. They will have no effect on the printed page and they will reset when the file is printed or saved and closed. To permanently affect the canvas, you must adjust the page size or grid spacing, or use page tiling.

NOTE: The date above the mouse cursor will automatically zoom in and out. To prevent this, uncheck the box under System Preferences in the Schedule Zoom & Stretch window. This setting will be saved with the computer.

Steps

By Percentage

1. Click the **Zoom/Stretch**  icon in the toolbar.
2. To adjust the **Zoom Factor**, click the up or down arrows, or enter a desired number.
3. To adjust the **Stretch Factor**, click the up or down arrows, or enter a desired number.
4. To return to defaults, click the **Reset** button accordingly.
5. When finished, click the red X in the upper right-hand corner. Any changes will be automatically saved.

NOTE: The Zoom Factor can be set from 25 to 300 percent of the default page layout. The Stretch Factor can be set from 25 to 1000 percent of the default page layout.

Using the Keyboard

1. To zoom, press the **SHIFT** key on the keyboard
 - a. Without letting go, scroll up (to zoom in) or down (to zoom out) with the mouse.
 - b. When satisfied, release the key.
2. To stretch, press the **CTRL** key on the keyboard.
 - a. Without letting go, scroll up (to increase) or down (to decrease) with the mouse.
 - b. When satisfied, release the key.

With the Mouse (stretch only)

1. Click the **Select Tool**  icon in the toolbar.
2. Move the cursor over the calendar strips at the top of the canvas.
3. Left-click with the mouse, noticing the percentage appear.
4. Without letting go, drag the mouse to the left (to decrease) or right (to increase).
5. When satisfied, release the mouse.

NOTE: If the stretch factor is reduced to a point where the width of a time unit is less than the width of a pixel on screen, the stretch factor text will turn red. Modifying the network under these conditions is not recommended, as input from the mouse could no longer translate accurately on the canvas. It is recommended to increase the stretch factor before proceeding to work on the network.

Filters and Searching

Filters allow all objects in the schedule to be searched for based on specific sets of criteria. Filters are often used to generate a list of objects that can then be bulk-edited. For example, to maintain GPM planned dates before exporting to P6, you could filter the schedule for all activities and milestones with drift buffer greater than 0 and apply start-no-earlier constraints to all of them.

The list of objects that results from a filter can also be highlighted or faded on the canvas, and filtering can also be used to find just a single object in the schedule. If you're just trying to find a specific activity and you know it's description, you can use the **Activity Quick Search** instead.

Filters can be saved with the schedule for future use as well as shared between schedules by exporting and importing.

Steps

Applying a Filter (Example: Objects on Planned Dates)

1. Click **Edit > Search and Filter Objects** from the menu bar.
2. Click the down arrow under **Objects to Include** and add or remove any objects by checking the boxes next to their names. *For this example, add Start Milestones and Finish Milestones*
 - a. Click outside of the list to close it.
3. Click the down arrow under **Options** and add or remove any criteria by checking the boxes next to them. *For this example, leave them blank.*
 - a. Click outside of the list to close it.
4. To activate criteria in the **Properties** section, click the down arrow next to a given property and change it from **Ignore**. *For this example, click the down arrow next to Drift Buffer and choose >. In the number control to the right of Drift Buffer, enter 0.*
5. To activate colors, resources, codes, or calendars as a criteria, select one or more items from their corresponding lists. *For this example, leave all as Ignore.*
6. When finished, click **Add to List**.
7. To modify the results, repeat steps 2-5 and click **Refine List**.
8. To add another search to the results, repeat steps 2-5 and click **Add to List** again.
9. When finished, click the red **X** in the upper right corner. The **List of Filtered Objects** window will remain open with the objects spotlighted in yellow on the canvas.

Working with Filter Results

1. Follow steps 1-8 above.
2. To reorder the list, click the down arrow next to **Order List By** and choose from the options.
3. To locate an object on the canvas, choose it in the list. The canvas will scroll to it and the object will be boxed in black.
4. To fade or highlight an object on the canvas, choose it in the list and press **Fade** or **Highlight** accordingly.
5. To reset any fades or highlights, press **Restore**.
6. To select one or more objects, mark the selection circle field in between the object icon and the description field, or use the **Select All** button to select every item in the list.
7. To edit selected objects in bulk, click **Objects > Set Properties for Selected Objects**. For more info, see [Bulk Editing of Objects](#).
8. When finished, click the red **X** in the upper right corner.

Importing a Filter

1. Click **Edit > Search and Filter Objects** from the menu bar.
2. Choose the radio button next to **Existing Filters**.
3. Click the **Import** button on the right.
4. Browse for the file and click **Open**.
5. Check the box next to the filter(s) desired for importing and click **Import**.

Exporting a Filter

1. Click **Edit > Search and Filter Objects** from the menu bar.
2. Click the **Options** button at the bottom of the window and choose **Export All Filters**.
3. Enter a filename and click **Save**.

Deleting a Filter

1. Click **Edit > Search and Filter Objects** from the menu bar.
2. Click the **Options** button at the bottom of the window and choose **Export All Filters**.
3. Enter a filename and click **Save**.

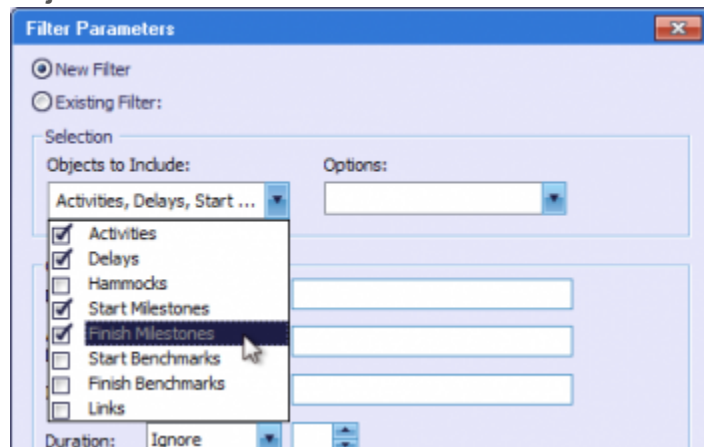
Finding a Single Activity

1. Click **Edit > Find an Activity...** from the menu bar.
2. Choose the radio button next to **Existing Filters**.
3. Click the dropdown and choose a desired filter.
4. Click the **Options** button at the bottom of the window and choose **Delete Filter**.
5. Click **Delete**.

Notes

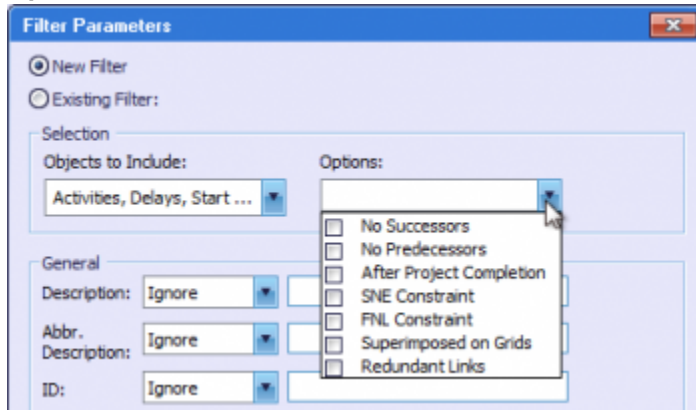
Search Criteria

Objects to Include



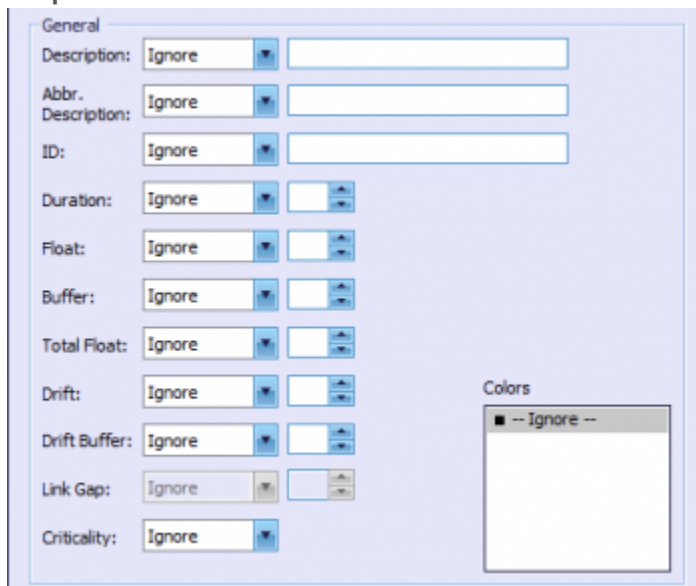
By default, activities and delays will be included in the search.

Options



Criteria	Definition
No Successors/ Predecessors	Check these boxes to find objects without any successors or predecessors. If both were checked, all unlinked or unconnected objects would result.
After Project Completion	Check this box to find any objects that finish after the project completion date.
SNE/FNL Constraints	Check these boxes to include any objects which have had their start or finish dates constrained accordingly.
Superimposed On Grids	Check this box to find any objects that are overlapping on the same horizontal grid line.
Redundant Link	Check this box to include any links that are redundant. For more information, see Treating Redundancies .

Properties



Criteria	Definition
Description Fields & ID	Search for objects that contain, start with, or match certain characters. Not case sensitive.
Duration, Float, Buffer, Total Float, Drift, Drift Buffer, Link Gap	These fields can be searched based on whether or not they are less than, greater than, equal to, or in between a given number.
Criticality	Search for critical or non-critical objects.
Color	Search for objects of a certain color. Multiple colors can be chosen.

Assignments

Criteria	Definition
Resources	Search for objects based on what resources they've been assigned. Any will yield objects with ANY resources assigned; None will yield objects with NO resources assigned. Multiple resources can be chosen.
Codes/ Values	Search for objects based on what values they've been assigned. Any will yield objects with ANY values assigned; None will yield objects with NO values assigned. Multiple values can be chosen.
Calendars	Search for objects based on what calendar they've been assigned.

Criteria Behavior

Parameters under **Properties** and **Options** behave as AND-CRITERIA, meaning that an object must match ALL of the parameters chosen to be added to the list. For example, if **No Successors** is chosen and **Superimposed on Grids** is chosen, only those objects which have both no successors AND are superimposed on grids will be added to the list.

Parameters under **Colors**, **Resources**, and **Codes/Values** behave as OR-CRITERIA, meaning that an object which matches ANY of the parameters will be added to the list. For example, if **Carpenters** is chosen and **Electricians** is chosen, objects which have been assigned carpenters OR electricians will be added to the list.

If you've selected **Activities**, **No Successors**, **Superimposed on Grids**, **Carpenters** and **Electricians** all together, then activities that have both no successors AND are superimposed on grids that have been assigned either carpenters OR electricians will be added to the list.

Bulk-Editing Objects

Multiple objects may be edited at the same time through the **Set Global Properties** window. Bulk-editing objects in this way can be instrumental in saving time. The process involves first selecting all objects you want to edit on the canvas and then making the desired changes in each relevant tab in the **Set Global Properties** window.

If you already know exactly which objects you want to modify, they can be selected one-by-one or in groups on the canvas. In other cases, you may not be able to identify or locate all of the objects on the canvas, and filtering may be required. For example, to maintain GPM planned dates before exporting to P6, you could search for all activities and milestones with drift buffer greater than 0 (via the [Apply Filters](#) window), and then apply start-no-earlier constraints in bulk.

Objects that can be bulk-edited include activities, delays, embeds, links, milestones, benchmarks, shades, and text boxes. Each object type has its own tab that contains the available properties that can be modified. Notable exclusions include properties unique to individual objects, such as descriptions and IDs. Any change made will override the individual settings for all objects selected, but only for the type of object in a given tab. For example, if the box next to **Float** were checked in the **Activity** tab, every activity selected would have its float displayed on the canvas, but milestones would not be affected. In order to display float for every milestone selected, the corresponding box would have to be checked in the **Milestone** tab as well.

Steps (Example: Applying SNE Constraints)

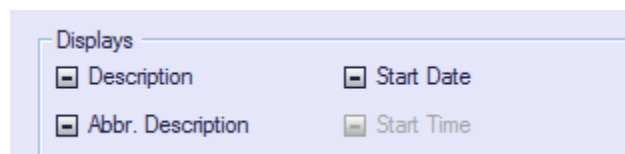
1. With the **Multi-Object Select Tool**  chosen in the toolbar, select the desired objects one-by-one on the canvas. Alternatively, you can click on a blank portion of the canvas and drag a rectangular marquee box around all of the desired objects. *For this example, see [Applying a Filter](#) to make the selection.*
2. With the objects still selected, click **Objects > Set Properties for Selected Objects** from the menu bar.
3. Choose the tab for the type of object you want to modify. *For this example, stay on the Activity tab.*
4. To make a change, click the control next to the property you want to modify. *For this example, under the Constraints section, click twice in the box next to Start no earlier than until a check appears, and then click once in the box next to Use Planned Dates until a check appears.*
5. If more than one type of object was selected, repeat steps 3-4 for each object type. *For this example, click the Milestones tab and repeat step 4.*
6. When finished, click **OK**. Changes from across all tabs will be applied at once.

NOTE: If the outcome is not as desired, click **Undo** or press **Ctrl+Z** on the keyboard to cancel the last set of changes.

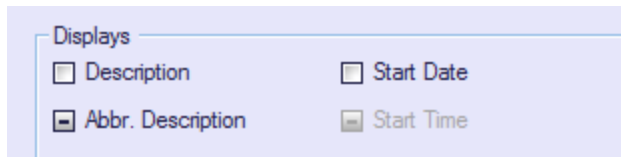
Notes

Setting Properties with Check Boxes

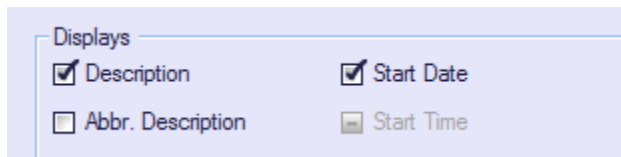
A property with a minus sign next to it indicates mixed assignment and will be left the way it was before the **Set Global Properties** window was opened.



If a check box showing a minus sign is clicked, it becomes blank. In the case of displays, a blank box means that the property will not be displayed. In the case of actualization, a blank box means that an actualized activity would be de-actualized.



If a blank check box is clicked, it becomes checked. In the case of displays this means the checked property will be displayed. Clicking once more will cycle the check box back to the mixed state, which will leave the property for all applicable objects the way it was prior to entering the **Set Global Properties** window.



Constraint Behavior

Applying no-earlier-than (NET) or no-later-than (NLT) constraints may have a different effect depending on whether or not any of the selected activities or milestones currently have or previously had constraints set in the past. For example, for NET constraints:

If an activity or milestone...	...and...	...then...
Has NEVER HAD a no-earlier-than constraint before...	Start-no-earlier-than is checked...	No constraint will be set.
Has NEVER HAD a no-earlier-than constraint before...	Both Start-no-earlier-than and Use Planned Dates are checked...	An NET constraint will be applied at its current start date.
Currently HAS a no-earlier-than-constraint...	Start-no-earlier-than is checked...	No change will be made.
Currently HAS a no-earlier-	Both Start-no-earlier-	The existing NET constraint will be reset to the current start date.

If an activity or milestone...	...and...	...then...
than-constraint...	than <i>and Use Planned Dates</i> are checked...	
Once HAD a no-earlier-than constraint...	Start-no-earlier-than is checked...	An NET constraint will be applied at whatever date the constraint was last set at <i>unless</i> the current start date is earlier than its last constraint date, in which case no constraint will be set.
Once HAD a no-earlier-than constraint...	Both Start-no-earlier-than <i>and Use Planned Dates</i> are checked...	An NET constraint will be applied at its current start date.

If both **Start-no-earlier-than** *and Use Planned Dates* are *unchecked*, the objects will be unconstrained but their most recent constraint dates will still be remembered internally.

Text Style

The description and the abbreviated description can be styled/formatted differently or the same. To make them different, choose **Description** from the dropdown, customize the options, and then choose **Abbr. Description** from the dropdown, and customize the options for it. The settings will be saved, and you can switch between the two to confirm.

To make them the same, choose **Both** from the dropdown, and then customize the options. Note that choosing both will only overwrite the options that receive a change. For example, imagine the description's font has been formatted to Arial and the abbreviated description's font has been formatted to Tahoma. If **Both** is chosen from the dropdown and the font is set to Calibri, then the font for both the description and the abbreviated description will be set to Calibri, but the color, alignment, style, and size will be unaffected, since they didn't receive any change. This can be confirmed by switching between the description and the abbreviated description and reviewing them without losing any of the changes.

NOTE: If both the description and the abbreviated description have been turned off but the ID is turned on, then the ID will inherit the formatting of the abbreviated description. Otherwise, it will inherit the formatting of whichever description field is being displayed along with it.

Activity Tab

For info on individual fields, see [➔Adding/Modifying Activities](#).

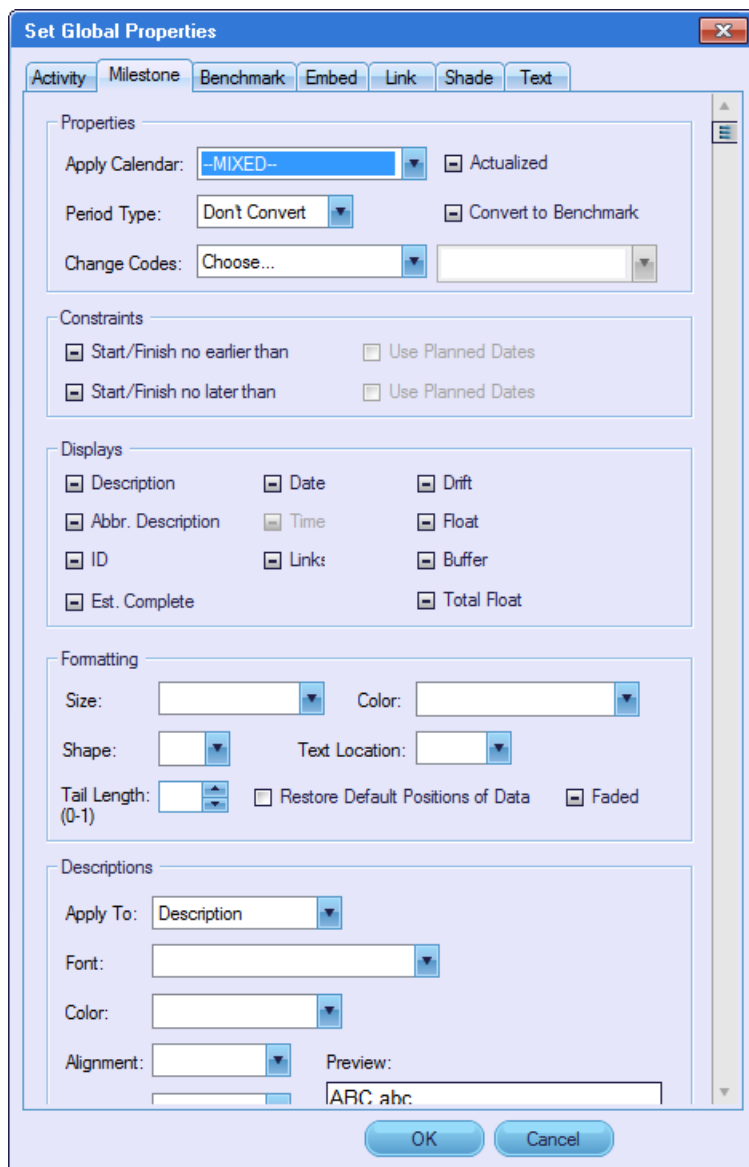
The screenshot shows the 'Set Global Properties' dialog box with the 'Milestone' tab selected. The dialog has a title bar with a close button. Below the title bar is a tabbed interface with tabs for 'Activity', 'Milestone', 'Benchmark', 'Embed', 'Link', 'Shade', and 'Text'. The 'Milestone' tab is active, showing several sections of settings:

- Properties:** Includes 'Apply Calendar' (a dropdown menu), 'Resources' (with an 'Assign...' button), 'Dates' (a dropdown menu set to '-No Action-' and an 'Actualized' checkbox), 'Change Codes' (two empty text boxes), and 'Work Type' (a dropdown menu).
- Constraints:** Includes 'Start no earlier than' and 'Finish no later than' (both with checkboxes), and 'Use Planned Dates' (two checkboxes).
- Displays:** A grid of checkboxes for various display options: Description, Start Date, Drift, Abbr. Description, Start Time, Float, ID, Finish Date, Buffer, Calc. Complete, Finish Time, Total Float, Est. Complete, Progressed Duration, Resource Profile, Duration, and Remaining Duration.
- Formatting:** Includes 'Size' and 'Color' (both dropdown menus), 'Pattern' and 'Nodes' (both dropdown menus), and checkboxes for 'Restore Default Positions of Data' and 'Faded'.
- Descriptions:** Includes 'Apply To' (a dropdown menu set to 'Description'), 'Font' (a dropdown menu), and 'Color' (a dropdown menu).

At the bottom of the dialog are 'OK' and 'Cancel' buttons.

Milestone Tab

For info on individual fields, see [Adding/Modifying Milestones/Benchmarks](#).



The image shows a software dialog box titled "Set Global Properties" with a close button (X) in the top right corner. The "Benchmark" tab is selected, indicated by a blue highlight. The dialog is organized into several sections: "Properties", "Constraints", "Displays", "Formatting", and "Descriptions". The "Properties" section includes dropdown menus for "Apply Calendar" (set to "--MIXED--"), "Period Type" (set to "Don't Convert"), and "Change Codes" (set to "Choose..."). It also has checkboxes for "Actualized" and "Convert to Benchmark". The "Constraints" section has checkboxes for "Start/Finish no earlier than" and "Start/Finish no later than", each with a "Use Planned Dates" checkbox. The "Displays" section contains a grid of checkboxes for various fields: Description, Date, Drift, Abbr. Description, Time, Float, ID, Link, Buffer, Est. Complete, and Total Float. The "Formatting" section includes dropdowns for "Size" and "Color", a "Shape" dropdown, a "Text Location" dropdown, a "Tail Length" spinner (0-1), and checkboxes for "Restore Default Positions of Data" and "Faded". The "Descriptions" section has dropdowns for "Apply To" (set to "Description"), "Font", and "Color", an "Alignment" dropdown, and a "Preview" area showing "ARC abc". At the bottom are "OK" and "Cancel" buttons.

Set Global Properties

Activity Milestone **Benchmark** Embed Link Shade Text

Properties

Apply Calendar: --MIXED-- ☐ Actualized

Period Type: Don't Convert ☐ Convert to Benchmark

Change Codes: Choose...

Constraints

☐ Start/Finish no earlier than ☐ Use Planned Dates

☐ Start/Finish no later than ☐ Use Planned Dates

Displays

☐ Description ☐ Date ☐ Drift

☐ Abbr. Description ☐ Time ☐ Float

☐ ID ☐ Link ☐ Buffer

☐ Est. Complete ☐ Total Float

Formatting

Size: Color:

Shape: Text Location:

Tail Length: (0-1) ☐ Restore Default Positions of Data ☐ Faded

Descriptions

Apply To: Description

Font:

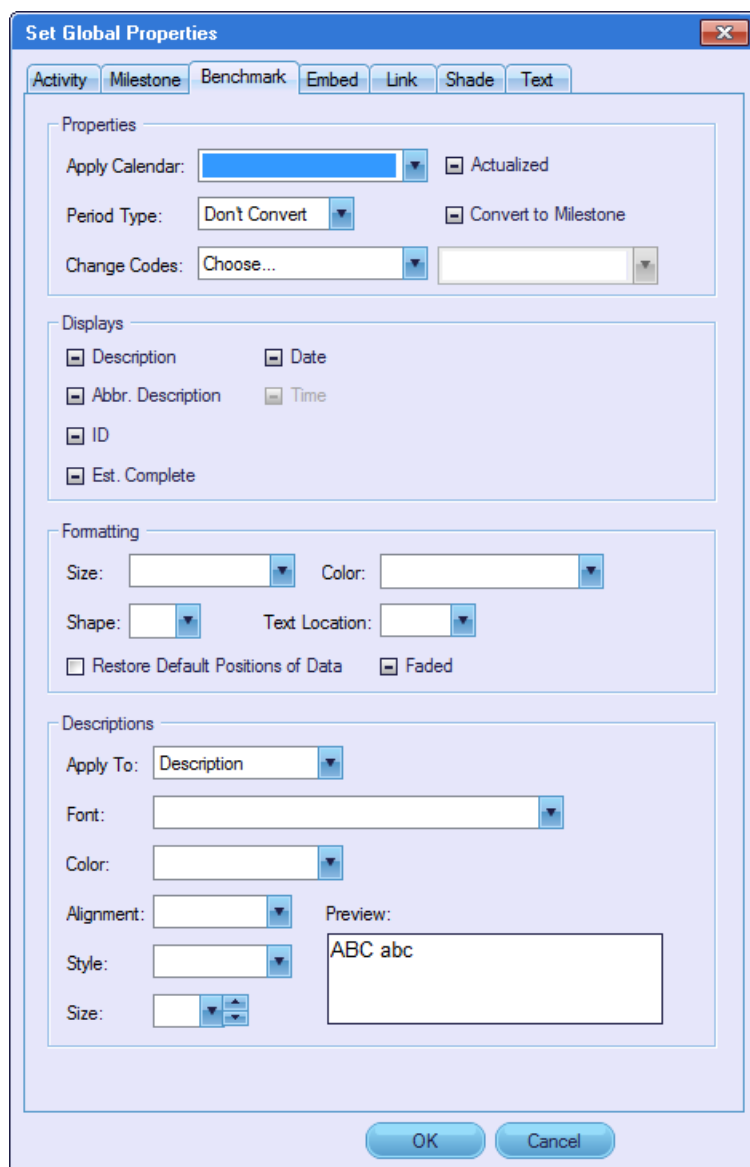
Color:

Alignment: Preview: ARC abc

OK Cancel

Benchmark Tab

For info on individual fields, see [Adding/Modifying Milestones/Benchmarks](#).



The "Set Global Properties" dialog box is shown with the "Embed" tab selected. It contains several sections for configuring the display and formatting of embedded data.

Activity | Milestone | Benchmark | Embed | Link | Shade | Text

Properties

- Apply Calendar: [Dropdown] ☐ Actualized
- Period Type: [Dropdown: Don't Convert] ☐ Convert to Milestone
- Change Codes: [Dropdown: Choose...] [Dropdown]

Displays

- ☐ Description ☐ Date
- ☐ Abbr. Description ☐ Time
- ☐ ID
- ☐ Est. Complete

Formatting

- Size: [Dropdown] Color: [Dropdown]
- Shape: [Dropdown] Text Location: [Dropdown]
- ☐ Restore Default Positions of Data ☐ Faded

Descriptions

- Apply To: [Dropdown: Description]
- Font: [Dropdown]
- Color: [Dropdown]
- Alignment: [Dropdown] Preview: [Text Area: ABC abc]
- Style: [Dropdown]
- Size: [Dropdown] [Up/Down Arrows]

OK Cancel

Embed Tab

For info on individual fields, see [➔Using Embeds to Create SS and FF Links](#).

Set Global Properties

Activity | Milestone | Benchmark | Embed | Link | Shade | Text

Apply Changes Below to: BOTH

Properties

Behavior: --MIXED--

☐ Display Lag Below

Displays

☐ Date ☐ Lag from Start

☐ Time ☐ Lag to Finish

Formatting

Color:

☐ Faded

Link Tab

For info on individual fields, see [Creating FS Links](#).

Set Global Properties

Activity | Milestone | Benchmark | Embed | Link | Shade | Text

Properties

Logic: --MIXED--

Displays

☐ Gap ☐ Lag Notation

Formatting

☐ Restore Default Positions of Data

☐ Faded

Shade Tab

For info on individual fields, see [Markup Tools](#).

Set Global Properties

Activity | Milestone | Benchmark | Embed | Link | Shade | Text

Properties

Orientation: --MIXED--

Edge Type: --MIXED--

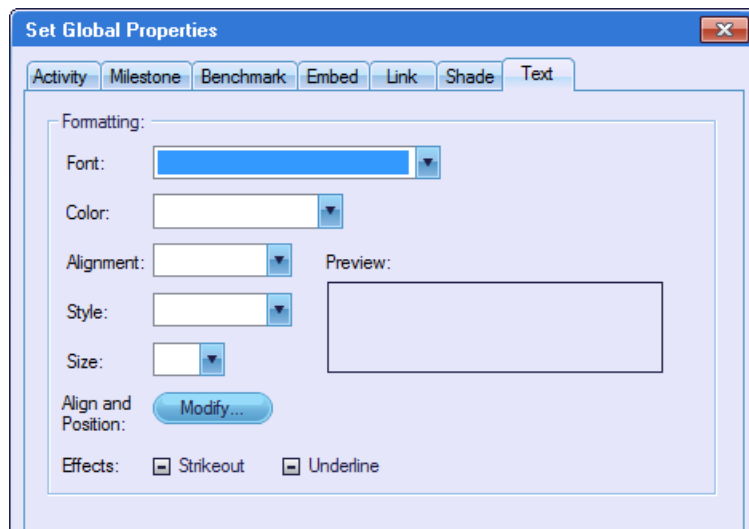
Behavior: --MIXED--

Formatting

Color:

Text Tab

For info on individual fields, see [Markup Tools](#).



The image shows a screenshot of the "Set Global Properties" dialog box, specifically the "Text" tab. The dialog has a title bar with a close button (X). Below the title bar are several tabs: "Activity", "Milestone", "Benchmark", "Embed", "Link", "Shade", and "Text". The "Text" tab is currently selected. The main area of the dialog is divided into two sections. The left section, labeled "Formatting:", contains several controls: a "Font:" dropdown menu, a "Color:" color picker, an "Alignment:" dropdown menu, a "Style:" dropdown menu, and a "Size:" spinner. Below these is a "Modify..." button. The right section, labeled "Preview:", contains a large rectangular text area. At the bottom of the dialog, under the "Effects:" label, there are two checkboxes: "Strikeout" and "Underline".

Schedule MD™

NOTE: The full chapter is available with the purchase of the license.

Schedule MD™ is a companion application to NetPoint for analyzing and determining the reliability of schedules. The Schedule IQ™ Score is a rating system based on the *Core Traits of a Reliable Schedule*, aka the *20-Trait Protocol*, as developed by PMA Consultants (for more info, see <http://pmaconsultants.com/publications/core-traits-of-reliable-schedule/>). Scores are reported from 0 to 100. The software provides IQ scores for the overall schedule as well as breaks them down into 4 categories (comprehensive, credible, well-built, and controlled) and into 20 “IQ Metrics”. As a result, reliability can be assessed at both macro and micro levels.

The comprehensive IQ provides insight into the reliability of activities and their durations; contract dates; and resource, code, and WBS assignments. The credible IQ provides insight into the reliability of the critical path, total floats, risk, and weather. The well-built IQ provides insight into the reliability of milestones, logic, constraints, and calendars. And the controlled IQ provides insight into the reliability of progress, forensics, out-of-sequence activities, and remaining durations. The four categories correspond with the *GAO Assessment Guide* and the *20-Trait Protocol*.

In addition to the Schedule IQ™ Score, the software also offers robust analytics: the visualization and filtering capability of over one hundred related metrics and their data.

This chapter details instructions for the following:

1. Launching and modifying inputs (.np4/.xer)
2. The Metrics Manager™
3. Stakeholders and the narrative checklist
4. Loading a base case file
5. Weather
6. Metrics libraries
7. Healing the schedule via the results browser
8. Exporting to Excel

Launching and Modifying Inputs (.np4/.xer files)

For running Schedule MD™ on an .np4 file, the **Metrics Manager** can be opened from directly within NetPoint.

For running Schedule MD™ on an .xer file, the .xer file can be opened outside of NetPoint, bypassing the import process. This is useful, because when an .xer file is imported into NetPoint, certain characteristics of the schedule may change to align the network to GPM principles. Opening it directly in the **Metrics Manager** will maintain the schedule in its original form.

One change will take place, however, which is to convert time units from hours into days (and rounded). If the .xer file was saved using version 7 or later, the conversion will be based on the settings found under the “Time Periods” button in P6 for each calendar. If the .xer file was saved using any earlier version, the conversion will use a constant 8 hours/day, 5 days/week, 22 days/month, and 12 months/year.

In addition, when opened outside of NetPoint, healing the schedule is not possible, since Metrics does not modify the original file. For more info see, [➔Healing the Schedule via the Results Browser](#).

NOTE: For an .xer file, the Schedule IQ™ Score is based on a maximum of 34 instead of 42 metrics since they do not contain the information needed to calculate the following: “Improper Detail”, “Subjective Durations”, “Missing Resources”, “Off-base Delays”, “Breaching Finish Dates”, “Breaching Start Dates”, “Abnormal Resource Logic”, and “Secondary Calendar Use”. These metrics will be displayed with an NA; any other data metrics not used in the Schedule IQ Score will be hidden.

Steps

NetPoint Files (.np4)

1. With NetPoint already open, click **Metrics > Open Metrics Manager** from the menu bar.
2. Fill in the inputs as needed and click **Save**.
3. Click the desired icon at the top.
4. When finished, click the **Close** button.

NOTE: When opening from within NetPoint, the manager will display a green dot  in the upper-right hand corner. This signifies that Schedule MD is communicating with NetPoint and healing is possible. For more info, see [➔Healing the Schedule via the Results Browser](#).

P6 Files (.xer)

1. Double-click the “ScheduleMD.exe” shortcut , typically located on the desktop or in the Windows Start screen. If not there, check “C:\Program Files (x86)\NetPoint 5”.
2. Browse for a desired .xer file and click **Open**.
3. Fill in the inputs as needed and click **Save**.
4. To save your work, click **Save As** in the subsequent notification. If you click **Don’t Save**, you can still do so upon closing the application or by returning to the inputs window.
5. To shut down the application, click the red **X** or the **Close** button from the manager.

NOTE: Metrics results may be inconsistent if the project was not properly scheduled and summarized. Therefore, always ensure that the project was scheduled and summarized AFTER the latest changes were made in P6 and BEFORE exporting it as an XER for metrics analysis.

Changing .xer files

1. To view metrics for a different .xer file, click the **List** icon  in the upper-right corner of the manager and choose **Configure Schedule MD Inputs**.
2. Click **Browse**.
3. Browse for the desired .xer file and click **Open**.
4. Fill in the inputs as needed and click **Save**.
5. To save your work, click **Save As** in the subsequent notification. If you click **Don't Save**, you can still do so upon closing the application or by returning to the inputs window.
6. To shut down the application, click the red **X** or the **Close** button from the manager.

Modifying MD Inputs

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. When finished, click **OK**.

Notes

Schedule MD Inputs

The window will always open upon opening the manager, until the **Industry**, **Budget**, and **Delivery** have been entered.

NOTE: Undo is not available when a file is opened outside of NetPoint.

Main

Metrics - IQ Inputs

Properties | Review & Sign | Base Case | Weather

Filepath:

P6 XER

List of Available Projects

General

? Level:

? Revision:

? Industry:

? Budget:

? Delivery:

? Completion Likelihood: %

Thresholds

? Criticality: ≤ % (0 Days)

? Near-criticality: ≤ % (42 Days)

? Criticality Factor:

Field	Definition
List of Available Projects	A list of all projects available in the XER file. Only one can be chosen for analysis at a given time.
Level	The level of a schedule can be approximated based on the majority of activity durations: 2-12 months for level 1; 6 weeks-6 months for level 2 (5-15% of project duration); 2-6 weeks for level 3 (1-3% of project duration); and ≤2 weeks for level 4 (≤1% of project duration). No IQ metrics are calculated for level 5.

Field	Definition
Revision	An unprogressed baseline shows no progress. A revised baseline may include changes to the contract scope after the project has started. An update shows actual progress. If a metric cannot be calculated due to revision, it will be displayed as NA.
Industry	The industry is a qualitative field only and is not used for calculation of the Schedule IQ Score.
Budget	The budget is a qualitative field only and is not used for calculation of the Schedule IQ Score.
Delivery	The delivery method will determine the roles available in the Stakeholders section of the Review & Sign tab.
Completion Likelihood	The completion likelihood is obtained by simulating the schedule as part of a risk analysis. If the schedule has not been simulated, leave this value blank.
Criticality	Determines at what point (e.g. days) an activity will be displayed as critical. % is calculated as the percent of remaining duration from the data date onwards. As a result, as the data date approaches the project finish date, the threshold will approach 0. If the data date equals the project finish date and percent is chosen, the threshold will equal 0.
Near-criticality	Determines at what point (e.g. days) an activity will be displayed as near-critical (excludes critical activities). % is calculated as the percent of remaining duration from the data date onwards. As a result, as the data date approaches the project finish date, the threshold will approach 0. If the data date equals the project finish date and percent is chosen, the threshold will equal 0.
Criticality Factor	Determines how criticality and near-criticality will be calculated. Float is defined as the amount of time an activity can be delayed before delaying the completion of the project. Drift is defined as the amount of time an activity can gain before advancing the start of the project. Total Float is defined as float + drift and is equal to the CPM value of Total Float. NOTE: Choosing Float will discount drift values. Choose Drift will discount float values.

The Metrics Manager™

At the top of the **Metrics Manager**, each category of schedule reliability is given an icon. Beneath the icons is the list of metrics, all related to a given category (e.g. comprehensive). On the bottom is the **Results Browser**, where you can choose between either a dashboard widget or a filter. As you click through the list on top, the **Results Browser** refreshes accordingly. This layout promotes a process of analyzing your schedule in a progressively narrowing manner, as you move from left to right and top to bottom until you get the information you need to take action on.

The window itself is non-modal, meaning that you can navigate to other menus or make modifications on the canvas without having to close it.

Lists can be filtered if a search field is provided above them, and all lists can be customized by having their columns sorted, added and removed. As you click through the lower list (the **Results Browser**), the canvas will be scrolled and the item spotlighted, but this can be disabled. In addition, all lists can be individually exported to Excel.

Steps

Viewing the Results Browser

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. Click the desired icon at the top.
3. Choose a desired metric.
4. Click the dropdown next to **Viewing** and choose **Matching Objects**.
5. To select multiple rows, click and hold the **Shift** key on the keyboard.
6. To select 1 or more items on the canvas, click the **Select** button. For more info, see [Bulk-editing Objects](#).
7. To highlight 1 or more items on the canvas, click the **Highlight** button.
8. To reset any faded objects, click **Restore All**.

Sorting, Adding, and Removing Columns

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. Click the desired icon at the top.
3. To sort by a certain field, click once on the column's label to sort from top to bottom or click twice to sort from bottom to top.
4. To add or remove a column, right-click on any column.
 - a. Choose the desired field. A checkmark will appear or disappear and the list will refresh.
 - b. To close the menu, click anywhere outside of it.
5. To remove a column, left-click on the desired column.
 - a. Without releasing the mouse, drag it up or down until a black "X" appears.
 - b. Release the mouse.
6. To rearrange a column, left-click on the desired column
 - a. Without releasing the mouse, drag it left or right until two red arrows appear to indicate where it will be placed.
 - b. Release the mouse
7. When finished, click **Close**.

NOTE: As new columns are added, a scroll bar will be added at the bottom. Any customization is tied to the computer (not to the schedule) and will be reset upon re-installation of NetPoint.

Exporting a List to Excel

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. In any view, right-click on the columns at the top of any list.
3. Choose **Export All Rows & Columns**, **Export Displayed Rows & Columns**, or **Export to Excel**, as needed.
4. Repeat one by one for all lists desired.

Filtering Lists

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. Click the desired icon at the top.
3. Click in the search field next to **Filter**.
4. Begin typing, and the list will adjust to display any matches.
5. Select an item in the list and it will be indicated on the canvas.
6. Click the **Highlight** button to fade all objects on the canvas that are not included in the filtered results.
7. Click the **Select** button to select it on the canvas for modification.
8. To return to the full list, delete all text from the search field.
9. When finished, click **Close**.

NOTE: Any alphanumeric characters can be entered, and the results will include anything that matches, starts with, or contains what has been entered, pulled from every column currently being shown. For example, if searching for a duration, make sure the **Duration column is being displayed.**

Enlarging a List

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. Click the desired icon at the top.
3. Click the **Expand** icon  above any list.

NOTE: Any customization of the columns will be lost upon closing and reset to that of the Metrics Manager

Enabling/Disabling Canvas Scroll

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. Click the **List** icon  in the upper-right corner.
3. Click **Scroll Canvas with Manager Selections**. A checkmark will appear or disappear accordingly.

Notes

The Manager

Metrics - Well Built

Summary Comp. Credible **Well-Built** Controlled

Type anything to filter...

S..	Item	Result	Tripwire
☒	Well Built IQ	57*	! ≥ 80
☒	+ Phased	×	0 ! ≥ 80
☒	+ Calendar-Fit	×	NA ! -
☒	- Logical	57*	! ≥ 80
☒	Redundant Logic	×	12 % ! ≤ 5%
☒	Activities with Constraints	57*	3 % ! ≤ 2%
☒	Non-FS Logic	✓	20 % ≤ 20%

Viewing: Matching Links Type anything to filter...

ID	Predecessor Des...	Sucessor Descrip...	Sucessor Sta...	Sucessor Fir
193	Purchase Demoliti...	Fire Department...	9/26/2013	10/1/2
194	BasementSubsta...	SubstantialCompl...	4/24/2014	4/24/2
195	SubstantialCompl...	SubstantialCompl...	4/24/2014	4/24/2
196	SubstantialCompl...	SubstantialCompl...	4/24/2014	4/24/2
197	1st FloorSubstan...	SubstantialCompl...	4/24/2014	4/24/2
198	FinishElectricalEx...	SubstantialCompl...	4/24/2014	4/24/2
199	ExteriorSubstanti...	SubstantialCompl...	4/24/2014	4/24/2

Select Highlight Restore All Close

Icon	Definition
 Summary	The IQ score for the overall schedule and categories.
 Comp.	Whether or not the schedule is aligned, complete, conforming, formulaic, and resourced. For example: activities and their durations, contract dates, resource, code, and WBS assignments, etc.

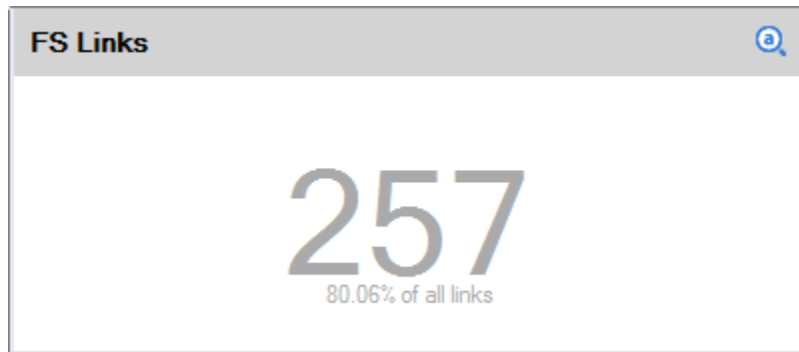
Icon	Definition
 Credible	Whether or not the schedule is predictive, risked, weather-fit, resource-flowing, and flexible. For example: critical activities, total float, risk, weather, etc,
 Well-Built	Whether or not the schedule is hierarchical, phased, logical, connected, and calendar-fit. For example: milestones, constraints, redundant logic, dangling and open ends, calendars etc.
 Controlled	Whether or not the schedule is statused, weathered, re-baselined, forensic, and trended. For example: activities completed, out-of-sequence progress, remaining duration, etc.

The Results Browser

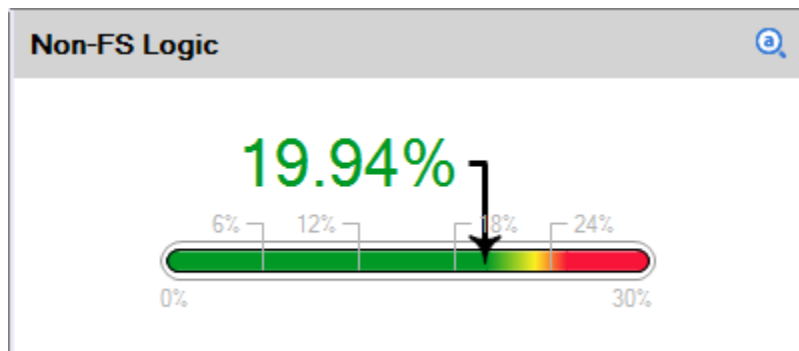
The **Results Browser** is how you dive even deeper into a given metric. You can choose between a dashboard widget, a matching objects filter, and a deficiency report (tornado diagram).

Dashboard Widgets

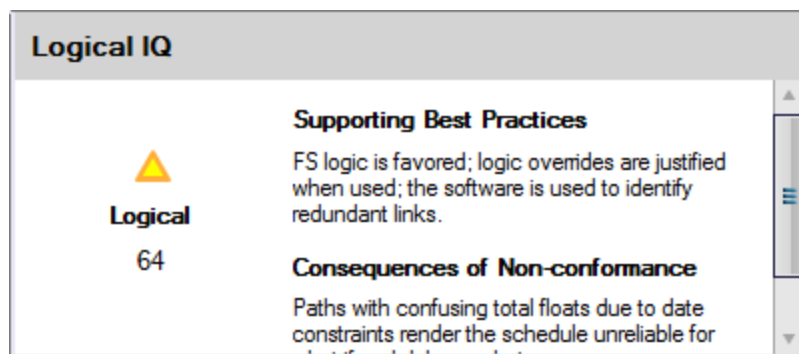
Dashboard widgets are visual representations of data that can be viewed for any metric. For simple data metrics, they take the form of just a number:



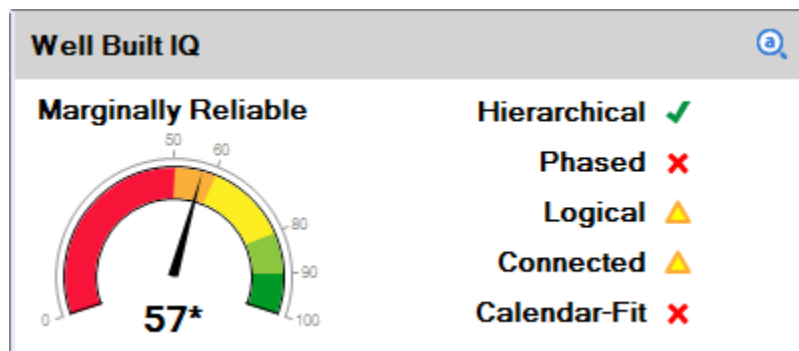
For indicators, they display bars with normative ranges:



For IQ metrics, they show their score and related info:



For categories, they show their IQ gauge and summary:



Matching Objects Filter

The matching objects filter displays a list of all matching objects (i.e. activities, or links, or resources) for a selected metric, which helps you find them on the canvas and make changes to improve your IQ scores. The filter can be viewed for most data metrics as well as indicators.

ID	Predecessor Desc...	Successor Descr...	Logic Type	Rela...	Lag
191	Draft/ExecuteLett...	Draft andExecu...	Hard	FS	0
202	SubstantialCompl...	CertificateofOc...	Hard	FS	0
210	Draft/ExecuteLett...	Purchase Demo...	Hard	FS	0
212	CompleteUtilitySu...	SewerLineRepairs	Hard	FS	0
213	Mobilization	StormDrainLine...	Hard	FS	0
214	CompleteUtilitySu...	StormDrainLine...	Hard	FS	0
215	StormDrainLineRe...	RainLeader & Si...	Hard	FS	0
216	RainLeader & Site...	RainLeader & Si...	Hard	FS	0

Deficiency Report (Tornado Diagram)

The deficiency report is a list of all activities sorted by how many indicators they are tripping (the deficiency count). One column is provided for each indicator and an "x" is placed where an indicator was tripped. It is viewable from the **Summary** tab, or when selected on one of the following rows: **Comprehensive IQ**, **Credible IQ**, **Well-Built IQ**, and **Controlled IQ**. When viewing from one of the rows, the total indicator count is limited to only those indicators within the same tab.

Description	Deficiency Count	A.	D.	C.	A.	A.	A.	A.	A.	A.	I.	D.	Z.
Draft/Execu...	7	X	X	X	X	X	X	X					
Certificateof...	7	X	X	X	X	X		X	X				
CompleteUtil...	6	X	X	X		X	X	X					
Purchase Bal...	6	X	X	X	X	X			X				
WindowWell...	6	X	X	X		X						X	X
1st FloorClo...	6	X	X	X		X		X		X			
2nd FloorRo...	6	X	X	X	X	X				X			
1st FloorToil...	6	X	X	X		X		X		X			

In addition, a column can be turned on in any list of **Matching Objects** to see the **Deficiency Count**.

T..	Description	Deficiency Count	Start Date	Duration
	Draft/ExecuteLetterof In...	7	9/9/2013	
	CertificateofOccupancy	7	4/25/2014	
	Purchase Balance of Trades	6	12/6/2013	
	CompleteUtilitySurvey	6	9/9/2013	
	WindowWellDrains	6	10/16/2013	
	1st FloorCloset Systems	6	4/21/2014	
	1st FloorToilet Accessories	6	4/10/2014	

The deficiency report can also be exported to Excel. For more info, see [➔Exporting to Excel](#).

Stakeholders and Narrative Checklist

The stakeholders section is for documenting who has signed off on the schedule being analyzed. The number of stakeholders that have signed off will positively or negatively affect the Schedule IQ Score. Signatures can be segmented into groups (revisions) so that the history of all signatures can be maintained throughout updates.

Below the stakeholders section is the narrative checklist. The narrative checklist includes questions to be answered with respect to the narrative document included with the schedule submittal.

Technically, adding stakeholders or reviewing the schedule narrative is optional; however, it is in the best interest of the analyst to do both in accordance with the *Core Traits of a Reliable Schedule*. For metrics info, see [Schedule Sign-offs](#) and [Narrative Score](#) metrics.

NOTE: If the schedule is modified after having signed-off, the signatures will be archived, a new revision will be created, and the schedule will need to be re-signed again.

Steps

Adding Stakeholders

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. In the **Properties** tab, choose an option next to **Delivery**.
4. Click the **Review & Sign** tab.
5. Click **Add Revision**.
6. Enter a name and click Enter on the keyboard.
7. Click **Add Stakeholder**.
8. Enter a name and click Enter on the keyboard.
9. Click **Undefined** underneath the **Role** column and choose an appropriate role.
10. When finished, click **Save** to save.

Signing-off on the Schedule

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. Click the **Review & Sign** tab.
4. Click the checkbox underneath the **Sign-off** column in the row of the desired stakeholder.
5. Add any comments and/or click **Continue**.
6. Read the message and click **Sign**.
7. When finished, click **Save** to save.

NOTE: Once signed, a stakeholder can no longer be edited, and all details will be disabled.

Adding Revisions

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. Click the **Review & Sign** tab.
4. Click the **Add Revision** button. Any stakeholders already signed-off will be duplicated as a starting point.

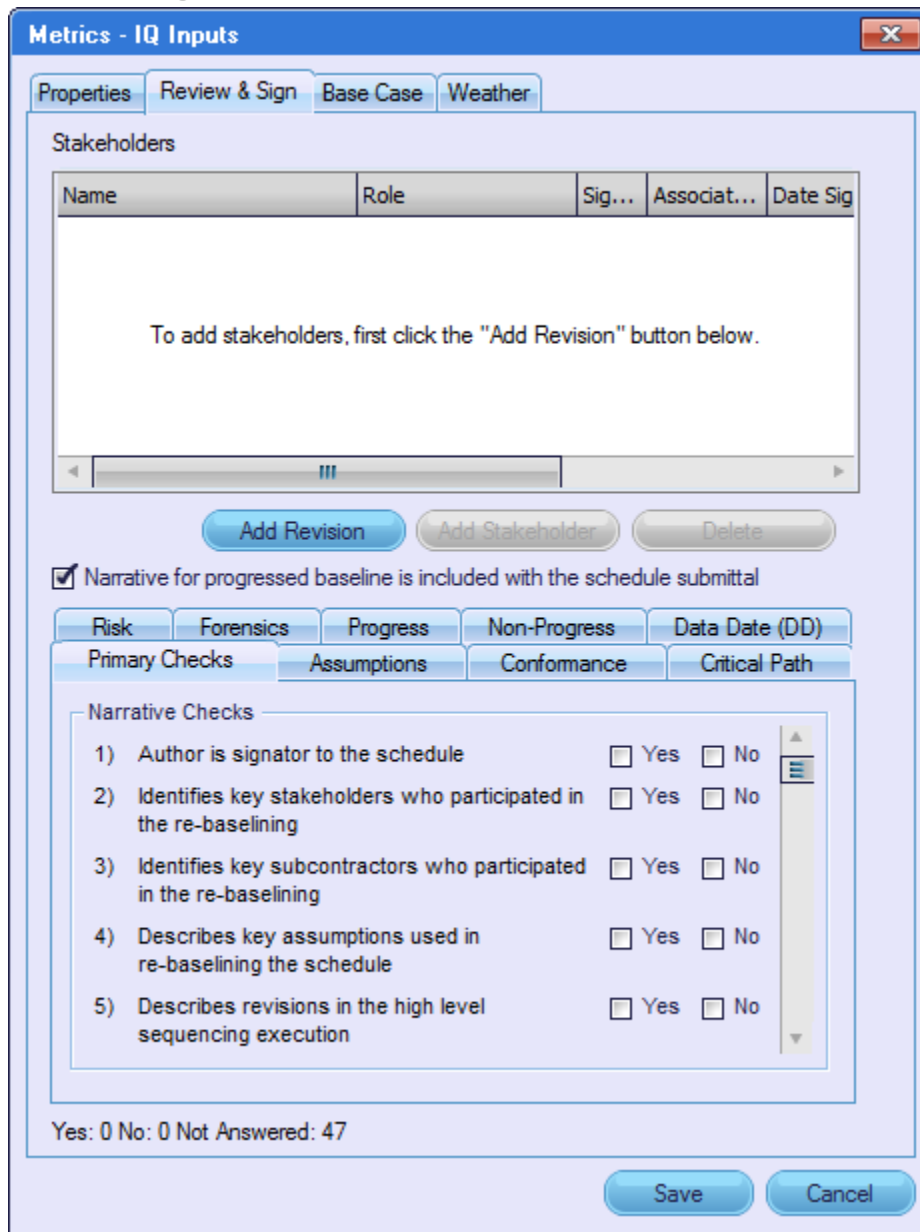
- When finished, click **Save** to save.

Reviewing the Schedule Narrative (Checklist)

- With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
- Choose **Configure Schedule MD Inputs**.
- Click the **Review & Sign** tab.
- Check the box next to **Narrative for [progressed baseline] is included with the schedule submittal**.
- Starting with the **Primary Checks** tab, answer each question as it relates to the narrative document.
- When finished, click **Save** to save.

Notes

Review & Sign tab



Metrics - IQ Inputs

Properties | **Review & Sign** | Base Case | Weather

Stakeholders

Name	Role	Sig...	Associat...	Date Sig
To add stakeholders, first click the "Add Revision" button below.				

☒ Narrative for progressed baseline is included with the schedule submittal

Risk | Forensics | Progress | Non-Progress | Data Date (DD)
 Primary Checks | Assumptions | Conformance | Critical Path

Narrative Checks

1) Author is signator to the schedule	☐ Yes	☐ No
2) Identifies key stakeholders who participated in the re-baselining	☐ Yes	☐ No
3) Identifies key subcontractors who participated in the re-baselining	☐ Yes	☐ No
4) Describes key assumptions used in re-baselining the schedule	☐ Yes	☐ No
5) Describes revisions in the high level sequencing execution	☐ Yes	☐ No

Yes: 0 No: 0 Not Answered: 47

Stakeholders Details

Field	Definition
Name	A text field for entering the first and last name of the stakeholder.
Role	The role of the stakeholder on the project team. Roles are dependent on upon the delivery method chosen. For Design-bid-build : Contractor, Project Engineer, Project Executive, Project Manager, Project Scheduler, and Project Superintendent. For CMAR : Construction Manager, Lead Estimator, Project Controls Engineer, Project Engineer, Project Executive, Project Manager, and Project Scheduler. For Design-build : Construction Manager, Design Manager, Lead Estimator, Project Engineer, Project Executive, Project Manager, and Project Scheduler. For EPC : Construction Manager, Design Manager, Lead Estimator, Project Controls Engineer, Project Engineer, Project Executive, Project Manager, Project Scheduler, and Project Superintendent.
Sign-off	A checkbox for initiating a signing.
Associated Target	The latest target that existed at the time of signing. If the latest target that existed at the time of signing different from the schedule at the time of signing, or if no target exists at the time of signing, a new target is created and associated to the signature, so that the state of the schedule when signed can always be retrieved.
Date Signed	The date the stakeholder was signed.
Comments	A text field for adding comments when signing.

Loading a Base Case

A base case schedule is required for some metrics to be calculated. Currently, there is 1 IQ metric and 2 DCMA metrics that require a base case schedule. For definitions of each metric and how the base schedule is used, see [Schedule Index](#), [Missed Tasks](#), and [Baseline Execution Index](#) accordingly. Choosing the base case schedule is done through the Schedule MD Inputs window. Instructions provided below.

Steps

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. Click the **Base Case** tab.
4. Underneath **Filtepath**, click **Browse**.
5. Choose the desired file and click **Open**.
6. If working with an .xer file, choose from the **List of Available Projects in XER** dropdown.
7. Click **Save** to save and close.

Notes

Base Case tab

The screenshot shows the 'Metrics - IQ Inputs' dialog box with the 'Base Case' tab selected. The dialog has a title bar with a close button. Below the title bar are four tabs: 'Properties', 'Review & Sign', 'Base Case', and 'Weather'. The 'Base Case' tab is active and contains the following fields:

- Base Case** (Section Header)
- Filepath**: A text input field with a 'Browse...' button and a warning icon.
- List of Available Projects in XER**: A list box with a dropdown arrow.
- Completion Likelihood**: A spin box with a '%' symbol and a warning icon.

At the bottom of the dialog are 'Save' and 'Cancel' buttons, both with warning icons.

Base Case

Field	Definition
Filepath	The path of the selected base case file.
List of Available Projects in XER	A list of projects contained within the .xer file. Only available for .xer files.

Field	Definition
Completion Likelihood	The completion likelihood is obtained by simulating the schedule as part of a risk analysis. For more info, see Completion Likelihood Index and Completion Likelihood .

Weather

Schedule MD™ includes a one-of-a-kind feature-set for planning and analyzing normal and actual weather, as well as two metrics for determining the reliability of the weather embedded in the schedule. The functionality is divided into four main areas: planned weather, anticipated weather, observed weather, and actual weather.

Planned weather is that which has been added to the schedule right of the data date. To import planned weather, non-working days must be added into calendars in the schedule using your preferred scheduling software, and then identified using the Schedule MD inputs.

Anticipated weather is the normal adverse weather, according to one or more weather stations, averaged over a 10-year period. To calculate this, an internet connection is required to connect to the National Oceanic and Atmospheric Administration (NOAA) database. If weather has yet to be added to the schedule, this data becomes an invaluable source for planning purposes and should be the first step undertaken. If weather has already been added to the schedule, the anticipated weather is still necessary for calculating the schedule reliability. For more info, see [Planned Weather Index](#).

Actual weather is that which has been added to the schedule left of the data date. The only method supported here is embedding the weather days into calendars. To import actual weather, field records and other sources should be consulted to determine actual work stoppages, and then non-working days must be added into calendars in the schedule using your scheduling software, before then being identified in Schedule MD™.

Observed weather is the actual weather that occurred according to the selected weather station. The same parameters used to define the anticipated weather and weather stations are used here.

NOTE: If weather is not applicable to a given project, it can be disabled for the schedule and removed from scoring. For more info, see the steps below.

Tip: ALWAYS use the full monthly estimate when entering planned nonwork days due to weather into Primavera P6. For example, if the NOAA average for January is 7 days, but 2 of those days fall on weekends (after using a random number generator), DO NOT enter only 5. Otherwise, the Planned Weather Index will score lower.

Steps

Enabling/Disabling Weather

1. With the **Metrics Manager** open, click the **Credible** icon.
2. Expand **Weather-Fit** and check or uncheck the box next to **Planned Weather Index** accordingly.
3. Click the **Controlled** icon.
4. Expand **Weathered** and check or uncheck the box next to **Actual Weather Index**.

Importing Planned Weather

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. Click the **Weather** tab.
4. If weather has been added to calendars, choose **Weather Calendars** next to **Method**.
1. Expand the calendar(s) which contain the desired non-working days.

2. Check the box in the **Planned Weather** column next to the desired non-working days.
5. Click **Save** to save and close.

NOTE: When working with a NetPoint schedule, any NetPoint weather days will already be checked and cannot be unchecked.

Downloading Anticipated Weather

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. Click the **Weather** tab.
4. Click the **Anticipated (NOAA)** tab.
5. Click the **Configure** button.
6. Fill in the parameters accordingly.
7. Click **OK**.
8. Click the dropdown next to **Select Stations** and check the box next to each station desired for use. If more than one station is selected, their totals will be averaged together.
9. Click outside the dropdown to close it.
10. Click **Load**.
11. Click **Save** to save and close.

Tip: ALWAYS use the full monthly estimate when entering planned nonwork days due to weather into Primavera P6. For example, if the NOAA average for January is 7 days, but 2 of those days fall on weekends (after using a random number generator), DO NOT enter only 5. Otherwise, the Planned Weather Index will score lower.

Importing Actual Weather

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. Click the **Weather** tab.
4. Click the **Actual** tab.
5. Expand the calendar(s) which contain the desired non-working days.
6. Check the box in the **Weather** column next to the desired non-working days.
7. Click **Save** to save and close.

Downloading Observed Weather

1. With the **Metrics Manager** open, click the **List** icon  in the upper-right corner.
2. Choose **Configure Schedule MD Inputs**.
3. Click the **Weather** tab.
4. Click the **Observed** tab.
5. Click next to **Stations** and select from the list.
6. Click **Save** to save and close.

Notes

Planned tab

Weather Calendars

Schedule MD Inputs

Properties | Review & Sign | Base Case | ⚠ Weather

Planned | Anticipated (NOAA) | Actual | ⚠ Observed (NOAA)

Method: Weather Calendars

Calendars & Non-work days

P..	Name	Type	Date	Duration
☐	BOS-BBU 5 D...			
☒	Non-work38	Fixed Date	5/29/2017	1
☒	Non-work39	Fixed Date	7/4/2017	1
☒	Non-work40	Fixed Date	9/4/2017	1
☒	Non-work41	Fixed Date	10/9/2017	1
☒	Non-work42	Fixed Date	11/23/2017	1
☒	Non-work43	Fixed Date	12/25/2017	1
☒	Non-work44	Fixed Date	1/1/2018	1
☒	Non-work45	Fixed Date	2/19/2018	1
☒	Non-work46	Fixed Date	4/23/2018	1
☒	Non-work47	Fixed Date	5/28/2018	1
☒	Non-work48	Fixed Date	7/4/2018	1
☒	Non-work49	Fixed Date	9/3/2018	1
☒	Non-work50	Fixed Date	10/8/2018	1
☒	Non-work51	Fixed Date	11/12/2018	1
☒	Non-work52	Fixed Date	11/22/2018	1
☒	Non-work53	Fixed Date	12/25/2018	1
☒	Non-work54	Fixed Date	1/1/2019	1

? Uncheck (1)

⚠ Save Cancel

Column	Definition
Planned Weather	Controls whether or not a non-working day will be considered as a weather day. NetPoint weather days will be automatically checked and cannot be unchecked. If a multi-day holiday is marked as planned weather, it will also be marked as actual weather, and vice versa. Only calendars with at least 1 assignment will show up.
Name	The name of the non-working day. Since non-working days in .xer files do not have names, a generic name will be displayed. If the entire day is non-working, its name will include "Non-

Column	Definition
	working". If part of the day is non-working, its name will include "Exception". For non-working days which repeat in NetPoint, all instances will be collected together and can only be designated as a weather day as an entire group.
Type	The type of non-working day: Fixed-Date, Annual, Floating-Date, and Easter-Relative. For more details, see ➔Configuring Holidays .
Date	The date the non-working day occurs.
Duration	The number of days the non-working day lasts. For .xer files, both the workweek and the shift times are used to calculate fractional days, if necessary. For NetPoint projects, the duration equals the total entered from the calendars interface minus any days that fall on weekends, and adjusted if the month is not complete (e.g. the project start, data date, or project completion has cut it short). As a result, the total may differ from the planned weather metric. For actual count, see metric Planned Weather .

Anticipated (NOAA) tab

10-year Averages

Metrics - IQ Inputs

Properties Review & Sign Base Case  Weather

Planned Anticipated (NOAA) Actual  Observed (NOAA)

Parameters: [Configure...](#)

Select Stations: WINAMAC 2 SSE, IN US - ... [Load](#)

10-year Averages (days)

Station	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
WINAMAC 2 ...	27	25	26	17	10	7	6	6	5	18	22
Precipita...	4	3	5	5	6	7	6	6	4	5	5
Low Temp.	27	24	25	14	4	0	0	0	1	14	21
High Temp.	0	0	0	0	0	0	0	0	0	0	0
Average	27	25	26	17	10	7	6	6	5	18	22

 [Save](#) [Cancel](#)

Column	Definition
Station	The name of the station, followed by the distance from the user-entered latitude and longitude, followed by the station's latitude and longitude.
Month	The average number of non-working days that occurred for the given month during a 10-year period, starting with the project start. If the project start is in the future or has not been reported yet, the latest complete 10-years of data are used. A dash symbol "-" means there was no data for that weather type at that station. Averages for the station row do not double count when more than one type of weather has occurred. Losses are considered in order of precipitation, wind, and temperature. For example, a day previously counted as lost due to

Column	Definition
	rain is not counted again as a day lost due to wind or temperature. As a result, the totals in the station row may differ from the sum of the individual rows.
Yearly Total	The sum of all 12 months.
Project Total	The sum of the months during which the project takes places, prorated accordingly.

NOAA Parameters

Field	Definition
Use default Token	A pre-programmed token shared among all installations of NetPoint. This option may become invalid if the number of requests for a given day has been exceeded.
Use my own token	A link to the NOAA web page for registering for access and a field for providing your own token after having registered. This option allows more requests to be made, as it is unique to each installation.
Units	An option for displaying units in standard US or metric.

Field	Definition
Latitude	The latitude of the project location in terms of degrees north. To enter degrees south, use a negative sign before the value.
Longitude	The longitude of the project location in terms of degrees east. To enter degrees west, use a negative sign before the value.
Radius	The distance from the latitude and longitude entered for determining the search area. The area is determined by taking the square around the circle resulting from the radius.
Low Temp	The threshold for determining when a day is lost due to low temperatures. If the lowest temperature recorded between the start and finish is below the threshold, the entire day is considered lost. Default 40 degrees fahrenheit.
High Temp	The threshold for determining when a day is lost due to high temperatures. If the highest temperature recorded between the start and finish is above the threshold, the entire day is considered lost. Default 100 degrees fahrenheit.
Precipitation	The threshold for determining when a day is lost due to precipitation (rain and water equivalent). If the threshold is crossed, the entire day is considered lost. Default 0.1 inches.
Recovery Precip.	The threshold for determining when the following day is lost due to precipitation (rain and water equivalent) on the previous day. If the threshold is crossed, the entire next day is considered lost. Default 1 inch.
Wind	The threshold for determining when a day is lost due to wind. If any 2-minute sustained speed is above the threshold, the entire day is considered lost. Default 20 mph.
Heavy Wind	A secondary threshold for determining when a day is lost due to wind. If any 2-minute sustained speed is above the threshold, the entire day is considered lost. Default 30 mph.

Actual tab

Schedule MD Inputs

Properties Review & Sign Base Case  Weather

Planned Anticipated (NOAA) Actual  Observed (NOAA)

Non-working Days

A...	Name	Type	Date	Duration
	BOS-BBU 5 Da...			
☒	Non-work24	Fixed Date	1/1/2016	1
☐	Non-work25	Fixed Date	1/18/2016	1
☐	Non-work26	Fixed Date	2/15/2016	1
☐	Non-work27	Fixed Date	4/18/2016	1
☐	Non-work28	Fixed Date	5/30/2016	1
☐	Non-work29	Fixed Date	7/4/2016	1
☐	Non-work30	Fixed Date	9/5/2016	1
☐	Non-work31	Fixed Date	10/10/2016	1
☐	Non-work32	Fixed Date	11/11/2016	1
☐	Non-work33	Fixed Date	11/24/2016	1
☐	Non-work34	Fixed Date	12/26/2016	1
☐	Non-work35	Fixed Date	1/2/2017	1
☐	Non-work36	Fixed Date	2/20/2017	1
☐	Non-work37	Fixed Date	4/17/2017	1
	HSEC - 5 Day ...			
☐	Non-work24	Fixed Date	1/1/2016	1
☐	Non-work25	Fixed Date	1/18/2016	1
☐	Non-work26	Fixed Date	2/15/2016	1

 Uncheck (1)

 Save Cancel

Column	Definition
Actual Weather	Controls whether or not a non-working day will be considered as a weather day. NetPoint weather days will be automatically unchecked and cannot be checked since they are merely randomly generated estimates. If a multi-day holiday is marked as planned weather, it will also be marked as actual weather, and vice versa. Only calendars with at least 1 assignment will show up.
Name	The name of the non-working day. Since non-working days in .xer files do not have names, a generic name will be displayed. If the entire day is non-working, its name will include "non-work". If part of the day is non-working, its name will include "Exception". For non-working

Column	Definition
	days which repeat in NetPoint, all instances will be collected together and can only be designated as a weather day as an entire group.
Type	The type of non-working day: Fixed-Date, Annual, Floating-Date, and Easter-Relative. For more details, see ➔Configuring Holidays .
Date	The date the non-working day occurs.
Duration	The number of days the non-working day lasts. For .xer files, both the workweek and the shift times are used to calculate fractional days, if necessary.

Observed (NOAA) tab

Only dates with losses are displayed.

Metrics - IQ Inputs

Properties Review & Sign Base Case  Weather

Planned Anticipated (NOAA) Actual  Observed (NOAA)

Stations: WINAMAC 2 SSE, IN US - 21.6 miles

Losses due to weather

Date	Precipi...	Low Tem...	High Tem...	Grand Total
2011				
Aug				
8/3/2011	1			
8/8/2011	1			
8/9/2011	1			
8/14/2011	1			
8/21/2011	1			
Total Observ...	5	0	0	5
Total Anticip...	6	0	0	6
Total Abnor...	-1	0	0	-1
Total Observed	5			5
Total Anticipated	6			6
Total Abnormal	-1			-1

 Save Cancel

Column	Definition
Weather Type	A column is displayed for each type of weather that was selected in the NOAA Paramters window along with the number of days lost (e.g. "1") for each date. If the number of days lost shows up in gray (e.g. "1"), then it passed the threshold but will not be counted since another type of weather was already observed.
Grand Total	The totals of rows Total Observed , Total Anticipated , and Total Abnormal .

NOTE: NCEI/NOAA data and products that contain international data can be used within the United States or for noncommercial international activities without restriction. Redistribution of these data

by others must provide this same notification. The non-U.S. data cannot be redistributed for commercial purposes. For details, please consult [WMO Resolution 40](#).

Metrics Libraries

Metrics are organized into a number of different libraries from which the user can browse. These include “IQ Metrics”, “DCMA Metrics”, and “Data Metrics”. By default, they are sorted by the **Result** column (i.e. bad to good for IQ Metrics), but they may be sorted by clicking on any column.

If an IQ metric/indicator cannot be calculated, IQ scores will be qualified with an asterisk (*). Any such indicator can be manually removed from scoring by the user. For a snapshot of any/all metrics removed from scoring, export the IQ Report to Excel and look for “Not Scored” in the **Rating** column.

Steps

Removing a Metric from Scoring

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. Click the desired icon at the top.
3. Make sure the **Scored** column is displayed.
4. Uncheck the box next to the desired metric.

NOTE: IQ metrics/indicators can only be removed when they cannot be calculated. Each time the Metrics Manager is closed, any metrics that were removed will be saved for the future.

Notes

IQ Metrics

Each IQ Metric corresponds with a trait from the 20-Trait Protocol. When expanded, a list of indicators from which the metric is calculated are shown. The IQ metrics within a given category are totaled to determine the category’s IQ score, and all the categories’ IQ scores are totaled to determine the overall IQ score for the schedule. In total, the Schedule IQ Score is derived based on 42 calculations. In some cases, one or more of these calculations may not be able to be carried out. For example, if the file being read is an .xer (P6) or if an there is no progress in the schedule. In these cases, the encompassing IQ score will be displayed with an asterisk.

NOTE: IQ indicators are rounded to the whole numbers. As a result, in some rare cases, the result may get a yellow triangle  even though it appears to be equal to the tripwire.

Comprehensive

Metric	Default Tripwire	Definition	Tips for Healing/Notes
Aligned	≥80	<u>Supporting Best Practices:</u> The schedule is both supported by a narrative and is signed off by a sufficient number of appropriate stakeholders. <u>Consequences of Non-conformance:</u>	

Metric	Default Tripwire	Definition	Tips for Healing/Notes
		Schedule without a narrative and/or not documenting involvement by key stakeholders may justify withholding acceptance.	
<i>Narrative Score</i>	≥12 (unprogressed baseline), 13 (revised baseline), 11 (update)	<i>The number of points (out of 12 for an unprogressed baseline, 13 for a revised baseline, and 11 for an update) given to the narrative checklist.</i>	<i>Input required in the Schedule MD Inputs window. For more info, see Stakeholders and Narrative Checklist.</i>
<i>Schedule Sign-offs</i>	≥5 (level 1/2), ≥4 (level 3/4)	<i>The number of stakeholders who have signed off on the schedule via the Schedule MD inputs.</i>	<i>Input required in the Schedule MD Inputs window. For more info, see Stakeholders and Narrative Checklist.</i>
Complete	≥80	<u>Supporting Best Practices:</u> Activities are assigned to a WBS element and are assigned codes useful to organize, filter, and roll up the activities. <u>Consequences of Non-conformance:</u> An incomplete schedule is susceptible to delay and manipulation; an unstructured schedule is cumbersome.	-
<i>Missing WBS</i>	≤5%	<i>The number of activities without a WBS element divided by the total number of activities. *Not calculated when the denominator equals 0.</i>	
<i>Missing Codes</i>	≤5%	<i>The number of activities without a code divided by the total number of activities. *Not calculated when the denominator equals 0."</i>	

Metric	Default Tripwire	Definition	Tips for Healing/Notes
Conforming	≥80	<u>Supporting Best Practices:</u> No delay exists in a baseline; every contractual start or finish milestone falls on/after or on/before the specified contract date. <u>Consequences of Non-conformance:</u> A non-conforming baseline, either impacted by delay and/or off spec, may contain false critical paths and be unacceptable.	-
Breaching Finish Dates	0	Finish milestones or benchmarks that fall after their respective contract dates. *Not calculated for .xer files or when zero contract finish dates detected.	To view the contract date, display the Contract Date column in the list of matching objects of the results browser.
Breaching Start Dates	0	Start milestones or benchmarks that fall before their respective contract dates. *Not calculated for .xer files or when zero contract start dates detected.	To view the contract date, display the Contract Date column in the list of matching objects of the results browser.
Off-base Delays	0	Delay activities that are NOT modeling weather, according to their property boxes, and that are present in the baseline schedule. *Not calculated for .xer files, revised baselines, or updates.	
Formulaic	≥80	<u>Supporting Best Practices:</u> Critical/near-critical physical work activities have formulaic durations and ≥ 80% of activities have durations within the norm. <u>Consequences of Non-conformance:</u> Subjective durations are unreliable for forensic analysis; insufficient/	-

Metric	Default Tripwire	Definition	Tips for Healing/Notes
		excessive granularity impairs progress measurement.	
Subjective Durations	≤20%	Activities designated as physical work whose durations are missing a formula and whose criticality factor is ≤ the near-criticality threshold (user defined), divided by all activities designated as physical work whose criticality factor is ≤ the near-criticality threshold. For current thresholds, see Schedule MD Inputs. *Not calculated for .xer files or when the denominator equals 0.	To indicate physical work activities, display the Work Type column in the list of matching objects of the results browser while viewing metrics “Critical Activities” and “Near-Critical Activities.” Then, change each activity in the Dates tab one-by-one or using Objects > Set Properties for Selected Objects . The Formulaic column will show activities designated as physical work but with an incomplete formula with an exclamation mark.
Improper Detail	≤20%	Incomplete activities designated as physical work with durations outside the recommended range, divided by all incomplete activities designated as physical work. The recommended range by level for each activity according to its calendar is: The recommended range by level for each activity according to its calendar is: 2-12 months for level 1, 6 weeks-6 months for level 2, 2-6 weeks for level 3, and ≤ 2 weeks for level 4.	Durations are first converted to weeks based on activity calendars and then converted to months using the average of 4.33 weeks/month. If an activity’s calendar contains NO workdays/week, the activity will be automatically marked as outside the norm (if physical work). Not available for NetPoint files using years or quarters.
Resourced	≥80	<u>Supporting Best Practices:</u> Physical work activities are loaded with resources; resource loading is consistent with activity duration predicates. <u>Consequences of Non-conformance:</u> Trade congestion, density, and resource limit overage may	-

Metric	Default Tripwire	Definition	Tips for Healing/Notes
		not be solvable without extending completion or acceleration.	
Missing Resources	≤5%	The number of activities designated as physical work without a resource divided by the total number of activities designated as physical work. *Not calculated for .xer files or when the denominator equals 0.	To indicate physical work activities, display the Work Type column in the list of matching objects of the results browser while viewing metrics “Critical Activities” and “Near-Critical Activities.” Then, change each activity in the Dates tab one-by-one or using Objects > Set Properties for Selected Objects .
Upcoming Overages	≤10%	The number of time units (e.g. days) that resource levels exceed limits within 13 periods (e.g. weeks) after the data date, divided by the total number of time units with resources assigned within 13 periods after the data date. *Not calculated when the denominator equals 0.	

Credible

Metric	Default Tripwire	Definition	Tips for Healing/Notes
Predictive	≥80	Supporting Best Practices: Critical path total float between 5%-10% of project length is generated through early completion vs. critical path breaks. Consequences of Non-conformance: Schedule with insufficient critical path margin is unlikely to correlate to a reasonable probability of on time completion.	

Metric	Default Tripwire	Definition	Tips for Healing/Notes
<i>Critical Activities</i>	15%-30%	<i>The number of incomplete activities whose criticality factor is $\leq$ the threshold (user defined), divided by the total number of incomplete activities. For current thresholds, see the Schedule MD Inputs. *Not calculated when the denominator equals 0.</i>	<i>Delays also included.</i>
<i>Inadequate Float</i>	$\leq 25\%$	<i>Activities with merge risk indices ≥ 0.5 divided by the total number of activities. Merge risk indices are calculated as an activity's merge float divided by its total float and indicates if the activity's total float is adequate to mitigate the risk of starting later due to overruns on long-duration paths that merge into the activity. *Not calculated when the denominator equals 0.</i>	<i>To view the merge risk indices for individually activities, display the Merge Risk Index column in the list of matching objects of the results browser. A value of infinity means the activity has 0 total float.</i>
<i>Schedule Margin</i>	5%-10%	<i>The project completion date minus the latest early finish date of all activities or milestones divided by the project completion date minus the data date. *Not calculated when the denominator equals 0.</i>	<i>If at least 2 stakeholders have signed off on the schedule, then a high CPI will not bring down the IQ score as much as it otherwise would.</i>
<i>Hidden Delay</i>	0	<i>Right of the date date, the gap sum of the longest path (i.e. the path with the lowest total float) divided by the duration of the longest path (including non-working). *Not calculated when the denominator equals 0.</i>	<i>Gaps before the first activity or after the last activity not included.</i>
Flexible	≥ 80	<u>Supporting Best Practices:</u> Total float (CPM schedules) and schedule gap index and float performance index (GPM schedules) are	-

Metric	Default Tripwire	Definition	Tips for Healing/Notes
		trended. <u>Consequences of Non-conformance:</u> Unknown, lopsided rate of gap and total float consumption undermine schedule credibility right of the data date.	
Schedule Gap Index	≥ 0.50	<i>Right of the data date, the sum of all link gaps (excluding redundant links) divided by the sum of remaining non-critical durations. *Not calculated when the denominator equals 0.</i>	
Resource-Flowing	≥ 80	<u>Supporting Best Practices:</u> If the project comprises repeating activities, resource flow links are FS and crew-flow charts are provided. <u>Consequences of Non-conformance:</u> The schedule completes unreasonably early as it requires unrealistic levels for resources that drive physical work.	-
Abnormal Resource Logic	$\leq 5\%$	<i>The number of non-finish-to-start links that have been designated as "Resource-Flow" in their property boxes divided by the total number of links designated as "Resource-Flow". *Not calculated for .xer files or when denominator equals 0.</i>	
Risk	≥ 80	<u>Supporting Best Practices:</u> Using risk analysis, the schedule reserves schedule margin sufficient to support a reasonable completion likelihood. <u>Consequences of</u>	-

Metric	Default Tripwire	Definition	Tips for Healing/Notes
		Non-conformance: Absent a risk analysis, activity durations and logic are likely to underestimate the true length of the critical path.	
Completion Likelihood	≥75%	The probability of completion as entered in the IQ inputs. *Not calculated for updates.	
Weather-fit	≥80	Supporting Best Practices: The schedule correctly integrates normal adverse weather according to the specifications/best practices. Consequences of Non-conformance: Schedule margin may be consumed and/or completion of the project may be delayed by normal adverse weather.	-
Planned Weather Index	≥0.95	Right of the data date, the number of planned weather days divided by the number of anticipated weather days as modeled in the Schedule MD inputs. *Not calculated when the denominator equals 0.	Input required in the Weather tab of the Schedule MD Inputs window. For more info, see Weather . If the same day has been designated as a weather day in more than one calendar (for example, in one calendar it's been marked as due to rain and in another calendar it's been marked as due to wind), it will only be counted once. For NetPoint weather days, the total number of days are used, regardless of whether or not they fall on weekends.

Well-Built

Metric	Default Tripwire	Definition	Tips for Healing/Notes
Connected	≥80	Supporting Best Practices: Activities have at least one FS or SS predecessor and	

Metric	Default Tripwire	Definition	Tips for Healing/Notes
		one FS or FF successor; PDM logic is used judiciously. <u>Consequences of Non-conformance</u> : Open/dangling ends corrupt total floats; an unconnected schedule is unreliable for what-if, risk, and delay analysis.	
Paired Activities	≤2%	Activities connected by both start-to-start and finish-to-finish links to the same predecessor or successor divided by the total number of activities. *Not calculated when the denominator equals 0.	
Open Ends	≤1%	Activities that have either no predecessors or no successors divided by the total number of activities. *Not calculated when the denominator equals 0.	Delays also included.
Dangling Ends	≤2%	Activities that have only finish-to-finish and/or start-to-finish predecessors or only start-to-start and/or start-to-finish successors, divided by the total number of activities. *Not calculated when the denominator equals 0.	
Logic Index	1.25-1.75 (Level 2), 1.5-2.5 (Level 3/4)	The number of valid activity-to-activity links (i.e. excluding redundant links) divided by the total number of activities. *Not calculated when the denominator equals 0.	

Metric	Default Tripwire	Definition	Tips for Healing/Notes
Inhibiting Logic	≤5%	<i>Finish-to-finish + start-to-finish links with activity successors and zero-gap divided by the total number of finish-to-finish, start-to-finish, and start-to-start links with activity successors (links with benchmarks excluded). *Not calculated when the denominator equals 0.</i>	
Questionable Logic	≤5%	<i>Start-to-start and finish-to-finish links with zero lags divided by all start-to-start and finish-to-finish links (activity-to-activity only). *Not calculated when the denominator equals 0.</i>	
Calendar-Fit	≥80	<u>Supporting Best Practices:</u> Calendars reflect work schedules; multiple calendars are used judiciously and minimized for crew-loaded activities. <u>Consequences of Non-conformance:</u> Unnecessary calendar variations exacerbate total float breaks and blur path patterns and critical path continuity.	
Secondary Calendar Use	≤5%	<i>The number of activities designated as physical work assigned to secondary calendars, divided by the total number of activities designated as physical work. A secondary calendar is any calendar other than that which has the most assignments (weather calendars</i>	<i>The primary calendar may include weather. In case there is a tie in the primary calendar, then any calendar with weather is removed, and one is randomly selected. In such cases, the list of matching objects will show ALL physical work/undefined activities on non-weather calendars. To view the assigned calendar, display the Calendar column in the list of</i>

Metric	Default Tripwire	Definition	Tips for Healing/Notes
		<i>excluded). *Not calculated for .xer files or when the denominator equals 0.</i>	<i>matching objects of the results browser.</i>
Hierarchical	≥80	<u>Supporting Best Practices:</u> Connected milestones and benchmarks are used to enhance vertical traceability and/or to integrate contractual dates. <u>Consequences of Non-conformance:</u> Loss of vertical traceability undermines use of the schedule; an unconnected milestone is not horizontally traceable.	
Missing Hammocks	≤5% (level 3/4)	<i>Activities that aren't associated with any hammock divided by the total number of activities. *Not calculated for level 1/2 schedules or when the denominator equals 0.</i>	<i>Delays also included.</i>
Milestone Density	5%-15% (Level 1/2), 2%-5% (Level 3/4)	<i>The number of milestones and benchmarks without contract dates divided by the total number of activities. *Not calculated when the denominator equals 0.</i>	For .xer files, all milestones and benchmarks are treated as if they are not contract dates.
Logical	≥80	<u>Supporting Best Practices:</u> FS logic is favored; logic overrides are justified when used; the software is used to identify redundant links. <u>Consequences of Non-conformance:</u> Paths with confusing total floats due to date constraints render the schedule unreliable for what-if and delay analysis.	

Metric	Default Tripwire	Definition	Tips for Healing/Notes
Constraints	≤2%	The number of activities with constraints divided by the total number of activities. *Not calculated when the denominator equals 0.	Delays also included. The Constraint column will list the constraint type and date. The following abbreviations are used for constraints created in P6 when viewing directly on .xer files: ALAP (As Late As Possible), MEO (Finish On), MEOA (Finish On or After), FNL (Finish On or Before), MS (Mandatory Start), MF (Mandatory Finish), MSO (Start On), SNE (Start On or After), and MEOB (Start On or Before).
PDM Logic	≤20%	The number of non-finish-to-start activity-to-activity links divided by the total number of links. *Not calculated when the denominator equals 0.	
Improper Leads	0	The number of negative lags on finish-to-start links that are longer than the predecessor duration.	
Redundant Logic	≤5%	The number of redundant links divided by the total number of links. *Not calculated when the denominator equals 0.	
Extreme Logic	0%	The number of start-to-finish links divided by the total number of at-to-finish, finish-to-finish, and start-to-start links. *Not calculated when the denominator equals 0.	
Phased	≥80	Supporting Best Practices: Construction phases are benchmarked; benchmarks are used if total float is allocated to phases of the work. Consequences of Non-conformance: Phase	

Metric	Default Tripwire	Definition	Tips for Healing/Notes
		durations are not coherent with the overall project duration; early trade contractors overconsume total float.	
<i>Benchmark Frequency</i>	1-2	<i>The number of benchmarks with at least one predecessor or successor divided by the total number of years in the schedule. *Not calculated for .xer files.</i>	<i>Years are calculated as the number of months from the project start to the project completion divided by 12.</i>

Controlled

Metric	Default Tripwire	Definition	Tips for Healing/Notes
Forensic	≥80	<u>Supporting Best Practices:</u> The as-built critical path is algorithmically identified left of the data date. <u>Consequences of Non-conformance:</u> Broken critical path and negative gaps left of the data date impair schedule reliability for forensic schedule analysis.	-
<i>Hidden Forensic Delay</i>	0	<i>Left of the data date, the sum of all gaps on the path with the lowest forensic total float. *Not calculated for unprogressed baselines.</i>	
<i>Breached Gaps</i>	≤2%	<i>Links that span a period of time less than 0 and which have been actualized, divided by all actualized links. An actualized link is defined as the link between two actualized or partially actualized activities, so long as the embedded node (GPM) falls before the data date</i>	

Metric	Default Tripwire	Definition	Tips for Healing/ Notes
		<i>(for non-finish-to-start links). *Not calculated for unprogressed baselines or when the denominator equals 0.</i>	
Re-baselined	≥80	Supporting Best Practices: To ensure a realistic forecast, the schedule is timely re-baselined to reflect replanning, time impacts, and schedule recovery. Consequences of Non-conformance: An obsolete baseline conveys an unreliable forecast. Schedule margin may not reflect updated risk profiles.	–
Completion Likelihood Index	>0.95	<i>The completion likelihood in the revised baseline divided by the completion likelihood in the unprogressed baseline. *Not calculated for unprogressed baselines, updates, or when the denominator equals 0.</i>	<i>The likelihood for the revised baseline is entered in the Properties tab of the Schedule MD Inputs window. The likelihood for the unprogressed baseline is entered in the Base Case tab of the Schedule MD Inputs window. For more info, see ➔Loading a Base Case.</i>
Statused	≥80	Supporting Best Practices: Imminent level 3 activities are resource leveled; out-of-sequence progress for critical/near-critical activities is repaired. Consequences of Non-conformance: Update with flawed actualized logic is unreliable for delay/disruption analysis, whether prospective or retrospective.	–

Metric	Default Tripwire	Definition	Tips for Healing/ Notes
<i>Improper Status</i>	0	<i>The number of completed activities right of the data date + the number of incomplete activities left of the data date. *Not calculated for unprogressed baselines.</i>	<i>Delays also included.</i>
<i>Impractical Starts</i>	$\leq 80\%$ (level 1/2); $\leq 60\%$ (level 3/4)	<i>For level 3 or 4 schedules, non-critical activities that start on planned dates within 8 periods (e.g. weeks for a schedule in days) after the data date, divided by the total number of activities within 8 periods after the data date. For level 1 or 2 schedules, within 26 periods after the data date. *Not calculated for .xer files or when the denominator equals 0.</i>	
<i>Out-of-sequence Work</i>	$\leq 2\%$	<i>Actualized critical and near-critical successor activities whose start dates breach the logic tie type, divided by the total number of actualized critical and near-critical activities. *Not calculated for unprogressed baselines or when the denominator equals 0.</i>	<i>Delays also included.</i>
Trended	≥ 80	<u>Supporting Best Practices:</u> Activity completion is sufficient so that scope of remaining activities is congruent with an achievable rate of progress. <u>Consequences of Non-conformance:</u> More activities/work effort than planned beyond the data date may render the balance of the schedule unrealistic.	-
<i>Earned Schedule Index</i>	≥ 0.95	<i>The project duration at which planned percent complete (i.e.</i>	<i>Percent complete uses duration, resource</i>

Metric	Default Tripwire	Definition	Tips for Healing/ Notes
		<i>resource consumption) should have occurred (in the selected base case file) divided by the elapsed duration from project start to data date. If > 1.0, progress is ahead of schedule. If < 1.0, progress is behind schedule. *Not calculated for unprogressed baselines or when the denominator equals 0.</i>	<i>assignments, and resource cost. If an activity has no resources, only duration is used. Current schedule must have a data date. Base case is entered in the Base Case tab of the Schedule MD Inputs window. For more info, see Loading a Base Case.</i>
Weathered	≥80	<u>Supporting Best Practices:</u> The schedule is used to evaluate weather delay/gain originating from actual weather conditions in the prior month. <u>Consequences of Non-conformance:</u> Updates that overlook actual weather are unreliable for delay/disruption analysis, whether prospective or retrospective.	-
<i>Actual Weather Index</i>	0.90-1.10	<i>Left of the data date, the number of actual weather days identified divided by the number of observed weather days according to the selected NOAA weather station(s). *Not calculated for unprogressed baselines or when the denominator equals 0.</i>	<i>Input required in the Weather tab of the Schedule MD Inputs window. For more info, see Weather.</i>

IQ Scoring

IQ Category Thresholds

Range	Rating
90-100	Highly Reliable
80-89	Reliable

Range	Rating
60-79	Somewhat Reliable
51-59	Marginally Reliable
≤ 50	Unreliable

IQ Metric Thresholds

Icon	Range	Rating
✓	80-100	Reliable
⚠	51-79	Somewhat Reliable
✗	≤ 50	Unreliable
–	–	Not Scored

NOTE: IQ indicators are rounded to the whole numbers. As a result, in some rare cases, the result may get a yellow triangle even though it appears to be equal to the tripwire.

DCMA Metrics

DCMA metrics key schedule parameters with tripwire thresholds, developed by the Defense Contract Management Agency. Together, they comprise the *DCMA 14-point assessment*.

NOTE: All DCMA metrics refer to only incomplete work (right of data date) and exclude milestones, benchmarks, summaries, and LOE's in the calculations. NetPoint delays are not excluded.

Comprehensive

Metric	Default Threshold	Definition
High Duration	≤5%	Incomplete activities with duration > 44 days divided by the total number of incomplete activities.
Resources	0%	Incomplete activities without resources divided by the total number of incomplete activities.

Credible

Metric	Default Threshold	Definition	Notes
Critical Path Test	PASS/FAIL	Whether or not the last activity on the critical path shifts in direct proportion (within 90%) to a delay to another activity on the critical path.	A shift of 600 days is applied at random to an activity in the critical path. A buffer of 90% is allowed to account for non-working windows or gap on the critical path in the case of a GPM schedule. Not available for .xer files.
Critical Path Length Index	≤5%	Critical path length + total float divided by the critical path length (in working days).	Not well-suited to GPM schedules with negative drift, as total float may be negative even though float is positive.
High Float	≤5%	Incomplete activities with total float > 44 days divided by the total number of incomplete activities.	

Well-Built

Metric	Default Threshold	Definition
Hard Constraints	≤5%	Manually scheduled incomplete activities, and/or incomplete activities with hard “Must Start On”, “Must Finish On”, “Start No Later Than”, or “Finish No Later Than” constraints divided by the total number of incomplete activities.
Lags	≤5%	Incomplete links with lags divided by the total number of incomplete links.
Leads	0%	Incomplete links with negative lags or negative gaps divided by the total number of incomplete links. A link is considered incomplete unless the successor is fully actualized.
Logic	≤5%	Incomplete activities with no predecessor and/or no successor divided by the total number of incomplete activities.
Relationships	≥90%	Incomplete finish-to-start links divided by the total number of incomplete links.

Controlled

Metric	Default Threshold	Definition	Notes
Baseline Execution Index	≥0.95	Completed activities divided by the number of activities that should have been completed according to the selected base case scenario.	Base case is entered in the Base Case tab of the Schedule MD Inputs window. For more info, see Loading a Base Case .
Invalid Dates	0%	Completed activities before the data date + incomplete activities after the data date divided by the total number of activities.	
Missed Tasks	≤5	Completed or incomplete activities with a baseline finish on or before the data date whose current or actual finish is later than their baseline finish, divided by the total number of activities with a baseline finish on or before the data date.	Base case is entered in the Base Case tab of the Schedule MD Inputs window. If an activity from the base case file is not found in the current file, it will be displayed in gray. For more info, see Loading a Base Case .
Negative Float	0%	Incomplete activities with negative float (GPM) or total float (CPM) divided by the total number of incomplete activities.	Not well-suited to GPM schedules with negative drift, as total float may be negative even though float is positive.

Data Metrics

Data metrics are simple counts and attributes that characterize the schedule, of which there are over a hundred.

Comprehensive

Metric	Definition	Tips for Repairing/Notes
Activities	The building blocks of a schedule. Activities span a duration from start date to a finish date, each date portrayed by an end node.	The Formulaic column in the list of matching objects of the results browser will show which activities have been checked as formulaic. If they are missing a quantity, rate, or duration, they will get an exclamation mark.
Activities with Normative Durations	Incomplete physical work activities with durations within the recommended range. The recommended range by level for each activity according to its calendar	Durations are first converted to weeks based on activity calendars and then converted to months using the average of 4.33 weeks/month. Will always return 0

Metric	Definition	Tips for Repairing/Notes
	is: 2-12 months for level 1, 6 weeks-6 months for level 2, 2-6 weeks for level 3, and ≤ 2 weeks for level 4.	for .xer files since all activities are by definition “undefined”. Not available NetPoint files using years or quarters.
Activities without a Description	Activities whose text field for naming or summarizing is empty.	
Activities without an ID	Activities without a user-input unique identifier.	
Closeout Activities	Activities that have been designated as “Closeout” in their property boxes.	Will always return 0 for .xer files since all activities are by definition “undefined”.
Codes	Applied to activities, milestones, and benchmarks for categorizing, organizing, sorting, filtering, or reporting more easily.	
Commissioning Activities	Activities that have been designated as “Commissioning” in their property boxes.	Will always return 0 for .xer files since all activities are by definition “undefined”.
Contract Dates	Dates according to a respective contract and entered in milestone or benchmark property boxes.	To view the contract date, display the Contract Date column in the list of matching objects of the results browser. Will always return 0 for .xer files since the option is not provided in P6.
Contractor Delays	Delays that have been designated as “Contractor” in their property boxes.	Not available for .xer files.
Delay Activities	Activities which have been designated as delays, or whose durations can collapse to zero.	Not available for .xer files.
Delivery Activities	Activities that have been designated as “Delivery” in their property boxes.	Will always return 0 for .xer files since all activities are by definition “undefined”.
Design Activities	Activities that have been designated as “Design” in their property boxes.	Will always return 0 for .xer files since all activities are by definition “undefined”.
Finish Benchmarks	Finish benchmarks which are finishing on or before the input contract date.	To view the contract date, display the Contract Date column in the list of matching objects of the results browser.

Metric	Definition	Tips for Repairing/Notes
on Contract Dates		Will always return 0 for .xer files since the option is not provided in P6.
Finish Milestones on Contract Dates	Finish milestones finishing on or before their input contract date.	To view the contract date, display the Contract Date column in the list of matching objects of the results browser. Will always return 0 for .xer files since the option is not provided in P6.
Non-weather Force Majeure Delays	Delays that have been designated as “Non-weather Force Majeure” in their property boxes.	Not available for .xer files.
Formulaic Activities	Activities that have had formulaic durations entered in their property boxes.	If an activity’s formulaic duration differs from its planned duration, it will get an exclamation mark in the Formulaic Duration column in the list of matching objects of the results browser. Will always return 0 for .xer files since all activities are by definition “undefined”.
Info Objects	External files or attachments embedded on the canvas or on activities within NetPoint.	
Other Work Type Activities	Activities that have been designated as “Other” in their property boxes.	Will always return 0 for .xer files since all activities are by definition “undefined”.
Owner Delays	Delays that have been designated as “Owner” in their property boxes.	Not available for .xer files.
Physical Work Activities	Activities that have been designated as “Physical Work” in their property boxes.	Activities designated as physical work but with an incomplete formula will get an exclamation mark in the Formulaic column in the list of matching objects of the results browser. Will always return 0 for .xer files since all activities are by definition “undefined”.
Procurement Activities	Activities that have been designated as “Procurement” in their property boxes.	Will always return 0 for .xer files since all activities are by definition “undefined”.

Metric	Definition	Tips for Repairing/Notes
Resources	Applied to activities, milestones, and benchmarks for categorizing, organizing, sorting, filtering, or reporting more easily.	
Start Benchmarks on Contract Dates	Start benchmarks which are starting on or after the input contract date.	To view the contract date, display the Contract Date column in the list of matching objects of the results browser. Will always return 0 for .xer files since the option is not provided in P6.
Start Milestones on Contract Dates	Start milestones starting on or before their input contract date.	To view the contract date, display the Contract Date column in the list of matching objects of the results browser. Will always return 0 for .xer files since the option is not provided in P6.
Submittal Activities	Activities that have been designated as "Submittal" in their property boxes.	Will always return 0 for .xer files since all activities are by definition "undefined".
Third Party Delays	Delays that have been designated as "Third Party" in their property boxes.	Not available for .xer files.
Unspecified Delays	Delays that have been designated as "Unspecified" in their property boxes.	Not available for .xer files.
Validation Activities	Activities that have been designated as "Validation" in their property boxes.	Will always return 0 for .xer files since all activities are by definition "undefined".
Weather Delays	Delays that have been designated as "Weather" in their property boxes.	Not available for .xer files.

Credible

Metric	Definition	Notes
100th Percentile Period Density	The highest number of activities taking place during any period (e.g. any week for a schedule in days). NOTE: Period is always defined as the next higher unit except for minutes (which uses hours), weeks (which	

Metric	Definition	Notes
	uses months), and years (which uses years).	
10th Total Float Percentile	10% of incomplete activities have this amount of total float or less.	
20th Percentile Period Density	80% of the weeks (for a schedule in days) from project start to project finish have at least this many activities taking place. NOTE: Period is always defined as the next higher unit except for minutes (which uses hours), weeks (which uses months), and years (which uses years).	
50th Percentile Period Density	50% of the weeks (for a schedule in days) from project start to project finish have at least this many activities taking place. NOTE: Period is always defined as the next higher unit except for minutes (which uses hours), weeks (which uses months), and years (which uses years).	
5th Total Float Percentile	5% of incomplete activities have this amount of total float or less.	
80th Total Float Percentile	80% of incomplete activities have this many days of total float or less.	
Activities on Weather Calendars	Activities assigned a calendar in their property boxes, which has been designated as including weather.	Includes input from the Weather tab of the Schedule MD Inputs window. For more info, see ➔Weather .
Activities with Buffer = 0	Activities which if delayed, delay at least one successor.	

Metric	Definition	Notes
Activities with Extreme Total Float	Activities whose total float is at or above the 80th total float percentile.	
Activities with total float $\leq$ 10th percentile	Incomplete activities whose total float is less than or equal to the 10th percentile. For more info, see metric "10th Total Float Percentile".	
Activities with total float $\leq$ 5th percentile	Incomplete activities whose total float is less than or equal to the 5th percentile. For more info, see metric "5th Total Float Percentile".	
Planned Weather	The number of days right of the data date due to weather as modeled in the Schedule MD Inputs.	Includes input from the Weather tab of the Schedule MD Inputs window. For more info, see Weather . If the same day has been designated as a weather day in more than one calendar (for example, in one calendar it's been marked as due to rain and in another calendar it's been marked as due to wind), it will only be counted once. For NetPoint weather days, the total number of days are used, regardless of whether or not they fall on weekends.
Average Period Density	The number of activities divided by the number of periods (weeks for a schedule in days) from the project start date to the project finish date. NOTE: Period is always defined as the next higher unit except for minutes (which uses hours), weeks (which uses months), and years (which uses years).	
Contractual Links	Relationships that are mandated or dictated by a contract, as set in their property boxes.	Not available for .xer files.
Critical Activities	Activities whose criticality factor is $\leq$ the threshold (user defined).	

Metric	Definition	Notes
	For current thresholds, see the Schedule MD Inputs.	
Critical Path Total Float	The total float of the last object on the longest path (excluding benchmarks).	
Anticipated Weather (NOAA)	The number of weather days for the project right of the data date based on a 10-year average observed by NOAA for the selected weather station(s).	Input required in the Weather tab of the Schedule MD Inputs window. For more info, see ➔ Weather .
External Links	Relationships that are mandated or dictated by an external source to the schedule, such as an imposed interface with another, unrelated project, as set in their property boxes.	Not available for .xer files.
Extreme Gap Links	Links whose gap is greater than or equal to the 80th percentile total float, multiplied by 80%	
Hard Links	Relationships inherent to the nature of the work being done, as set in their property boxes.	Not available for .xer files.
Links with Gap $\geq$ 20 Periods	Links which span a period of time more than 20 periods (e.g. weeks for a schedule in days). NOTE: Period is always defined as the next higher unit except for minutes (which uses hours), weeks (which uses months), and years (which uses years).	
Mean Total Float	The sum total float of all activities, divided by the number of activities in the schedule.	

Metric	Definition	Notes
Median Total Float	The 50th total float percentile, after ordering all activities from least total float to largest total float.	
Near-critical Activities	Activities whose criticality factor is $>$ the criticality threshold, but $\leq$ the near-criticality threshold (user defined). For current thresholds, see the Schedule MD Inputs.	
Negative-gap Links	Links which span a period of time less than 0.	
Negative-total-float Activities	Activities that, due to a delay or advance, are affecting the project completion or start, or interim completion event (benchmark).	
Normal Weather Days per Year	The number of planned weather days right of the data date divided by the number of years from the data date to the project finish date.	Includes input from the Weather tab of the Schedule MD Inputs window. For more info, see ➔ Weather . If the same day has been designated as a weather day in more than one calendar (for example, in one calendar it's been marked as due to rain and in another calendar it's been marked as due to wind), it will only be counted once. For NetPoint weather days, the total number of days are used, regardless of whether or not they fall on weekends.
Observed Weather (NOAA)	The number of weather days observed by NOAA left of the data date for the designated weather station(s).	Includes input from the Weather tab of the Schedule MD Inputs window. For more info, see ➔ Weather .
Positive-gap Links	Links which span a period of time greater than 0.	
Resource-flow Links	Relationships that are based on resource usage, as set in their property boxes.	Not available for .xer files.

Metric	Definition	Notes
Soft Links	Relationships that are a preferred choice, as set in their property boxes.	Not available for .xer files.
Targets	A snapshot of a schedule at a given moment in time used for comparing updates to one another.	
Weather Calendars	Calendars that have been designated as including weather.	Includes input from the Weather tab of the Schedule MD Inputs window. For more info, see ➔Weather .
Zero-gap Links	Links which do not span any period of time.	

Well-Built

Metric	Definition	Tips for Repairing/Notes
Activities on Constraints	Constraints are applied to dates to prevent activities from being moved earlier or later.	<i>The Constraint column will list the constraint type and date. The following abbreviations are used for constraints created in P6 when viewing directly on .xer files: ALAP (As Late As Possible), MEO (Finish On), MEOA (Finish On or After), FNL (Finish On or Before), MS (Mandatory Start), MF (Mandatory Finish), MSO (Start On), SNE (Start On or After), and MEOB (Start On or Before).</i>
Activities on FNL Constraints	A finish-no-later-than constraint is applied to an activity late finish date and prevents it from being moved later in time.	
Activities on Paired SS & FF Logic	Activity and successor activity (or activity and predecessor activity) connected by both start-to-start and finish-to-finish logic ties.	
Activities on SNE Constraints	A start no-earlier-than constraint is applied to an activity early start date	

Metric	Definition	Tips for Repairing/Notes
	and prevents it from being moved earlier in time.	
Activities with Dangling Finish	Activities with at least one predecessor and at least one successor that have only start-to-start and/or start-to-finish successors.	
Activities with Dangling Start	Activities with at least one predecessor and at least one successor that have only finish-to-finish and/or start-to-finish predecessors.	
Activities with FS/SS & FF Predecessor(s)	Activities that have at least one finish-to-start OR start-to-start predecessor AND at least one other finish-to-finish predecessor.	
Activities with Open Finishes	Activities without any successors.	
Activities with Open Starts	Activities without any predecessors.	
Activity-to-activity Links	Links between any two activities only.	
Benchmarks	Zero-duration events that are used to represent key dates or points of progress in the schedule. Benchmarks remain fixed at whatever date positioned, regardless of whether or not they've been linked.	
Calendar Periods	The number of periods (e.g. weeks for a schedule in days) from the activity with the earliest early-start date to the activity with the latest early-finish date. NOTE: Period is always defined as the next higher	

Metric	Definition	Tips for Repairing/Notes
	unit except for minutes (which uses hours), weeks (which uses months), and years (which uses years).	
Calendars In-use	The number of calendars with at least one or more activities, milestones, or benchmarks assigned.	
Constant Embeds	An embedded node whose date changes to maintain its lag from either the start or finish of its host activity.	Not available for .xer files.
FS Leads > Predecessor Duration	A negative lag between a predecessor-successor finish-to-start link that exceeds the duration of the predecessor.	
FS Links	Relationships where the successor cannot start before the predecessor has finished (finish to start).	
FS Links with Lag	FS logic tie datum that denotes the least time between the finish of the predecessor and the start of the successor.	
FS Links with Lead	Negative lag modeling an overlap between two finish-to-start-connected activities.	
Finish Benchmarks with Open End	Finish benchmarks without any predecessors.	
Finish Milestones with Open End	Finish milestones without any predecessors.	
Hammocks	A type of activity used to represent or summarize a group of other	

Metric	Definition	Tips for Repairing/Notes
	activities, milestones, and/or benchmarks. Hammocks cannot be linked or assigned resources and are depicted with different sizes and patterns.	
Logic Loops	A circular relationship causing a single activity to be both a predecessor and a successor to the same activity or chain. (NOTE: Redundant link metrics will not be able to be calculated until all logic loops have been removed).	To guarantee the loop is eliminated, delete only the link whose row shows up in red. If the link in red must be conserved, multiple other contributing links will have to be deleted for the loop to be eliminated.
Milestone/ benchmark Density	The number of milestones and benchmarks divided by the total number of activities in the schedule.	
Milestones	Zero-duration events used to represent key dates or points of progress in the schedule. Similarly to activities, they will track along with the network, once they've been linked to any predecessors or successors.	
Object-to-object Links	Relationships between any two objects (activities, milestones, or benchmarks).	
Paired SS/FF Links	Start-to-start and finish-to-finish links with the same predecessor and successor.	
Proportionate Embeds	An embedded node whose lag changes (depending if it's a start or finish) to maintain its percentage offset from either the start or finish of its host activity.	Not available for .xer files.
Redundant Links	An activity-successor link that cannot acquire zero gap under any combination of durations and lags,	

Metric	Definition	Tips for Repairing/Notes
	because the successor is preceded by another successor to the activity.	
Reverse Embeds	An embedded node whose date changes (depending if it's a start or finish) to maintain its lag from the opposite end of its host activity.	Not available for .xer files.
SS Links + FF Links	The subset of all logic ties comprising start-to-start or finish-to-finish links.	
Start Benchmarks with Open End	Start benchmarks without any successors.	
Start Milestones with Open End	Start milestones without any successors.	
Variable Embeds	An embedded node whose date changes or doesn't change (depending if it's a start or finish) to maintain its lag from either the start or finish of its host activity.	Not available for .xer files.
Zero-lag SS + Zero-lag FF Links	Start-to-start links with zero lag plus finish-to-finish links with zero lag (activity-to-activity only).	

Controlled

Metric	Definition	Notes
Activities Completed	Activities which have had both their start and finish dates actualized. Outlined in blue, they can no longer be modified.	
Activities In-progress	Activities which have begun but have not been finished. Intersecting with the data date, these activities are partially outlined in blue.	

Metric	Definition	Notes
Activities Remaining	Activities which have not yet begun (which have not yet been actualized), and which take place right of the data date.	
Activities Starting on Data Date	Activities whose start dates coincide with the data date.	
Activities on Early Dates	Activities whose drift is less than or equal to 0.	
Activities on Late Dates	Activities whose float is less than or equal to 0 and drift is greater than 0.	
Activities on Planned Dates	Activities whose float and drift is greater than zero.	Not available for .xer files.
Actual Weather	The number of weather days identified left of the data date according to all calendars.	Input required in the Weather tab of the Schedule MD Inputs window. For more info, see Weather . NetPoint weather days are NOT counted, since they are merely randomly generated estimates.
Actualized Zero-gap Links	Links which do not span any period of time between two completed activities or between a completed predecessor and an in-progress successor (if finish-to-start).	
Completed Benchmarks	Benchmarks which have had their dates actualized. Outlined in blue, they can no longer be modified.	
Completed Critical Activities	Completed activities whose criticality factor is $\leq$ the criticality threshold (user defined). For current thresholds, see the Schedule MD Inputs.	
Completed Links	Links between two completed or partially completed activities, so long as the the embedded node (GPM) falls	

Metric	Definition	Notes
	before the data date (for non-finish-to-start links).	
Completed Milestones	Milestones which have had their dates actualized. Outlined in blue, they can no longer be modified.	
Completed Near-critical Activities	Completed activities whose criticality factor is $>$ the criticality threshold, but $\leq$ the near-criticality threshold.	
Completed Negative-gap Links	Links that span a period of time less than 0, and which have been completed. A completed link is defined as the link between two completed or partially completed activities, so long as the the embedded node (GPM) falls before the data date (for non-finish-to-start links).	
Completed Zero-gap Links	Links that do not span any period of time, and which have been completed. A completed link is defined as the link between two completed or partially completed activities, so long as the the embedded node (GPM) falls before the data date (for non-finish-to-start links).	
Completion Likelihood Index	The completion likelihood in the current schedule divided by the completion likelihood in the base case schedule (regardless of revision or progress).	The likelihood for the current schedule entered in the Properties tab of the Schedule MD Inputs window. The likelihood for the base case schedule is entered in the Base Case tab of the Schedule MD Inputs window. For more info, see ➔Loading a Base Case .
Critical Path Length from Project Start	The interval (including non-working) between the project start and the early finish date of the last network object on the path with the lowest total float.	
Critical Path Length from Data Date	The interval (including non-working) between the data date and the early	

Metric	Definition	Notes
	finish date of the last network object on the path with the lowest total float.	
Gap Sum Embedded in the Longest Path Left of the Data Date	For the path with the longest duration (excluding link gaps), the sum of the link gaps from the first object in the path to the data date.	
Gap Sum Embedded in the Longest Path Right of the Data Date	For the path with the longest duration (excluding link gaps), the sum of the link gaps from the data date to the last object in the path.	
Holidays per Year	The number of holidays from project start date to project finish date divided by the number of years between the project start date and the project finish date (assigned calendars only).	Duplicates excluded. For schedules with time unit greater than days, holidays per year will be 0.
Progressed Activities Right of the Data Date	Activities right of the data date which have already been actualized.	
Unprogressed Activities Left of the Data Date	Activities left of the data date which have not been actualized yet.	
Remaining Schedule Duration (Calendar Days)	Duration (in calendar days if the schedule is in days) from the data date to project completion.	
Remaining Schedule Duration (United States)	Duration according to the default calendar for activities (in days if the schedule is in days) from the data date to project completion.	
Schedule Duration (Calendar Days)	Duration (in calendar days if the schedule is in days) from the project start to project completion.	
Schedule Duration (United States)	Duration according to the default calendar for activities (in days if the	

Metric	Definition	Notes
	schedule is in days) from the project start to project completion.	

Healing the Schedule via the Results Browser

The **Metrics Manager** interface makes it incredibly easy and intuitive to find, analyze, and take action on a given metric. For this reason, NetPoint espouses a manual approach to healing the schedule. As a result, the planner is able to maintain ownership of all changes to the schedule, resulting in a better understanding as well as more control of the result.

As the user moves from left to right and from top to bottom within the manager interface, the information becomes more and more specific. To see how this works, and for an actual example, see the instructions below.

The manager can be used to identify and heal any single one of the indicators that comprise the IQ metrics, including but not limited to redundant logic, dangling or open ends, activities on constraints, or out-of-sequence critical and near-critical activities.

NOTE: Healing the network can only be done when the Metrics Manager is launched from within NetPoint. As a result, healing an .xer file is not possible at this time.

Steps (Example: Redundant Links)

1. Click **Metrics > Open Metrics Manager** from the menu bar. If the manager is already open, click the **Summary** icon. *The IQ is displayed for the entire schedule.*
2. To drill down deeper, click an icon at the top. *For example, Well Built. The IQ is displayed for the entire category.*
3. To drill down deeper, choose an IQ metric. *For example, Logical. The IQ is displayed for the metric.*
4. To drill down deeper, choose a related indicator. *For example, Redundant Logic. The indicator bar and range is displayed.*
5. To drill down deeper, choose a related data metric. *For example, Redundant Links. The count of all links is displayed as a dashboard widget.*
6. To drill down deeper, view the **Results Browser** by clicking the dropdown next to **Viewing** and choosing **Matching Links**.
7. To identify actual redundant links, click through the list to see them spotlighted on the canvas.
8. To delete some or all redundant links, select multiple rows while holding the **Shift** or **Alt** key on the keyboard.
9. Click the **Select** button.
10. Click the **Cut** icon in the toolbar.
11. Click the **Metrics Manager** to give it focus. If the manager is behind NetPoint, find its icon in the taskbar to bring it in front.
12. Repeat steps 6 through 1 in reverse order to zoom out and see the effect on the schedule's IQ score.

Exporting to Excel

The **Metrics Manager** comes pre-configured with 5 reports: the summary report, which includes *all* metrics but not their matching objects. It's best for an overall picture of the schedule or for executive reporting.

The IQ report, which includes *only* IQ metrics, as well as their matching objects. It's best for drilling down into the IQ scores and individual activities.

The DCMA report, which includes *only* the DCMA 14-point assessment metrics, as well as their matching objects. It's best for only focusing on the DCMA metrics and nothing else.

The deficiency report (tornado), which includes all activities organized by how many indicators they've tripped. It does not include any Schedule IQ Score information or any metrics definitions. It is best for focusing on activities only and their potential impact on the schedule.

And the raw metrics report, which includes every metric displayed on a single sheet.

In addition, any individual list can be exported one-by-one as needed. Instructions for both are provided below.

Steps

Exporting a pre-configured Report

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. Click the **List** icon  in the upper-right corner.
3. Choose **Export to Excel**.
4. Select the desired report.

Tip: IQ Metric rows can be expanded and collapsed by using the + and – icons to the left of column “A”. Any metric that is underlined and blue contains a hyperlink to its sheet of matching objects.

Exporting List-by-list

1. Click **Metrics > Open Metrics Manager** from the menu bar.
2. In any view, right-click on the columns at the top of any list.
3. Choose **Export All Rows & Columns**, **Export Displayed Rows & Columns**, or **Export to Excel**, as needed.
4. Repeat one by one for all lists desired.

Split-Screen/Multiple Canvases

NetPoint allows you to divide each window into more than one canvas for the same schedule. This can be useful when you have a large schedule that can't fit within view all at once. The window can be split vertically, horizontally, or both, the latter of which produces four canvases.

Each canvas can be independently zoomed and scrolled, or the zoom can be linked via the menu option in the **View** menu.

Each canvas can have the resource plots and **Date Lock** mode turned on and off independently. Everything else applies to all canvases, including the **GEM**, **Logic** and **Planning** modes, canvas formatting, grid spacing, and stretch factor. Any modifications to the network performed in one of the canvases will be immediately reflected in the others. Please refer to the following steps.

NOTE: if the header or footer have been turned on, the horizontal and vertical splits will be disabled.

Steps

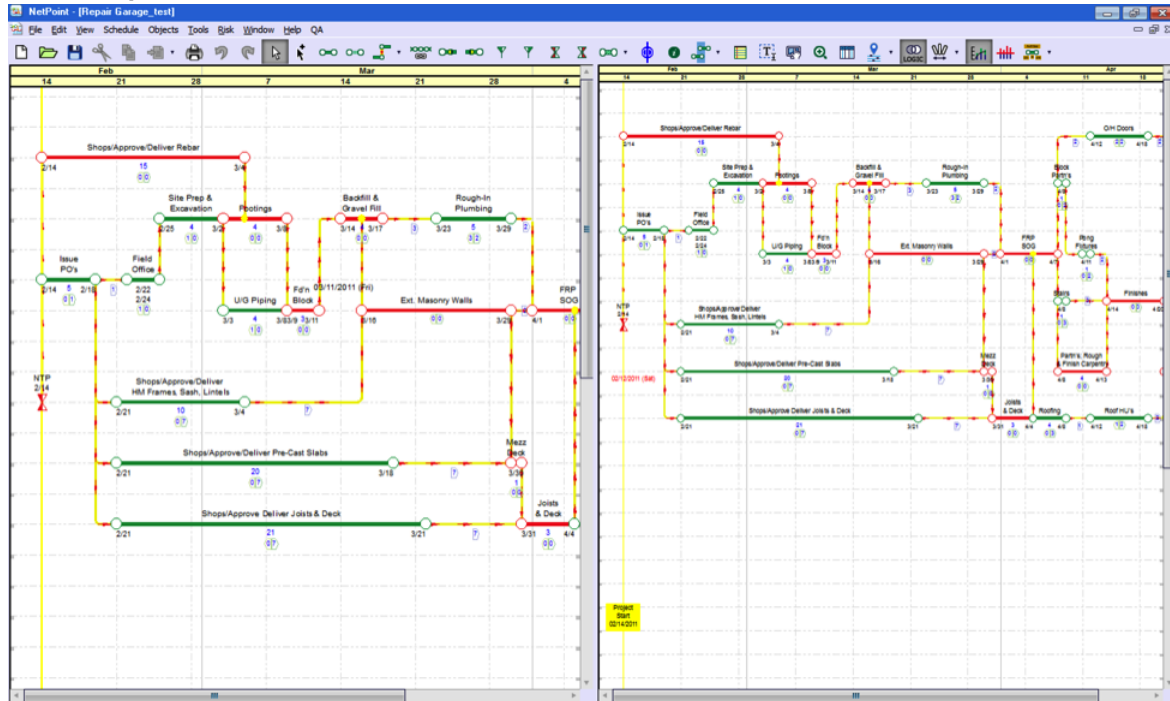
1. Open the desired project via **File > Open Project** from the menu bar.
2. To split the window into two side-by-side canvases, choose **View > Vertical Split**.
3. To split the window into two canvases stacked on top of each other, choose **View > Horizontal Split**.
4. To split the window into four canvases, apply steps 2 and 3 (in either order).
5. To switch focus and control from one canvas to another, click the **Select Tool**  icon from the toolbar and left-click with the mouse on an empty portion of the canvas.
6. To resize the canvases, move the cursor over the dividing line between canvasses.
 - a. Click-and-drag the double sided arrow that appears in the desired direction.
7. To remove the split, move the cursor over the dividing line between canvasses. .
 - a. Click-and-drag the double sided arrow that appears to *any* edge of the application window. Alternatively, click **View** from the menu bar and uncheck the split you wish to remove.

Notes

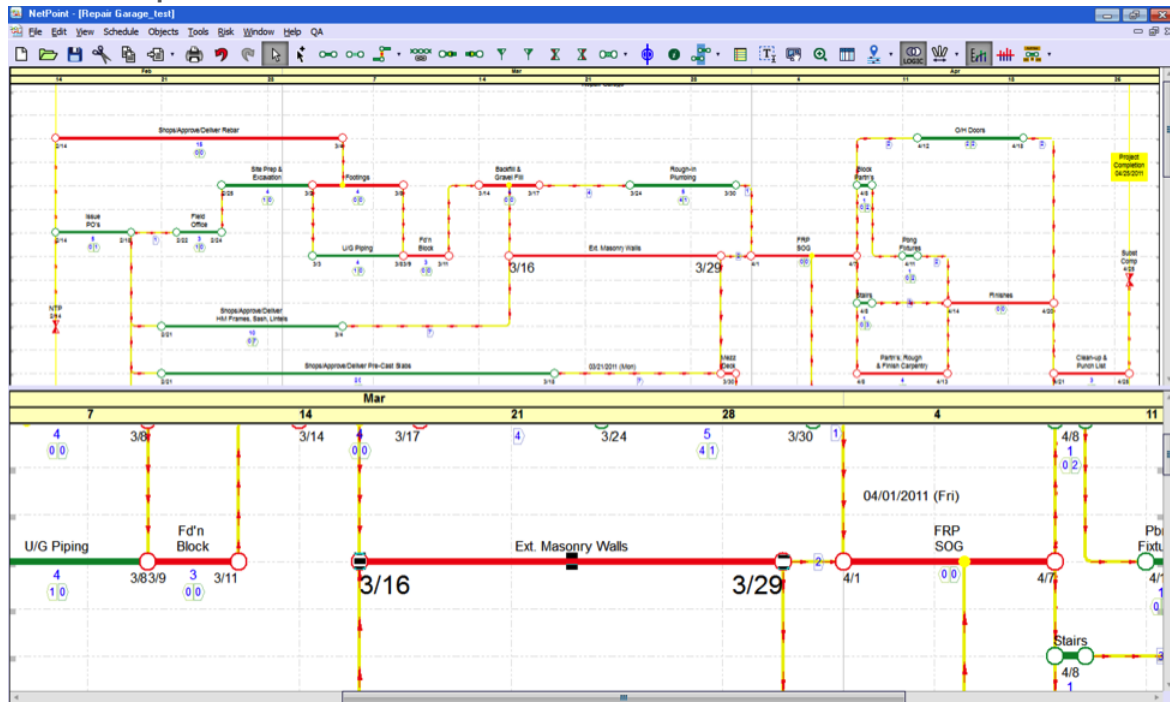
The Active Canvas

At any given time, only one canvas can be active per window. Changes made in the active canvas will be reflected in the other canvases for the same window; the time it takes will depend on the size of the schedule. Always change the active canvas before adding or making any modifications to it; otherwise, the cursor may not update and the date will not follow along.

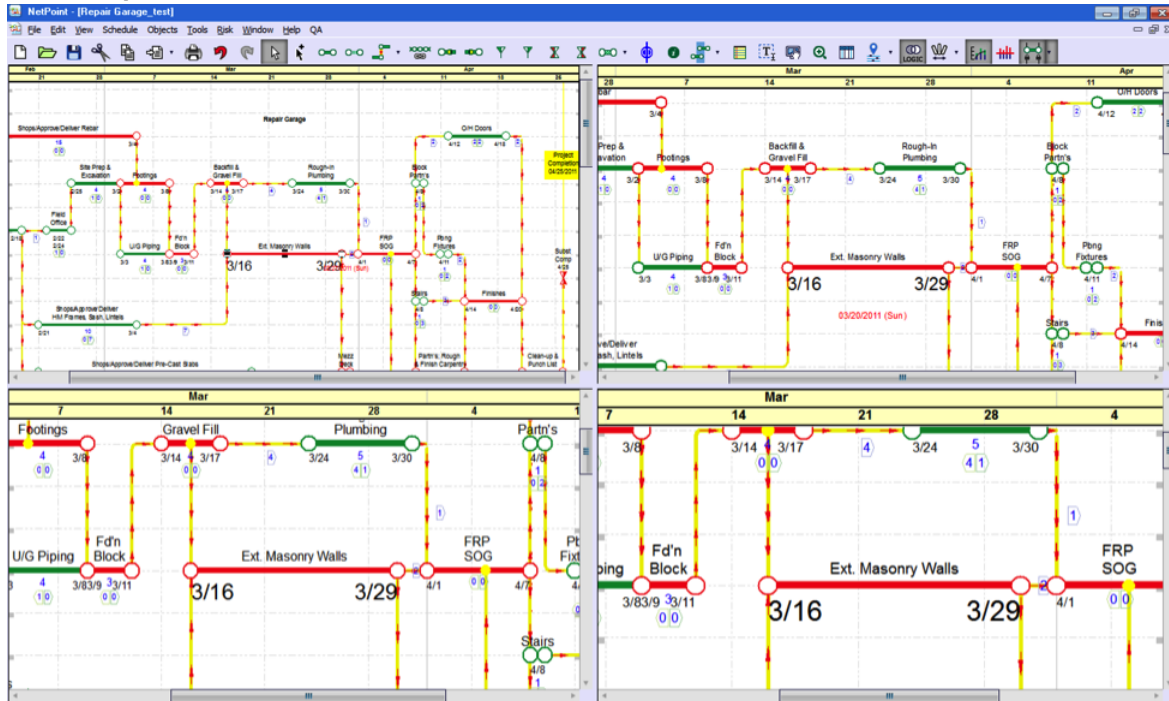
Vertical Split:



Horizontal Split:



Double Split :



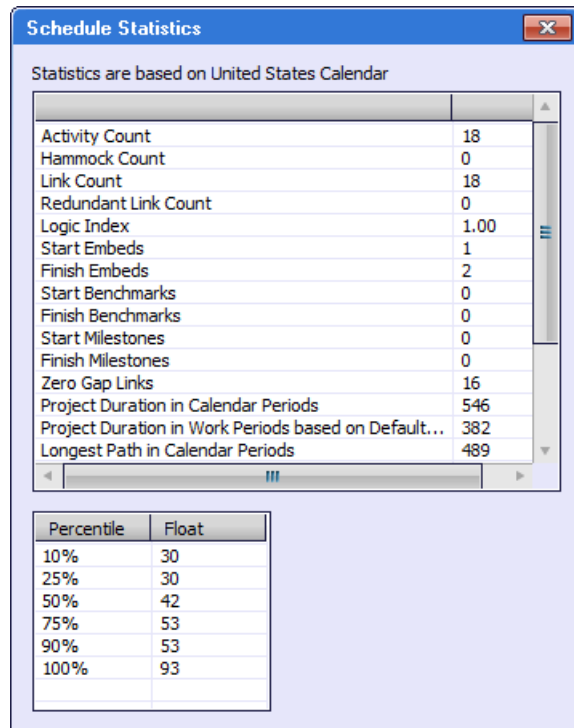
Schedule Statistics

NetPoint tracks a number of statistics for each plan, such as the number of activities, types of links, and float distributions.

Steps

1. Click **Schedule > View Statistics...** from the menu bar. The **Schedule Statistics** window will open.
2. When finished, click the red **X** in the upper right to close the window.

Notes



Metric	Definition
Activity Count	The number of activities in the schedule.
Hammock Count	The number of hammocks in the schedule.
Link Count	The number of links in the schedule.
Redundant Link Count	The number of redundant links in the schedule.
Logic Index	The number of valid links divided by number of activities in the schedule.

Metric	Definition
Start Embeds	The number of start embeds in the schedule.
Finish Embeds	The number of finish embeds in the schedule.
Start Milestones	The number of start milestones in the schedule.
Finish Milestones	The number of finish milestones in the schedule.
Start Milestones	The number of start benchmarks in the schedule.
Finish Milestones	The number of finish benchmarks in the schedule.
Zero-gap Links	The number of zero-gap links in the schedule.
Project Duration in Calendar Periods	The number of days (or, if PolyTime is enabled, number of minutes, hours, etc.) between the project start and project completion, including holidays, non-working periods, and weekends.
Project Duration in Work Periods	The number of days (or, if PolyTime is enabled, number of minutes, hours, etc.) between the project start and project completion, excluding holidays, non-working periods, and weekends (based on whichever calendar has been selected as default).
Longest Path in Calendar Periods	The chain of linked objects that lasts the most number of days (or, if PolyTime is enabled, number of minutes, hours, etc.) including holidays, non-working periods, and weekends.
Longest Path in Work Periods	The chain of linked objects that lasts the most number of days (or, if PolyTime is enabled, number of minutes, hours, etc.) excluding holidays, non-working periods, and weekends.
Hard-Links Count	The number of hard links in the schedule.

Metric	Definition
Soft-Links Count	The number of soft links in the schedule.
Resource-Links Count	The number of resources links in the schedule.
External-Links Count	The number of external links in the schedule.
Contractual-Links Count	The number of contractual links in the schedule.
Float Distributions	The percentile rank for different float values.

Predecessors and Successors

As objects become linked, they form chains. For any object in a chain, its predecessors are those that are connected by a link that *terminates at* the given object, and its successors are those that are connected by a link that *emanates from* the given object. A list of an object's predecessors and successors can be viewed in NetPoint, and predecessor/successor data can also be exported to excel. Instructions for both are provided.

Steps

Viewing and Breaking Links

1. Right-click on a desired object and choose **View Predecessors and Successors** from the menu. The **Predecessor and Successors** window will open.
2. To remove the link between the selected object and a given predecessor or successor, select it in the list and click **Break Link**. The link will be removed upon closing the window.
3. When finished, click the red **Exit** icon.

Comparing Resources

1. With the **Select Tool**  chosen in the toolbar, right-click on a desired object and choose **View Predecessors and Successors** from the menu.
2. To view assignments for a given resource, click the dropdown button next to **Compare Resources** and choose from the list.
3. To add, modify, or remove a resource assignment, select the desired activity in the list.
4. Under **Rate**, click in the box and enter, edit, or remove the value. **NOTE: If the rate column is left blank, no assignment will take place.**
5. To distribute the assignment over the entire duration of the activity, click the drop down button under **Distribution** and choose **/Duration**. For more info, see [Assignment Options](#).
6. When finished, click the red **Exit**  icon to close the window. Changes will be saved automatically.

Exporting to Excel

WARNING: When a template is exported to Excel, the new Excel file will automatically overwrite the last-exported Excel file unless the last one was renamed or moved. To avoid losing the previous Excel file, rename it or move it to a different location *before* doing the next export.

1. With the **Select Tool**  chosen in the toolbar, right-click on a desired object and choose **View Predecessors and Successors** from the menu. If desired, the predecessor/successor data that's exported can be customized first. To do so, skip to step 1 of [Modifying/Exporting an Existing Template](#) instead and choose "Defaults – Logic, Predecessors & Successors" from the dropdown.
2. Click the **Export** button. Excel will open with the exported data. **NOTE: On export, the Excel file will be automatically saved in the schedule's project folder and become linked to the schedule for updating. This new file will automatically overwrite the last-exported Excel file (for that schedule) unless it was renamed or moved.**
3. If you plan on exporting additional templates in the future, then we suggest renaming the file or moving it to a different location to avoid having it being overwritten. Click **File > Save As** from the menu bar, enter a name, and click **Save**. **NOTE: If you DO rename or move the file, you can ONLY initiate an update from it from within Excel. For more info, see [Updating from Linked Reports](#).**

NOTE: For more info, see [Predecessor and Successor Data](#) in the [Exporting Linked Reports](#) section.

Notes

Predecessors and Successors Window

Predecessors and Successors

Compare Resources: Carpenter(Commodity)

Predecessors

T...	Description	ID	Rate	Distribut...	Float	Relation	Gap
	Erect Walls				104	FS	0
	Steel Fab				159	FS	55

Selected **ID:** **Rate:** **Distribution:** **Float:**

SteelDeck 104

Successors

T...	Description	ID	Rate	Distribut...	Float	Relation	Gap
	Pour Decks				104	FS	0
	Masonry Walls				113	SS35	0
	Interior MEP				126	FS	0

[Break Link](#) [Export...](#)

Column	Definition
Type	The type of object, such as activity, link, milestone, benchmark, delay, or hammock.
Description	The object's description field, used to describe the object.
ID	The object's ID. The ID is a unique identifier.
Rate	The amount or number of the resource required.
Distribution	How the resource is used by an activity. Resources may be assigned per time period or duration.
Float	The float of the object. The float is the amount of time an object can be delayed before delaying the completion of the project.
Relation	The relationship with the selected object, such as start-to-start (SS), finish-to-finish (FF), finish-to-start (FS), or start-to-finish (SF). If the relationship is SS or FF, it's followed by the lead or lag of the embedded node.
Gap	The duration of the link to the selected object.

6) Updating a Schedule

Once a schedule has been created, it may be updated to track and monitor progress. In NetPoint, entering actual dates and durations (also known as “statusing”) is done just like drawing activities or leveling resources—graphically on the canvas. The entire process involves setting a data date, storing a target, statusing activities, and then actualizing statused activities. An update is complete once all activities left of the data date have been actualized.

Unique to NetPoint is the ability to calculate float attributes forensically. As the schedule is revised right of the data date, drifts, floats, and total floats continue to refresh for actualized activities left of the data date, reflecting any re-planning of the network.

Tip: To maintain a record of date-specific as-built forensic float attributes, save each update individually. Also, the user may choose to further secure the data. See [➔Setting a Password](#).

Data Dates

A data date is an object on the canvas used to represent the date to which the project has been statused. Data dates help provide a clear visual indicator of the progress of the schedule. The data date appears as a blue circle with a blue line extending the height of the canvas on that date.

By default, the data date is set to the project start. When equal to the project start, however, it cannot be displayed or modified. Once a new data date is inserted, the prior data (if any) date will become faded, and if any un-actualized objects cross the data date, a red counter will appear in the upper-right corner displaying the exact number.



A data date cannot be moved on the canvas. If needed, it may be modified using the arrow keys on the keyboard or the **Gestural Interface**. For more info, see [Pushing or Pulling Objects](#). Likewise, a data date cannot be deleted. However, the most current data date may be reset to the project start. Instructions for this, as well as inserting, hiding, or showing the data date, are provided below.

Steps

Inserting a Data Date

1. Click the Data Date icon  in the toolbar.
2. Move the cursor over the canvas and stop at the desired date, indicated by the text above the crosshairs.
3. Left-click with the mouse and release.
4. In the confirmation, choose to either capture or not capture a corresponding target. For more info, see [The Targets Manager](#).

NOTE: If working in **CPM Mode**, all activities to the left of the data date (or crossing through it) must have start and finish constraints applied before the data date is placed. Otherwise, the network (i.e. all connected activities and milestones) will jump to the data date (since this is where drift is calculated to) and be unable to be actualized. To set constraints globally for all necessary objects, see [Bulk-Editing Objects](#).

Resetting the Data Date

1. Click the **Select Tool**  icon in the toolbar.
2. Right-click on the data date and select **Reset to Project Start**.

NOTE: This will have the effect of deleting all previous data dates and de-actualizing all objects.

Hiding/Showing Data Dates

1. Click **Schedule > Modify Schedule Properties** and choose the **Displays** tab.
2. Check or uncheck the boxes next to **Current Data Date** and/or **Old Data Dates**, as desired.

NOTE: Old data dates can only be displayed if the current data date is displayed.

Statusing and Entering Percent Complete

Once the data date has been set and a target has been captured, actual dates and durations can be entered (also known as “statusing”). In NetPoint, actual dates and durations are one in the same with planned dates and planned durations. As a result, statusing individual activities is best done graphically on the canvas. However, if multiple activities haven’t yet started, they can be delayed to the data date as a group automatically. In addition, percent of work completed can be entered using the **Estimated Complete** field. Instructions for each are provided below.

Steps

Statusing Manually

1. Click the **Select Tool** icon  in the toolbar.
2. Select the desired object on the canvas.
3. If the object has been delayed or started earlier, drag it to the right or left respectively and release the mouse when finished.
4. To change the duration of an activity, hover the mouse over the start or finish node respectively until a double-arrow appears.
5. Click and drag the node earlier or later until it lands on the desired date.
6. Release the mouse. This will also update the duration.

NOTE: Dates and durations can also be statused graphically using the Gestural Interface or numerically through an object’s property window. For more info, see [↗Fine-Tuning/Gestural Controls](#).

Statusing to the Data Date

1. With the **Select Tool**  chosen in the toolbar, double-click the red counter on the data date, or click **Schedule > Manage Schedule Progress...** from the menu bar.
2. Select a desired activity in the list.
3. Press the **CTRL** key on the keyboard
4. Without releasing the **CTRL** key, select the next desired activity in the list, scrolling if necessary.
5. Repeat step 4 as many times as needed.
6. Click **Slide to DD**.
7. Click the red “X” to close the window.

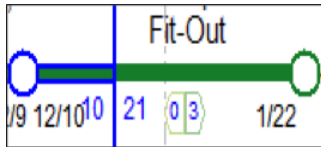
NOTE: If an attempt is made to slide an activity to the data date but that activity has any successors that have already been actualized (i.e. blocking objects), the activity will only jump to the blocking objects. A notification will appear and the blocking objects will be spotlighted in yellow until the user makes a change on the canvas, such as moving an object. The yellow box can also be cleared by choosing **Objects > Restore Faded Colors for All Objects**.

Entering Percent Complete

1. With the **Select Tool**  chosen in the toolbar, double-click an activity or right-click and choose **Properties**.
2. Enter a number next to **Estimated Complete**.
3. Click **OK**.

Actualizing Objects

Once dates and/or durations have been updated on the canvas (i.e. statused), objects must be actualized to complete the update process. Once actualized, the object becomes outlined in blue, and its dates and dependencies remain fixed unless purposely de-actualized. However, if a portion of an activity is finishing after the data date (i.e. a partially actualized or in-progress activity), the remaining duration will not be outlined in blue and may still be modified.



In addition, for activities, delays and hammocks, a new field **Calculated Complete** will appear in the property window showing the percent of duration completed (progressed duration divided by total duration). This field cannot be edited and will only display for actualized activities. To manually enter the percent of work completed, see [➔Statusing and Entering Percent Complete](#). Instructions for actualizing are provided below.

Steps

Actualizing a Single Object

1. With the **Select Tool**  chosen in the toolbar, right-click on the desired object and choose **Actualize**.
2. To de-actualize, repeat step 1 and choose **Deactualize**.

Actualizing Selected Objects

1. With the **Select Tool**  chosen in the toolbar, select the desired objects on the canvas.
2. Click **Objects > Set Properties for Selected Objects...** from the menu bar.
3. In the **Properties** section, click twice to check the box next to **Actualized**.
4. Click **OK**.
5. To deactualize, repeat steps 1-4 clicking once to uncheck the box.

NOTE: Each type of object has an individual tab and must be actualized within their respective tabs as noted above. For more info, see [➔Bulk-Editing Objects](#).

Actualizing all Pending Objects

1. With the **Select Tool**  chosen in the toolbar, double-click the red counter on the data date, or click **Schedule > Manage Schedule Progress...** from the menu bar.
2. Select the first row in the list on top and scroll to the bottom of the list.
3. Press the **SHIFT** key on the key board and select the last row in the list.
4. Click **Actualize**.
5. Click the red "X" to close the window.

NOTE: The **CTRL** key can be used to select non-adjacent activities.

Targets and Comparisons

A target is a snapshot of a schedule at a given moment in time. Creating a target captures the start date, finish date, duration, drift, float, and total float of all activities, milestones, benchmarks, delays, and hammocks. As the schedule progresses, current activity data may then be compared to earlier target data

NetPoint generates comparisons by using targets that correspond with different states of a schedule. This is done by first designating one target as the base and then another as the alternate. At any given time, only two targets can be compared to one another, but any target can be the alternate and any target can be the base. NetPoint includes both a tabular comparison and a visual comparison.

This chapter includes details for the following:

1. The Targets Manager
2. Capturing targets
3. Tabular comparisons
4. Visual target mode
5. Exporting targets/comparisons

The Targets Manager

The **Targets Manager** is where all the target functionality is located, such as capturing targets, comparing targets to one another, and exporting targets. The window itself is non-modal, meaning that you can navigate to other menus or make modifications on the canvas without having to close it.

Both lists can be customized by having their columns sorted, added and removed. In the bottom list, the canvas will be scrolled and the item spotlighted as you select different items, but this can be disabled. In addition, both lists can be individually exported to Excel. Instructions provided below.

NOTE: Sorting is not saved and will refresh when changing views or leaving and coming back.

Steps

Sorting, Adding, and Removing Columns

1. Click **Tools > Open Targets Manager** from the menu bar.
2. To sort by a certain field, click once on the column's label to sort from top to bottom, or click twice to sort from bottom to top.
3. To add or remove a column, right-click on any column.
 - a. Choose the desired field. A checkmark will appear or disappear and the list will refresh.
 - b. To close the menu, click anywhere outside of it.
4. Repeat step 3 as desired.
5. To remove a column graphically (upper list only), left-click on the desired column.
 - a. Without releasing the mouse, drag it up or down until a black "X" appears.
 - b. Release the mouse.
6. Repeat step 5 as desired.
7. To rearrange a column, left-click on the desired column
 - a. Without releasing the mouse, drag it left or right until two red arrows appear to indicate where it will be placed.
 - b. Release the mouse
8. When finished, click **Close**. Any changes will be automatically saved.

NOTE: As new columns are added, a scroll bar will be added to make more space. Any customization is tied to the computer (not to the schedule) and will be reset upon re-installation of NetPoint.

Exporting a List to Excel

1. Click **Tools > Open Targets Manager** from the menu bar.
2. Right-click on the columns at the top of either list.
3. Choose either **Export Displayed Rows & Columns** or **Export All Rows & Columns**.

Enabling/Disabling Canvas Scroll

1. Click **Tools > Open Targets Manager** from the menu bar. The **Targets Manager** will open.
2. Click the **List** icon  in the upper-right corner
3. Click **Scroll Canvas with Manager Selections**. A checkmark will appear or disappear accordingly.

NOTE: This setting will also apply to the Codes and NetRisk managers, and will be saved after the manager is closed.

Notes

Manager Views

Icon	Definition
 Targets	The Targets view is for capturing targets and comparing variances between them. For instructions, see ➔Capturing Targets .

Capturing Targets

By default, the **Targets Manager** comes pre-loaded with a permanent target for the current state of the schedule. This can not be renamed or deleted and will update in real-time with the network, regardless of whether or not the schedule has been saved. This section provides guidance on capturing and deleting targets. Instructions are provided below.

Steps

Capturing a Target

1. Click **Tools > Open Targets Manager** from the menu bar. The **Targets Manager** will open.
2. Beneath the list, click **New**. A target will be added to the list, and the timestamp and most current data date (if any) will populate.
3. Begin typing to rename the target and press **ENTER**.
4. When finished, click **Close**. Any changes will be automatically saved.

NOTE: Targets can also be captured automatically by dropping a data date and following the prompt. In these cases, the data date displayed will be the one already existing NOT the one being added.

Renaming a Target

1. With the **Targets Manager** open, select the desired target in the list.
2. Click the **Rename** button. You may also double-click the target to enter editing mode.
3. Begin typing to and press **ENTER**.
4. When finished, click **Close**. Any changes will be automatically saved.

Deleting a Target

1. With the **Targets Manager** open, select the desired target in the list.
2. Click **Delete**.
3. In the confirmation, click **Yes**.
4. When finished, click **Close**. Any changes will be automatically saved.

Notes

The Targets View

The **Targets** view contains two list boxes. The main list on top displays all captured targets; the secondary list on bottom displays a tabular comparison.

Upper List

B..	A..	Description	Data Date	Timestamp
☐	☐	Current State		
☐	☐	Baseline	06/30/2007 23:59	01/23/2015 08:57
☐	☐	Rev1	07/29/2007 23:59	01/23/2015 08:57
☐	☐	Rev2	08/25/2007 00:00	01/23/2015 08:58

Buttons: New, Rename, Delete

Icon	Description	Base Drift	Alternate...	Drift Vari...	Base Float
------	-------------	------------	--------------	---------------	------------

Column	Definition
Base	Controls whether or to use the target as the base for comparison purposes.
Alternate	Controls whether or not to use the target as the alternate for comparison purposes.
Description	A text field for describing the target.
Data Date (DD)	The date of the most recent data date, if any, on the canvas when the target was captured. NOTE: For targets that are generated by dropping a data date, the most current data date will be the one already existing NOT the one being added.
Timestamp	The date and time when the target was captured.

Tabular Comparison

Once a target has been captured in the **Targets Manager**, a base and an alternate must be designated accordingly. Any target can be the base and any target can be the alternate, but only two targets can be compared at a time. Start dates, durations, finish dates, floats, drifts, and total floats can be compared to determine the variance, if any, between the two targets. At least one target must be captured to generate the comparison.

Variances are calculated as *base minus alternate* for start date, duration, and finish date. To put it simply, a negative variance can be interpreted as a delay, and a positive variance can be interpreted as a gain. To the contrary, for drift, float, and total float, variances are calculated as *alternate minus base*. In this case, a negative variance can be interpreted as a decrease, and a positive variance can be interpreted as an increase.

NOTE: If an object is deleted from the schedule, it will no longer show up in the comparison, even if it existed when the target being compared was captured. If an object was added after the target was captured, or if info was added (e.g. a contract date) after the target was captured, it will not have any data and “None” will be displayed.

Steps

Viewing the Comparison

1. Click **Tools > Open Targets Manager** from the menu bar. The **Targets Manager** will open.
2. In the first column on the left (base), check the box next to a desired target.
3. In the second column (alternate), check the box next to a different target.

Enlarging the Comparison

1. With the comparison being viewed, click the **Enlarge**  icon above the tabular comparison. The **Expanded Report** window will open.
2. When finished, click the **Exit** icon.

NOTE: Any customization of the columns will lost upon closing and reset to that of the **Targets Manager**.

Notes

Targets View

The list on the bottom displays the tabular comparison. As you click through the list, the canvas will be scrolled and the item spotlighted. For more info on how to turn off scrolling, see [➔Enabling/Disabling Canvas Scroll](#) in *The Targets Manager* section.

Lower List

NEW Rename Delete

Comparison: "Rev1" (alternate) to "Baseline" (base)

Icon	Description	Base Dur...	Alternate...	Duration
✕ ✕	cure	28	28	
○ ○	Valve Access Platform Steel/Pi...	101	101	
○ ○	U/G Sewer for Permit	13	13	
○ ○	U/G Piping	46	46	
○ ○	TAR	61	61	
○ ○	Steel Shop Dwgs	32	32	
○ ○	Seal Drum Fab/Del [25%]	183	183	
○ ○	Seal Drum &Set Steel [26%]	12	12	
○ ○	Punch[47%]	17	17	

Highlight Restore Close

Column	Definition
Type	An image representing the type of object.
Description	The text field for describing the object.
ID	The user-entered unique identifier
Abbr. Description	A text field limited to 48 characters to be displayed on the canvas if the description is too long.
Base Start Date	The object's start date when the base target was captured.
Alternate Start Date	The object's start date when the alternate target was captured.
Start Date Variance	The base minus the alternate, in time units (i.e. days), according to calendar for each object.
Base Duration	The object's duration when the base target was captured.
Alternate Duration	The object's duration when the alternate target was captured.

Column	Definition
Duration Variance	The base minus the alternate, in time units (i.e. days), according to calendar for each object.
Base Finish Date	The object's finish date when the base target was captured.
Alternate Finish Date	The object's finish date when the alternate target was captured.
Finish Date Variance	The base minus the alternate, in time units (i.e. days), according to calendar for each object.
Base Drift	The object's drift when the base target was captured.
Alternate Drift	The object's drift when the alternate target was captured.
Drift Variance	The alternate minus the base, in time units (i.e. days), according to calendar for each object.
Base Float	The object's float when the base target was captured.
Alternate Float	The object's float when the alternate target was captured.
Float Variance	The alternate minus the base, in time units (i.e. days), according to calendar for each object.
Base Total Float	The object's total float when the base target was captured.
Alternate Total Float	The object's total float when the alternate target was captured.
Total Float Variance	The alternate minus the base, in time units (i.e. days), according to calendar for each object.
Base Contract	The object's date according to a contract as entered in its property box when the base target was captured. Only applies to milestones and benchmarks.
Alternate Contract	The object's date according to a contract as entered in its property box when the alternate target was captured. Only applies to milestones and benchmarks.

Column	Definition
Contract Variance	The alternate minus the base, in time units (i.e. days), according to calendar for each object.

NOTE: Durations and their variances for milestones and benchmarks will show up as empty. Drift, float, total float, and their variances for benchmarks will show up empty. Contract dates and their variances for activities will show up as empty.

Expanded Report

Any customization of the columns will lost upon closing and reset to that of the **Targets Manager**.

Expanded Report (Read Only)

☒ Fit Columns to Width Search:

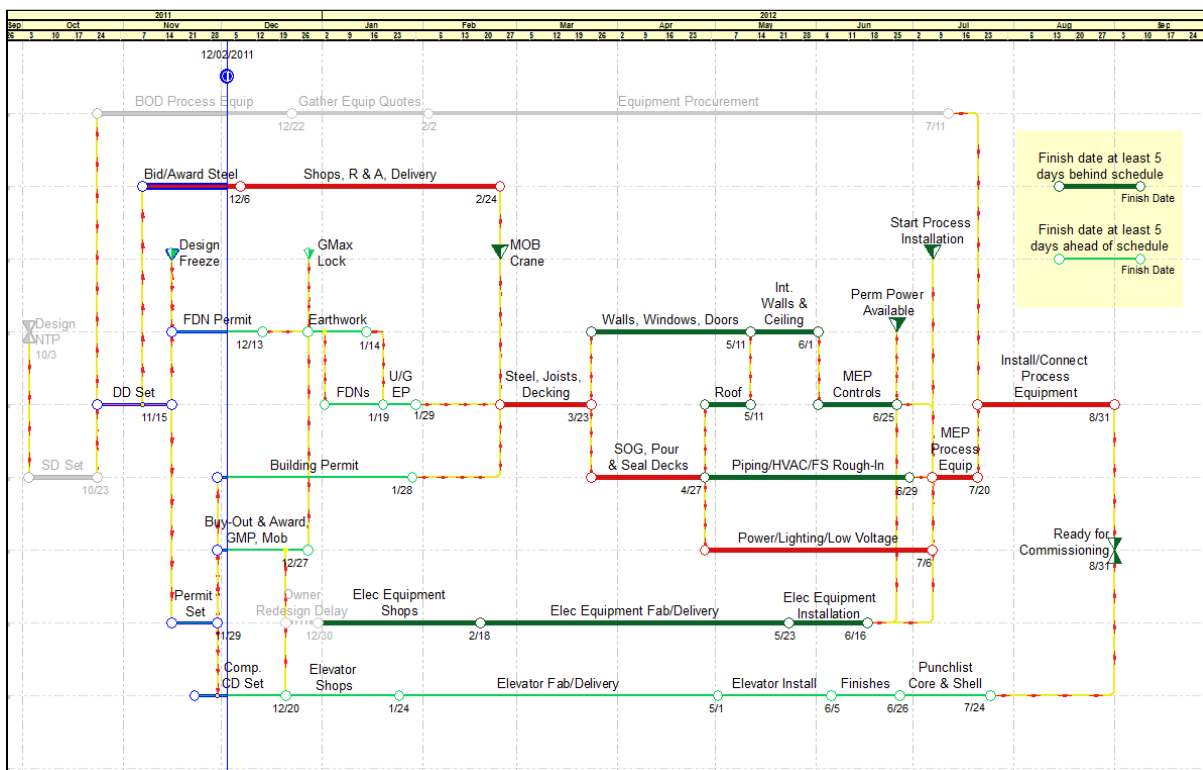
T...	Description	Base Duration	Alternate Dur...	Duration Varia...	Base Finish	Alternate Finish	Finish Variance	Base Total Float	Alternate Tota...	Total Float Var...
	Project Completion				11/20/2017	12/6/2017	-11			
	Start Design				2/2/2015	1/28/2015	3			
	Blue Tag 1				9/14/2016	9/16/2016	-2	0	-186	-186
	Red Tag 2				12/7/2016	12/2/2016	3	0	-186	-186
	Green Tag				7/18/2016	7/15/2016	1	0	-186	-186
	NTP				11/11/2015	11/9/2015	2	0	-186	-186
	Construction Permit	5	5	0	12/9/2015	12/7/2015	2	0	-186	-186
	Conceptual Design (BOD A)	40	40	0	3/27/2015	3/25/2015	2	0	-186	-186
	Preliminary Design (BOD B)	50	50	0	6/8/2015	6/4/2015	2	0	-186	-186
	Detail Design (BOD C)	70	70	0	9/15/2015	9/11/2015	2	0	-186	-186
	Solicit & Contract W/General C...	40	40	0	11/10/2015	11/6/2015	2	0	-186	-186
	Contractor Mobilizes & Takes Co...	15	15	0	12/2/2015	11/30/2015	2	0	-186	-186
	Construction Commissioning	42	45	-3	9/14/2016	9/16/2016	-2	0	-186	-186
	Qualification (IQ/OQ) Cleaning ...	54	54	0	12/7/2016	12/2/2016	3	0	-186	-186
	Develop FL URS	10	10	0	9/29/2015	9/25/2015	2	9	-155	-164
	Solicit FL Proposals	15	15	0	10/20/2015	10/16/2015	2	9	-155	-164
	FL Vendor Selection	10	10	0	11/3/2015	10/30/2015	2	9	-155	-164
	FL Supplier Agreement & PO	5	5	0	11/10/2015	11/6/2015	2	9	-155	-164
	Develop & Approve FRS & SRS f...	55	47	8	1/29/2016	1/21/2016	6	9	-155	-164
	FL Design Qualification	15	15	0	2/19/2016	2/11/2016	6	9	-155	-164
	Fabricate Filling Line	55	44	11	6/6/2016	6/1/2016	3	9	-155	-164
	FAT Filling Line	40	40	0	8/2/2016	7/28/2016	3	9	-155	-164
	Ship Filling Line	10	10	0	8/16/2016	8/11/2016	3	9	-155	-164
	Install & SAT Filling Line	25	25	0	9/21/2016	9/16/2016	3	9	-155	-164

Export to Excel...

Visual Target Mode

In **Visual Target Mode**, activities, milestones, and benchmarks that are ahead of or behind schedule can be made either smaller, larger, or lighter or darker, resulting in a visual comparison. The definition of ahead of schedule and behind schedule can be customized through a filter to include variances in start dates, durations, finish dates, floats, drifts, total floats, or any combination desired. And the criteria of the filter can be defined as “and”, in which case an activity must satisfy ALL of the criteria, or the criteria can be defined as “or”, in which case an activity can satisfy ANY of the criteria.

Once a filter has been applied and the **Visual Target Mode**  icon selected, any activity which matches the filter will be formatted on the canvas. This formatting will have NO effect on colors or sizes specified outside of **Visual Target Mode** and the two will be stored separately. If an object is modified within **Visual Target Mode** such that it no longer matches filter (e.g. its duration is crashed and its variance reaches zero), its formatting will refresh dynamically, resulting in kinetic, real-time feedback. **Visual Target Mode** requires at least one target to have been captured by the user and a base and alternate designated.



Activities whose finish dates are at least 5 days behind schedule have been made darker and bigger; activities whose finish dates are at least 5 days ahead of schedule have been made lighter and smaller; activities which match neither are faded.

NOTE: Whereas the tabular comparison can be generated between any two targets, the visual comparison can **ONLY** be generated if one of the two targets is the **Current State**. If two targets have been selected but neither is the **Current State**, the formatting on the canvas will be cleared and the **Apply** button will be disabled.

Steps

Viewing the Comparison

1. Click the **Visual Target Mode**  icon in the toolbar. The **Visual Target Mode Filters** window will open.
 - a. If a base and/or alternate have not yet been designated, the **Targets Manager** will open. In the first column on the left (base), check the box next to a desired target.
 - b. In the second column (alternate), check the box next to **Current State**.
2. If no criteria have been added yet, click the **Add** button, and customize as desired. Otherwise, skip to step 3.
3. Click the **Apply** button.
4. When finished, click the **Exit** icon in the upper right.

Clearing the Comparison

1. Click the **Visual Target Mode**  icon in the toolbar.

Adding Filter Criteria

1. Click **Tools > Modify Visual Target Filters** from the menu bar. The **Visual Target Mode Filters** window will open.
2. Click the **Add** button.
3. Customize as desired.
4. Click **Apply** to save the changes and display them on the canvas. **NOTE: The Apply button will be disabled unless Target Mode has been entered. To enter, click the Target Mode icon  in the toolbar.**
5. When finished, click the **Exit** icon in the upper right.

Modifying Filter Criteria

1. Click **Tools > Modify Visual Target Filters** from the menu bar. The **Visual Target Mode Filters** window will open.
2. Select the desired cell in the list.
3. Customize as desired.
4. Click **Apply** to save the changes and display them on the canvas. **NOTE: The Apply button will be disabled unless Target Mode has been entered. To enter, click the Target Mode icon  in the toolbar.**
5. When finished, click the **Exit** icon in the upper right.

Deleting Filter Criteria

1. Click **Tools > Modify Visual Target Filters** from the menu bar. The **Visual Target Mode Filters** window will open.
2. Select the desired row in the list and click the **Remove** button.
3. Click **Apply** to save the changes and display them on the canvas. **NOTE: The Apply button will be disabled unless Target Mode has been entered. To enter, click the Target Mode icon  in the toolbar.**
4. When finished, click the **Exit** icon in the upper right.

Notes

Bar Shadows and Selecting Activities

In **Visual Target Mode**, selecting an activity, milestone, or benchmark on the canvas will trigger a light grey shadow to appear beneath its bar, indicating its position and dates from the base target. In addition, dates, durations, etc. will no longer reset or enlarge.

Visual Target Mode Filters

The **Visual Target Mode Filters** window can be left open and worked with side-by-side with the **Targets Manager**, if desired.

Visual Target Mode Filters

Base Target: Baseline
Alternate Target: Current State

Filter Criteria Match: or

Variance in	Operator	Thr...	Unit	Status	Size	Color
Finish Date	at least	1	Day(s)	Behind	Larger	Dark Green

Add Remove

Non-matching Objects

☒ Reset Sizes to Normal ☒ Reset Colors to Green ☒ Fade

Matching Objects

☒ Display Shadows

All Objects

☒ Display Variances

Apply Cancel

Field	Definition
Base Target	The description of the target currently selected as the base.
Alternate Target	The description of the target currently selected as the alternate.

Field	Definition
Match	Controls how to aggregate multiple criteria. The criteria can either be considered as “and”, in which case an activity must satisfy ALL of the criteria, or the criteria can be considered as “or”, in which case an activity can satisfy ANY of the criteria.

Filter Criteria

Alternate Target: Current State

Filter Criteria Match:

Variance in	Operator	Thr...	Unit	Status	Size	Color
Finish Date	at least	1	Day(s)	Behind	Larger	Dark Green

Column	Definition
Variance in	A field for determining what the given criteria is based on. Options include start date, duration, finish date, float, drift, and total float.
Operator	A field for determining how the variance relates to the threshold. Options include at least, up to, and equals. For greater than, choose at least ; for less than, choose up to .
Threshold	A field for entering the degree of variance.
Unit	A field for choosing the threshold's unit. Options include in days (or time unit) or percent.
Status	A field for choosing between positive or negative variance. Options include ahead of schedule, behind schedule, and on schedule. NOTE: When choosing On Schedule, the Operator will automatically be taken to be Equals and any value in the Threshold will be ignored.
Size	A field for formatting the size of any matching activities. Options include no change, small, normal, and large.
Color	A field for formatting the color of any matching activities. Options include no change, light green, green, and dark green.

Non-matching Objects

This formatting will have NO effect on colors or sizes specified outside of **Visual Target Mode** and the two will be stored separately.

Non-matching Objects

☒ Reset Sizes to Normal ☒ Reset Colors to Green ☒ Fade

Field	Definition
Reset Sizes to Normal	Controls whether or not all objects not captured by the filter will be reset to their default normal size.
Reset Colors to Green	Controls whether or not all objects not captured by the filter will be reset to their default green color.
Fade	Controls whether or not all objects not captured by the filter will be faded on the canvas.

Matching Objects

Matching Objects

☒ Display Shadows ▼

Field	Definition
Display Shadows	Displays the shadows for all objects matching the filter (activities, milestones, or benchmarks). When selected, selecting an object will no longer trigger the shadow.
Color	The color for the shadow. Also used in the case where the checkbox is unchecked.

All Objects

All Objects

☒ Display Variances

Apply Cancel

Field	Definition
Display Variances	When selected, variances will be displayed for all objects, but only if the parent field is displayed in the object's property window. By default, variances will show up beneath the parent field for start date, finish date, and all float values, but to the right for the duration.

Field	Definition
	Variances cannot be repositioned separately from their parents but are controlled by them. If the activity is sufficiently short, variances will be displayed to the right for all fields and repositioned with their parents according to NetPoint's clashing algorithm (this does not occur if the positions have already been customized by the user). The user is reminded that the float block and duration cannot be displayed at the same time in these cases. If both are selected, only the float block will show up until the activity is stretched back out. Variance values are always calculated according to the object's calendar.

Exporting Targets/Comparisons

Comparisons can be exported as part of a linked Excel report or as a single standalone spreadsheet. When exporting the linked Excel report, one sheet exports for each target, using the current state as the alternate. When exporting the standalone spreadsheet, any target can be the base and any target can be the alternate (as chosen in the manager). Instructions for both are provided below.

Steps

Exporting the Linked Excel Report

1. Click **Tools > Open Targets Manager** from the menu bar. The **Targets Manager** will open.
2. Click the **List** icon  in the upper-right corner of the **Targets Manager**.
3. Choose **Export Linked Excel File (Default Report)**. Excel will open.

NOTE: For a detailed explanation of this report, see [Target Data](#) in the section *Exporting Linked Reports*.

Exporting a Standalone Spreadsheet

1. Click **Tools > Open Targets Manager** from the menu bar. The **Targets Manager** will open.
2. Click the **Enlarge**  icon above the tabular comparison. The **Expanded Comparison** window will open.
3. Click the **Export to Excel** button, or right-click on the columns at the top.
4. Choose either **Export Displayed Rows & Columns** or **Export All Rows & Columns**.

Displaying Percents Complete and In-Progress Durations

If desired, the **Estimated Complete** or **Calculated Complete** can be displayed on the canvas in place of the duration; however, only one can be displayed at a time. Turning any one of them on will automatically turn the others off.

When an activity is in-progress, the planned duration will be split into the progressed duration and remaining duration. By default, these will show up to the left and right of the data date line but can be customized manually. Each field can be turned off individually; however, the planned duration will not be able to be displayed while in progress.

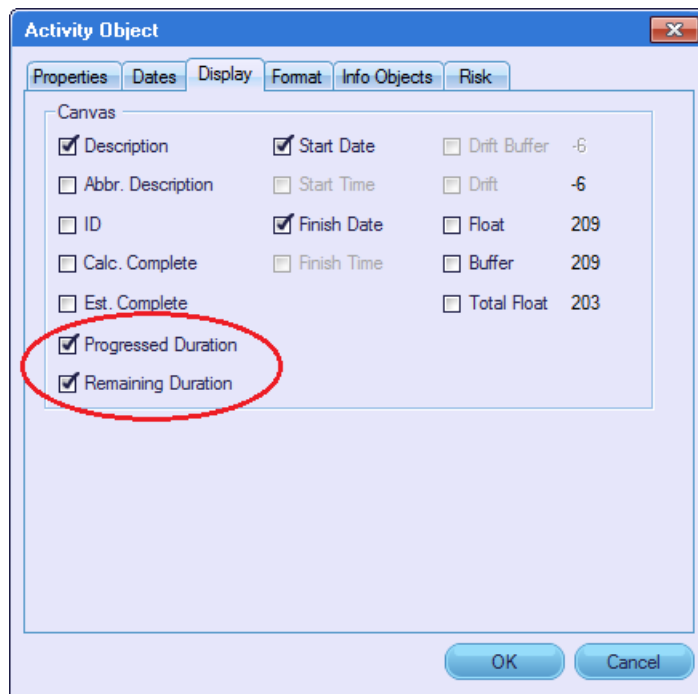
Likewise, the **Estimated Complete** and **Calculated Complete** will be split into progressed completes and remaining completes. However, these can only be turned on and off as a pair.

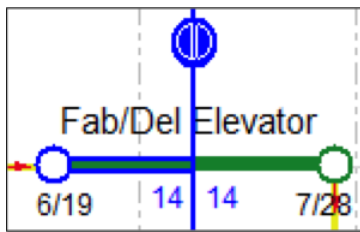
Steps

1. With the **Select Tool**  chosen in the toolbar, double-click an activity or right-click and choose **Properties**.
2. Click the **Displays** tab.
3. Check the box next to the desired field.
4. Click **OK**.

NOTE: These fields can also be turned on or off for multiple objects at a time. For more info, see [Bulk-Editing Objects](#).

Notes





Forensic Floats for Actualized Objects

Since the Graphical Path Method (GPM) doesn't depend on early and late-dates to calculate float, GPM float attributes continue to refresh for actualized activities left of the data date. This is called forensic float, and it can be particularly useful when analyzing a schedule for delays after the project has started or finished.

By default, forensic float is not turned on, meaning when a data date is set and activities are actualized, float attributes will disappear and be disabled in the property boxes. Instructions for enabling forensic floats are provided below.

NOTE: When a data date is dropped, the drift of an un-actualized or partially-actualized activity will go negative until it's actualized, since drift is measured from the data date. In addition, for in-progress activities, float and drift calculations ignore links to and from embeds on the left side of the data date.

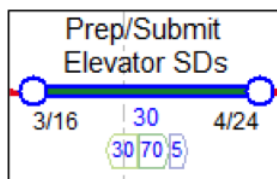
Steps

1. Click **Objects >** from the menu bar and choose **Display Float and Drifts for Actualized Objects.**

NOTE: If float attributes still are not showing up, make sure they're turned on for the desired objects. This can be done one-by-one in their property windows or one-at-a-time. For more info, see [Bulk-Editing Objects](#).

Notes

Drift, float, buffer, and total float are displayed in that order below the activity bar on the same line. They may be distinguished from one another by the shape of their bounding boxes: drift points to the left, or project start; float and buffer point to the right, or project completion; and total float points in both directions.



In this example, drift is equal to 30, float is equal to 70, and buffer is equal to 5. Either buffer or total float may be displayed, but not both at the same time.

7) Inputs and Outputs

NetPoint schedules can be imported from and exported to Primavera P6, Deltek Open Plan, and Microsoft Project, and exported to Synchro Professional. Activities and resources can also be defined in Excel and then imported into NetPoint, and NetPoint can export reports to Excel. Instructions for all, in addition to printing, presenting, and capturing snapshots, are provided in this chapter.

Preparing to Export from NetPoint (GPM)

When exporting to P6, Open Plan, or MS Project, any activities or milestones placed on planned-dates (between their early-start and late-start) will be pulled back to their traditional CPM early-start dates. In order to maintain a comparable network, constraints must be applied to any activities and milestones starting on planned-dates. This can be accomplished using a filters.

Steps

Constraining Activities and Milestones

1. Click **Edit > Search and Filter Objects...** from the menu bar.
2. Click the down arrow under **Objects to Include** and add **Start Milestones** and **Finish Milestones** by checking the boxes next to their names. Make sure to click outside of the list to close it.
3. In the **Properties** section, click the down arrow next to **Drift Buffer** and choose **>**.
4. In the number control to the right of **Drift Buffer**, enter **0**.
5. At the bottom, click **Add to List**.
6. In the **List of Filtered Objects**, click **Select All**.
7. To close both windows, click the red **X** in the upper right corner of the **List of Filtered Objects** window.
8. To begin bulk-editing, click **Objects > Set Properties for Selected Objects** from the menu bar.
9. Under the **Constraints** section, click twice in the box next to **Start no earlier than** until a check appears, and then click once in the box next to **Use Planned Dates** until a check appears.
10. Click the **Milestones** tab and repeat step 9.
11. When finished, click **OK**. Changes from across all tabs will be applied at once.

Notes

For more info, see [➔Constraining Activities](#).

Primavera Project Management

NetPoint features importing and exporting with Primavera P6. This is done by first exporting an .XER file from one application and then importing it into the other. NetPoint can write XER files versions 6.2.1, 7.0, 8.1, 8.2, and 8.3 and read XER files versions 6 and onward.

Primary information such as activities, logic, grid position, and assignments are all transferred, while some formatting characteristics such as link geometries, shades, page size, and canvas color are not. For a full list of supported fields, see the notes at the end of the following sections.

If you're importing/exporting "round-trip", pay special attention to the notes in **bold**. A "round-trip" has occurred if the same schedule ends up back in the application it originated in, e.g. *NetPoint -> P6 -> NetPoint* or *P6 -> NetPoint -> P6*.

This section details instructions for the following:

1. Importing from P6
2. Exporting to P6

Importing from P6

To import a schedule from P6 into NetPoint, you must first go to P6 and export it as an XER file. While in P6, a couple of important steps should be taken to prevent any potential conflicts. For example, in NetPoint, the relationship calendar is based on the predecessor activity. This should have been set in P6 from the onset, but if not, it should be reviewed and updated beforehand (see “in P6” steps below). In addition, the project should be “scheduled” to avoid any potential terminations and “summarized” to ensure that the project start and end dates are correct. And finally, all activities should be of the “task-dependent” type.

Upon importing, a new NetPoint file will be created and the data from the XER file brought in. At this time, you cannot update or import into an existing NetPoint schedule. If the schedule was created in P6, or if it was created in NetPoint using a version prior to 4.1, release 25 Feb 2013, the user should run the automated layout engine to optimize the network. For more information, see [Running the Engine](#). Instructions for importing are provided below.

If you’re importing/exporting “round-trip”, pay special attention to the notes in **bold**. A “round-trip” has occurred if the same schedule ends up back in the application it originated in, e.g. *NetPoint -> P6 -> NetPoint* or *P6 -> NetPoint -> P6*.

NOTE: Any start dates later than noon will be adjusted to the beginning of the next day; any finish dates earlier than noon will be adjusted to the end of the previous day. This will result in dates that are different from those in P6. This can be avoided by making sure all activities start before noon and end after noon.

Steps

In P6

1. While viewing the desired project, click **Tools > Schedule** from the menu bar. If this project has already been imported before, skip to step 1.4.
 - a. Click the **Options** button.
 - b. Under **Calendar for Scheduling Relationship Lag**, make sure **Predecessor Activity Calendar** is selected*.
 - c. Click **Close**.
 - d. Click the **Schedule** button.
2. Click **Tools > Summarize > Open Projects** from the menu bar.
3. Click **OK**.
4. Click **File > Export** from the menu bar. With **Primavera PM / MM – (XER)** selected, click **Next**.
5. With **Project** selected, click **Next**.
6. Check the box next to the desired projects and click **Next**.
7. Name the file and click **Finish**.

***NOTE: Changing the calendar for the relationship lag can potentially affect activity dates. If the calendar is changed, please review the schedule afterward.**

In NetPoint

1. Click **File > Import/Export** from the menu bar and choose **Import from Primavera (XER)**.
2. Browse for the desired XER file and click **Open**.

- If the file contained more than one project, choose the desired one from the dropdown above the list. Otherwise, skip to step 4.
- To exclude an item from being imported, uncheck the box under the **Import** column next to its name in the list. When finished, click **Import**. A report will open showing a summary of any incompatibilities that were found
- When ready, click **Finalize Import**. The schedule will then open in NetPoint.

Notes

P6 Import – Configure Window

Columns can be sorted, added, or removed by left-clicking or right-clicking accordingly. At this time, only one project can be imported at a time.

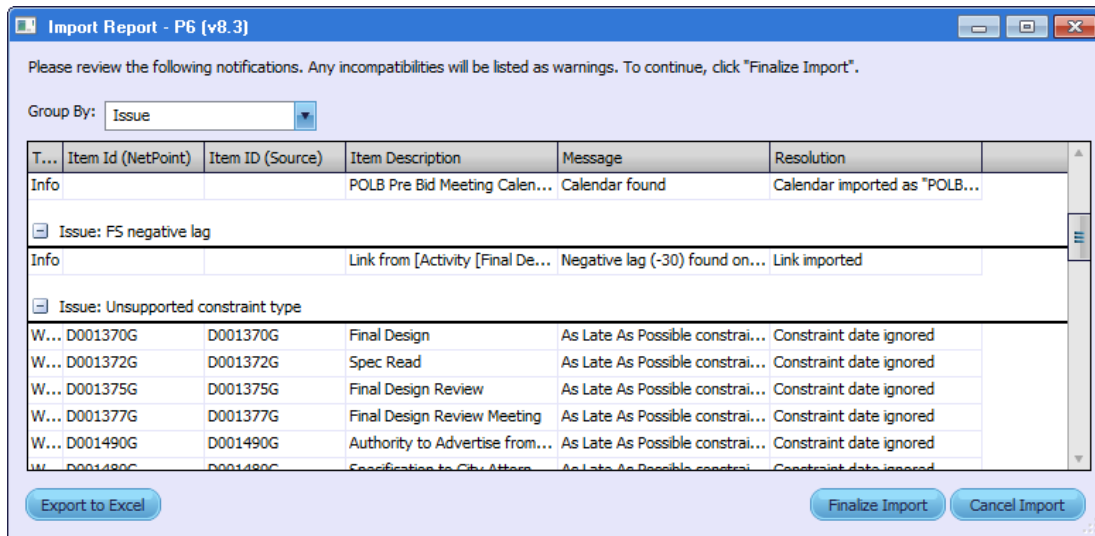
Import	Activity ID	Activity Type	Activity Name	Start	Finish	Original Dur...
☒	D001300A	Level of Effort	Design Summary	10/1/2009	8/11/2011	680
☒	D001345A	Task Dependent	100% Design Review	8/12/2010	8/26/2010	15
☒	D001325A	Task Dependent	50% Design Review	4/11/2010	4/30/2010	20
☒	D001370A	Task Dependent	Final Design	9/3/2010	11/2/2010	61
☒	D001315A	Task Dependent	15% Design Review	11/13/2009	12/11/2009	29
☒	D001310A	Task Dependent	15% Design (Conceptual Design)	10/1/2009	11/12/2009	43
☒	D001317A	Finish Milestone	15% Design Review Meeting	11/12/2009	11/12/2009	1
☒	D001320A	Task Dependent	50% Design (Preliminary Design)	1/16/2010	4/5/2010	80
☒	D001327A	Finish Milestone	50% Design Review Meeting	5/19/2010	5/19/2010	1
☒	D001340A	Task Dependent	100% Design	5/29/2010	8/12/2010	76
☒	D001373A	Task Dependent	Spec Read	11/18/2010	11/18/2010	1

Column	Definition
Import	Controls whether or not the item will be imported.
Activity ID	The identifier used in P6.
Activity Type	The type of activity as set in P6.
Activity Name	The description of the activity in P6.
Start	The date and/or time the activity begins in P6.
Finish	The date and/or time the activity finishes in P6.
Original Duration	The amount of time it takes the activity to complete in P6.
Total Float	The total float according to P6.

Column	Definition
Calendar	The calendar assigned in P6.
WBS	The WBS node assigned in P6.
WBS Path	The full WBS path assigned in P6.
Assigned Codes	The code assignments from P6.

Import Report – P6

Summarizes any discrepancies found during import. The report can be grouped by any field at the top. Columns can be sorted, added, or removed by left-clicking or right-clicking accordingly.



Column	Definition
Type	Type of notification. Info notifications include confirmations and minor incompatibilities, such as when items are converted to be compatible with NetPoint. Warning notifications include major incompatibilities, such as items that are changed, which affect logic, or items not being imported altogether.
Issue	A general category for the notification, such as calendars, resources, data dates, etc.
Item Id (NetPoint)	The identifier as found in NetPoint.
Item ID (Source)	The identifier as found in P6.

Column	Definition
Item Description	The name or description of the item in question.
Message	The notification itself.
Resolution	How (if possible) the issue has been resolved.

Supported Fields

P6 Objects and Fields

The following P6 objects and fields are brought into NetPoint. If a field is not listed, then it does NOT transfer. Some examples of fields that do not transfer are identified where relevant.

This field in P6	Becomes this field in NetPoint	Notes
Task Dependent	Activity	
Start/Finish Milestone	Start/Finish Milestone	
Level of Effort	Hammock	Only LOE's which are connected to their fragnet by start-to-start AND finish-to-finish relationships will be imported as hammocks. If an object is found under multiple LOE's, it will only be brought in beneath the first one. Any LOE's left without any fragnets will not be brought into NetPoint.
Lag	Embedded node or link lag	If positive, the lag will be captured in the embed unless it is larger than the duration of the activity, in which case it will be captured in the link lag. If negative, it will be captured in the link lag for FS relationships or otherwise ignored.
Documents	Info Objects	Documents associated with activities, milestones, or benchmarks will be assigned to the appropriate object. Project documents will become info objects on the canvas.
Activity Name	Description	
Activity ID	ID	

This field in P6	Becomes this field in NetPoint	Notes
Start	Start Date	Start dates later than noon are adjusted to the beginning of the next day.
Finish	Finish Date	Finish dates earlier than noon are adjusted to the end of the previous day.
Actual Start	Start Date	
Actual Finish	Finish Date	
Original Duration	Duration	
Remaining Duration	Remaining Duration	
Actual Duration	Duration	
Free Float	Buffer	
Total Float	Total Float	
Finish On/ Mandatory Finish (milestones)	Finish benchmark	On activities, ignored.
Finish On Or After (finish milestones)	Finish-no-earlier than (FNE)	On activities, ignored.
Finish On Or Before	Finish-no-later than (FNL)	
Start On/ Mandatory Start (milestones)	Start benchmark	On activities, ignored.

This field in P6	Becomes this field in NetPoint	Notes
Start On Or After	Start-no-earlier than (SNE)	
Start On Or Before (start milestones)	Start-no-later than (SNL)	On activities, ignored.
Relationships	Relationships/ Links	P6 allows FS relationships for start and finish milestones; however, in NetPoint a link with a start milestone as predecessor can only be SS or SF; a link with a finish milestone as successor can only be FF or SF. After a round trip, such links on milestones using FS links will be changed to match NetPoint requirements
Calendars	Calendars	Any calendars at the project level will transfer, regardless of whether or not they've been assigned; any calendars at the global level will transfer, as long as they have been assigned to at least one activity; and any global calendars from which a project calendar was created will transfer. Any calendar exceptions are NOT imported; therefore, any activities assigned a calendar with an exception should be reviewed after importing. NOTE: If you plan on going from P6 -> NetPoint -> P6 and updating an existing project, use the same number of working hours/day for all calendars in the schedule. When exporting out of NetPoint, this can only be set for all calendars globally. In addition, each time a schedule is imported into NetPoint, two calendars will be added for "United States" and "Calendar Days". Review the calendars and delete any duplicates.
Codes	Codes	This includes all codes from the project level plus any codes from the global level that have had at least one of their values assigned. Unassigned values of global codes will not be brought into NetPoint. NOTE: If you plan on going from P6 -> NetPoint -> P6 and updating an existing project, don't use global codes if you plan on making changes to them in NetPoint, since NetPoint cannot export them as global. If exported out of NetPoint, you will have duplicate assignments: one from the global level version of the code and one from the project level version of the code.
Activity Code Name	Name	

This field in P6	Becomes this field in NetPoint	Notes
Description	Value	
Code Value	NA	Transferred into NetPoint but cannot be edited.
Resources	Resources	Any resource that has been assigned, regardless of which project it is under, will be brought into NetPoint; however, any resources that haven't been assigned, even if within the same project, will not transfer. In addition, NetPoint only supports resources on activities. This means any activities with 0 duration (delays), milestones, or levels of effort (hammocks) will not have their resources transfer. NOTE: If you plan on going from P6 -> NetPoint -> P6 and updating an existing project, make sure all your resources are from the same project. Once imported into NetPoint, resources are given a unique ID. When updating back into P6, you will then have duplicate resources and assignments.
Resource Name	Name	The "Resource ID" field from P6 does not transfer.
Price/Unit	Cost/Unit	P6 stores the price/unit using hours (e.g. \$0.125/hour). When the XER file is exported from P6, this value is rounded to 2 decimal places (e.g. \$0.13), which is then converted to cost/unit, resulting in a different overall value. After importing from P6, it is recommended to review the cost/unit of each resource to correct for any significant differences due to rounding.
Resource Type	Category	
Budgeted Units	Rate/Dur or Rate/Day	An activity with duration type of "fixed units" will have its resources end up as rate/duration. An activity with duration type "fixed units/time" will have its resources end up as rate/day. When the XER file is exported from P6, this value is rounded to 1 decimal place (e.g. 0.15/day would become 0.2/day), resulting in a different overall value. After importing from P6, it is recommended to review the assignments of each resource to correct for any significant differences due to rounding.

NetPoint Fields

If the schedule was exported from NetPoint already, everything will be brought in EXCEPT for the following:

Characteristic	Result
Layouts	Gone
Delays	Become activities
Shades	Gone
Notepads	Gone
Link Geometries	Become default
Embed Behaviors	Become constant
Custom Positioning	Becomes default
Resource Limits	Gone
Resource Formatting	Become default (color, histogram patterns, color intensities)
Page Size	Becomes letter
Page Tiling	Becomes 1×1
Grid Spacing	Becomes default
Canvas Color	Becomes default
Calendar Strips	Become default
Calendar Sightlines	Become default
Text Size	Becomes default
Text Objects	Gone

Exporting to P6

When exporting, keep in mind that there are many differences between NetPoint and P6. Items like resources, calendars, codes, and constraints, for example, are all handled slightly differently. In addition, some characteristics are not maintained during round-trip import and export. For a full list of supported fields, see the notes at the end of this section.

Before exporting, if any activities or milestones have been scheduled on planned-dates, make sure to follow the instructions in [➔Preparing to Export from NetPoint \(GPM\)](#). Once exported, the XER file must be imported into P6 to complete the process.

NOTE: Before attempting to update an existing project in P6, make sure to see the notes in bold in the supported fields table below.

Steps

In NetPoint

1. Click **File > Import/Export** from the menu bar and choose **Export to Primavera (XER)**.
2. Fill in the properties as desired.
3. When finished, click **Next**.
4. If satisfactory, click **Export**. Otherwise, click **Configure** and return to step 2.
5. Enter a name and click **Save**.

In P6

1. Click **File > Import** from the menu bar.
2. In the **Import Format** screen, with **Import PM / MM (XER)** selected, click **Next**.
3. In the **Import Type** screen, with **Project** selected, click **Next**.
4. In the **File Name** screen, click **...** to the right of the file path, browse for the desired XER file, and click **Open**.
5. In the **File Name** screen click **Next**.
6. In the **Import Project Options** screen, if the Project ID that was entered on export from NetPoint already exists in the P6 database, the check box under **Match** will be selected. Even though match is checked, you'll have the option under **Import Action** to update the existing project, create a new one, add to, ignore, or replace. Once chosen, click **Next**.*
7. In the **Update Project Options** screen, click **Next**.
8. In the **Finish** screen, click **Finish**.
9. In the confirmation, click **OK**.
10. With the schedule open, click **Tools > Schedule** from the menu bar (or press **F9** on the keyboard).
11. Click **Schedule**. All activities will move to their early dates.

***NOTE: Before attempting to update an existing project, see the notes in bold in the supported fields table below.**

Notes

P6 Export – Configure Window

P6 Export - Configure

P6 Project ID: XER Version:

Calendars
☒ Create Calendars as Global

Project Codes
☒ Export as P6 Activity Codes ?

Project Resources
☒ Export
☐ Convert Mixed Assignments to: ?
☒ Convert All Assignments to:

Working Hours/Day
☒ 8am To 12pm and 1pm to 5pm (8 hours)
☐ 7am To 5pm (10 hours)
☐ Custom Work Periods

Start time: to End time:

Included Periods:

☒ Always open first

Field	Definition
P6 Project ID	The ID once exported to P6. NOTE: If you want to update an existing project in the P6 database, make sure you enter the same P6 Project ID. Before attempting to update, see the notes in bold in the supported fields table below.
XER Version	Controls what version of XER to export.
Calendars	Controls whether or not calendars will export as global. NOTE: NetPoint global calendars will only export if they have at least one assignment.
Project Codes	Controls whether or not codes will be exported. Any codes which allow more than one of their values to be assigned to the same activity will not transfer. To review, click Tools > Open Codes Manager and review their behavior. NOTE: If you want to update an

Field	Definition
	existing project in the P6 database, checking this box will result in duplicate codes and values. For more info, see the notes in bold in the supported fields table below.
Project Resources	Controls whether or not resources will be exported and how. NOTE: If you want to update an existing project in the P6 database, checking this box will result in duplicate resources and assignments. For more info, see the notes in bold in the supported fields table below.
Working Hours/Day	Specifies the working hours for all calendars once in P6. NOTE: In a day-based schedule, NetPoint does not distinguish between which hours are working and which hours are not. These options allow you to configure how P6 will handle resources and other properties, before it's even opened in P6.

Supported Fields

NetPoint Objects and Fields

The following NetPoint objects and fields are brought into P6. If a field is not listed, then it does NOT transfer. Some examples of fields that do not transfer are identified where relevant.

This field in NetPoint	Becomes this field in P6	Notes
Activity, Delay	Task Dependent	
Milestones and benchmarks (both start and finish)	Start/Finish Milestone	Benchmarks in NetPoint become milestones with Mandatory Start or Mandatory Finish constraints in P6 accordingly.
Hammock	Level of Effort	Any unconnected milestones or benchmarks will be connected to the LOE with a start-to-start or finish-to-finish relationship accordingly.
Embed, link lag	Lag	If both exist in the same relationship, they will be combined. For example, an SS relationship with a link lag of 5 days and an embed lag of 5 days will transfer as an SS with 10 day lag in P6.
Info Object	Documents	Info objects that have been attached to activities, milestones, or benchmarks will be assigned to the appropriate object. Info

This field in NetPoint	Becomes this field in P6	Notes
		objects drawn on the canvas will also be exported and associated with the exported project
Description	Activity Name	Any description over 120 characters will be truncated.
ID	Activity ID	If empty, IDs are created in P6 with prefix “~”.
Start Date	Start, Actual Start	
Finish Date	Finish, Actual Finish	
Duration	Original Duration, Actual Duration	
Remaining Duration	Remaining Duration	
No-earlier-than Constraint	Start On Or After (for activities and start milestones), Finish On Or After (for finish milestones)	
No-later-than Constraint	Finish On Or Before (activities and finish milestones), Start On Or Before (start milestones)	
Buffer	Free Float	
Total Float	Total Float	
Grid Position	—	This field is exported for round-trip purposes only and isn’t used in P6.

This field in NetPoint	Becomes this field in P6	Notes
Color	—	This field is exported for round-trip purposes only and isn't used in P6.
Shape (milestones/benchmarks)	—	This field is exported for round-trip purposes only and isn't used in P6.
Text Formatting	—	This field is exported for round-trip purposes only and isn't used in P6. Includes font, color, alignment, style, size, and line breaks.
Text Display	—	This field is exported for round-trip purposes only and isn't used in P6. Includes whether text fields, date fields, float fields, etc. are turned on or off.
Calendars	Calendars	If Create Calendars as Global is checked but a calendar created in P6 with the same name already exists in the global record, it will NOT be overwritten or replaced. Instead, another calendar will be created with its own properties and assigned to the corresponding activities. The latest one will always show up last behind any other identically named calendars. NetPoint global calendars will only export if they have at least one assignment. Recurring holidays ("fixed-date" holidays in NetPoint with a repeat factor of 1 year) will transfer individually, with each case being a separate holiday. NOTE: If you plan on importing back into NetPoint, avoid using the default "United States" or "Calendar Days" calendars. After a round trip (NetPoint -> P6 -> NetPoint), a new set of default calendars are created, resulting in duplicates.
Codes	Codes	Codes in NetPoint transfer as activity codes in P6. NetPoint cannot export codes as global, meaning all codes will end up at the project level. NOTE: If the schedule was exported from P6 already, and some of your codes were global, either create a new project instead of updating the existing one OR don't export them at all. Otherwise, you will have duplicate assignments.
Name	Activity Code Name	The "Code Value" field in P6 will be filled in with a unique ID generated by NetPoint.
Value	Description	In NetPoint, multiple values per code may be assigned to the same activity. In P6, however, only one value per code may be

This field in NetPoint	Becomes this field in P6	Notes
		assigned to the same activity. Project codes are not exported if they are set to have multiple values in NetPoint (even if only one value is actually defined for the code or assigned to an activity).
Resources	Resources	Limits, Histogram Pattern, Color, and Color Intensity are not exported at this time. NOTE: If the schedule was exported from P6 already, and some of your resources were assigned at the global level in P6 and others at the project level in P6, either create a new project instead of updating the existing one OR don't export them at all. Otherwise, you will have duplicate resources and assignments. In addition, the primary resource setting is not maintained, which will need to be reset manually after the round trip.
Name	Resource ID, Resource Name	The "Resource ID" field is filled in with the name of the resource as well.
Cost/Unit	Price/Unit	P6 stores the cost per hour. For example, \$1 dollar per day will be transferred to P6 as \$0.125/hour.
Category	Resource Type	Any category other than material, equipment, and labor is converted to material, equipment, or labor.
Rate/Dur, Rate/Day	Budgeted Units & Remaining Units	In NetPoint, an activity may have some resources assigned to it "per day" and other resources assigned to it "per duration". This is called a <i>mixed assignment</i> . In P6, the resource assignments for a given activity must be all of the same type. If an activity has its resources exported per day, that activity will have its duration type set as "fixed units/day" in P6. If an activity has its resources exported per duration, that activity will have its duration type set as "fixed units" in P6. For more on distribution types, see Assignment Options . If the activity is in progress or actualized, resource data will also be mapped to the "Actual Units".

Deltek Open Plan

NetPoint features round-trip importing and exporting with Deltek Open Plan. This works by asking the user for a username and password and then establishing a connection with the Open Plan database located on the same machine as NetPoint. Schedules can be exported from NetPoint, updated externally, and imported back in.

Primary information such as activities, logic, grid position, and assignments are all transferred, while some formatting characteristics such as link geometries, shades, page size, and canvas color are not. For a full list of supported fields, see the notes at the end of the following sections.

This section details instructions for the following:

1. Importing from Open Plan
2. Exporting to Open Plan

Importing from Open Plan

To import a schedule, the user must have a username and password to access the Open Plan database, and the project must be stored on the same machine as NetPoint.

After importing, the plan will vary slightly depending on whether or not the schedule was originally created in NetPoint or Open Plan. If the schedule was originally created in NetPoint, subsequently exported, and is now being imported back, the look and layout of the schedule will be the same. If the schedule was created in Open Plan, or if it was created in NetPoint using a version prior to 4.1, release 25 Feb 2013, NetPoint will layout the network avoiding clashes, but the user should run the automated layout engine. For more information, see [Running The Engine](#). If further manual repositioning of data elements is desired, see [Repositioning Fields \(GEM\)](#) ➔ .

NOTE: The calendar for a relationship (link) cannot be defined as that of the predecessor. As a result, dates and float values in NetPoint will not align with those in Open Plan in the case where the calendar chosen for relationships in Open Plan has different holidays over the course of a link than those of its predecessor's calendar.

Steps

In Open Plan

1. In Open Plan, click **Project > Time Analysis** from the menu bar (or press **F9** on the keyboard), enter a **Time Now**, and click **OK**.
2. Close the project to ensure that NetPoint can access it properly.

In NetPoint

1. In NetPoint, click **File > Import/Export...** from the menu bar and choose **Import From Open Plan**.
2. If Open Plan is already open, skip to step 3. If Open Plan isn't open, you'll be prompted to enter your Open Plan credentials and click **OK**.
3. Browse for the desired project, select it in the list, and click **Import**. A report will open showing a summary of any incompatibilities that were found.
4. When ready, click **Finalize Import**. The schedule will then open in NetPoint.

Notes

Import Report – Open Plan

Import Report - Open Plan

Please review the following notifications. Any incompatibilities will be listed as warnings. To continue, click "Finalize Import".

Group By:

Type	Item Id (NetPo...	Item ID (Source)	Item Description	Message	Resolution
☐ Issue: Calendar definition					
Info			< Default >	Calendar found	Imported as "< Default >"
Info			Std_Holidays	Calendar found	Imported as "Std_Holidays"
Info			Std_Holidays.7Day	Calendar found	Imported as "Std_Holidays.7Day"
☐ Issue: Zero duration activity					
Info	B0001S	B0001S	Plants & Placements	Zero-duration activity found	Activity converted to delay activity
Info	B0001X	B0001X	Final Completion	Zero-duration activity found	Activity converted to delay activity
Info	B0002A	B0002A	Proposal #93L002	Zero-duration activity found	Activity converted to delay activity

Export to Excel Finalize Import Cancel Import

Column	Definition
Import	Controls whether or not the item will be imported.
Activity ID	The identifier used in Open Plan.
Activity Type	The type of activity as set in Open Plan.
Activity Name	The description of the activity in Open Plan.
Start	The date and/or time the activity begins in Open Plan.
Finish	The date and/or time the activity finishes in Open Plan.
Original Duration	The amount of time it takes the activity to complete in Open Plan.

Supported Fields

Open Plan Objects and Fields

The following Open Plan objects and fields are brought into NetPoint. If a field is not listed, then it does NOT transfer. Some examples of fields that do not transfer are identified where relevant

This field in Open Plan	Becomes this field in NetPoint	Notes
Activity	Activity	
Start/Finish Milestone	Start/Finish Milestone	For benchmarks, see Constraints below.
Hammock	Hammock	
Lag	Embed or link lag	If positive, the lag will be captured in the embed unless it is larger than the duration of the activity, in which case it will be captured in the link lag. If negative, it will be captured in the link lag for FS relationships, otherwise ignored.
Description	Description	
ID	ID	
Start	Start Date	
Finish	Finish Date	
Duration	Duration	
Free Float	Buffer	
Total Float	Total Float	
Fixed Target (milestones)	Benchmark	On activities, ignored.
Not Earlier Than	Start/finish no-earlier-than (SNE/ FNE)	On activity finish date, ignored.
Not Later Than	Start/finish no-later-than (SNL/ FNL)	On activity start date, ignored.

This field in Open Plan	Becomes this field in NetPoint	Notes
On Target (milestones)	Benchmark	On activities, ignored
Calendar	Calendar	Holidays will only be included if they fall between the start and end dates of the project. If a holiday coincides with a non-working day, it will not be transferred into NetPoint.
Resources	Resources	
Description	Name	
Unit Cost	Cost/Unit	
Category	Category	
Resource Level	Rate/Day	Open Plan only supports units/day. When importing, this value is rounded to 1 decimal place (e.g. 0.15/day would become 0.2/day), resulting in a different overall value. After importing from Open Plan, it is recommended to review the assignments of each resource to correct for any significant differences due to rounding.
Codes	Codes	
Code	Name	
Value	Value	

NetPoint Fields

If the schedule was exported from NetPoint already, everything will be brought in EXCEPT for the following:

Characteristic	Result
Layouts	Gone
Delays	Become activities
Shades	Gone

Characteristic	Result
Notepads	Gone
Link Geometries	Become default
Embed Behaviors	Become constant
Custom Positioning	Becomes default
Resource Limits	Gone
Resource Formatting	Become default (color, histogram patterns, color intensities)
Page Size	Becomes letter
Page Tiling	Becomes 1×1
Grid Spacing	Becomes default
Canvas Color	Becomes default
Calendar Strips	Become default
Calendar Sightlines	Become default
Text Size	Becomes default
Text Objects	Gone

Exporting to Open Plan

To export a schedule, the user must have a username and password to access the Open Plan database. The exported schedule will be stored on the same machine as NetPoint.

When exporting, keep in mind that there are many differences between NetPoint and Open Plan. Items like resources, calendars, codes, and constraints, for example, are all handled slightly differently. An explanation of these difference can be found in the notes below. If any activities or milestones have been scheduled on planned-dates, make sure to follow the instructions in [Preparing to Export from NetPoint \(GPM\)](#) before exporting.

Steps

In NetPoint

1. Click **File > Import/Export** from the menu bar and choose **Export to Open Plan**.
2. Enter your Open Plan credentials and choose a **Project Name**.
3. Fill in the properties as desired.
4. When finished, click **OK**. The schedule will appear under the **Projects** node in the **Open Plan Explorer** window.
5. Click **OK** to confirm.

In Open Plan

1. With the schedule open in Open Plan, click **Project > Time Analysis** from the menu bar (or press **F9** on the keyboard).
2. Specify current time and click **OK**.

Notes

Export to Open Plan Window

Export to Open Plan

Open Plan Login Details

Username:

Password:

Project Name

Calendars

☐ Create Calendars as Global

Project Codes

☐ Export Activity Codes [?](#)

Project Resources

☐ Export

☒ Convert Mixed Assignments to: [?](#)

☐ Convert All Assignments to:

Field	Definition
Username	The username for logging into to Open Plan. NOTE: If Open Plan is already open, the username will already be known and disabled accordingly.
Password	The password for logging into to Open Plan. NOTE: If Open Plan is already open, the password will already be known and disabled accordingly.
Project Name	The name once exported to Open Plan.
Calendars	Controls whether or not calendars will export as global.
Project Codes	Controls whether or not codes will be exported. Any codes which allow more than one of their values to be assigned to the same activity will not transfer. To review, click Tools > Open Codes Manager and review their behavior.
Project Resources	Controls whether or not resources will be exported and how.

Supported Fields

NetPoint Objects and Fields

The following NetPoint objects and fields are brought into P6. If a field is not listed, then it does NOT transfer. Some examples of fields that do not transfer are identified where relevant

This field in NetPoint	Becomes this field in Open Plan	Notes
Activity, Delay	Activity	
Milestones and benchmarks (both start and finish)	Start/ Finish Milestone	Milestones that were benchmarks in NetPoint are given Fixed Target dates Open Plan.
Hammock	Hammock	Each hammock must include at least one member activity. Otherwise, the schedule may not export or network logic may not transfer. Furthermore, we recommend that hammocks contain a <i>continuous</i> network of member objects. If present in hammocks, a start milestone or benchmark must have at least one successor within the hammock and a finish milestone or

This field in NetPoint	Becomes this field in Open Plan	Notes
		benchmark must have at least one predecessor within the hammock. Any unconnected milestones or benchmarks will be connected to the LOE with a start-to-start or finish-to-finish relationship accordingly.
Embedded nodes and link lags	Lag	If both exist in the same relationship, they will be combined. For example, an SS relationship with a link lag of 5 days and an embed lag of 5 days will transfer as an SS with 10 day lag in P6.
Description	Description	Any description over 100 characters will be truncated.
ID	ID	If empty, IDs are created in Open Plan with prefix “NP”.
Start Date	Start	
Finish Date	Finish	
Duration	Duration	
No-earlier-than Constraint	Not Earlier Than	Constraints transfer to “Target Dates” in Open Plan
No-later-than Constraint	Not Later Than	Constraints transfer to “Target Dates” in Open Plan
Buffer	Free Float	
Total Float	Total Float	
Calendar	Calendar	Recurring holidays (“fixed-date” holidays in NetPoint with a repeat factor of 1 year) will transfer individually, with each case being a separate holiday.
Grid Position	—	This field is exported for round-trip purposes only and isn’t used in Open Plan.
Color	—	This field is exported for round-trip purposes only and isn’t used in Open Plan.

This field in NetPoint	Becomes this field in Open Plan	Notes
Shape (milestones/benchmarks)	—	This field is exported for round-trip purposes only and isn't used in Open Plan.
Text Formatting	—	This field is exported for round-trip purposes only and isn't used in Open Plan. Includes font, color, alignment, style, size, and line breaks.
Text Display	—	This field is exported for round-trip purposes only and isn't used in Open Plan. Includes text fields, date fields, float fields, etc.
Resources	Resources	In NetPoint, an activity may have some resources assigned to it "per day" and other resources assigned to it "per duration". This is called a <i>mixed assignment</i> . In Open Plan, all resource assignments are assigned per day. For more on distribution types, see Assignment Options . NOTE: Histogram, Color, and Color Intensity are not exported at this time.
Name	Description	
Cost/Unit	Unit Cost	
Category	Category	"Material" and "commodity" become "material", "dollars" becomes "other direct cost", and every other type becomes "labor".
Rate/Dur, Rate/Day	Resource Level	Upon export, all resource assignments are converted to units per day.
Codes	Codes	In NetPoint, multiple values per code may be assigned to the same activity. In Open, however, only one value per code may be assigned to the same activity. Project codes are not exported if they are set to have multiple values in NetPoint (even if only one value is actually defined for the code or assigned to an activity).
Name	Code	
Value	Value	

Microsoft Project

NetPoint features round-trip importing and exporting with MS Project 2007 and 2010. This is done by first exporting an XML file from one application and then importing it into the other.

Primary information such as activities, logic, grid position, and assignments are all transferred, while some formatting characteristics such as link geometries, shades, page size, and canvas color are not. For a full list of supported fields, see the notes at the end of the following sections.

This section details instructions for the following:

1. Importing from MS Project
2. Exporting to MS Project

Importing from MS Project

An imported plan will vary slightly depending on whether or not the schedule was originally created in NetPoint or MS Project. If the schedule was originally created in NetPoint, subsequently exported, and is now being imported back, the look and layout of the schedule will be the same. If the schedule was created in MS Project, or if it was created in NetPoint using a version prior to 4.1, release 25 Feb 2013, NetPoint will layout the network avoiding clashes, but the user should run the automated layout engine. For more information, see [Running the Engine](#). If further manual repositioning of data elements is desired, see [Repositioning Fields \(GEM\)](#). Instructions are provided below.

Steps

In MS Project

1. While viewing the desired schedule, click **File > Save As** from the menu bar.
2. Click next to **Save as type** and choose **XML Format (*.xml)**.
3. If desired, change the **File name** and click **Save**.

In NetPoint

1. Click **File > Import/Export** from the menu bar and choose **Import from MS Project XM**.
2. Browse for the desired XML file and click **Open**. A report will open showing a summary of any incompatibilities that were found.
3. When ready, click **Finalize Import**. The schedule will then open in NetPoint.

NOTE: A non-MS Project-based xml file can technically be chosen but will not properly import.

Notes

Import Report – MS Project

Please review the following notifications. Any incompatibilities will be listed as warnings. To continue, click "Finalize Import".

Group By:

Type	Item Id (NetPo...	Item ID (Source)	Item Description	Message	Resolution
Issue: Calendar definition					
Info	3		Standard	Calendar found	Calendar imported
Info		5	RES DOTS RED DUR		Calendar imported
Issue: Conversion to benchmark					
Info	5	5	Unlinked Flag Finish Be...	Must Start/Finish On constraint foun...	Milestone converted to benchmark
Issue: Invalid calendar definition					
Warning			Calendar Days	Calendar name already exists	Calendar renamed "Calendar Days1"
Warning			United States	Calendar name already exists	Calendar renamed "United States2"

Export to Excel Finalize Import Cancel Import

Column	Definition
Type	Type of notification. Info notifications include confirmations and minor incompatibilities, such as when items are converted to be compatible with NetPoint. Warning notifications

Column	Definition
	include major incompatibilities, such as items that are changed, which affect logic, or items not being imported altogether.
Issue	A general category for the notification, such as calendars, resources, data dates, etc.
Item Id (NetPoint)	The identifier as found in NetPoint.
Item ID (Source)	The identifier as found in MS Project.
Item Description	The name or description of the item in question.
Message	The notification itself.
Resolution	How (if possible) the issue has been resolved.

Supported Fields

MS Project Objects and Fields

The following MS Project objects and fields are brought into NetPoint. If a field is not listed, then it does NOT transfer. Some examples of fields that do not transfer are identified where relevant

This field in MSP	Becomes this field in NetPoint	Notes
Task	Activity	
Task (with 0 duration)	Start/ Finish Milestone	If the task has no predecessor or starts before noon, it becomes a start milestone. Otherwise, it becomes a finish milestone.
Summary Task	Hammock	Only one level (the lowest) is brought in.
Lag	Embedded node or link lag	If positive, the lag will be captured in the embed unless it is larger than the duration of the activity, in which case it will be captured in the link lag. If negative, it will be captured in the link lag for FS relationships, otherwise ignored.

This field in MSP	Becomes this field in NetPoint	Notes
Name	Description	
NetPoint Activity ID (custom field)	ID	If the NetPoint Activity ID field isn't available, then the information in the ID field will transfer over.
Start	Start Date	
Finish	Finish Date	
Actual Start	Start Date	
Actual Finish	Finish Date	
Duration	Duration	
Remaining Duration	Remaining Duration	
Actual Duration	Duration	
Free Slack	Buffer	
Slack	Total Float	
Calendar	Calendar	Repeating holidays that do not occur within the entire range of the project will transfer as individual holidays to NetPoint.
Must Start On	Start benchmark	On activities, ignored.
Must Finish On	Finish benchmark	On activities, ignored.

This field in MSP	Becomes this field in NetPoint	Notes
Start No Earlier Than	Start No Earlier Than	
Start No Later Than	Ignored	
Finish No Earlier Than	Ignored	
Finish No Later Than	Finish No Later Than	
Resources	Resources	MS Project only supports units/day; any schedule that was exported from NetPoint and then brought in should be have its resource assignment distributions reviewed. Furthermore, assignments will only be brought in for activities.
Resource Name	Name	
Std. Rate	Cost/Unit	
Type	Category	
Work (also Units)	Rate/Day	MS Project only supports units/day.
Codes	Codes	Codes that have been entered in the Custom Fields tab of a task will become project codes in NetPoint. The following fields are imported.
Custom Field Name	Name	
Value	Value	

NetPoint Fields

If the schedule was exported from NetPoint already, everything will be brought in EXCEPT for the following:

Characteristic	Result
Layouts	Gone
Delays	Become activities
Shades	Gone
Notepads	Gone
Link Geometries	Become default
Embed Behaviors	Become constant
Custom Positioning	Becomes default
Resource Limits	Gone
Resource Formatting	Become default (color, histogram patterns, color intensities)
Page Size	Becomes letter
Page Tiling	Becomes 1×1
Grid Spacing	Becomes default
Canvas Color	Becomes default
Calendar Strips	Become default
Calendar Sightlines	Become default
Text Size	Becomes default
Text Objects	Gone

Exporting to MS Project

When exporting, keep in mind that there are many differences between NetPoint and MS Project. Items like resources, calendars, codes, and constraints, for example, are all handled slightly differently. An explanation of these difference can be found in the notes below.

Before exporting, if any activities or milestones have been scheduled on planned-dates, make sure to follow the instructions in [➔Preparing to Export from NetPoint \(GPM\)](#). Once exported, the XML file must be imported into MS Project to complete the process. Instructions for both are provided below.

Steps

In NetPoint

1. Click **File > Import/Export** from the menu bar and choose **Export to MS Project XML**.
2. Enter a new name and click **Save**.

In MS Project

1. Click **File > Open** from the menu bar.
2. Click the button next to **File name** and choose **XML Format (*.xml)**
3. Browse for the desired XML file and click **Open**.
4. Choose how to import the project and click **Finish**.

Notes

Supported Fields

NetPoint Objects and Fields

The following NetPoint objects and fields are brought into P6. If a field is not listed, then it does NOT transfer. Some examples of fields that do not transfer are identified where relevant.

This field in NetPoint	Becomes this field in MSP	Notes
Activity, Delay	Task	
Milestones and benchmarks (both start and finish)	Task (with 0 duration and set as milestone)	Benchmarks in NetPoint become milestones with Must Start On or Must Finish On constraints accordingly.
Hammock	Summary Task	

This field in NetPoint	Becomes this field in MSP	Notes
Embedded nodes and link lags	Lag	If both exist in the same relationship, they will be combined. For example, an SS relationship with a link lag of 5 days and an embed lag of 5 days will transfer as an SS with 10 day lag in P6.
Description	Name	Any description over 255 characters will be truncated.
ID	NetPoint Activity ID (custom field)	ID's in MS Project are numeric only.
Start Date	Start, Actual Start	
Finish Date	Finish, Actual Finish	
Duration	Duration, Actual Duration	
Remaining Duration	Remaining Duration	
No-earlier-than Constraint	No Earlier Than Constraint	In MS Project, only one constraint per task is supported. If an activity or milestone in NetPoint has 2 constraints, only the start constraint will be transferred.
No-later-than Constraint	No Later Than Constraint	In MS Project, only one constraint per task is supported. If an activity or milestone in NetPoint has 2 constraints, only the start constraint will be transferred.
Buffer	Free Slack	
Total Float	Slack	
Calendar	Calendar	Holidays only export within the project start and completion.

This field in NetPoint	Becomes this field in MSP	Notes
Grid Position	—	This field is exported for round-trip purposes only and isn't used in MS Project.
Color	—	This field is exported for round-trip purposes only and isn't used in MS Project.
Shape (milestones/benchmarks)	—	This field is exported for round-trip purposes only and isn't used in MS Project.
Text Formatting	—	This field is exported for round-trip purposes only and isn't used in MS Project. Includes font, color, alignment, style, size, and line breaks.
Text Display	—	This field is exported for round-trip purposes only and isn't used in MS Project. Includes text fields, date fields, float fields, etc.
Resources	Resources	In NetPoint, an activity may have some resources assigned to it “per day” and other resources assigned to it “per duration”. This is called a <i>mixed assignment</i> . In MS Project, all resource assignments are assigned per day. For more on distribution types, see Assignment Options . NOTE: Histogram, Color, and Color Intensity are not exported at this time.
Name	Resource Name	
Cost/Unit	Std. Rate	
Category	Type	“Material” and “commodity” become “material”, “dollars” becomes “cost”, and every other type becomes “work”. KNOWN ISSUE: The unit cost for a resource with category “dollars” in NetPoint exports as 0.
Rate/Dur, Rate/Day	Work (also Units)	Upon export, all resource assignments are converted to units per day.
Codes	Codes	In NetPoint, multiple values per code may be assigned to the same activity. In MS Project, however, only one value per code may be assigned to the same activity. Project codes are not exported if they are set to have multiple values in NetPoint (even if only one value is actually defined for the code or assigned to an activity). Single-assignment codes export to the Custom Fields tab.

This field in NetPoint	Becomes this field in MSP	Notes
Name	Custom Field Name	
Value	Value	

Microsoft Excel

NetPoint features round-trip importing and exporting with Microsoft Excel 2010 onwards. This is useful for spreadsheet-style data entry, building custom reports, working with raw schedule output, and for performing custom analyses or computation methods otherwise not offered by NetPoint.

This section details instructions for the following:

1. Importing generic spreadsheets
2. Exporting linked reports
3. Updating from linked reports

Importing Generic Spreadsheets

NetPoint allows you to import information from user-created spreadsheets. For example, you may have a list of activities you've compiled in Excel based on previous projects or collected from different team members. Upon importing, NetPoint will identify the fields that it recognizes and map them automatically; however, some manual correction and mapping may be necessary. If any fields remain unmapped, or if the same field is mapped to more than one column, the **OK** button will become disabled and warning icons will appear. For a full list of supported fields, see the notes at the end of this chapter.

Before importing, make sure the Excel file has been saved, as the content from the most recently saved file will be imported. If Excel is open, do not close it during any step of the import process, and vice versa. Also, make sure that the sheet being imported isn't protected, as this may interfere with the import process. Data can be imported into new or previously-saved schedules.

NOTE: Upon import, any comments on cells in Excel will be removed. To avoid loss of any comments, duplicate the Excel file and import the duplicate.

Steps

1. With NetPoint open, click **File > Import/Export** from the menu bar and choose **Import From Excel File**.
2. Browse for the desired Excel file and click **Open**.
3. To import a risk register or resources, choose **Risk Register** or **Resources** next to **Data Type**. Otherwise, leave it on **Schedule Objects**.
4. To change to a different sheet, click the down arrow next to **Sheet Name** and choose accordingly.
5. To set the data area, click the arrows up or down under **Data Range**.
6. To include or exclude data from a given column, check or uncheck the box next to its name.
7. To map a new field or correct a default mapping, click the down arrow button to the right and choose accordingly.
8. Once all fields are mapped successfully, click **OK**. A report will open listing any warnings found (all warnings will be added to the cells in Excel if the file was closed beforehand).
9. To correct any invalid data, click **Cancel Import**.
10. To continue with the import, click **Finalize Import**.

Notes

Import from Excel Window

Import from Excel

Data Type: Risk Register

Sheet Name: Sheet1

Data Range

Header Row: 1 First Row: 2 Last Row: 11 First Column: A Last Column: J

Column-to-Field Mapping

Excel Columns	NetPoint Fields
☒ Risk ID	ID
☒ Risk Description	Risk Description
☒ Probability	Duration Probability
☒ Impact	Duration Impact
☐ Rating	
☐ Column F	
☒ Status	Status
☒ Owner	Owner
☒ Comments	Comments

☐ Select All Import Cancel

Field	Definition
Data Type	The type of data desired for import, whether Schedule Object , Resources , or Risk Register . Each type will determine which fields are available for mapping below.
Sheet Name	The name of the sheet desired for import.
Header Row	The row in Excel being used as column headers. NetPoint will attempt to automatically detect this.
First Row	The first row of data desired for import. NetPoint will attempt to automatically detect this. Any row above will not be brought in.
Last Row	The last row of data desired for import. NetPoint will attempt to automatically detect this. Any row below will not be brought in. If 10 empty rows are detected, the import will stop, regardless of whether or not it's reached the last row.

Field	Definition
First Column	The first column of data desired for import. NetPoint will attempt to automatically detect this. Any column before will not be brought in.
Last Column	The last column of data desired for import. NetPoint will attempt to automatically detect this. Any column after will not be brought in.

Schedule Object Fields

Field	Requirements
Object Type	For an activity, enter as “activity,” “task,” “task dependent,” “fixed units,” “fixed duration,” or “fixed work”. For a milestone, enter as “milestone,” “start milestone,” “sm,” “finish milestone,” or “fm”. For a benchmark, enter as “benchmark,” “start benchmark,” “sb,” “finish benchmark,” and “fb”. For delay, enter as “delay”. If missing or invalid, it will be imported as an activity.
ID	Can be any combination of letters and numbers (case sensitive). Characters such as “-” are supported but others such as “#” will be dropped.
Description	Can be any combination of letters, numbers, and characters.
Calendar	Must be defined in the file you are importing into. For example “United States” or “Calendar Days”. Case sensitive. To assign a global calendar, make sure to include [space][parenthesis]Global[parenthesis], e.g. “Calendar (Global)”. If missing or invalid, it will be set to the default calendar for the schedule you’re importing into.
Duration	Must be greater than 0 for activities and equal to 0 for milestones and benchmarks. If missing or invalid, it will be set as 1 for activities and 0 for milestones, benchmarks, and delays.
Start Date	Can be entered in any date format recognized by Excel, including but not limited to: 3/14/2012; 3/14/12; 03/14/12; 14-Mar-12; 14-Mar-2012; March 14, 2012; Wednesday, March 14, 2012. If a date is entered that is not in that object’s calendar’s work days, the object goes to the next work day for that calendar. If missing or invalid, it may be set to its early date. NOTE: If a cell hasn’t been formatted as a date, then the date will be brought in according to the date format of the operating system. For example, 1/2/13 will be brought in as February 1st if Windows is using a UK format but January 2nd if Windows is using a US format.
Finish Date	Can be entered in any date format that is recognized by Excel, including but not limited to: 3/14/2012; 3/14/12; 03/14/12; 14-Mar-12; 14-Mar-2012; March 14, 2012; Wednesday, March 14, 2012. If a date is entered that is not in that object’s calendar’s work days, the object goes to the next work day for that calendar. If missing or invalid, it may be set to its

Field	Requirements
	early date. NOTE: If a cell hasn't been formatted as a date, then the date will be brought in according to the date format of the operating system. For example, 1/2/13 will be brought in as February 1st if Windows is using a UK format but January 2nd if Windows is using a US format.
Predecessor	Enter as EITHER [Row #][Relationship][Lag] OR if ID's are being defined, [ID][Relationship][Lag], e.g. "12FS1" or "ABFS1". In either case "[Lag]" is optional, e.g. "12FS" or "ABFS". If ID's are being defined, row numbers can NOT be used. Possible relationships include SS, FS, FF, or SF. If the relationship or row #/ID is not recognized, the link will not be imported.
Successor	Enter as EITHER [Row #][Relationship][Lag] OR if ID's are being defined, [ID][Relationship][Lag], e.g. "12FS1" or "ABFS1". In either case "[Lag]" is optional, e.g. "12FS" or "ABFS". If ID's are being defined, row numbers can NOT be used. Possible relationships include SS, FS, FF, or SF. If the relationship or row #/ID is not recognized, the link will not be imported.
Color	Enter as EITHER R,G,B (for example, 255, 50, 50) OR by name. A list of supported names can be found at http://www.rapidtables.com/web/color/RGB_Color.htm . If invalid or not recognized, it will be set as the default color for that object.
Grid	Enter any number greater than 1. Unrecognized input (negative numbers or characters) will result in being placed on the first grid. An object placed on a grid number greater than the highest being displayed will still be successful, but will only be visible after increasing the page tiling.

Resource Fields

Field	Requirements
Resource Name	The description of the resource. Can be any combination of letters, numbers, and characters.
Category	The type of resource. Can be "Commodity," "Construction Equipment," "Dollars," "Equipment," "Labor," "Manpower," "Material," or "Professional." If empty or invalid, set as equipment.
Cost/Unit	The cost per unit of resource, used when exporting to Excel or viewing the WBS roll-up. If empty or invalid, set as 1.00.
Pattern	A visual cue for identifying resource bars on the histogram. Can be "Solid," "Dots," "Dk Horizontal," "Dk Vertical," "Dk Trellis," "Dk Grid," "Dk Up Diagonal," "Dk Down Diagonal," "Dk

Field	Requirements
	Horizontal," "Lt Vertical," "Lt rellis," "Lt Grid," "Lt Up Diagonal," or "Lt Down Diagonal," If empty or invalid, set as solid.
Color	The color of the resource bars on the histogram. Enter as EITHER R,G,B (for example, 255, 50, 50) OR by name. A list of supported names can be found at http://www.rapidtables.com/web/color/RGB_Color.htm . If empty or invalid, set as default.
Intensity	The darkness of the resource bars on the histogram. If empty or invalid, set as 100%.

Risk Register Fields

Field	Requirements
ID	The reference number for the risk.
Risk Description	A short explanation of the risk
Owner	The person responsible for tracking a risk, and for ensuring an appropriate response is selected and implemented if it occurs.
Status	The current state of the risk in the project. Can be "New," "Open," "In-Progress," or "Closed".
Category	A designation of type. Can be "Threat" or "Opportunity".
Strategy	The type of response to a risk, depending on if it's a threat or opportunity. For threats, it can be "Accept," "Mitigate," "Transfer," or "Avoid". For opportunities, it can be "Accept," "Enhance," "Share," or "Exploit".
Duration Probability	The chance that a risk could occur, as defined in the probability/impact matrix. Can be "Ultra High," "Extremely High," "Very High," "High," "Moderate," "Low," "Very Low," "Extremely Low," or "Ultra Low," as well as their abbreviations: "UH," "EH," "VH," "H," "M," "L," "VL," "EL," "UL".
Duration Impact	The degree to which the risk could affect activity durations. Can be "Ultra High," "Extremely High," "Very High," "High," "Moderate," "Low," "Very Low," "Extremely Low," or "Ultra Low," as well as their abbreviations: "UH," "EH," "VH," "H," "M," "L," "VL," "EL," "UL".
Response	Notes for detailing the approach for handling the risk.
Comments	Notes for summarizing team discussions or including additional information about a risk.

Exporting Linked Reports

The export to Excel feature is designed to provide a more flexible way to build reports and update schedules. Custom spreadsheets can be configured based on any number and combination of objects and their data, allowing you work with as much raw data as desired in order to build custom reports and templates for updating schedules. Note that the resulting Excel files contain minimal formatting and presentation.

A few common combinations of data to export come pre-saved as templates to help get you started. Both default and user-created templates are shared at the system-level, meaning that a template created for one schedule will be available when working with any other schedule. You can choose to export data from an existing template as-is, reconfigure an existing template and then export the data, or create a new template altogether to export from. In addition, the export process can be automated and triggered directly from your desktop by specifying the desired schedule and report ahead of time. Instructions for each are provided.

If you're exporting data to Excel to edit and update NetPoint, the fields that can be edited are identified in **bold** in the **Configure Rows & Columns** window in NetPoint and in the corresponding cells in Excel. All other columns will be *protected*, except for the code table, resource table, and target sheets (for more on protected sheets, see the notes at the end of this section). For more info on updating, see [➔Updating from Linked Reports](#).

WARNING: When a template is exported to Excel, the new Excel file will automatically overwrite the last-exported Excel file (for that schedule) unless the last one was renamed or moved. To avoid losing the previous Excel file, rename it or move it to a different location *before* doing the next export.

NOTE: The schedule will be saved each time a template is exported. To avoid having changes permanently saved, save the schedule as a separate document first, or use Undo to roll back the changes before closing the schedule.

Steps

Creating a New Template

1. With NetPoint open, click **File > Import/Export** from the menu bar and choose **Export to Linked Excel File**.
2. Click the **New** button.
3. Make sure the circle is checked next to **New Configuration**.
 - a. To remove or add object, check the boxes next to their labels.
 - b. To include a field, select it in the **Available** list on the left and click the **>>** button.
 - c. To remove a field, select it in the **Included** list on the right and click the **<<** button.
4. Once configured, click the **Save** button.
5. Enter a description and click **Save**.
6. To finish the export process, click the **Export** button. Excel will open with the exported data, and the file will be automatically saved in the schedule's project folder.
7. If you plan on exporting additional templates in the future, then we suggest renaming the file or moving it to a different location to avoid having it being overwritten. Click **File > Save As** from the menu bar, enter a name, and click **Save**. **NOTE:** If you **DO** rename or move the file, you can **ONLY** initiate an update from it from within Excel. For more info, see [➔Updating from Linked Reports](#).

NOTE: When adding/removing fields, fields within Object Data, Links Data, Resource Data, and Target Data will re-order themselves automatically. Fields within Predecessor Data, Successor Data, Tables, and Hammocks will depend on the order that they're added.

Modifying/Exporting an Existing Template

1. With NetPoint open, click **File > Import/Export** from the menu bar and choose **Export to Linked Excel File**.
2. Choose a desired template from the list.
3. To export right away, click the **Export** button and skip to step 4. To modify first, click the **Modify** button. **NOTE: Only user-created templates can be modified. If you reconfigure a default template, you will not be able to export it unless you first save it as a new template.**
 - a. To remove or add objects, check the boxes next to their labels.
 - b. To include a field, select it in the **Available** list on the left and click the **>>** button.
 - c. To remove a field, select it in the **Included** list on the right and click the **<<** button.
 - d. Once configured, click **Save** and continue to step 2e. To save it as a new template, click **Save As**.
 - a. Enter a description and click the **Save** button.
 - e. To finish the export process, click the **Export** button. Excel will open with the exported data, and the file will be automatically saved in the schedule's project folder.
4. If you plan on exporting additional templates in the future, then we suggest renaming the file or moving it to a different location to avoid having it being overwritten. Click **File > Save As** from the menu bar, enter a name, and click **Save**. **NOTE: If you DO rename or move the file, you can ONLY initiate an update from it from within Excel. For more info, see [➔Updating from Linked Reports](#).**

NOTE: When adding/removing fields, fields within Object Data, Links Data, Resource Data, and Target Data will re-order themselves automatically. Fields within Predecessor Data, Successor Data, Tables, and Hammocks will depend on the order that they're added.

Automating the Export Process

1. In windows explorer, navigate to the location where NetPoint has been installed
2. Right-click on "NetPoint.exe" and click **Create Shortcut**.
3. In the notification, click **Yes**.
4. On the desktop, find the new shortcut and right-click and choose **Properties**.
5. Next to **Target**, add the following *to the end, without the quotation marks*: (space) "path" (space) "/" "Report" (space) "template\$1" (space) "template\$2" (space) "etc.", where (space) shouldn't be spelled out but just a single space from your spacebar, "path" is equal to the filepath for your schedule, "Report" is just a required variable, and "template\$n" is equal the name of the template(s) you want to export. Be sure to substitute any spaces in your template name with dollar signs (\$), and don't forget to remove the quotation marks above. For example, if you want to export the templates called "Default – All Objects" and "Default – Targets" for the schedule "Tech Transfer", the target would look like the following (with the bold being what you would add): "C:\Program Files\NetPoint\NetPoint.exe" **C:\Users\Administrator\Documents\NetPoint Projects\Tech Transfer\NetPointFile.np4 /Report Default\$-\$All\$Objects Default\$-\$Targets**
6. When finished, click **OK**.
7. To initialize the export, double-click the shortcut or right-click and choose **Open**. The schedule will open in NetPoint and the report will open in Excel.

8. If you plan on exporting additional templates on the same day, then we suggest renaming the file or moving it to a different location to avoid having it being overwritten. Click **File > Save As** from the menu bar, enter a name, and click **Save**.
9. If desired, you may rename the shortcut by right-clicking on it and choosing **Rename**.
10. Enter a name and click outside to finish.

NOTE: Upon export, the Excel file will be automatically saved in the schedule's project folder and named after the template, with the month and day appended to the end. Excel files that are generated automatically can only have their changes pushed from within Excel. For more info, see [➔Updating from Linked Reports](#).

Notes

Working with Multiple Schedules

Only one linked Excel file can be open at a time. This means that when using **File > Import/Export > Export to Linked Excel File**, or **Open Linked Excel File**, any Excel files that were open for other schedules will be closed, and any unsaved changes will be lost. To have linked Excel files open for more than one schedule at a time, open them directly from their project folders or through Excel instead of through NetPoint.

Filename Incompatibility

Once exported, the Excel file will be automatically named using the same name as the project. However, Excel does not allow filepaths over 218 characters or which contain any of the following symbols: < > ? [] : | *. Any templates that are exported from NetPoint projects that do not satisfy these criteria will have to be renamed in Excel. As a result, you'll only be able to initialize an update from such files from within Excel.

Tips for Excel

Field-to-Column Mapping

Each field that is exported will become a column in Excel, organized horizontally in the order that they appear in the list on the right of the **Configure Rows & Columns** window. At this time, fields/columns cannot be reordered after being added. For example, to move a desired field higher, remove the fields above it and re-add them back below.

Column Descriptions

The names of the column headers in Excel will be derived directly from how they appear in the list on the right of the **Configure Rows & Columns** window.

Sheet Structure

Depending on which information is chosen for exporting, the resulting Excel file may contain multiple sheets. Object data, including predecessor and successors, will be located on a sheet with the same name as the template itself. For target data, each target will get its own sheet. For resource data, each resource will also get its own sheet (or pair of sheets, if consumption and/or cost is exported per day or per month, the second sheet being reserved for a graph).

By default, all fields under the **Object Data** node in the **Configure Rows and Columns** window will be repeated on all other eligible sheets. This can be customized by right-clicking and choosing accordingly.

Unprotecting Excel Sheets

Any column that cannot be edited will be protected in Excel, except for the code table, resource table, and target sheets. When protected, some features in Excel, such as Find and Replace, may not be available. To

access these features, unprotect the sheet in the Review tab or menu option using “NetPoint” as the password.

Cut/Copy/Paste

Only cells which can be edited will allow cut, copy, or paste, unless the sheet has been unprotected, as described above. In any case, cut and paste should be strictly avoided, as it will alter the underlying formula of the cells. As an alternative, you can right-click on a cell and choose clear contents. Copy and paste may be used; however, when pasting, it is recommend to paste only values. This can be accessed via the paste options.

Project Info and Spreadsheet Key

The last sheet of every report contains information specific to the project:

- Project filepath
- Date exported
- Name of the template
- Time unit of the schedule
- Project Start Date
- Project Completion Date
- Data Date
- Asterisk Key

In addition, at the top of every sheet is a key explaining various aspects of the report.

Available Fields

Object Data

If exported, each object that is included (activities, links, milestones, benchmarks, delays, hammocks) will become a row in Excel, organized in the order they were added to the network (system ID). The system ID will always automatically be exported, and will be the first column on the left, after the change log. After exporting, these fields will be located on a sheet with the same name as the template itself.

Field	Definition
Object Type	The type of object, whether activity, link, milestone, benchmark, delay, or hammock.
Description	The object's description. The description is for describing the object.
Abbr. Description	The object's abbreviated description. This abbreviated description is better for displaying on the canvas and is limited to 48 characters.
Start Date	The object's start date.
Finish Date	The object's finish date.
Duration	The object's duration.

Field	Definition
Progressed Duration	For in-progress activities, the amount of duration actualized up to the data date.
Remaining Duration	For in-progress activities, the amount of duration unactualized after the data date.
Calculated Complete	For actualized activities, the amount of progressed duration divided by total duration. If an activity or hammock contains an un-actualized portion before the data date, the calculated complete will export as NA.
Estimated Complete	A user-entered estimate of work completed.
Calendar	The object's calendar. This calendar dictates on which days work may take place on the object.
ID	The object's ID. The ID is a unique identifier.
Start Constraint	If applied, a constraint on the start date that prevents the object from starting earlier.
Finish Constraint	If applied, a constraint on the finish date that prevents the object from finishing later.
Float*	The float of the object. The float is the amount of time an object can be delayed before delaying the completion of the project. *An asterisk denotes that the activity is actualized, and that the value is forensic.
Total Float*	The total float of the object. Total float is equal to float + drift. *An asterisk denotes that the activity is actualized, and that the value is forensic.
Buffer*	The buffer of the object. The buffer is the amount of time an object can be delayed before delaying the start of its nearest successor. *An asterisk denotes that the activity is actualized, and that the value is forensic.
Drift*	The drift of the object. The drift is the amount of time an object can gain before advancing the start of the project. *An asterisk denotes that the activity is actualized, and that the value is forensic.
Drift Buffer*	The drift of the object. The drift buffer is the amount of time an object can gain before advancing the finish of its nearest predecessor. *An asterisk denotes that the activity is actualized, and that the value is forensic.

Field	Definition
Grid	The gridline or row number that the object is drawn on.
Codes	The codes and values assigned to the object. Where an assignment exists, the cell underneath the value's column will get an "x". Export this field for modifying the codes assigned to an activity or doing global assignments.
Code Summary	A comma-separated list of all values (if any) of a particular code assigned to the object. Where an assignment exists, the cell underneath the code's column will contain the value.

Predecessor and Successor Data

If chosen for exporting, predecessor and successor data will be added to the main sheet (the sheet with the same name as the template itself). Note that for every object, all predecessors and all successors are grouped together in one cell for each type. For example, if "Activity A" is a predecessor to "Activity B" and "Activity C," then both "Activity B" and "Activity C" will show up under the "Successor Description(s)" column, separated by commas. The same goes for their IDs and/or finish date(s).

Field	Definition
Predecessor(s) Description- Relationship	All, if any, of its predecessors' descriptions follow by their relationships. Relationships are displayed as either FS (finish-to-start), SS (start-to-start), or FF (finish-to-finish) followed by the link gap, e.g. "Activity1-SS0".
Predecessor(s) ID- Relationship	All, if any, of its predecessors' given IDs followed by their relationships. Relationships are displayed as either FS (finish-to-start), SS (start-to-start), or FF (finish-to-finish) followed by the link gap, e.g. "12345-SS0."
Predecessor System ID(s)	All, if any, of its predecessors' system IDs.
Predecessor ID(s)	All, if any, of its predecessors' given IDs.
Predecessor(s) Description	All, if any, of its predecessors' descriptions.
Successor(s) Description- Relationship	All, if any, of its successor's descriptions follow by their relationships. Relationships are displayed as either FS (finish-to-start), SS (start-to-start), or FF (finish-to-finish) followed by the link gap, e.g. "Activity1-SS0".
Successor(s) ID- Relationship	All, if any, of its successor's given IDs followed by their relationships. Relationships are displayed as either FS (finish-to-start), SS (start-to-start), or FF (finish-to-finish) followed by the link gap, e.g. "12345-SS0."

Field	Definition
Successor System ID	All, if any, of its successors' system IDs.
Successor ID(s)	All, if any, of its successors' given IDs.
Successor(s) Description	All, if any, of its successors' descriptions.

Link Data

If exported, link data will show up in its own sheet called "Links".

Field	Definition
Predecessor	The beginning object in the relationship.
Successor	The ending object in the relationship.
Type	The type of relationship of the link. Relationships can be either FS (finish-to-start), SS (start-to-start), or FF (finish-to-finish).
Lag	The lag, if any, of the relationship. The lag is the amount of time after the start of a predecessor and the start of its successor or the finish of a predecessor and the finish of its successor in a start-to-start relationship or finish-to-finish relationship respectively.
Gap	The gap value of the link. The gap is the duration of a link.
Logic Type	The logic type of the link. For more info, see ➔Creating FS Links .
Shape	The geometry of the link. For more info, see ➔Optimizing Link Geometries .

Resource Data

When exporting resource data, most of the fields will become columns in Excel, organized horizontally in the order that they appear in the list on the right of the **Configure Rows & Columns** window. Specifically, **Rate/Dur**, **Rate/Day**, **Total Consumption**, **Total Cost**, **Consumption**, and **Cost** are repeated for every resource and put on their own sheet with **Consumption** and **Cost** actually spanning several columns, one column for every unit of time during which the project takes place. If multiple units of time are chosen (e.g. days and months), then all of the sheets for the first time unit will show up followed by all of the sheets for the next time unit. A resource's sheet can be identified by its name, which follows as "Res-[description]-[internal ID]-[time unit]". For example, if resources are exported by day and the schedule contains a resource named "Plumber" with an internal ID of "5," then the sheet would be titled "Res-Plumber-5-day". At the end of the Excel file, a resource "summary" sheet will be added for each time unit exported.

Graph Consumption and **Graph Cost** are also repeated for every resource and put on their own sheet at the beginning of the Excel file, with all of the consumption sheets showing up first followed by all of the cost sheets. If multiple time units are chosen (e.g. days and months), then all of the sheets for the first time unit will show up followed by all of the sheets for the next time unit. For example, if **Graph Consumption** and **Graph Cost** were exported by days and months for a schedule with 3 resources, there would be 3 sheets for consumption/day and 3 sheets for consumption/month followed by 3 sheets for cost/day and 3 sheets for cost/month. These sheets follow a similar naming convention, which follows as “type-description-ID-time unit”. For example, if consumption and cost are exported by day, then the consumption and cost graphs for the same Plumber resource would be located on the sheets titled “Cons-Plumber-5-by-day” and “Cost-Plumber-5-by-day”, respectively. Before each of the consumption and cost sheets, a “total” graph will be added for each time unit exported.

NOTE: For graphs to be exported, their data must be included too. For example, for Graph Cost to be exported, Cost must be included for at least one time unit; for Graph Consumption to be exported, Consumption must be included for at least one time unit. If an attempt is made to add a Graph Cost or Graph Consumption without at least one time unit for Cost or Consumption respectively, NetPoint will choose a default time unit (year) and add it with the graph. Likewise, if Cost or Consumption are completely removed from a template, their graphs will be removed as well.

NOTE: Depending on your machine, choosing to export the graphs may add a significant amount of processing time.

Field	Definition
Rate/Day	The quantity of this resource assigned to that activity on a per day basis. Export this field for modifying the resources assigned to an activity or doing global assignments. To assign a resource per day, enter a value in the Rate/Day column, AND make sure there is a no value or a “0” in the Rate/Dur column. As soon as there is a value in the Rate/Dur column, any value in the Rate/Day column will not be used.
Rate/Dur	The quantity of this resource assigned to that activity on a per duration basis. To assign a resource per duration, enter a value in the Rate/Dur column. As soon as there is a value in the Rate/Dur column, any value in the Rate/Day column will not be used. If this column contains a “0”, the value from Rate/Day will be used instead.
Total Consumption	The rate/day times the duration of the activity (or the rate/dur).
Total Cost	The cost/day time the duration of the activity.
Consumption	A per time unit breakdown of the consumption of this resource over the duration of the schedule. Consumption will span several columns, one column for every unit of time during which the project takes place. Right-click in the list on the left to add individual

Field	Definition
	time units; right-click in the list on the right to remove them. If added without choosing, years will be the default.
Cost	A per time unit breakdown of the cost of this resource over the duration of the schedule. Cost will span several columns, one column for every unit of time during which the project takes place. Right-click in the list on the left to add individual time units; right-click in the list on the right to remove them. If added without choosing, years will be the default.
Graph Consumption	Whether or not to include graphs of all the consumption data. The consumption field must be included for at least one time unit.
Graph Cost	Whether or not to include graphs of all the cost data. The cost field must be included for at least one time unit.

Target Data

When exporting target data, the reported fields will be repeated for every target, with each target getting its own sheet named after the number of the target followed by the description. For example, the first target's sheet could be named "Target1-Baseline" followed by "Target2-Update" and so on. Like with object data, each field that is exported will become a column in Excel, organized horizontally in the order that they appear in the list on the right of the **Configure Rows & Columns** window.

NOTE: If no targets have been stored in NetPoint, then no sheets will actually show up in Excel for targets.

Field	Definition
Target Start Date	The object's start date when the target was captured.
Start Date Variance	The target minus the current, in time units (i.e. days).
Target Finish Date	The object's finish date when the target was captured.
Finish Date Variance	The target minus the current, in time units (i.e. days).
Target Duration	The object's duration when the target was captured.
Duration Variance	The target minus the current, in time units (i.e. days).
Target Float	The object's float when the target was captured.

Field	Definition
Float Variance	The current minus the target, in time units (i.e. days).
Target Total Float	The object's total float when the target was captured.
Total Float Variance	The current minus the target, in time units (i.e. days).
Target Drift	The object's drift when the target was captured.
Drift Variance	The current minus the target, in time units (i.e. days).

Tip: At this time, variances are only calculated to the current state, not between any two targets, but such reports can be exported from the Target Manager by clicking Tools > Open Targets Manager, right-clicking on the columns, and choosing Export all Rows & Columns.

Tables

Tables are another way for adding or deleting codes and resources from the schedule. When exported, they will be put in their own separate sheets. For more instructions, see [➔Adding and Modifying Codes via Excel](#) and [➔Adding and Modifying Resources via Excel](#).

Field	Definition
Code Table	An editable table for adding, modifying, and deleting codes. Export this table for managing codes and values.
Resource Table	An editable table for adding, modifying, and deleting resources. Export this table for managing resources and cost.

Hammocks

Hammocks will show up on the main data sheet.

Field	Definition
Min Member Drift	The lowest drift value of all a hammock's members.
Min Member Float	The lowest float value of all a hammock's members.
Min Member Buffer	The lowest buffer value of all a hammock's members.
Min Member Total Float	The lowest total float value of all a hammock's members.
Member List	A list of all members of the hammock.

Updating from Linked Reports

With the Excel data transfer feature, you can export a schedule to Excel, update its activity IDs or assign new resources, and then push those changes back into NetPoint without closing or leaving either application. In this scheme, Excel serves as a seamless data entry and updating method. Other uses for the Excel data transfer feature include spell checking or changing line breaks in activity descriptions, for example.

Fields that can be edited are identified in **bold** in both the **Configure Rows & Columns** window as well as in Excel itself. All other columns will be *protected*, except for the code table, resource table, and target sheets (for more on protected sheets, and for a full list of supported fields, see the notes at the end of this section).

In order to execute the Excel data transfer, the file from which you want to update must be generated by an export from NetPoint. For example, to update activity IDs, first export activities and their ID fields to Excel. For more information, see [Exporting Linked Reports](#).

NOTE: NetPoint must remain open at all times during exporting and/or updating. If NetPoint is ever closed, you must close Excel and then open NetPoint again to refresh the link between applications before you can initiate an update. Excel, on the other hand, can be closed and opened as many times as desired without interrupting the connection.

Steps

Editing in Excel

1. Make sure the Excel file you want to edit and update from was generated by an export from NetPoint. For more information, see [Exporting Linked Reports](#).
2. If the Excel file you want to edit was just exported from NetPoint and opened, skip to step 4. If not, click **File > Import/Export** from the NetPoint menu bar.
3. Choose **Open Linked Excel File**. **NOTE: If you open the desired Excel file from within Windows Explorer or Excel itself, you can still edit and save it; however, to initiate an update, you must close it and open it from the NetPoint menu bar.**
4. Edit any cells whose columns are identified in bold. Once modified, a cell will be colored in blue and its row will be populated with a star under the **Changed** column.
5. When finished, click outside of the cell being edited or press **Enter**.
6. To update, see [Initiating an Update from within Excel](#).

Initiating an Update from within Excel

1. If you're coming from [Editing in Excel](#), skip to step 3. Otherwise, click **File > Import/Export** from the NetPoint menu bar.
2. Choose **Open Linked Excel File**. **NOTE: If trying to update from an Excel file that was renamed, moved, or received from another user, first put it back in its project folder and then ensure its filename matches its project folder. If the schedule has been edited since the Excel file was originally exported, changes will be reconciled as follows: If an activity or other object was changed in NetPoint AND changed in Excel, the state from Excel will prevail. If an activity or other object was changed in NetPoint but NOT changed in Excel, the state from NetPoint will prevail. If an activity or other object in Excel is not found in NetPoint, it will NOT be added in NetPoint and it will be removed from Excel. If an activity or other object in NetPoint is not found in Excel, it will be added in Excel. WARNING: Compatibility cannot be guaranteed if a schedule is updated from an unrelated Excel file.**

3. In Excel, choose the **ADD-INS** tab and click the **Update Schedule**  icon in the **Menu Commands** section.
4. To review the changes first, click **Review Changes**. To accept without reviewing, click **Accept All** and skip to step 7.
 - a. To deny a change, uncheck the box next to it under the **Accept** column.
 - b. When finished, click the **Continue** button.
 - c. When ready, click the **Accept All** button.
5. Excel will open and any changes from NetPoint will be brought in as well.

NOTE: To initiate an update, you must open the file via the NetPoint menu bar, NOT from Windows Explorer or Excel itself.

Adding and Modifying Codes via Excel

1. First, export the “Default – Code & Code Assignments” template from NetPoint (or any other template as long as it includes the **Code Table**). For detailed steps, see [Modifying/Exporting an Existing Template](#).
2. In the linked Excel file, click the **Code Table** tab.
3. To add a new code, click in the first empty row under the **Name** column and enter a description.
 - a. Next to the name under the **Value** column, enter a value. **NOTE: Once the name and value have been entered successfully, ADD will appear under the Changed column.**
 - b. To allow more than one value of the same code to be assigned to the same activity, leave an “X” under the **Multi Assignments** column.
 - c. To enter another value, repeat steps 3-3b using the same description under the **Name** column for every row.
 - d. To enter another code, repeat steps 3-3b using a different description under the **Name** column.
4. To delete a value, enter any text (for example, an “x” or “DEL”) under the **Delete** column.
5. To delete a code, enter any text under the **Delete** column for every one of its values.
6. To modify a code or value, select its description and change it accordingly.
7. When finished, choose the **ADD-INS** tab and click the **Update Schedule**  icon in the **Menu Commands** section.
8. To review the changes first, click **Review Changes**. To accept without reviewing, click **Accept All** and skip to step 9.
 - a. To deny a change, uncheck the box next to it under the **Accept** column.
 - b. When finished, click the **Continue** button.
 - c. When ready, click the **Accept All** button.
9. Excel will open and any changes from NetPoint will be brought in as well.

Adding and Modifying Resources via Excel

1. First, export the “Default – Cost, Resources & Assignments” template from NetPoint (or any other template as long as it includes the **Resource Table**). For detailed steps, see [Modifying/Exporting an Existing Template](#).
2. In the linked Excel file, click the **Resource Table** tab.
3. To add a new resource, click in the first empty row under the **Resource Name** column and enter a description.
 - a. Next to the name under the **Cost Per Unit** column, enter a value. Once the name and cost per unit have been entered successfully, ADD will appear under the **Changed** column.
 - b. To specify a category, choose from the list.
 - c. To enter another resource, repeat steps 3-3b.

4. To delete a resource, enter any text (for example, an “x” or “DEL”) under the **Delete** column.
5. To modify a resource, select one of its cells and change it accordingly.
6. When finished, choose the **ADD-INS** tab and click the **Update Schedule**  icon in the **Menu Commands** section.
7. To review the changes first, click **Review Changes**. To accept without reviewing, click **Accept All** and skip to step 8.
 - a. To deny a change, uncheck the box next to it under the **Accept** column.
 - b. When finished, click the **Continue** button.
 - c. When ready, click the **Accept All** button.
8. Excel will open and any changes from NetPoint will be brought in as well.

NOTE: Colors for resources will be set automatically and can be modified from within NetPoint. For more info, see [Modifying an Existing Resource](#).

Notes

View Excel Updates Window

Columns can be sorted, added, or removed by left-clicking or right-clicking accordingly.

View excel updates

To exclude a change, uncheck the box under the "Accept" column in the desired row. When ready, click "Continue".

Accept	System ID	Description	Change Type	Field Type	Old Value	New Value
☒	53	User Relocation	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	54	Mobilization	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	55	Deenergize/TempPower/Lighting	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	56	Temp Partitions	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	57	Environmental Study	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	58	Asbestos Abatement	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	59	FD #1/RFP #3--RemoveCase...	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	60	Plumbing Cut & Cap	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	61	Mechanical Cut & Cap	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	62	General Demolition	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	63	FD #1/RFP #3--RemoveDI Wa...	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	64	MEP Demolition	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	65	Layout & Top Track	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	66	MEP Shaft Work	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	67	Sprayed Fireproofing	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	68	Electric Ceiling Rough In	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	69	Ductwork & Heat Pumps Ceilings	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	70	Piping Ceilings	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	71	Sprinkler Ceiling Rough In	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On
☒	72	Plumbing Ceiling Rough In	Activity	Assign Codes	Location(4th Floor) Off	Location(4th Floor) On

[Continue](#)

Column	Definition
Accept	Controls whether or not the change will be included.
System ID	An internal ID used by NetPoint
Change Type	The type of object or item where the change was made.
Field Type	The type of field where the change was made.

Column	Definition
Old Value	The original value before the change.
New Value	The new value after the change (if accepted).

Supported Fields

Field	Definition
Description	The object's description. The description is for describing the object.
Abbreviated Description	The object's abbreviated description. This abbreviated description is better for displaying on the canvas and is limited to 48 characters.
Estimated Complete	The estimated complete field is a user-entered field for estimating the percent of work completed.
ID	The object's ID. The ID is a unique identifier.
Codes	Add or remove codes.
Code Names	A short description of the code.
Code Values	A short description of the value.
Code Assignment Behavior	Whether or not multiple values can be assigned to the same activity.
Code Assignments	The codes and values assigned to the object. Where an assignment exists, the cell underneath the value's column will get an "x"
Resources	Add or remove resources.
Resource Names	A short description of the resource.
Resource Categories	A label for describing the type of resource.
Resource Cost	The cost of the resource per unit of time.
Resource Assignments (Rate/Day or Rate/ Duration)	The quantity of a resource assigned to an activity on a per day or per duration basis.

Tips

Forcing two-line Descriptions

Long descriptions can be forced to two lines in Excel by pressing **ALT + Enter**.

Unprotecting Excel Sheets

Any column that cannot be edited will be protected in Excel, except for the code table, resource table, and target sheets. When protected, some features in Excel, such as Find and Replace, may not be available. To access these features, unprotect the sheet in the Review tab or menu option using "NetPoint" as the password.

Cut/Copy/Paste

Only cells which can be edited will allow cut, copy, or paste, unless the sheet has been unprotected, as described above. In any case, cut and paste should be strictly avoided, as it will alter the underlying formula of the cells. As an alternative, you can right-click on a cell and choose clear contents. Copy and paste may be used; however, when pasting, it is recommend to paste only values. This can be accessed via the paste options.

Updating from Earlier Excel Files

The following paragraph discusses what happens if a schedule has been edited in NetPoint and it is updated from an earlier Excel file. If an activity or other object was changed in NetPoint AND changed in Excel, the state from Excel will prevail. If an activity or other object was changed in NetPoint but NOT changed in Excel, the state from NetPoint will prevail. In short, only rows with an asterisk in the **Changed** column in Excel will have their data pushed into NetPoint. Lastly, if an activity or other object in Excel is not found in NetPoint, it will NOT be added in NetPoint and it will be removed from Excel. If an activity or other object in NetPoint is not found in Excel, it will be added in Excel.

Native XML

For integrating NetPoint schedule data with applications other than Primavera P6, Deltek Open Plan, MS Project, or MS Excel, NetPoint can import and export native XML files. Activity ID's, descriptions, planned dates, and durations; predecessor/successor logic; resources; codes; and calendars are included in the export. Instructions are provided below.

NOTE: MS Project XML files will not be able to be imported using this option.

Steps

Importing

1. Click **File > Import/Export > Import From Native XML...** from the menu bar. An explorer window will open.
2. Browse for the desired file and click **Open**.

Exporting

1. Click **File > Import/Export > Export to Native XML...** from the menu bar. An explorer window will open.
2. Browse for the desired location and click **Save**.

Synchro Professional

Synchro Professional is a 4D building information modeling (BIM) application that allows project planners to better simulate and model complex construction sequences by providing enhanced visualization and animation capabilities. Synchro Professional controls the display of 3-dimensional objects, like building components and other physical structures, based on a timeline which can be defined by the start and finish dates of NetPoint activities. Before utilizing NetPoint schedule data in Synchro Professional, the NetPoint schedule must first be exported as a native XML file. The export will include activity ID's, descriptions, planned dates, and durations; predecessor/successor logic; resources; codes; and calendars. Please consult Synchro's documentation regarding which data they import. Instructions are provided below.

Steps

Exporting from NetPoint

1. Click **File > Import/Export > Export to Native XML...** from the menu bar. An explorer window will open.
2. Browse for the desired location and click **Save**.

Importing an XML from NetPoint into Synchro

1. Click **File > Import > PMA NetPoint** from the main menu. An explorer window will open.
2. Click **Browse**, locate the desired file, and click **Open**.
3. Select **Next**, and then click **Import** to import all available attributes.
4. An import report will open. Click **Finish** to complete the import.

To utilize NetPoint's schedule data in Synchro Professional, follow the instructions provided by Synchro for linking 3D objects with activities. Once linked, you will be able to simulate and visualize the construction sequence in 3D, driven by a NetPoint schedule.

Capturing Snapshots

A snapshot may be captured of the entire schedule or from the user's selection.

Steps

Entire Schedule

1. Click **File > Take Snapshot > Of Entire Schedule (as printed)** from the menu bar. The **Screenshot Settings** window opens.
2. Confirm the width.
3. To save to your computer, choose **File** next to **Save To**. Otherwise, leave as **Clipboard**.
 - a. Confirm the **Image Format**.
 - b. Click the **Choose** button.
 - c. Browse for the desired location, enter a filename, and click **Open**.
4. When finished, click **OK**. If clipboard was chosen, you may now paste the image in the desired application.

From Selection

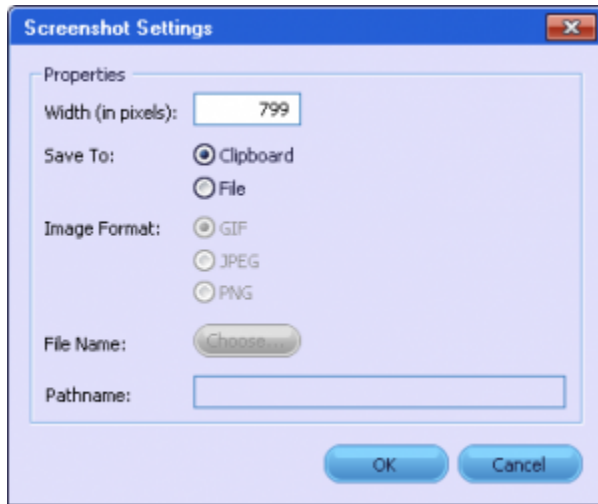
1. Click **File > Take Snapshot > From Selection (as printed)** from the menu bar. The cursor will change from the **Select Tool**  to the crosshairs **+**.
2. Left-click with the mouse on the canvas and without letting go, drag the cursor around the desired area.
3. Release the mouse. The **Screenshot Settings** window will open.
4. See steps 2-3 above.
5. When finished, click **OK**. If clipboard was chosen, you may now paste the image in the desired application.

*Tip: A screen shot may also be taken from selection by pressing the **CTRL** key while using the **Select Tool** .*

NOTE: When capturing a snapshot from selection, the captured image won't necessarily include everything as it's displayed. This is because the snapshot corresponds with how the canvas is printed, not how it's rendered on-screen. For example, if the schedule has been scrolled to a particular section of the network and a snapshot is taken from selection, the calendar strips won't actually show up, because their location when printed is at the top of the canvas. The same applies for the holiday line and the resource profile. Additionally, elements like dialog windows (e.g. the **Resource Key**) cannot be captured, as they only display on-screen.

Notes

Screenshot Settings



Field	Definition
Width (in pixels)	Displays the size of the screenshot to be captured. By default, the size will match the size of the canvas. It may be scaled, however, by adjusting the width (the height will adjust proportionally).
Save To	The destination of the screenshot. Images saved to the clipboard for pasting are only temporary.
Image Format	The format of the image.
File Name	For naming and choosing the destination.
Pathname	Displays the destination chosen above.

8) Troubleshooting

In this chapter, you'll find information on how to resolve issues that haven't been covered elsewhere in the user guide.

Error Reporting

In the event of an error, information can be gathered and sent to NetPoint staff to assist with support. This can be done on demand through the **Help** menu by generating a log file or after a crash by opening a support ticket.

This chapter includes details for the following:

1. Generating log files
2. Opening support tickets
3. Error Reporting Confidentiality

Generating Log Files

Log files can be useful in the event that a ticket has been opened with NetPoint support, and a support team member is requesting more information. Log files include general system information – such as the type of processor, memory, and screen resolution; application information, such as the current version and installation directory; license information, such the type and key; and a list of recent actions taken. Together with the schedule, this information can greatly assist NetPoint support in recreating any errors encountered. Once generated, the log file is stored in MyDocuments\NetPointAppData as SupportLog.html.

NOTE: Only ONE log file can be stored at a time and will overwrite any pre-existing log file.

Steps

1. Click **Help** from the menu bar and choose **Collect Information for NetPoint Support**. A confirmation will open.
2. To attach the log to a new message using your default email client, click **Attach to New Email**. Otherwise, click **Close**.

Opening Support Tickets

In the event that NetPoint terminates unexpectedly, the user may choose to open a support ticket. The ticket will be opened by sending an error report in the background using their default email client.

By default, the report will include a screenshot of NetPoint at the time of the error, the NetPoint log file, an XML file, a DMP file, and the project .nzp file for the schedule that caused the termination. Any of these files can be removed via right-click, and other files may added in the same manner. The information sent to NetPoint support will be kept strictly confidential and used solely for the purposes of repairing the error or otherwise improving NetPoint. For more information, see [➔Error Reporting Confidentiality](#).

Once submitted, NetPoint will reopen automatically, and the file recovery process will launch if applicable. For more information, see [➔Understanding File Recovery](#).

Steps

1. The **NetPoint Helpdesk** window will open automatically.
2. To view or modify included files, click **What will be included?**
 - a. To preview a file, select it in the list and scan the preview below.
 - b. To open a file, double-click it in the list.
 - c. To remove a file, right-click it in the list and click **Delete Selected File(s)**.
 - d. To add a file, right-click on any item and choose **Attach More File(s)**.
 - e. To export a ZIP of all files, click the **Export** button at the bottom and save it accordingly.
 - f. When finished, click **Close**.
3. To add a comment, click **Add a comment**.
 - a. Type in the text area provided.
4. When ready, click **Open Ticket**. A confirmation will open.
5. Click **OK** to complete the process.
6. Depending on your security, another confirmation may open. Click **Allow**.

NOTE: If a default email client is not found, the error report will be saved to MyDocuments\NetPointAppData\CrashReports as a zip file, which can be emailed directly to support.

Notes

By default, the following files are included in the error report. Your privacy is taken very seriously; for more information, see [➔Error Reporting Confidentiality](#).

Field	Definition
SupportLog.html	An HTML file that includes system information as well as a list of recent actions taken. Once generated, the log file is stored in MyDocuments\NetPointAppData as SupportLog.html. For more information, see ➔Generating Log Files .
ProjectFile.nzp	The project folder and its contents, including the schedule and other supporting files. All information will be kept strictly confidential and used solely for the purposes of

Field	Definition
	repairing the error or otherwise improving NetPoint. For more information, see Error Reporting Confidentiality .
crashdump.dmp	A Windows file generated by dumping the memory at the time of the crash.
crashrpt.xml	An XML file that includes system information, such as the operating system, version number, and memory usage.
screenshot0.jpg	A screenshot taken at the time of the error, just before NetPoint closes. NOTE: The screenshot ONLY includes the NetPoint window; any other windows, including the desktop, will be blacked out.

Error Reporting Confidentiality

In consideration for providing PMA Technologies, LLC ("PMA TECH") with files and or other requested material or information ("CONFIDENTIAL INFORMATION"), PMA TECH agrees, except as required otherwise by applicable law, court, governmental order, or to the extent reasonably necessary to prosecute or defend PMA TECH from litigation related to the License Agreement, to hold such CONFIDENTIAL INFORMATION in strict confidence and to use such CONFIDENTIAL INFORMATION only as it deems necessary to assist PMA TECH with repairing or otherwise improving NetPoint or its other software products ("SOFTWARE"). Provider of such CONFIDENTIAL INFORMATION represents that it owns or has authority to provide such CONFIDENTIAL INFORMATION to PMA TECH for the purpose(s) provided herein, and such use by PMA TECH shall not constitute an infringement of any party's ownership or proprietary rights. The terms found online under http://pma.sourcedigitalmarketing.com/user_guide/troubleshooting/error-reporting/error-reporting-confidentiality/ are the most up-to-date and in case of discrepancy supersede terms found printed.

Last updated: 06/28/15.

Understanding File Recovery

NetPoint® makes use of a file recovery system to prevent the loss of data in the case of an unexpected crash. This works by checking to see, every 30 seconds, if any *covered* unsaved changes have been made. These include most mouse and keyboard inputs that relate to menus, the toolbar, and the canvas (some actions and intermediate steps are not covered). If a covered unsaved change has been made, the state of the document *just prior* to that change is stored in separate recovery file (identified with a .NREC extension). For now, this means that the very last change will not make it into the recovery file. This also means that the first change made to a document will never trigger the file recovery system.

For new or unsaved projects, recovery files are stored in MyDocuments\NetPointRecovery. In the event of a crash, if any recovery files existed for new or unsaved schedules, they will be opened automatically when NetPoint® is launched. For existing projects (projects which have already been saved), recovery files are stored in their corresponding project folders. The recovery files for existing projects will open automatically when the original project is opened. Each case is slightly different.

Steps

New or Unsaved Projects

1. Run NetPoint. The **File(s) Recovered!** and **Remaining File(s)** windows will open.
2. In the **File(s) Recovered!** window, click **Review File(s)**. At this point, the toolbar, menu bar (except for Window), and keyboard/mouse input will all become disabled.
3. To view or switch to another open file, click **Window** from the menu bar and choose accordingly. The **Remaining File(s)** window will open and close automatically when switching between recovered and non-recovered schedules.
4. To discard the recovery file without storing it as a new project, click the **Delete** button.
5. To keep the recovery file as a new project, click the **Save As** button. The **Save Project As** window will open.
 1. Browse for a desired location on the left in which you want the project to be saved and select the folder from the list.
 2. Enter a description next to **Save Project As**. This will be the name of the new project.
 3. Click **Save**. **NOTE: The schedule itself will be named “NetPointFile.np4”. Upon clicking Save, a folder will be created using the description entered and the schedule file will be automatically placed in it.**
6. If there is another unsaved recovery file, NetPoint will automatically switch focus to it. When there are no more recovery files to handle, NetPoint will exit read-only mode and the **Remaining File(s)** window will close automatically.

Existing Projects

1. Click **File > Open Project** from the menu bar.
2. Browse for a project that was open during a crash and click **Open**. The **File(s) Recovered!** window will open.
3. To abort the operation, click **Cancel**. The **File Recovered!** window will open again the next time the project is opened.
4. To keep *only* the original file without reviewing the recovery, click **Discard Recovery**. The original schedule will open.
5. To preview the file, click **Review Recovery**. Both the recovered version and the last saved version will open, as well as the **Remaining File(s)** window.

1. To view the original file, click **Window** from the menu bar and choose accordingly.
2. To keep *only* the original file, click the **Delete Recovery** button.
3. To keep *only* the recovery file, click the **Replace Original** button.
4. To keep *both* files, click the **Save As** button. See step 5 under *New or Unsaved Projects* above.
5. The **Remaining File(s)** window will close automatically.

Version Compatibility

NetPoint schedules are not guaranteed to be backwards-compatible. That is, for example, a file created in a more recent release of NetPoint will not necessarily be able to be opened in an older release of NetPoint. Schedules created in NetPoint 4 and above will never open in NetPoint v3.0, as they have different file extensions.

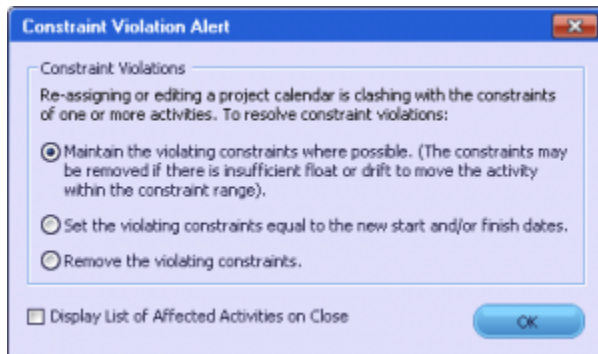
File Extensions

All schedules created in NetPoint 4 and above are given an .np4 extension. All schedules created in versions prior to NetPoint 4 are given a .npt extension. Versions 4 and above can read both .np4 and .npt schedules; however, NetPoint v3.0 cannot read .np4 files. If an attempt is made to open an older NetPoint file (.npt), it will be converted all the same to a project folder. The original .npt file will continue to exist as a backup.

Constraint Violations Due to Calendar Change

If the calendar of an activity is edited or a new calendar is assigned such that it elongates or reduces the length (not duration) of an activity, then the activity's dates will be adjusted accordingly. However, some activities may have no-later-than (NLT) or no-earlier-than (NET) constraints that become violated due to the new dates falling outside the constraint range.

After the calendar is changed, NetPoint automatically looks for activities that violate NLT or NET constraints. If NetPoint finds violations, a window displays prompting input for a resolution. There are three options:



Maintaining Constraints

If this option is selected, NetPoint will automatically correct the violating constraints for the activity by moving it so the finish date is the same as the NLT date. If the activity cannot be moved due insufficient float or drift or due to an NET constraint restriction, then the NLT constraint will be silently removed.

Resetting Constraints

If this option is selected, then the violating constraints are automatically changed to the new start and finish dates of the activity. This will correct any violations.

Removing Constraints

If this option is selected, the violating constraints will be silently removed.

For all options, a checkbox at the bottom may be clicked to display a list of the affected activities along with the information about the constraint, the constraint type that has been fixed, and the constraint dates.

END OF USER GUIDE